Environmental History 11

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Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century

Springer

Environmental History

Volume 11

Series Editor

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ISSN 2211-9019 ISSN 2211-9027 (electronic)
Environmental History
ISBN 978-3-030-33478-9 ISBN 978-3-030-33479-6 (eBook)
https://doi.org/10.1007/978-3-030-33479-6

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

Series Editor Preface

This book addresses a theme of not only European but global relevance. The theme of the relationship between nature and culture has, in fact, become central in a historical moment in which concerns about the influence of humans on the environment and its consequences, both with regard to climate change and with regard to biodiversity, do not address only scientific studies, but also conservation and development policies. In this sense, the history of the Białowieża Forest is an example of great symbolic meaning, because it concerns one of the places considered as a “sanctuary” of nature, the place that preserves the natural habitat of the last specimens of the European bison. The Forest has been considered perfect example of the UNESCO World Heritage List, in which the Białowieża Forest, defined as “primary forest”, was registered in 1979.

In 2016, the Forest rose to the fore in the media and scholars' attention because the logging activities carried out by the Polish authorities leading to an intervention of the European Court of Justice in 2018. The court has announced in their ruling that the logging activities in the UNESCO World Heritage site and Natura 2000 Białowieża Forest violated the EU laws. The relevance of the events described and the debate aroused in public opinion adds particular interest to this study not only to deepen the history of the relationship between man and nature in Białowieża. In fact, it is a matter of correctly interpreting the relationship that binds historical reality with natural dynamics and the initiatives implemented worldwide for the conservation of nature, including the perception that the public and many scholars have about what is natural and what is not.

For these reasons, the book lends itself to multiple readings, the first of which can refer to historical investigations. The authors’ work, as a method, falls in the tradition of forest history studies, which initially developed in Germany in the eighteenth century, following an approach straddling the institutional history of forests and the social history of the woodland, as defined by von Hornstein in 1951. The sources used are therefore written sources, organized in primary sources, such as archive documents and secondary sources, linked to printed works.

From this point of view, the authors' work appears meticulous and effective in reconstructing the historical events that determined Białowieża's evolutionary dynamics. The reconstruction offered clarifies how, over many centuries, all the characteristics of the forest such as extension, density, structure and specific composition have been altered by anthropic activity. Both traditional agricultural and forestry practices carried out by local populations, which also included grazing, the production of resin from pine trees and the production of honey, as well as those deriving from hunting and management activities, give the measure of intensity and the quality of human interventions. According to the "FAO Forest Resource Assessment 2015—Terms and Definitions", pristine are forests where there are no clearly visible indications of human activities, while primeval, according to Cambridge English Dictionary, means ancient or is referred to something existing since a very long time. In this respect, we can say that the forest may be not "pristine" but it surely a primeval forest, as indicated in the title of the book.

Naturally, a management approach aimed at recreating an "ideal" relationship between predators, herbivores and primary producers, as the one we find in pristine environments, can be developed, but this relationship has never been stable in the first place, at any time in the history of the planet, even before the appearance of humans. Secondly, this approach represents, in any case, a "cultural" operation, which leads to a landscape that must be kept in balance artificially, with continuous management interventions. The real naturalness of many forest ecosystems defined as such is a theme already addressed not only in the context of forest history studies but also in historical ecology.

It is important to underline that there are often questionable scientific approaches at the base of the indications aimed at attributing naturalness to a given vegetative system or to direct its management towards this objective. At the base of these approaches, allegedly there is geobotany, whose roots can be traced back to Alexander von Humboldt. This science indicates the plant community as a set of species that occupies a defined space and to which Clements' theories then applied a theoretical evolutionary model aimed at achieving the maximum naturalness stage called "climax". The system determined by the interaction between plant species in an area corresponds to the plant association, which is the subject of phytosociology, developed following the studies of the Swiss Braun-Blanquet (1884–1980) and today of general use for interpreting vegetation communities at any scale. This vision is based on the assumption that a certain environment, for example, a mountain slope in a certain climatic zone, corresponds to a certain community of plant species and, conversely, to a certain type of plant species corresponds to a given environment.

The result is that the plant community present at the time of observation becomes a "biological test" of the natural characteristics of that environment. According to this approach, among the various factors that influence vegetation are also included the "anthropic" ones, understood, however, as factors that "degrade" the vegetation, interrupting the natural evolution, or bringing it back to primitive stages (for example, from forest to scrub using fire). Each individual vegetation survey is then brought back to a theoretical natural reference model, identified as potential vegetation, that is, a higher order plant population to which the vegetation should tend, according to repeatable models on a geographical scale, wherever equivalent environmental conditions occur. Unfortunately, this vision does not consider the historical processes that took place locally, which are instead studied by forest history and which influence the local environmental conditions, for example, by modifying the structure of the soil and therefore the development of some species instead of others.

The difference between the two approaches consists in the fact that the phytosociological approach tends to abstract the place studied from the historical–topographical context to bring it back to a model of theoretical plant association towards which management is aimed. Unfortunately, the models created by phytosociology, when defining the evolutionary or involutionary history of forest landscapes, through the comparative analysis of current and potential ecosystems, hardly ever manage to take into account the complexity of the historical dynamics that have affected current ecosystems. These are historical dynamics that do not necessarily coincide with the theoretical natural dynamic series. In other words, the concepts of “potential vegetation”, “climax”, “evolutionary/regressive series” and “degree of naturalness” often obscure the reality of historical processes behind theoretical schemes, which are yet to be demonstrated, but above all deny historical–cultural values combined with vegetation structures. This scientific approach, therefore, in some cases develops ineffective interpretative models to describe landscapes influenced by man’s work, also influencing management strategies, the media and institutions.

Considering the above, many woods classified today as “natural” or “semi-natural”, according to a rigorous study, could instead be classified as cultural or semi-cultural. In the final analysis, whatever the management direction, be it for a naturalistic or productive purpose, it is always an action motivated by cultural phenomena, regardless of whether it resembles more what we think is the ideal natural model. Specifically, the conservation of the European bison does not exclude anthropic activities, just as its survival is linked also to the presence of predators, also part of the ecosystem. In any case, the book offers a broad knowledge base for general public and scholars alike to understand the history of one of the most interesting forest landscapes on the European continent.

Florence, Italy
18 March 2020

Mauro Agnoletti

Acknowledgements

Preparation of this book was possible thanks to the funds of the Mammal Research Institute, Polish Academy of Sciences in Białowieża, Poland, which supported the work of Tomasz Samojlik and partially covered the cost of archival surveys and obtaining copies of documents. The last stage of work on the book was supported by the grant "Perception of European bison and primeval forest in the 18th–19th century: shared cultural and natural heritage of Poland and Lithuania" (UMO-2017/27/L/HS3/031870) financed by National Science Centre, Poland.

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Chapter 1
Introduction

Abstract This chapter sets the scene for the volume. Białowieża Primeval Forest (BPF) is a biological, ecological, historical and cultural phenomenon, with continuous vegetation history reaching back to the last glaciation (about 11–12 thousand years ago). This landscape and its ecology persisted until modern times with richly diverse animal and plant species, diverse forest environments with abundance of dead and decaying trees, and above all, its ecological systems driven by largely natural processes. Divided between Poland and Belarus, BPF is a trans-boundary UNESCO World Heritage Site "Białowieża Forest" (established in 1992), with the Polish part being a UNESCO Biosphere Reserve since 1976, and covered by a Natura 2000 network of nature protection areas (since 2004). BPF is a reference point in research on temperate forests and is a forest biodiversity "hot-spot". However, BPF is also a historical phenomenon. The first identifiable traces of human presence in the forest go back several thousand years. Scattered throughout the forest there are remnants of past settlements, cemeteries, and agricultural activities from ancient and early mediaeval periods. Furthermore, these are preserved in a very good state due to BPF's conservation as a royal hunting ground since the fourteenth century. Finally, BPF is also a cultural phenomenon and noted to be outstanding and important already by the eighteenth century. At this time, it was recognised as one of the last truly untamed and wild forest in Europe holding one of the only two remaining populations of European bison. The forest became a study area, source of inspiration, and travel destination, playing a major role in the history and development of relevant science, especially the natural sciences. The focus of our book is the process through which this cultural role of BPF evolved. To answer this question, we consider the history and environmental history of the forest through the "long nineteenth century" (confined between two dates: 1795 and 1915). In parallel with this exploration, we follow the cultural significance of the forest as manifested in material imprints of management and conservation, and in the cultural heritage and ultural recognition of the forest among naturalists, travellers, writers and artists. The first chapter closes with some thoughts on the current issues and challenges for the forest.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_1

Fig. 1.1 Map of contemporary Białowieża Primeval Forest, divided between Poland and Belarus (drawn by Tomasz Samojlik)

Białowieża Primeval Forest (BPF) is a biological, ecological, historical and cultural phenomenon. As the only forest in the European lowlands with continuity back to vegetation established after the last glaciation (about 11–12 thousand years ago; Latałowa et al. 2016), the site is unique. This landscape and its ecology persisted until modern times with a plethora of animal and plant species, diverse forest environments with abundance of dead and decaying trees, and above all, its ecological systems driven by largely natural processes. At present, BPF covers 1,450 km^2 and straddles the border between Poland and Belarus, with 600 km^2 on the Polish side (Fig. 1.1). The latter incorporates the Białowieża National Park (105 km^2 with 47.5 km^2 of old-growth forest strictly protected since 1921), with no human intervention allowed, and managed commercially under state forestry (with 21 smaller nature reserves amounting to a little over 120 km^2) scattered within them. The Belarussian part of the Forest was proclaimed a reserve (*zapovednik*) in 1944 and since the end of the

Soviet period has been the Belarusian State National Park "Belovezhskaya Pushcha" (Jędrzejewska and Jędrzejewski 1998; Kavalenia et al. 2009). The entire BPF is a trans-boundary UNESCO World Heritage Site "Białowieża Forest" (established in 1992), whereas the Polish part has been a UNESCO Biosphere Reserve since 1976, and is covered by a Natura 2000 network of nature protection areas (since 2004) (Pabian and Jaroszewicz 2009). Even a brief look at species numbers recorded for the Forest justifies all the various conservation measures. So far, there have been records of around 12 thousand species of animals, including 59 mammals, 250 birds, and almost 10 thousand insects (Gutowski and Jaroszewicz 2001, 2004). This is along with over 4 thousand (other estimates point to even 5 thousand) species of fungi (Gierczyk et al. 2018; Wesołowski et al. 2018). BPF is a reference point in research on temperate forests and is a forest biodiversity "hot-spot". It holds almost complete sets of ungulate species, predators, and old-growth specialist fauna (including woodpeckers, saproxylic and old-forest relict insect species), all typical for natural forests.

From a bird's-eye perspective, BPF is a closed-canopy forest with a small amount of open areas made up of forest gaps, river valleys and marshes. The Forest itself is a mosaic of different habitats, commonly grouped into five main forest types: coniferous (with *Pinus sylvestris* and *Picea abies* dominating), mixed coniferous (*Pinus sylvestris*, *Picea abies* and *Quercus robur*), deciduous (*Quercus robur*, *Tilia cordata* and *Carpinus betulus*), mixed deciduous (*Picea abies*, *Quercus robur*, *Tilia cordata* and *Carpinus betulus*) and wet black alder bog forest and streamside alder-ash forest (*Alnus glutinosa* and *Fraxinus excelsior*). In the Polish part of BPF, coniferous forest covers 52% of the wooded area, whereas in Belarussian part, coniferous stands straddle over 69% of the area. The climate is transitional between Continental and Atlantic, with mean annual temperatures reaching 7.0 °C (−4.8 °C for January and 18.4 °C for June) and average annual precipitation of circa 640 mm (Jędrzejewska and Jędrzejewski 1998). It is worth noting that BPF is a border zone of the geographical ranges of species like Norway spruce and sessile oak *Quercus petraea* (Faliński 1986).

According to the European Environment Agency, an undisturbed forest shows natural forest dynamics (natural tree composition, occurrence of dead wood, differentiated age structure and natural regeneration processes), sufficient to maintain its natural characteristics, and it has not been heavily influenced by people. If the latter is not the case, then sufficient time has passed since significant human disturbance for natural processes to re-establish (EEA 2007). This is a definition that applies well to BPF. Along with natural succession governing most of the forest, there are several other natural processes that persist. These include cycles of tree masting years which affect rodent and predator populations (Cornulier et al. 2013), and recurring outbreaks of bark beetle (*Ips typographus*) and other insects that trigger cascade effects of natural forest regeneration (Müller et al. 2008). There are also influences sometimes called "landscape of fear" in which the presence of natural predators inhibits herbivores' browsing pressure on regenerating trees (Kuijper et al. 2013) and may lead to patterns of tree and shrub regeneration.

Fig. 1.2 Early mediaeval cemetery with 79 burial barrows in Jelonka Range in the Polish part of BPF (photo by Tomasz Samojlik)

However, BPF is also a historical phenomenon. The first identifiable traces of human presence in the Forest go back several thousand years (Latałowa et al. 2015, 2016), and the earliest settlements with established agriculture, husbandry and tool production have been evidenced by archaeology to the first century BC (Samojlik 2007; Olczak et al. 2018). Remnants of past settlements, cemeteries (Fig. 1.2), and agricultural activities from ancient and early mediaeval periods are scattered throughout the forest and usually preserved in a very good state (Krasnodębski et al. 2005, 2008, 2011). This situation gives archaeologists a rare opportunity to study waves of human settlement in a landscape that has not subsequently been significantly disturbed, ploughed or converted to other land-use. The lack of macro-disturbance is due to BPF's conservation as a royal hunting ground since the fourteenth century. At that time, BPF was a hereditary property of the Lithuanian grand dukes (Sahanowicz 2002), and then, after the Polish-Lithuanian union in 1385 it became a royal forest. Its main role for the next four centuries was to serve as a royal hunting ground and this proved crucial for the Forest's survival until the end of the eighteenth century. This was enough to counter the potential impacts of continuously rising demand for wood and timber in Europe during times of rapid industrialization (Williams 2003; Samojlik 2006; Samojlik et al. 2013).

Finally, BPF is also a cultural phenomenon. Already by the eighteenth century, it was noted to be outstanding and important. Not only was it one of the last truly untamed and wild forests in Europe (which was anyway mostly deforested by that time), but also it held one of the last two populations of European bison. Furthermore, the second one, in the Caucasus Mountains, was long considered a myth (Daszkiewicz and Samojlik 2004). First mention of BPF and its bison population by European naturalists dates back to the early eighteenth century. But it was only in the nineteenth century that the Forest was fully incorporated into the circulation of thoughts between travellers, naturalists, hunters, writers and various artists. The Forest became a study area, source of inspiration and travel destination. BPF played a major role in the history and development of relevant science, especially the natural sciences. This was from the earliest studies on European bison and its prehistory (Daszkiewicz et al. 2004, 2012a), through to controversies about distinguishing bison from aurochs *Bos primigenius*, and experiments with crossbreeding bison with domestic cattle (Daszkiewicz et al. 2012b). The Forest was also important in the development of the emerging science of phytosociology (Maycock 1967). All these matters originated in BPF and were present in nineteenth-century scientific debates. In this century, BPF became a cultural designatum of the wild and natural, often dangerous and unpredictable forest, a role that it still serves today. The main focus of our book is the process through which this cultural role of BPF evolved. To answer this question, we consider the nineteenth-century history and environmental history of the Forest. In parallel, we follow the cultural significance of the Forest as manifested in material imprints of management and conservation, and the cultural heritage and the cultural recognition of the Forest among naturalists, travellers, writers and artists. Our book covers the "long nineteenth century" and is confined between two dates: 1795 and 1915. In 1795, the third partition of Poland resulted in BPF and a part of former Grand Duchy of Lithuania (in union with Poland since the fourteenth century) being annexed by the Russian Empire. In 1915, the approaching military front forced the Russian administration to abandon BPF, effectively ending the Russian Imperial period of Forest's history. It was a time that proved crucial for the Forest's fate. Freshly deprived of its royal hunting ground status and turned into a regular state forest within the expanding Russian Empire, BPF could have fallen victim to chaotic redistribution of the estates (one of BPF's thirteen districts was given to count Rumyantsev and was felled completely within just a few years, Hedemann 1939), to pan-European processes of severance from traditional forms of land-use, and the consequent introduction of modern, "rational" forestry. Last but not least, there was the ever-present demand for timber in the Empire. Yet, a coincidence of circumstances prevented the Forest following the patterns of most other European sites. Firstly, the vast, natural forest with several different forest habitats and abundant dead wood was extremely hard to "tame" and manage even for the best forest specialists from St. Petersburg. Secondly, it was soon observed that BF was the last place where European bison were still present and therefore all management decisions were made with the conservation of bison a priority. Thirdly, the traditional perception of the forest and forest uses, culturally rooted in this area and originating from mediaeval (or older) practices, was an obstacle. Finally, since 1860 the Forest was gradually transformed into tsars' private hunting reserve, pushing further away the attempt at reducing the role of BPF to timber production. The forest administration eventually had to accept and incorporate all these factors into the management plans.

In this book, we focus on tracking all cultural imprints, both material (artificial landscapes, introduced alien species, human-induced processes) and immaterial ("traditional" and "scientific" knowledge of forest and use of forest resources, the political and cultural significance of the forest), that shaped BPF's current state and picture. Furthermore, we aim to present a picture of this crucial moment in forest history, relevant not only to Central Europe, but to the continent in general. This is the moment of transition from a royal hunting ground, a traditional type of use widespread throughout Europe at the time, to a modern, managed forest. Considering the main obstacles to the management shift, the essential difference in perceptions of the forest and of the goods it provided (in both modes of management), and the implications of management change during the long nineteenth century helps our understanding. This in turn can provide insight into the changes that European forests underwent in the wider landscape.

There are dozens of publications devoted to the history of BPF in the long nineteenth century, although only a small number written by professional historians. In many cases, the sources of information were not known, were they reliable enough, and if authors used a critical approach to study them. The best example is a book by Karcov (1903), which is widely considered a classic monograph on the subject. The author refered to several archival documents, however he only rarely gave their exact archival citation or or any critical assessment. Therefore, one of the key tasks for our study was the search for reliable documentary sources. We believe that we have achieved significant success in this regard discovering huge number of documents on BPF in international archives (see Sect. 2.2).

In recent years, BPF has grown its reputation as an exemplar site for an ancient European lowland forest. However, it has also found itself in the spotlight due to national- and then European-scale conflicts over the management and conservation measures applied in the Polish part of the managed forest. This has been in part in reaction to a bark beetle (*Ips typographus*) outbreak. In fact, debates on the appropriate management of the forest have been on-going at higher or lower intensity for decades, and can be traced back to the creation of the first nature reserve in Białowieża in 1921. Nevertheless, in 2016 and 2017, bark beetle-infested Norway spruce tree-stands were felled using heavy forestry vehicles (harvesters) and on a scale not seen in the forest for decades. According to research based on satellite images at least 675 ha of the forest were felled (including 229 ha of old-growth), and the negative impact of such felling was estimated to cover over four thousand hectares. In less than two years, logging increased the fragmentation of the forest by 26%, which can in turn affect landscape-scale ecosystem functioning (Mikusiński et al. 2018). Shamefully, the conflict triggered one of the main events to shape relevant scientific debate in 2017 (Callaway et al. 2017) and was taken to the Court of Justice of the European Union. That body issued a verdict in April 2018, requesting all logging to cease (CJEU 2018). Nevertheless, it seems that the conflict is far from being over and both the supporters of non-intervention and active "conservation" (e.g. Wesołowski et al. 2018; Hilszczański and Jaworski 2018) use various arguments, including historical ones, to defend their stance. While our book does not address the current controversies, it can be a valuable baseline to help determine the chronology and diversity of anthropogenic disturbances in the forest in the nineteenth century. It also demonstrates the rise of the cultural significance of BPF throughout the period, which directly ties into the status the Forest enjoys nowadays.

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Chapter 2
Sources and Methods

Abstract This second chapter addresses the archival, literature, and map-based research, which underpins our investigation. The materials for the study of environmental history of Białowieża Primeval Forest (BPF) used in this book were mostly written sources (published and archival manuscripts), along with archival maps and artistic depictions of the Forest and its dwellers (both animals and people). The written sources were obtained during archival surveys in archives and libraries of Russia, Belarus, Poland, France and Lithuania. The greatest number of documents on the management of BPF in the long nineteenth century is kept at the Russian State Historical Archive in St. Petersburg (RSHA), with the National Archive of the Republic of Belarus in Grodno (NHARBG) being the second largest collection of files connected with BPF. Unfortunately, both NHARBG and RSHA were substantially damaged during the Russian Revolution and both World Wars of the twentieth century. Only the last, appanage period in BPF's history (1889–1915) is present in the archive in a continuous way, while all other periods of the nineteenth century—are only fragmentary. Other surveyed archives included the Russian State Archive of Navy in St. Petersburg, the Archives of the Russian Academy of Sciences in Moscow and its St. Petersburg branch, the State Archive of the Russian Federation, Moscow, the Central State Historical Archive in St. Petersburg, the Central Archive of Historical Records in Warsaw, the National Library of Poland in Warsaw, the Polish State Archive in Białystok, the Archive of the Polish Academy of Sciences in Poznań, the Polish Library in Paris, the Muséum National d'Histoire Naturelle in Paris and the Museum's archives stored at Archives Nationales, the Defence Historical Service (Service Historique de la Défense) in Vincennes, the City Archive in Strasbourg (Archives de la Ville et de l'Eurométropole de Strasbourg), the Lithuanian State Historical Archives in Vilnius and the Manuscript Department of the Vilnius University Library, the National Archives in Tartu, Estonia and the Archives of Natural History Museum in Berlin. We also conducted a thorough search of literature concerning BPF and forests of the Grand Duchy of Lithuania published in second half of the eighteenth century through until the early twentieth century. Indeed, the bibliography of the nineteenth-century publications (books and articles) connected with the subject amounts to several hundred items. Every forest has its own history and in this case, most information came from different kinds of written sources (a) published and generally available to the public and (b) archival manuscripts. The latter

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_2

category includes meticulous accounts, tables and inventories. Archival maps that help identify nineteenth-century interactions between man and forest supplemented this knowledge. Artistic depictions of the Forest and its dwellers (both animals and people) were important in tracing the cultural significance of BPF in the nineteenth century. Through analysis of what was depicted, in what form, and where was the work published, it was possible to identify the significance and perception of BPF in European culture.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

2.1 Written Sources

2.1.1 Archival Surveys

In the search of historical data on BPF in the long nineteenth century, we undertook an archival survey of archives and libraries in Russia, Belarus, Poland, France and Lithuania.

The largest number of documents concerning the management of BPF in the long nineteenth century is kept by the Russian State Historical Archive in St. Petersburg (RSHA). RSHA is one of the largest archives in the world, collecting mainly documents of governmental agencies and institutions of Imperial Russia. From the perspective of our book, the most important part of the collection are the files on forest and game management in BPF during the period when it belonged to the Russian state (1795–1888) and the Tsar's family (1889–1915). It includes the files of the Forestry Department of the Admiralty Board/Ministry of Finance (1798–1811), Fond (1594, at least 25 files somehow discuss BPF management), the State Domains Department of the Ministry of Finance (Fond 379, 1802–1837, 50 files), the Second Department of the Ministry of State Domains (Fond 384, 1831–1865, at least 35 files), the Agricultural Department of the Ministry of State Domains/Ministry of Agriculture (Fond 398, 1837–1915, 20 files), the Forestry Department of the Ministry of State Domains (Fond 387, 1843–1888, more than 260 files), the Chancery of the Minister of State Domains/Ministry of Agriculture (Fond 381, 1837–1915, 10 files), the Department of Appanages/the Main Directorate of Appanages (Fond 515, 1888–1917, more than 500 files). To be able to compare the game management and organization of hunts in BPF with other hunting grounds of the Tsar we also worked with documents on Russian imperial hunting (Fond 478, 70 files on Białowieża and Gatchina game reserve) and the Chancery of the Minister of the Imperial Court (Fond 472, 35 files). Separate files were identified in the fonds of the Ministry of Education, the agencies of the Ministry of Internal Affairs, the Cabinet of Ministers, etc.

The National Archive of the Republic of Belarus in Grodno (NHARBG) holds at least three hundred files connected with BPF (including juridical cases of poaching, illegal cutting and other misdemeanors connected with the Forest). Unfortunately, both NHARBG and RSHA were substantially damaged during the Russian Revolution and both World Wars of the twentieth century. A significant part of archival collections of NHARBG that included documents connected with BPF were lost in evacuation of archive during WWI and Revolution, part of documents of RSHA was destroyed during WWII bombardment: the roof of the archive was destroyed directly above the collection of documents of the Ministry of Finances. This is one of the main reasons why only the last, appanage period in BPF's history (1889–1915) is present in the archive in a continuous way, while all other periods of the nineteenth century—only fragmentary. The most informative part of NHARBG's surviving collection was the set of documents of Grodno governor's chancellery (Fond 1).

Other Russian archives proved to be useful in answering specific questions, e.g. in the Russian State Archive of Navy in St. Petersburg (RSAN) 6 files on logging in BPF for the Navy were discovered. We also managed to identify a significant number of interesting documents in the Archives of the Russian Academy of Sciences in Moscow (ARAS) and its St. Petersburg branch (SPB ARAS). These documents are mainly related to the history of scientific research on BPF and BPF's European bison. The most informative files are collected in fonds of Professor Nikolai Kulagin, the organizer of the "European bison" expedition in 1906–1908 (ARAS, Fond 445, 40 files) and Friedrich Brandt, the Head of the Zoological Museum of the Academy of Sciences in St. Petersburg (SPB ARAS, Fond 51, 5 files, 1848–1863) and of the Zoological Museum itself (SPB ARAS, Fond 55, 6 files). There was also information in the Archive of the Museum (nowadays Zoological Institute of the Russian Academy of Sciences) and in its catalogues, as well as in the catalogues of the Zoological Museum of the Moscow State University. About a dozen of documents concerning BPF were also identified in State Archive of the Russian Federation (SARF, Moscow). The Central State Historical Archive in St. Petersburg (CSHAStPb) helped to find some documents on stuffed bison in zoological collections.

In Poland, we surveyed the collections of the Central Archive of Historical Records in Warsaw (CAHR), National Library of Poland in Warsaw (NLP), the Polish State Archive in Białystok (PSAB) and the Archive of the Polish Academy of Sciences in Poznań (APASP). The Polonica Collection in the Polish Library in Paris (PLP) was surveyed for documents connected with foresters from BPF who emigrated to France after the failure of the Polish national uprising in 1831. This included documents about the BPF prepared by Polish emigrants.

The manuscripts at the main library of the Muséum National d'Histoire Naturelle in Paris and Museum's archives stored at Archives Nationales (archival set AJ15) constitute one of the world's largest collections of natural history manuscripts and prints. We have surveyed the official museum documentation (e.g. books of purchases) in search of specimens from BPF and to check the origin of European bison (both live and dead) that were added to the collection in the eighteen and nineteenth centuries. We also reviewed the correspondence and manuscripts of naturalists connected with research on Białowieża, e.g. Jean-Etienne Guettard, Jean-Emmanuel Gilibert, Georg Forster, Georges Cuvier, Ludwig Heinrich Bojanus, Bory St Vincent, Jacques Boucher de Perthes and also Polish naturalists—Józef Jundziłł, Władysław

Taczanowski, Jan Sztolcman, Stanisław Batys Górski. Additionally, we surveyed documents of Scientific Societies, e.g. La Société Zoologique d'Acclimatation, interested in the introduction of Białowieża's European bison in France in the nineteenth century.

Military archives held at the Defence Historical Service (DHS, Service Historique de la Défense) in Vincennes near Paris were assessed for Napoleon's army reports prepared before the planned campaign of 1812 and French offensive in Lithuania. Special attention was paid to the information on forests and their fauna contained in those reports. In the City Archive in Strasbourg (CAS, Archives de la Ville et de l'Eurométropole de Strasbourg), we found a collection of documents connected with Natural History Cabinet of Jean Hermann and with efforts to acquire European bison for the Strasbourg's natural history museum in the second half of the nineteenth century. Single documents concerning bison specimens from Białowieża were also discovered in museums in Rouen and Elbeuf.

We also managed to find individual documents in several other archives: the Lithuanian State Historical Archives in Vilnius (LSHA) and the Manuscript Department of the Vilnius University Library (MDVUL), the National Archives in Tartu (Estonia, NAT), Archives of Natural History Museum, Berlin (Museum für Naturkunde).

2.1.2 Literature Surveys

We conducted a thorough search of the literature concerning (at last partially) BPF and forests of the Grand Duchy of Lithuania published in second half of the eighteenth century until the early twentieth century. The bibliography of the nineteenth-century publications (both books and articles) connected with the subject amounts to several hundred items—and below we list only the most prominent ones, in chronological order:

(1) "Indagatores naturae in Lithuania, seu Opuscula varii argumenti quae historiam animalium, vegetabilium, in magno ducatu Lithuaniae, et morborum quibus in hac provincia homines vel maxime obnoxii sunt, illustrare possunt, authore aut redactore Joan.-Emmanuele Gilibert" and "Flora lithuanica inchoata seu enumeratio plantarum quas circa Grodnam collegit et determinavit Iooannes Emmanuel Gilibert" by J. E. Gilibert, both published in 1781 (Gilibert 1781a, b), and "Opuscula phytologico-zoologica prima, autore J.E. Gilibert, in Universitate Vilnensi olim Historiae naturalis, nec-non Botanices Professore ordinario", published in 1782 (Gilibert 1782). These books, published in Vilnius and Grodno, and written by a prominent naturalist were brought to Grodno (*circa* 100 km in a straight line from BPF) by the Polish King. They contain descriptions of European bison and other animals from Lithuanian forests (including BPF), and flora characteristic for Polish-Lithuanian forests. Both were a major source of information for wide audiences in the early nineteenth century

("Flora" being the reference for all nineteenth-century botanical research in BPF).

Gilibert also published a collection of information about Lithuania's (including Białowieża's) fauna "Abrégé du Système de la nature, de Linné, histoire des mammaires ou des quadrupèdes et cétacées" in 1805 in Lyon (Gilibert 1805)—it was most probably the most cited of his works, with much wider impact than books published in Vilnius and Grodno. It was also a compendium of the era's knowledge on mammals.

(2) "Mémoire descriptif sur la forêt impériale de Bialowieza, en Lithuanie" (Description of the Imperial Białowieża Forest in Lithuania), is a book by Julius Karol Holte von den Brincken (1790–1846), published in 1826 in Warsaw (Brincken 1826). Raised and trained as a forester in Brunswick-Bevern, in 1818 he was appointed as the head forester of the Kingdom of Poland, a puppet state of the Russian Empire. He visited BPF in 1821 and 1823, hunting and collecting observations for his book. Dedicated to Tsar Nicholas I, the "description of the Imperial Białowieża Forest…" is in part the summary of Brincken's knowledge on BPF, also his plan for proper future management of the forest, and in no lesser part an act of his prostration towards the Tsar. While the book holds many interesting first-hand observations, a large portion is based on previous general knowledge of forests in this part of Europe, and there are also some significant errors Brincken in his characterization of Białowieża's woods (these were mostly rectified by other authors soon after Brincken's publication. A modern critical evaluation of Brincken's book was published in Polish in Daszkiewicz et al. 2004). Nevertheless, "the Description…" is still a valuable source, and the depictions of BPF and its dwellers it holds is an undoubted positive aspect of the book (by Jakub Sokołowski, see Chap. 4).

(3) The first botanical work specifically concerning BPF, published in 1829 by Stanisław Górski: "O roślinach Żubrom upodobanych, jakoteż innych w Puszczy Białowiezkey" (On plants preferred by European bison and others in Białowieża Primeval Forest) (Górski 1829). Górski visited BPF and studied its plants following Jan Fryderyk Wolfgang's—professor from Vilnius—Botanical Instruction for Białowieża Primeval Forest (Grębecka 1983).

(4) "O Puszczy Białowieżkiej i o celnieyszych w niej zwierzętach, czyli zdanie sprawy z polowania odbytego w dniach 15 i 16 lutego r.b. na dwa Żubry, które Nayjaśnieyszy Mikołay Iszy, Cesarz Wszech Rossyi, Król Polski etc. etc. dla gabinetu zoologicznego królewskiego aleksandrowskiego warszawskiego uniwersytetu nayłaskawiey darował" (On Białowieża Forest and most prominent animals, i.e. report from the hunt on 15th and 16th February this year on two bison which His Highness Nikolai I, Russian Tsar and Polish King, etc., generously donated to the nature cabinet of the Warsaw University) by F. P. Jarocki, published in 1830 (Jarocki 1830). The book contains the description of the forest and its animals but also various cultural remarks, e.g. on traditional beekeeping in the forest.

(5) "Rys historyczny powstania w Puszczy Białowieskiej w roku 1831" (The historical characteristic of the uprising in Białowieża Forest in 1831) published by Piotr Szretter in 1893 contains an entire section on the nature and traditional life in BPF (Szretter 1893). It is also one of the sources of traditional names for different forest habitats, which were later incorporated by the modern phytosociology.

(6) "Istorija zubra ili tura, vodjashhegosja v Belovezhskoj Pushche Grodnenskoj gubernii" (The history of a European bison or an aurochs, that dwells in BPF of the Grodno province) by Dmitry Dolmatov published in 1849 in Forestry Journal. It is an extract from the Dolmatov's large manuscript the later edition of which is kept now in the SARF (SARF 1860). It contains description (natural history) of the European bison, along with plans for forest management and exploitation.

(7) "Opisanie lasów Królestwa Polskiego i gubernij zachodnich cesarstwa rossyjskiego pod względem historycznym, statystycznym i gospodarczym" (Historical, statistical and economical description of the forests of the Kingdom of Poland) by Aleksander Połujański published in 1854 contains mainly economic point of view on plans of exploitation of BPF and is an interesting evidence of the possible fate of the Forest just six years prior to the first tsars' hunt in the Forest and subsequent change in the direction of forest management (Połujański 1854).

(8) "Materialy dla geografii i statistikii Rossii, sobraniye ofitserami generalnogo shtaba. Grodnenskaya Guberniya" (Data for geography and statistics of Russia, collected by the officers of General Staff. The Grodno Province). The series presented detailed information about the nature, economy and population of the province in the middle of the nineteenth century, collected by high-rank military officers. The volume devoted to the Grodno Province was compiled by lieutenant colonel Peter Bobrovskii and published in 1863 (Bobrovskii 1863). Apart from a general description of the region, it included a detailed article on BPF and European bison (mostly repeating already published information, but including some original data, obtained from BPF forestry officers).

(9) "Kharakteristika Belovezhskoi Pushchi i istoricheskiya o nei dannya" (Description of Białowieża Forest and its history), published in 1903 by Nestor Genko (Genko 1903), chief of the forestry taxation team that prepared the forest management plan in 1889. The book is a summary of this plan but also gives summaries about previous forest management plans as well as describes the history of forestry management of BPF.

(10) "Belovezhskaya Pushcha. Ee istoricheskii ocherk, sovremennoe okhotniche khozaistvo i vysochaishe okhoty v Pushche" (Białowieża Primeval Forest. Its historical description, contemporary game management and monarchical hunts in the forest) by Georgii Karcov, published in 1903 (Karcov 1903). This volume shows the state of BPF by the early twentieth century and sums up the knowledge on the management of the Forest. To this day, it is the biggest monograph of BPF. Since its publication, it was the starting point for the majority of

later works. During his work, Karcov used many archival documents (RSHA 1902–1904), lacking however professional historical research methodology.

(12) "Teoreticheskaja differencirovka nekotoryh zhvachnyh na drevesnojadnyh (Fruticivora) i travojadnyh (Herbivora) i prakticheskoe ee znachenie" (The theoretical differentiation of some ruminants on the Fruticivora and Herbivora and its practical significance) by Konrad Wróblewski published in *Arkhiv veterinarnykh nauk*, in 1912 (Wróblewski 1912). This paper discusses the diet of wild ungulates (European bison, red, roe and fallow deer) and effect of their huge population on the BPF vegetation.

(13) "Pasterelloz Bollingera u dikih i domashnih zhivotnyh v rajone Belovezhskoj pushchi" (Bollinger pasteurellosis in wild and domestic animals in BPF vicinity) by veterinarians Nikolay Ekkert and Vladimir Fedders published in 1912 discussed epizootic in BPF in 1910 (Ekkert and Fedders 1912). The paper contains a variety of data on wild and domestic ungulates in BPF vicinity, on peasant animal husbandry, pasturing for cattle and wild ungulates, on watering places and game mortality causes in the early twentieth century.

2.2 Maps and Graphical Depictions

A search for historical maps of BPF and artistic depictions of the Forest, its animals, dwellers of villages inside and in closest vicinity to BPF was conducted in the archives and libraries described earlier. The research also considered on-line repositories and collections (e.g. https://polona.pl, Polish Digital Libraries Federation—http://fbc.pionier.net.pl). This study yielded several previously unknown maps, drawings, paintings and sketches. Based on their remarkable contribution to knowledge, these were selected to be presented in the book.

(1) Map drawn by Michał Połchowski in 1784 "Location Białowieża with two hunting gardens" (CAHR 1784).

(2) Map of Białowieża Primeval Forest at the end of the eighteenth century, showing the division of the forest into thirteen forest districts ("straże"): "Map of Mieleczyce Province with Białowieża Primeval Forest" (CAHR 1796).

(3) Earliest known map of BPF drawn after the third partition of Poland, most probably prepared in 1795–1796 (not dated), based on an older map from the Polish period: "Map of Brześć District currently lying within the boundaries of the Russian Empire" (CAHR undated). The map shows all thirteen forest districts of BPF, as in the royal period.

(4) Map dated 1799 and entitled "Plan of Białowieża Primeval Forest showing which districts are within" (RSHA 1808), without one of thirteen forest districts from the royal period—already removed from state forest administration and deforested in following years.

(5) Map published in 1830 by E. K. Eichwald in "Naturhistorische Skizze von Lithuanien, Volhynien und Podolien" (Eichwald 1830).

(6) Map of BPF dated to 1847—the first map showing the division into rectangular compartments (1 × 2 *versta*), with five projected forest districts marked (RSHA 1840–1849).
(7) Map of BPF prepared by the forest inventory team in 1861, published in 1863, the first known map showing the forest composition (coniferous, deciduous and wet forest distinguished) and division of the forest into five projected forest districts (from: Bobrovskii 1863).
(8) Map of Białowieża Primeval Forest and Świsłocz Forestry in Grodno Province, showing the pasturing range in 1897, 1904 and 1905 (RSHA 1905A).
(9) The forest management map prepared under the supervision of Nestor Genko, dated 1890 (RSHA 1889–1890).

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Chapter 3
Traditions of a Royal Period (Until 1795)

Abstract This chapter covers the period when Białowieża Primeval Forest (BPF) served as a hunting reserve for the Grand Dukes of Lithuania and the Polish Kings from the late fourteenth century until 1795. At the same time, along with grand hunting, a various traditional uses were allowed associated with villages, towns, and churches. The most significant of these were haymaking and forest beekeeping. The anthropogenic environmental impact of this royal period can be seen in the creation of the cultural landscape of a hunting garden, in the evolution of stands of pure pine resulting from centuries of fire use, and the establishment of the conservation and winter supplementary feeding system for European bison. The cultural heritage of the period, incorporated traditions connected with European bison, small-leaved lime, Scots pine, and the local knowledge of forest herbs and their uses. All these traditions persisted here because of the royal status of the Forest and the longevity of its customary management. This entailed legal protection over game species and the forest itself ensured by customary laws, and later codified in three editions of Statutes of Lithuania. The history, impacts, and roles of royal hunting are considered along with the traditional uses of the forest by various groups of local people. The two aspects are closely intertwined as the overall usage was overseen by the designated authorities charged with responsibility for the Forest.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

3.1 Summary

From the late fourteenth century until 1795, Białowieża Primeval Forest (BPF) served as a hunting reserve for the Grand Dukes of Lithuania and Polish Kings. This entailed legal protection over game species and the forest itself ensured by customary laws, and later codified in three editions of Statutes of Lithuania dated 1529, 1566, and 1588. The direct impact of royal hunts was less important than the other consequence of BPF's royal status: its legal protection against settling, unauthorized hunting, and tree felling. At the same time, a variety of traditional uses was allowed

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_3

for villages, towns and churches, most prominently haymaking and forest beekeeping. The royal forest service protecting and managing the forest were entitled to similar uses, along with access to small timber and a brushwood quota for personal uses. The resources used included bark and bast stripping, collection of firewood and resinous chips from old pine trees for kindling, along with mushrooms, nuts, forest fruits and edible and medicinal herbs. The King's men had also the right to pasture their cattle in forests adjacent to their villages. From the seventeenth century, new types of uses were gradually introduced: bog iron ore extraction, production of potash, wood-tar, birch-tar and charcoal. Attempts at introducing commercial timber felling and floating were undertaken for short periods in the second half of the seventeenth and then the eighteenth century. Despite these, BPF survived until the end of this period with the majority of its forests preserved in close to natural state, with almost all big game—*animalia superiora*—preserved (European bison, moose and brown bear—but without red deer that went extinct in the late seventeenth century). The anthropogenic environmental impact of the royal period can be seen in creation of a cultural landscape of a hunting garden, the evolution of pure-pine stands as a result of centuries of fire use and establishment of the European bison conservation and support system. The cultural heritage of the period, on the other hand, incorporated traditions connected with European bison, small-leaved lime and Scots pine, and also the local knowledge of forest herbs and their uses. All these traditions were able to persist here because of the royal status of the Forest and the longevity of its customary management.

The international interest of naturalists, travellers and artists helped to shape the cultural significance of BPF in the period. From the seventeenth-century description of the Forest by Gonzalez (1646), through the eighteenth century writings of Rzączyński (1721), Guettard (MNHN 1760–1762) or Gilibert (1781a, 1796) to Jan Henryk Müntz's depictions (LWU 1780–1783)—all of those made the Forest well-known in scientific circles as the last refuge of the ancient species of European bison.

3.2 The Historical Background of BPF as a Royal Forest

Białowieża Primeval Forest (BPF) first appeared in written historical sources as hereditary private property (called "*votchina*", from the word meaning father) of the Grand Dukes of Lithuania (Sahanowicz 2002). The Grand Duchy of Lithuania or *Magnus Ducatus Lithuaniae*, was an ethnic mixture of Baltic and Slavonic tribes that rose to prominence during the reign of Grand Duke Mindaugas in the thirteenth century and was the last European state to convert to Christendom at the end of fourteenth century (Davies 2011). The political and legal practices and especially the cultural traditions of Lithuania survived here after its union with Poland and subsequent Christianisation. The 1385 union between the Kingdom of Poland and the Grand Duchy of Lithuania brought BPF and other hereditary properties as a dowry to this newly appointed commonwealth (Stone 2001). This also included a wide variety of traditions connected with the forest: the special status of "oak groves" as places of pagan sacred flames (Davies 2011), the tradition of the Grand Dukes' ownership of the large part of their state (in the sixteenth century estimated to cover around 30% of the territory of the entire Grand Duchy; Sahanowicz 2002), a reign characterised by regular tours between manors erected in different parts of the country, especially in forests, and the special status of forests and big game inhabiting them. The latter were expressed later in three editions of code of laws; the so-called Statutes of Lithuania. Legal protection of the sovereign's forests and big game ensured by those statutes eventually outlasted the Grand Duchy itself, as the Third Statute of Lithuania was still in effect almost until mid-nineteenth century, long after the country itself was erased from the map of Europe. The series of territorial partitions of the Commonwealth of Both Nations between Prussia, Austria and Russia ended in 1795; the also marked the end of the royal status of BPF. Until that time, the Forest's main purpose was to preserve big game, so called *animalia superiora* (in this case the group incorporated European bison *Bison bonasus*, brown bear *Ursus arctos*, red deer *Cervus elaphus* and moose *Alces alces* (Samsonowicz 2011)) for royal hunts—*venatio magna*. Only monarchs were privileged to hunt in BPF—and exceptions from that rule were very rare: in 1562 and 1577 Kings Zygmunt II August and Stefan Batory allowed other persons to hunt in BPF. Poaching, especially in the case of most valuable animals like European bison, was punished severely—firstly with the death penalty, and then later with substantial fines (Samojlik 2006a).

The role of monarchical hunts shifted from victualling royal court (and in some cases—the army) with venison, through special diplomatic meetings (Samsonowicz 2011), to purely symbolic reference to old traditions and entertainment. Also the hunting practice evolved, from kings risking their lives and hunting side by side with hunters in the forest (for example King Władysław II Jagiełło breaking his leg during bear hunt in 1426 shows this risk was real), to organizing much safer hunts in enclosures inside the forest, called "hunting gardens" (Fig. 3.1). Those enclosures were used to gather significant numbers of different games before the start of the hunt, and during the event internal barriers allowed chase animals towards royal visitors sheltered in hunting bower (Fig. 3.2). This ensured not only the safety of country's most important persons but also guaranteed a positive outcome of such hunt (Samojlik 2006a; Samojlik et al. 2013c).

Despite some popular beliefs, BPF was not frequently used for monarchical hunts—on the contrary, historical sources confirmed fewer than twenty cases of royal stays in BPF, from the first hunt of King Władysław II Jagiełło in 1409, to the last royal hunt of the last King of Poland, Stanisław II August, in 1784 (Samojlik 2005a, 2006a). The remains of two hunting manors erected for the purposes of royal stays in the Forest were discovered: the first, dated to the sixteenth century was located in "Old Białowieża", approx. 5 km from current location of Białowieża village, and the second one, dated to the seventeenth century, in the centre of present-day Białowieża (Samojlik et al. 2014). These settlements by royal mansions were the only exceptions from the rule of no habitation inside the Forest until the eighteenth century (Samojlik 2007). It is also worth noting that a royal visit to the Forest was

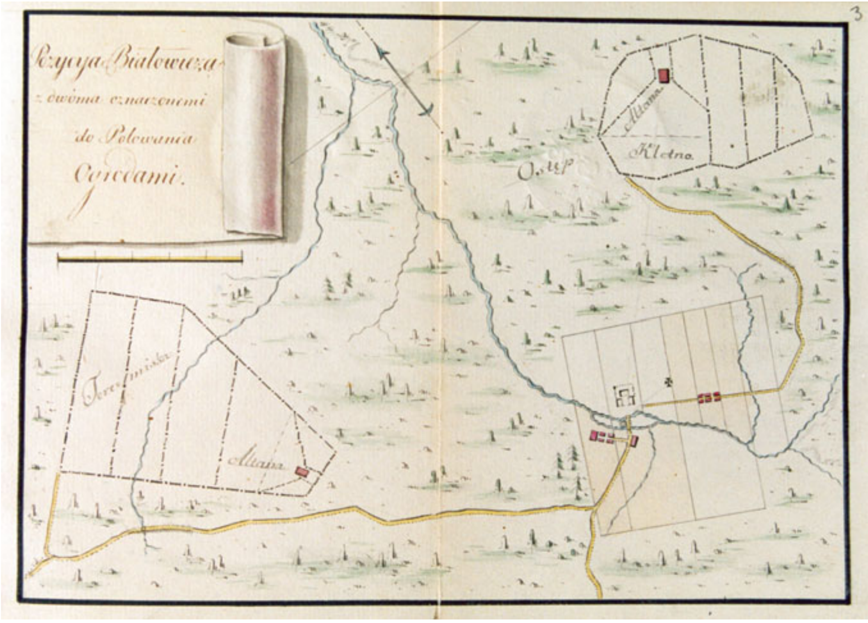

Fig. 3.1 Map "*Location Białowieża with two hunting gardens*" drawn by Michał Połchowski in 1784 (CAHR 1784)

not always connected with hunting, as Białowieża's hunting mansion was a convenient stop on the shortest route between two capitals of the Commonwealth of Both Nations—Cracow and Vilnius. Information about the outcome of those hunts is rarely available, and the biggest one in the entire royal period of BPF's history, the 1752 August III Wettin hunt, yielded 57 animals killed with 42 bison among them (von Brincken 1826; Samojlik 2005a). The direct impact of royal hunts on the forest's game was less important than the other consequence of BPF's royal status: its legal protection. The Forest was guarded by an approximately 300-person-strong royal forest guard settled in the ring of villages located on the edge of wooded area (Hedemann 1939). The forest guard included several different groups of servicemen, each with their own scope of authority. On the local level, forest administration consisted of deputy masters of the hunt ("podłowczy"), foresters ("leśniczy") and deputy foresters ("podleśniczy"), each with different levels of responsibility for organization of hunts, protection of game and conservation of the forest and management of local uses of forest resources. Below them in the hierarchy were guards ("strażnicy"), each with their own district of the forest to oversee (by the end of the eighteenth century the number of districts, "straże", was set to thirteen, Fig. 3.3) by the means of weekly tours around the district and constant contact with two lower levels of servicemen: riflemen ("strzelcy") and beaters ("osocznicy"). The riflemen were an armed game service that served also as local police (e.g. in cases of fights with poachers). Beaters, the most numerous category of servicemen, were also the oldest, as

Fig. 3.2 Watercolour by Jan Lindsay, *Hunting bower in Białowieża Primeval Forest erected by Enlightened Prince Stanisław Poniatowski, Lord Treasurer of the Grand Duchy of Lithuania for His Majesty King Stanisław Augist Poniatowski, hunting there in the way to Seym in Grodno in 1784* (LWU 1784)

first records of their service in the forests of the Grand Duchy of Lithuania date back to the fourteenth century (Stryjkowski 1582), and the first legal description of their duties to the early sixteenth century (Hedemann 1939). Entire families of beaters—originating from peasants—were obliged to participate in royal hunts, during which their main function (and the source of their traditional name “osocznik”—from verb “soczyć”, meaning tracking animals in the forest) was to track animals and drive them towards hunters. In the periods between hunts, their duty was to employ the daily routine of patrolling the forest and protecting it from unauthorized use, observing the whereabouts of animals, selecting parts the forest for royal hunts and creating and maintaining hunting gardens there. They repaired bridges and forest roads as might

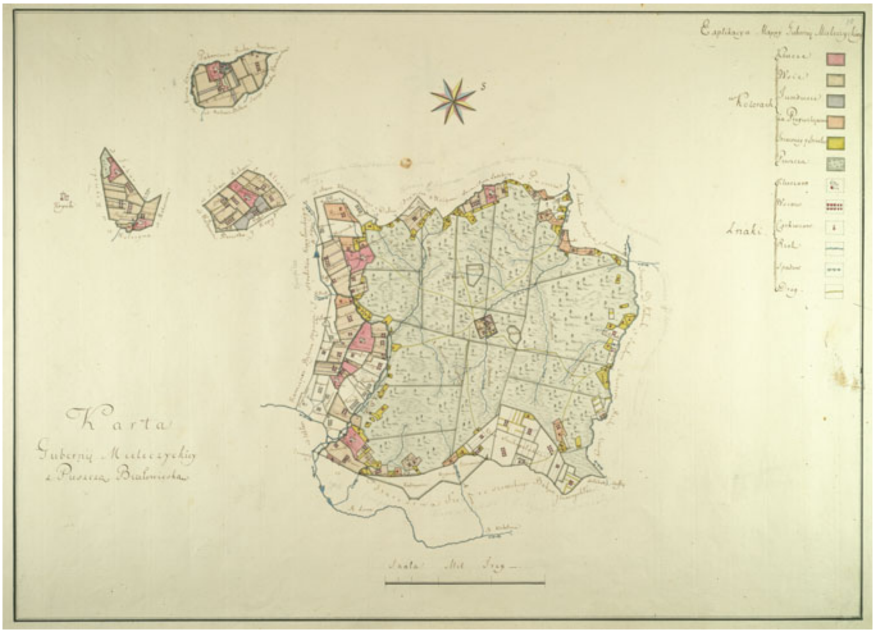

Fig. 3.3 Map of Białowieża Primeval Forest at the end of the eighteenth century, showing the division of the forest into thirteen forest districts ("straże"): "Map of Mieleczyce Province with Białowieża Primeval Forest" (Karta Guberni Mieleczyckiej z Puszczą Białowieską), dated 1796 (CAHR 1796)

be required. For their efforts, they were endowed with a patch of land and freed from any other peasant duties. In the seventeenth century BPF, beaters were the most numerous among forest service: in 1639, there were 277 beaters, 4 senior beaters, 2 hunters and a forester in service in BPF. In 1764, forest service incorporated 213 beaters, 16 guards, 5 riflemen and one deputy forester, whereas in 1795—102 riflemen, 16 beaters, 13 guards, 2 deputy foresters and one forester (Hedemann 1939; Samojlik and Jędrzejewska 2010).

An additional category of forest personnel were the "bobrownicy", i.e. beaver hunters, whose duty was to provide beaver tails (regarded as meatless dish or honorary fish, for periods of fasting), furs and castoreum, a sought-after secretion with medicinal and cosmetic use (Hedemann 1939; Samsonowicz 2002).

Protection of BPF meant that settling, commercial timber felling and hunting (apart from royal hunts) were prohibited, nevertheless a variety of traditional uses were allowed (Hedemann 1939). Grand Dukes and later Polish Kings granted local villages, towns and both Catholic and Orthodox churches special rights to use a strictly delimited part of the forest in a defined way. Those privileges, called "access rights", were hereditary, but did not change the ownership rights to the forest, which was still royal. Furthermore, those utilizing access areas were obliged to pay a rent (in goods or money) to the forest administration, which became important after the

Forest became a Royal Economy in 1589, with all income from the Forest allocated to the royal court only (Samojlik 2005a; Samojlik et al. 2013b).

Over forty access areas were described in BPF in the sixteenth century and over eighty in the seventeenth century (Samojlik 2007). These were most probably connected with the hereditary status of access rights and their division among heirs. Usually, more than one type of use was allowed on a single access area. Among those types of use the three most common were (1) haymaking, (2) traditional beekeeping and (3) fishing in forest rivers.

Haymaking was the right to scythe meadows inside the forest, both alongside river valleys ("*marshy*" haymaking) and in glades inside deciduous forest ("*oak wood*" haymaking), erect haystacks and then to transport collected hay to villages for winter fodder of domestic animals. The "*marshy*" haymaking had usually larger area of the two (usually area was expressed by the number of haystacks, and the biggest haymaking areas had around fifty of them). Sometimes the haymaking access area was so remote (up to 55 km from the village) that an additional permission was necessary—to allow building of small shelters for peasants making hay (Samojlik and Jędrzejewska 2004).

Traditional beekeeping incorporated carving artificial cavities (beehives) in Scots pine (and occasionally in oak, Norway spruce and lime), caring for nesting bees and then collecting honey and wax. Beekeepers were a closed corporation with their specific code of law and hereditary specialized knowledge, and their products were important sources of income from forests since mediaeval times (Hedemann 1939). Majority of beehive trees were located in coniferous, pine-dominated forests. In 1792, there were 7,155 beehives in BPF (out of which 936 were settled by bees), and in 1795: 6,851 and 600, accordingly (Samojlik 2007). Beekeepers needed smoke in their work (to handle bees), and were often accused of unintentionally starting forest fires: "*Frequent fires happen in the forest because of beekeepers' carelessness, when they—passing from place to place after their beehives with lit torches—start, even unwillingly, a fire*" (LSHA 1764). However, it is also possible that the fires were intentionally started and controlled by beekeepers, who could take advantage of them in various ways: from clearing the surroundings of beehive pine trees from any saplings and young trees that could potentially compete with those old pines, to enriching the herbaceous vegetation of the forest floor for the benefit of bees. The rise in frequency of small-scale and low-intensity forest fires in this period was confirmed by dendrochronological research (Niklasson et al. 2010), which could be a cumulative effect of beekeeping and other new types of forest production requiring fire introduced to BPF since the end of the seventeenth century.

The right to fish in forest rivers also included permission to build fishing weirs from wood posts with wattle fences and to create small fish ponds near rivers. In the sixteenth century, access rights with fishing weirs existed on six out of nine forest rivers.

In the royal period (fourteenth–eighteenth centuries), other types of utilization had only a marginal significance or at least were rarely mentioned in sources: right to maintain small houses inside the forest, occasional right to fell timber, right to keep cattle on an access area inside the forest during winter (Hedemann 1939; Samojlik 2007, 2010).

Apart from villages, towns and churches enjoying the access to forest resources, a similar scope of utilization prerogatives belonged to the royal forest service protecting and managing the forest. They were entitled to extract timber for purposes of building and rebuilding their houses, and brushwood for fences (without the right to sell collected wood), to collect firewood, strip tree bark and tear bast from lime (see Point 3.3 below for details), to collect mushrooms, nuts and other forest fruits, and various edible and medicinal herbs (including hogweed—see Box 3.3 and goutweed—see Box 4.3), and also to obtain resinous chips from old pine trees for use as kindling. The King's men also had the right to pasture their cattle in forests adjacent to their villages and use access areas in exchange for annual rent paid to the forest administration (Hedemann 1939; Samojlik 2005a, 2010).

From the seventeenth century, new types of uses were gradually introduced to BPF. The first was the exploitation of bog iron ore on the basis of contracts signed by the Polish Kings for four bog iron ore extraction sites located on forest rivers on the outer borders of BPF. Historical sources allow the dates for these activities to be narrowed down to the years from 1639 to 1747 (Samojlik 2009). The exploitation of bog iron ore was restricted to extraction (digging), drying and transportation of the ore. The actual production of iron took place in specialized smelting areas outside the BPF.

In the second half of the seventeenth century, the Commonwealth of Both Nations suffered from a series of devastating military conflicts that decimated the population, drained the national treasury, and led to severe economic crisis (Malinowski 2016). Under these circumstances, in an attempt at replenishing the treasury, King Jan III Sobieski allowed commercial timber felling and the production of potash in BPF in 1675 (Jędrzejewska and Samojlik 2004). Four years later, timber extraction was restricted only to oak trees and only to 5% of the forest area, and in 1691 excluded entirely. Potash production, on the other hand, persisted in the forest from 1675 until 1701, and then again in the period 1765–1792 (Samojlik 2016). Potash (potassium carbonate—K_2CO_3), was manufactured by reducing broadleaved trees and other vegetation to ashes, leaching and vaporizing water from them, and was one of the most sought-after forest products in Europe at that time. Used for fabric bleaching, production of soap, dye, glass, and porcelain, it was widely used in household and manufactures and played a crucial role in several branches of economy in the seventeenth and eighteenth centuries. Two types of potash were produced in BPF. In the first period, a more contaminated, cheaper type of a "trough" potash was produced directly inside the forest by slowly burning broadleaved trees, carefully watching them for the entire time and smothering the fire with lye if it became too strong. The result of burning—ashes—were carefully checked for unburned wood and charcoal, and loaded to barrels. The other type, produced in the latter period, was more complicated to obtain and required calcination with the use of cauldrons (hence the name "cauldron" potash). The production was abandoned in 1792 due to enormous cost of such production, falling prices of potash in European markets, the discovery in

Germany of mineral potash, and local malpractices in potash production (Samojlik 2016).

Another type of product manufactured in the forest was tar and this came in two forms, wood-tar and birch-tar. Wood-tar was a resinous tar that served as a popular glue, grease, insulating material, skin and wood preservative and medicine. In BPF, it was obtained almost exclusively from Scots pine (Samojlik 2006b; Samojlik et al. 2013a). A wood-tar kiln was a pit surrounded by soil embankment with sloping bottom and a wooden or clay-made tar drain with a container for tar at the end. Birch-tar, on the other hand, was used as a natural preservative, disinfectant and medicine. It was produced from birch bark in kilns similar to ones used for wood-tar production. Wood-tar and birch-tar production was for the first time mentioned in sources concerning BPF in 1696, when eight wood-tar and five birch-tar distillers were active here. A century later, in 1795, 82 wood-tar distillers were inventoried in BPF, but only 15–16 kilns were working—based on the total income to the treasury from wood-tar makers (Hedemann 1939).

Charcoal, i.e. part-combusted wood, was also manufactured. In the seventeenth and eighteenth centuries, charcoal was in high demand in connection with the rapid development of the iron industry. Its production demanded a specific technical knowledge—charcoal was produced in hearths located inside the forest, where piled wood was covered by an insulating layer of moss, litter and earth, then slowly burnt and closely watched to avoid cumulation of gases beneath insulation (which could even lead to an explosion). Study of the remains of eighteenth-century charcoal hearths in BPF revealed that charcoal burners most often utilized hornbeam *Carpinus betulus*, birches *Betula pendula* and *Betula pubescens*, and small-leaved lime *Tilia cordata*.

Timber was another potential product. An attempt at introducing commercial timber felling and floating was undertaken in the second half of the eighteenth century, during the reign of the last King of the Polish-Lithuanian Commonwealth, Stanisław II August. It was a part of economic reforms of the Grand Duchy of Lithuania designed by Antoni Tyzenhaus, Treasurer of the Grand Duchy of Lithuania and administrator of royal estates (of which BPF was a prominent part). Scarce historical sources do not contain detailed information on timber production in this period, allowing only for a general calculation of timber felling from 1779 to 1782 at 0.05–0.3 m^3/ha annually (Samojlik 2007).

Three partitions of the Polish–Lithuanian Commonwealth in 1772–1795 were a decisive point in the BPF's history. Russian Empire, Kingdom of Prussia and Habsburg Austria erased the Commonwealth from the map of Europe, with the largest part of the area, including BPF, falling under Russian rule (Davies 2005). Initially, BPF has lost its status of royal forest, as well as the centuries-old system of protection. The villages inhabited by riflemen and beaters were distributed among favoured aristocracy by Empress Catherine the Great. Riflemen and beaters were disconnected from their former duties and were now no more than ordinary state peasants ascribed to their land. One of the thirteen forest districts of BPF was given to Count Piotr Aleksandrovich Rumyantsev, one of the long-time favourites of Catherine II (Hedemann 1939). After Rumyantsev's death in 1796, his descendants managed to clear-cut the entire forest district, leaving a characteristic triangle on the eastern border of the Forest (see Figs. 1.1 and 4.2).

3.3 Environmental Impact of Royal Hunts and Centuries of Traditional Use of BPF

The material history of a certain place is often a palimpsest of layers overlapping one another. For example, one type of land management changes into other, where there was a building, another is erected or the entire construction is brought down, etc. In the case of BPF, looking for material imprints of the royal period in form of specific man-made objects persisting in the forest, anthropogenic changes visible in forest ecosystems, or even particular species entangled in the history of human intervention, requires the uncovering of those layers in a search for forms that survived here long enough to play a role in the Forest's history. Three such imprints will be discussed here: (1) the cultural landscape of a hunting garden, (2) the evolution of pure-pine stands as a result of centuries of fire use and (3) the establishment of the European bison conservation and support system.

(1) After a period when monarchs chased game in royal forests risking failure in obtaining venison and putting themselves to a hazard not lesser than their companion hunters (e.g. King Władysław Jagiełło, see Sect. 3.1), a more efficient and safer way of hunting was introduced. So-called "ogrody do polowań" (hunting gardens) were large sections of the forest surrounded by a wooden fence. They were present in many royal forests of Grand Duchy of Lithuania: "there were many hunting gardens both in the contemporary Grodno District and in the entire Lithuania. In each forest a convenient place was selected for an animal passage, enclosed with a fence, so that no animal could escape" (Połujański 1854). For the first time, they were mentioned in the literature in the beginning of the sixteenth century, when poet Nicolaus Hussowski (Lat. Hussovianus) described the method of hunting European bison in the Grand Duchy of Lithuania: "*We fell trees to encircle wide space.|And keep the beast trapped within.|Twelve miles and more counts the perimeter of the fence,|If we keep to the Latin measures,|The guards then surround the place,|So that the quarry will not escape.|Not always is that labour new to us;|Old enclosures wait open in woods,|The beast, which used to freely enter them| For grazing, easier gets entrapped.|In such parks we hunt* (…)" (Hussowski 1523). Hussowski described this method as already "well-established" but also quite dangerous, as entrapped animals could attack and harm hunters armed only with side arms—such an accident took place in a hunting garden in the royal forest most probably near Kraków in 1504, during the hunt of King Aleksander I Jagiellonian, when bison attacked hunting bower and almost caused Queen Helen and her maidservants to fall (Samojlik 2006).

In the royal period, there were at least three hunting gardens in BPF: "Augustowska", Kletna and Teremiska. The first one was visited by King August III Wettin during his hunt in 1752, yet apart from the information that it was located at a distance of a mile (approx. 7.4 km) from Białowieża village, its location is unknown (von Brincken 1826). The latter two were used during the last royal hunt of Stanisław August Poniatowski in 1784 (in fact, Teremiska hunting garden was erected especially for the purposes of this hunt (Samojlik 2005b; Samojlik et al. 2013c)). Based on historical maps, the area of those two gardens was estimated at 5 km^2(Kletna) and 8 km^2 (Teremiska) (Samojlik et al. 2013c). Inside, hunting gardens incorporated various forest habitats, with abundance of food for ungulate species such as European bison, moose, red and roe deer and wild boar and a source of water—in this case streams. The idea was to allow animals to acquaint themselves with the fence and habitat inside, and also to allow their survival when the fence would be eventually closed before the arrival of royal hunters. Before the hunt, beaters drove animals towards openings in the garden's fence to gather a significant number of large game inside, and then closed all gaps in the barrier and guarded it, as observed by king Stanisław August Poniatowski heading to Białowieża for his hunt in 1784: "in the forest, one mile from Białowieża, we met a group of peasants, guarding the backwood, that is the area fenced for a half mile around to keep various animals" (Naruszewicz 1784). Inside the enclosure, there were interior fences or nets dividing it into smaller compartments. When the hunt started, beaters moved in one direction, flushing out game and closing section after section, so that animals could not return. In effect, the animals were driven towards nets to be captured alive or towards a glade where the King and his hunters waited for a chance to affect a deadly blow or have a clear shot (after the introduction of firearms—in BPF evidenced for the first time the context of royal hunt in the mid-sixteenth century (Samojlik 2006a)).

Long-lasting interactions of humans with the natural environments can lead to the creation of cultural landscapes combining man-made and natural components, involving both local biotic and abiotic conditions and local historical, cultural, political influences (Berglund 1991). These landscapes are "eco-cultural" (Rotherham 2015) and their heritage assets are "biocultural" (Agnoletti and Rotherham 2015). Examples in the BPF include the erection of an enclosure inside the forest, watching over and mending the outer fence and inner sections for long periods between royal visits to BPF, and occasionally using gardens during monarchical hunts can be recognized as sufficient to develop the specific cultural landscape of a hunting garden. These activities and others modified the landscape through an interaction with its natural resources. The natural elements consisted of forest habitats that were selected as preferable for the big game. From the perspective of current ecology, ungulate abundance is correlated with the presence of deciduous forests with mature tree stands (Jędrzejewska et al. 1994), probably with enough open spaces to enable herbaceous vegetation to develop. The anthropogenic components, on the other hand, were wooden outer constructions, inner compartments and nets and hunting arbours or bowers erected on a glade (probably kept open by forest service responsible for maintaining the garden between royal hunts). The human-influenced aspects include the modified vegetation and the altered faunal populations. Not excessively intrusive and designed not to distress animals, the constructions were probably in relative harmony with the surrounding forest.

After BPF lost its status as a royal forest in 1795, the two known hunting gardens of Białowieża suffered different fates. The ending of traditional utilization meant a slow demise for the Kletna garden. Its fences were apparently still visible in the first half of the nineteenth century, as the 1846 description of the former lands of Grand Duchy of Lithuania stated, "Grand Kletna. It is a spacious and dry backwood overgrown with ancient oaks coiled by ivy. It is enclosed by fences as animals were driven there to be killed by Polish kings" (Baliński and Lipiński 1846). Not used and not maintained, the garden slowly disappeared in the natural forest, and is now commemorated only by the traditional name of the area, Grand Kletna (since 1921 this area is a part of a strictly protected nature reserve within the Białowieża National Park). The second hunting garden, Teremiska, has not fallen into demise, and was used as a basis of hunting enclosure created (or recreated) for the purposes of the first Tsar's hunt in BPF organized for Alexander II in 1860 (Fuchs and Zichy 1862; Daszkiewicz et al. 2012; more details in Chap. 5). The enclosure was later transformed into a game park, which played an important role in game management. The new role of Teremiska hunting garden lasted until the outbreak of World War I. After the war, part of the former royal enclosure became the European bison breeding centre—crucial in the process of restitution of the species that was exterminated in the wild in 1919. After ten years of absence, bison from European zoos were brought to an enclosure located in the part of Teremiska hunting garden (22 ha in 1929, enlarged to 59 ha in 1931). This function of the place was maintained until today—as a reserve with the pure lowland (Białowieża) line of European bison kept in case of epizootic or maybe other possible events that would require the restitution process to start over again (Krasińska and Krasiński 2013).

(2) Frequent fires of low intensity and small spatial scale are evidenced for the period between the sixteenth and eighteenth centuries by both dendrochronological studies (Niklasson et al. 2010) and palynological research (Latałowa et al. 2015). Their occurrence can be explained by the traditional types of forest use already present (such as beekeeping), or those introduced to BPF during the royal period of its history (e.g. potash burning, wood-tar and birch-tar smelting, charcoal production). All of these involved handling fire in the forest, and some of them, as beekeeping, even profited from introducing controlled burning. This was because small but frequent forest floor fires in coniferous and mixed coniferous forests eliminated underbrush, Norway spruce and young deciduous trees. Among all trees present in BPF, only Scots pine, due to its features such as thick bark and elevated crowns shows relative resistance to low and moderate-intensity fires (Goldammer and Furyaev 1996; Fernandes et al. 2008). Centuries of continuous fire management in areas of BPF that were used for beekeeping could have led to the creation of an anthropogenic type of forest and a strongly eco-cultural landscape with pure, single-species, pine stands. Such stands were locally known as "*lado*" forests (from Polish description of an area cleared for by burning; Samojlik 2007; Niklasson et al. 2010), and were reported to cover 39% of the forest area in 1889 (Genko 1903). After introduction of strict fire protection and gradual suspension of fire-demanding types of use in the beginning of the nineteenth century, the number of forest fires dropped (but their intensity grew

significantly; Niklasson et al. 2010). As a result, this type of forest started to dissolve. In the 1920s, Józef Paczoski observed “clean and beautiful pine forest” only in three separate places in BPF (Paczoski 1930), and since then coniferous forests were dominated by Norway spruce (Latałowa et al. 2015).

(3) Based on the archaeological evidence, it is safe to assume that European bison was present in the remote forests of Central and Eastern (and possibly also South-Eastern) Europe in the Middle Ages (Benecke 2005; Onar et al. 2017), but already in the fifteenth and sixteenth century, its range had rapidly shrunk. In the second half of the eighteenth century, apart from the population in Caucasus Mountains, the existence of which was reliably proven only in the 1830s (Baer 1836; Daszkiewicz and Samojlik 2004), the last known remaining population of free-living European bison was in Białowieża (Bojanus 1825). This particular forest became the last refuge of the species before its eventual extermination in 1919 was probably an effect of several fortunate circumstances overlapping here. The key factors were the legal protection of the species in the Grand Duchy of Lithuania, the special status of BPF itself as a royal hunting reserve, and the unintentional but positive outcome of traditional forest utilization during the royal period. Finally, from the eighteenth century onwards there were active conservation efforts employed to sustain the regional bison population.

The legal protection of European bison both in royal forests, as well as in private woodlands was codified in the first edition of the Statute of Lithuania in 1529. However, it was in already effect earlier, the legal practice including it in the group of *animalia superiora* and excluding it from most hunting privileges given by Grand Dukes and Polish Kings. In the first and second edition of Statutes of Lithuania, a poacher caught near a dead bison faced the death penalty; a punishment later modified to prison and large fine. The fine for poaching a bison was the highest among those listed in the Statute, and its value of 1440 grosz was equal to the price of 12 horses or 14 cows (Samojlik and Jędrzejewska 2010). The legal protection encompassed also the place where bison dwelled—the royal forest of Białowieża. As mentioned in Sect. 3.1, there was a specialized service of the Kings’ men to protect the Forest from unauthorized access, but also to guard animals from any harm. Protection of BPF as the habitat of bison was even strengthened in 1589, when the Forest was allotted to serve as the Kings’ private appanages. The appanages served as a source of income and goods (in this case—venison) for the King and his court. While other royal estates were frequently distributed among nobility, the Kings were not interested in giving away the rights to appanages. This allowed BPF to avoid the fate of several other Grand Ducal forests which were soon overexploited and deforested (Hedemann 1934).

The long-lasting effects of haymaking—the most widespread and common way of traditional utilization of BPF—were beneficial for European bison, too. Regular mowing of forest meadows led to increase in their area, creating favourable conditions for the bison, with abundance of grasses and herbs, their preferred food (Krasińska and Krasiński 2013). Collected hay was stored in haystacks on meadows until winter, when it was transported to villages outside BPF on carriages, but in practice it offered an easily accessible source of additional fodder for European bison after the vegetation period had ended. The local knowledge of bison taking advantage of hay stored in haystacks was probably widespread. Even the royal commissioners sent to BPF in 1700 to evaluate the state of the forest after a series of wars (but also after the coldest period of the so-called "Little Ice Age") were aware of it. The commissioners turned this unintentional system of supplementary winter feeding of bison into a binding legal regulation. From 1700, part of the hay produced in the forest was reserved to feed "animals" in winter (Samojlik and Jędrzejewska 2010).

This active conservation measure in the eighteenth century was accompanied by monitoring of the species' numbers. Annual winter counting of bison was recorded from the oldest known report on European bison numbers (dated 1783, with 284 bison counted after heavy snowfall (Samojlik and Jędrzejewska 2010).

Until the end of the Polish-Lithuanian Commonwealth, both the forest and the bison population inhabiting it survived and were in good condition (Samojlik et al. 2013b). The visible legacy of the system of European bison protection developed during the royal period of BPF's history, the last free-living population of lowland bison, was originally unrecognized. This was largely because it was a period of turmoil that started began after 1795. Fortunately, however, this legacy gained recognition and became the decisive factor in BPF's subsequent fate.

3.4 Cultural Heritage of the Royal Period in BPF

The royal status of BPF significantly influenced not only the forest's protection and eventual survival in excellent condition until the end of the eighteenth century, it also helped shape the cultural significance of BPF. The Forest has been inseparably intertwined with the history of the Polish-Lithuanian Commonwealth (see Schama 1996), and almost every literary or scientific work in the long nineteenth century referred to its royal past. But the royal period also left a more specific cultural heritage—one that lasted long after the demise of 1795. The traditional, local knowledge and customs developed in BPF (and Grand Duchy of Lithuania in general) throughout ages were connected with specific plants or animals. These traditions were able to persist only because of the royal status of the Forest and the longevity of the customary management maintained here. Examples of such traditions are connected with (1) European bison and (2) two tree species: small-leaved lime and Scots pine.

(1) Traditions connected with European bison pre-date the legal protection of the species, and originate probably in the period of abundance of those animals. One of those traditions, mentioned in the sixteenth-century chronicle, saw bison pelts used for building of small and light boats (Stryjkowski 1582). Another belief, still present at the end of the eighteenth century, considered bison pelts (especially from bison's forehead) to be helpful during childbirth (Gilibert 1781b). The majestic European bison was also traditionally treated as a defender of the homeland—in a symbolic and quite practical way. Pieces of bison skin with words or syllables written on theme were worn as talismans that were supposed to protect the bearer from injury and death on the battlefield. Jucewicz (1846) noted that the amulets he knew of were all very old, as in his time and age it was not easy either to lay hands on bison skin, or

to find a person who knew what words should be written to grant protection. The same author noted a legend in an old song referring to the Tatar invasion (probably in the thirteenth century). According to the song, all the bison helped to fight off the invaders, and the Tatars did not find a way to get around those defenders. Legend about brave bison from Białowieża was spread in Western Europe by Baron Joseph de Baye, a French archaeologist and ethnographer, who published it after his journey to Lithuania: "*The Lithuanians respected the bison also because they proved their patriotism and contributed to the defence of the country. It is said that when the Tartar horde approached Podlasie (Pinsk Province) all bison left the forest and marched towards the enemy roaring* [authors: European bison of course do not roar] *so terribly that invaders were forced to retreat*" (Baye 1905). The legend about brave bison had an exceptionally long life and unexpectedly returned in the twentieth century in the context of French and Russian war propaganda (see Sect. 8.4 for more on this subject).

(2) Small-leaved lime *Tilia cordata* and Scots pine *Pinus sylvestris* are two tree species strongly bonded with centuries-long, traditional utilization of BPF. In the royal period, small-leaved lime was treated differently than other trees, and the benefits it offered were far more diverse than merely the lime wood—as Połujański (1854) noted: "Lime is a very useful tree and should be spread as wide as possible, because all parts of that tree can be useful in the rural economy". Traditional uses of lime were listed among other prerogatives of local dwellers, like collection of forest mushrooms and fruits, herbs or firewood (Hedemann 1939; Samojlik and Jędrzejewska 2004). The variety of those uses was explained by Szretter (1893): "Its bark is used to make elegant bast shoes, ropes, baskets, punnets, mats, etc.; its soft wood is used tables, stools, hulls, grain-barrels, troughs, dugouts etc.; its flower heals diseases, and is used by the bees to collect the best white honey, called July". The first lime material mentioned by Szretter was bark and bast, used for production of countless items necessary for the daily functioning of forest farms. The lime bark was peeled off in May and June, in big wide sheets or narrow and long strips, and then treated for several weeks or even months (soaking, drying, straightening; Samojlik 2005c). After this treatment the bast was ready for use, e.g. to weave mats to transport venison during hunts (Jarocki 1830), to make ropes to pack hay (Połujański 1854) and to make nets used for live-capturing of animals (von Brincken 1826) (worth noting, nets for capturing wolves were at that time prepared in BPF with the use of hemp; RSHA 1827–1833). A prominent role among those uses was played by bast shoes, the most popular footwear not only among peasants, but also forest service: "no peasant, being in the forest, would miss the opportunity to make a few pairs of such shoes, and even beaters during hunts make them with great agility" (Brincken 1826).

Bast tearing from living trees was considered an essential right of dwellers of villages surrounding royal forests, guaranteed by King Zygmunt August in his 1557 "Law of Wołoka" (all the King's subjects had the right to enter royal forest, albeit not deep, to collect bast for their own purposes, not for sale (Hedemann 1939). In the sixteenth century, any attempts at prohibiting this prerogative were met with strong local resistance. Nevertheless, by 1700 the royal commissioners already observed the negative impacts of this activity on lime trees. In the seventeenth and eighteenth centuries, regulations were introduced so that villages paid an annual fee for bast tearing, collection of mushrooms and nuts. Nevertheless, the scale of continuous bast tearing became a problem as observed by the forest service. This was both because of its devastating effect on lime trees, but also the numbers of people involved which were high enough to trample new pathways throughout the forest. In 1795, it was proposed to put guards on these pathways (Hedemann 1939), yet the activity was not halted and it survived until the end of the nineteenth century and into the early twentieth century. Some of the early twentieth-century photographs depicting residents of Białowieża, Stoczek and Budy villages located inside BPF, locals still wear traditional bast shoes (Gloger 1903).

Lime wood was not considered suitable for construction, but was widely used for carving and woodwork. Manufacturing wooden spoons, cups, bowls, plates, troughs, buckets and other household tools out of lime was an established tradition. “Lime wood is necessary for various crafts, and, when dried, is almost eternal” (Marcin z Urzędowa 1595).

Lime-tree honey has a unique aroma. It has been (and still is) considered as one of the best types of honey, and parts of the forest abundant with lime are considered one of the most valuable for beekeeping (Jabłoński 1991). Although beehives in BPF were mainly carved in pine trees, the traditional forest beekeeping was also connected with lime trees: “the best honey is collected in the beginning of July, mainly in places where many lime trees grow” (von Brincken 1826). This notion survived long into the nineteenth century. In 1864 for example, Stralborn, the head forester of BPF, noted that lime honey was over three times more expensive than other kinds of honey (Stralborn 1864).

Different parts of lime tree were also traditionally used in medicine. This was from chewed bark that was supposed to heal wounds, through mashed leaves that helped against leg swelling, boiled leaves that assuaged mouth diseases, and had a diuretic effect (Marcin z Urzędowa 1595). Oil from lime seeds was specifically mentioned in connection to Białowieża as useful in healing lung diseases, and brew made of lime flowers as perfect for common cold (Bobrovskii 1863).

Scots pine was a species deeply rooted in traditional uses of forest resources. It was the main building material in the royal period (evidenced for Białowieża’s royal hunting manor—out of 23 buildings with identified timber, 21 were constructed with pine wood; Samojlik 2006b), valued also in nineteenth-century shipbuilding. Pines reached enormous sizes in BPF and were therefore considered an ideal material for ship masts (for more details see Chap. 5). Scots pine was also the main tree utilized by beekeepers to carve artificial cavities for beehives (recorded from sixteenth century to the nineteenth century, but probably practised earlier), and the sole species used for wood-tar production (from the seventeenth until the beginning of the twentieth century). Pinewood, especially the resinous parts of the trunk and roots, was also sought-after as an efficient fuel, tinder and a source of light. It was obtained by chopping off pieces of trunks of fire-scarred trees, in parts where the wood was saturated with resin. In the eighteenth century, among different pays for collection of forest products, there was a price for a cart of "resinous torch wood" (Hedemann 1939).

Medicinal uses of pine included the beneficial properties of pine bark, which was used to dry out wounds and help during labour, pine needles that were supposed to soothe inflammations, and, when boiled in vinegar, act as painkillers in toothaches. Pine resin was used against sore throat, discharging eyes and rough nails (Marcin z Urzędowa 1595).

3.5 BPF in the Works of Naturalists, Travellers and Artists Until the End of the Eighteenth Century

Information about BPF and its environmental uniqueness began to enter the awareness of naturalists only at the turn of the eighteenth century and into nineteenth. Prior to that, the Forest appeared in chronicles, travel diaries and other types of literature mainly in connection with royal hunts and European bison (although the fame of BPF being the last species' refuge came only at the end of the eighteenth century). A perfect example of the first type of account is the book by Spanish adventurer, diplomat and traveller Estebanillo Gonzalez, who visited Białowieża and participated in royal hunt of King Władysław IV Vasa in 1643. Since Gonzalez's book became one of the most important works of seventeenth-century Spanish literature, with several editions and translations, this description of BPF reached wide audiences in Western Europe (Gonzalez 1646; Daszkiewicz and Samojlik 2016). The second thread of literary and scientific works mentioning BPF in connection with European bison became visible only in the early eighteenth century. Earlier works, including one of the most important sources of knowledge about European bison, Sigismund Herberstein's "*Rerum Moscoviticarum Commentarii*", did not refer to BPF itself. This was because at the time of its publication bison still occurred in various forests of Central, Eastern, and Southern Europe. The first known scientific reference to BPF as bison refuge is Gabriel Rzączyński's work "*History of naturalis curiosa Regni Poloniae, Magni Ducatus Lituaniae…*". The book, published in 1721, had an ambitious goal of describing the natural history of the Commonwealth of Both Nations, and in the description of European bison simply noted: "European bison dwell in Lithuania in the forests of Białowieża" (Rzączyński 1721).

BPF also appeared in the writings of Jean-Etienne Guettard, who served as a physician of French ambassador in the Commonwealth of Both Nations from 1760 to 1762. During his stay in the Commonwealth of Both Nations, he conducted geological, meteorological, and botanical studies, some of which he published later (Daszkiewicz 1998). Some of his manuscripts and reports, however, remained unpublished and relatively little known. One particular handwritten note from this period is important in the context of BPF—it is a list of plants to find in the forests of Białowieża (Daszkiewicz et al. 2004). The list contains species or generic names of plants that had medicinal or other uses, sometimes extended with a description of their characteristic features. Out of 29 plants noted by Guettard, nine have the plus sign, and fie—minus, probably denoting the species found in the forest by author. It is the oldest proof of scientific approach to BPF's natural wealth and application of Linnaeus' systematics and nomenclature in Polish-Lithuanian Commonwealth, and probably also an evidence of the first scientific excursion to the Forest. Unfortunately, it remained unpublished and unknown to scientific community of Europe, therefore had no scientific repercussions or impact on further development of research in BPF.

Other Guettard's observation indeed played a role in the scientific discourse—his report on European bison seen during his stay in Poland being incorporated into the eighteenth-nineteenth century discussion on the extinction of species. A significant number of scholars did not consider the possibility of the extinction of species and claimed that all fossil forms still lived somewhere on Earth—Guettard's observation was important in their argument that species known in France only from fossils and excavations, continued to live in the forests of the Commonwealth of Both Nations (Buffetaut 1991).

BPF was also entangled in another turbulent scientific dispute that lingered on in the eighteenth and nineteenth centuries, and this related to whether the aurochs and European bison were one species or two different ones. Undoubtedly, this discussion played an important role in the recognition of BPF among European travellers and naturalists, as their attention turned to one of the few well-preserved forests in Europe. The dispute itself started from the first natural history encyclopaedia "Histoire naturelle" edited by Georges Leclerc Buffon and published in Paris since 1749 (Buffon 1764). It was both a widely used source of knowledge and the object of numerous polemics for several dozen years after its publication. The disputes made a huge impact on eighteenth- and nineteenth-century zoology. The dispute on the species separation between aurochs and European bison (and also on relation of both to domestic cattle) lasted for more than a century, with more than one hundred scholars taking part in it (Łukaszewicz 1952). Before reaching the final conclusion, this discussion helped to highlight a very limited range of European bison, making BPF well known in scientific circles as the last refuge of this ancient species. The name Białowieża was continuously written with different spelling mistakes. Currently, a new idea emerged from the study of ancient DNA and cave art: that the American bison was a direct descendant of an extinct steppe bison, and European bison the result of the secondary hybridization of the steppe bison with the aurochs (Soubrier et al. 2016).

Box 3.1 J. E. Gilibert's studies of BPF's animals and Lithuanian forests

Jean Emanuel Gilibert (1741–1814), professor of anatomy, surgery, and natural history at the College de Médecine in Lyon and the creator of the botanical garden in that city, was invited to the Commonwealth of Both Nations by King Stanisław August Poniatowski in 1775. This was to create a veterinary and medical school in Grodno. He was recommended to the Polish King by eminent scholars like Albert von Haller, Antoine Gouan, and François Boissier de

Sauvages de Lacroix. Royal patronage and care from Antoni Tyzenhauz, Treasurer of the Grand Duchy of Lithuania and the Governor of Grodno, allowed Gilbert to successfully create the Royal Medical School. He actively engaged in the development of this school, where he lectured on medicine, surgery, veterinary medicine and natural history. Apart from teaching duties, his botanical passion expressed itself in his floristic research and botanical collection. Thanks to his initiative, a Royal Botanic Garden was established near Grodno. The Garden's assembly encompassed about 2,000 species of plants. After six years spent in Grodno, Gilibert moved to Vilnius, where he was appointed as a professor at the Natural History Department at the Principal School of the Grand Duchy of Lithuania (later the Imperial University of Vilnius; Kamińska 2015). There, he continued his scientific work on botany, zoology, mineralogy and medicine, and he also founded a botanical garden and natural history cabinet in Vilnius. Finally, after personal turbulences, Gilibert decided to leave Vilnius in 1783.

During his stay in the Commonwealth of Both Nations, Gilibert's research and observations made him a precursor in studying and describing the Lithuanian flora (Gilibert 1781a) and fauna (Gilibert 1781b). Among many topics of his lively scientific work conducted both in Grodno and in Vilnius, an important place was occupied by his observations of wild mammals. The King's favour allowed Gilibert to keep wild fauna captured in royal forests, including BPF, and to observe their behaviour, feeding preferences, activity patterns, and anatomy. Gilibert observed animals in their natural habitat and in captivity: European bison, moose (*Alces alces*), brown bear (*Ursus arctos*), lynx (*Lynx lynx*), beaver (*Castor fiber*), European badger (*Meles meles*), wolf (*Canis lupus*) and hybrids of wolves and dogs, red fox (*Vulpes vulpes*), and European hedgehog (*Erinaceus europaeus*). He also had the western capercaillie (*Tetrao urogallus*), European pond turtle (*Emys orbicularis*), and even white mice. In most cases, Gilibert's observations entered the mainstream of European science as the first reliable and first-hand information on given species. The lack of previous knowledge resulted in many obvious mistakes and errors in his descriptions and handling of animals. However, while these are obvious from our contemporary perspective, they should not diminish the value of his pioneering works. Such was the status of his description of European bison in the 1781 volume "Indagatores naturae in Lithuania", published in Vilnius (Gilibert 1781b), that it was used in French natural history encyclopaedias of the time (first of all Georges Buffon's "Histoire naturelle"—Buffon 1764) and works by Georges Cuvier (Cuvier 1812). In this way, Gilibert's account became the main source of information on European bison anatomy, food preferences, and behaviour for several decades. Initially, royal hunters captured four bison calves, two males and two females, to bring them to Grodno. Unfortunately, both males died and eventually only one female calf was sent to Gilibert. After rearing her for four years, Gilibert managed to only partially tame her, as she still occasionally raged, e.g. when domestic cows were nearby or when Gilibert attempted to cross-breed her with a bull (such experiments became reality when conducted by Leopold Walicki around half a century later—see Chap. 5).

Gilibert noted the anatomical details of bison and compared them with those of domestic cattle, supplementing them with many observations of bison behaviour, made by himself or royal hunters, part of which was later refuted. For example—after being caught, bison calves refused to suckle on domestic cows, forcing the royal hunters to use goats to feed the bison. Size difference between the species was solved by putting goats on a table, which then led to next problem—after feeding on goats, bison used to shake their heads, flinging the poor goats from the table (Gilibert 1781b). Bison caught in the 1840s, on the contrary, rather willingly sucked cows (see Sect. 5.3). Some of Gilibert's information was still based in colloquial knowledge or even legends, like the bison's furious reactions to people wearing red, lying on the ground as the supposed method of defence against charging bison, or skin from the bison forehead as a medical aid for facilitating women's deliveries of babies.

Gilibert's observations of other wild mammals inhabiting the forests of the Grand Duchy of Lithuania also played an important role in building knowledge base and overthrowing superstitions, e.g. connected with moose. Before Gilibert's research, moose was known mainly from Renaissance fantastic descriptions and from the dubious medicinal tradition of using moose hoof as the treatment for epilepsy (Podgórny 2018). Similarly, Gilibert's first-hand information debunked the widespread legends concerning the brown bear (e.g. the belief that white bears, belonging to other species than polar bears, occurred in Lithuania).

On the margins of his animal observations, Gilibert described also the place of their occurrence, extensive forests of the Grand Duchy of Lithuania. Read from the contemporary perspective, some of his thoughts are surprisingly relevant. In his praise of "primeval nature, free from human actions and not disturbed by accident or by the impatience of human desires" (Gilibert 1796) he sounds very similar to today's eulogists of the primeval forest of Białowieża, but also very different from the mainstream eighteenth-century lines of thought about forests. A context for the forest dialogue was the excessive exploitation of forests for the needs of industrialization and urbanization of Western Europe. This caused rapid deforestation of large parts of the continent. A universally agreed "remedy" for this was increased human intervention: managing forests, replacing spontaneous regeneration with precisely measured and economically motivated planting. Natural forests were from this point of view "less useful". Gilibert also pointed out that "original forest" is characterized by the presence of some species absent or rare in forests overexploited by humans. He was also one of the few, if not the only, eighteenth-century author who noticed the ecological role of dead trees, commonly considered a sign of a "wild" forest which does not bring the expected benefits to man (Bobiec et al. 2005; Deuffic 2010;

Daszkiewicz 2018). Gilibert was also ahead of his time opposing the concept of nature which would degenerate and gravitate towards self-destruction without the care of man: "*In the refuge of these vast desolated areas away from human pursuits and almost free from humanity claims, many species of animals find survival and ensure the species lives on (…). But a researcher delving deep into this forest as old as the ground he covers unsuccessfully looks for the traces of degeneration of nature, which the scientist, the author of a general and detailed natural history drew from his first glance at his boards. Almost no trees are seen disturbing each other, no dead, decomposing plants burdening the soil and stifling seeds ready for germination, keeping the decay and sheltering poisonous and terrible animals. On the contrary, everything here seems to be full of movement and life, trunks overturned by the wind or old age* [on which] *parasitic plants: mosses, lichens, fungi suck the substance, numerous insects, many species of beetles, many kinds of longhorn beetles and bees attack the wood layer and quickly destroy it*" (Gilibert 1796). Gilibert also underlined the difference between the managed and natural forest, giving preference to the latter and denying any signs of degeneration: "*On the contrary, if you want to see traces of disorder, degradation and damage caused by man in nature, visit forests often wandered by people; traces of fire blacken the bases of the trunks, resinous juices removed, the trees wounded according to their age to extract a few litres of resin or tar from each trunk. It is here that the prematurely fallen trees cover the ground and show the image of destruction to the traveller; here the demands of crafts—depriving trees of leaves or bark, or cutting them with deep incisions—force nature to present a spectacle of weak and imperfect vegetation. By continuous felling, the destructive human hand partially reveals the soil and destroys the connections in this entire system, and the rarer, more timid animals seem to flee from humans. You will not hear these numerous and repeated concerts of singing birds that nature sent to the backwoods. The spectacle of life and movement weakens and moves away from the frequented forests which will become similar to vast deserts*" (Gilibert 1796).

While there is no evidence that Gilibert personally visited BPF, all his works on bison and the natural forest reference it in an obvious way. His praise of the natural, pristine, primeval forest, the description of European bison, and numerous observations of Lithuanian forests and their characteristics made him a pioneering scholar ahead of his time. He should be also treated as the creator and the first representative of the Vilnius-Krzemieniec school of natural sciences (Grębecka 2005), which, until its abrupt closure in 1831, brought a lot of attention to BPF. Additionally, Gilibert succeeded in transferring the leading French eighteenth-century natural sciences methodologies to Poland, and thus bringing its science closer to the leading schools of natural science in Europe.

Fig. 3.4 A sketch and a watercolour by Jan Henryk Müntz "*Białowieża Primeval Forest—the bear hunt*" (LWU 1780–1783)

Box 3.2 J. H. Müntz's depiction of an arboreal apiary in BPF

The sketch and watercolour by Jan Henryk Müntz entitled "Białowieża Primeval Forest—the bear hunt" (Fig. 3.4) are one of the oldest depictions of the forests of Białowieża. Created in 1780 (sketch) and 1783 (watercolour), are not only an interesting works of art in themselves, fitting into the discussion on evolution from Neoclassicism towards Picturesque in late eighteenth- and early nineteenth-century painting, but also serve as historical sources. Both works contain several aspects of information enriching our understanding of the environmental history of BPF. This is especially in connection with the description of the depicted situation as written by Müntz himself. Jan Henryk Müntz (1727–1790), a cosmopolitan man of many talents served as painter to Prince Stanisław Poniatowski, nephew of King Stanisław August Poniatowski. During his service, he made several trips in the eastern regions of the Commonwealth of Both Nations, illustrating and describing the most important and scenic places (Budzińska 1982).

Müntz visited BPF at least twice, in September 1780, when he made a drawing showing a beekeeper on a tree, another with a young European bison, and finally a sketch—"Białowieża Primeval Forest—the bear hunt". His next visit was in 1783, also the year when he made a watercolour with an identical title as the sketch (Budzińska 1982). Both works with shared title depict a unique situation, described by the artist himself (original in French): "*You can see here a bear that climbed an oak tree with tightly covered 24 beehives, broke the passage but after eating and destroying the entire content of the apiary, was not able to climb back down. It was killed with many bullets fired by peasants, owners of beehives. It was a specimen of the largest brown species. It was kept alive for a few days, well-guarded by a group of peasants like a real criminal. On the last day, I saw it alive on the tree making faces to all bystanders. All residents came from several miles to see this honey thief and enjoy their catch. (…) The apiary on a tree still existed in 1783, when we were passing nearby. (…) Drawn in 1780 and 1783 in the afternoon looking North*". Both drawing are the only known source of information on this unique form of beekeeping in a form of arboreal apiary. Previously, only beehives carved in trees were known from BPF (Karpiński 1948), but similar arboreal constructions were mentioned for other regions of the Grand Duchy of Lithuania. Reportedly, they survived in the Polesie region until the second half of the nineteenth century (Hedemann 1934; Blank-Weissberg 1937). Both drawings also show a very interesting system of protection of the apiary from brown bears, the biggest animal pests of traditional forest beekeeping. The tree with an apiary had a protective wooden collar embracing the trunk to prevent the bear from getting to the platform with beehives. In case the collar made the bear fall, there were sharpened poles placed at the base of the oak. Also, the diameter of the platform itself made it extremely difficult for any bear to climb over it. The one depicted by Müntz was obviously smarter than all those protective tricks, but not smart enough to make it out alive; or maybe it was too fixated on the stash of honey.

Box 3.3 The local knowledge on forest herbs and their uses

Among several plants traditionally collected in BPF in the royal period, "barshch"—hogweed (*Heracleum sphondylium*)—has probably had the most numerous uses in the rural household. According to Józef Rostafiński, naturalist and ethnologist who devoted several publications to the historical importance of this plant, its main use was a soup also called "barshch". To cook it, hogweed leaves were first pickled, to give the soup its sour taste. It was only in the eighteenth century, that the recipe for the sour soup changed and beets or rye were used more often than hogweed leaves. Consequently, "barshch" plant and "barshch" soup drifted apart linguistically and by the twentieth century, it was generally forgotten where the name of the soup originated (Rostafiński 1880).

The plant was also gathered to be eaten raw, as greens, especially important in the pre-harvest period and in times of famine. Hogweed was also dried for future use. The plant is relatively rich in sugar: "*The stalk of the hogweed is hollowed out, and it is covered with bark. The bask, and similarly, the root applied on human skin acts as an irritant. However, if young stems are scraped from bark ad dried, they exude yellow sugar with the taste of liquorice. This is a unique delicacy among domestic plants*" (Rostafiński 1911). This significant amount of sugar also meant that the plant was often used to produce alcohol. Alcoholic distillate was even more intoxicating when the fermenting juice was supplemented with the leaves of bog blueberry (*Vaccinium uliginosum*).

For centuries, hogweed was also a highly valued medicinal plant. Its root was consumed as an antipyretic and the juicy rootstock was used as an anti-epileptic and anti-diarrhoea medication. The fruits were believed to heal "spasms and delirium". Likewise, hogweed, sometimes called a "bear paw", was credited with magical powers; worn as a charm around the neck, it was supposed to make children "obedient and pleasing", and to protect them from "all beasts". Furthermore, even hanging seasoned hogweed over a table helped to remove venom from all dishes served on the table underneath. Last but not least, there was the magical application of hogweed as magical rings worn on fingers to give the ring bearer "happy thoughts" (Rostafiński 1893; Maurizio 1932).

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Chapter 4
The Beginning of the Imperial Period (1796–1837)

Abstract The fourth chapter addresses the period when Białowieża Primeval Forest (BPF) was taken into the custodianship of the Russian State. By 1795, after the third partition of Poland, BPF was incorporated into the Russian Empire, to be the property of the Russian state treasury. Following the 1795 partition, the administrative system that had previously protected the Forest partly collapsed. One of the thirteen forest districts of BPF given to a favourite of Russian Empress Catherine II was soon sold and clear-cut. This left a distinctive deforested triangle on the Forest's map. However, in 1802 and 1803 Tsar Alexander I reinstated all the former protection afforded to the European bison and BPF, and since 1820 any timber felling or hunting in the forest was prohibited. In the meantime, the Russian imperial administration sought to improve the forestry management based on "scientific" and "rational" principles, but it encountered several problems in achieving their ambitions.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

4.1 Summary

In 1795, after the third partition of Poland, BPF was incorporated into the Russian Empire, becoming the property of the state treasury. One of the thirteen forest districts of BPF given to a favourite of Empress Catherine II was soon clear-cut, leaving a characteristic deforested triangle on the map of the Forest. In the first years after the 1795 partition, management of new state property shifted between different offices and villages of beaters were donated to different people. The entire system which previously protected the Forest had collapsed leaving the European bison herd in danger. Fortunately, in 1802 and 1803, Tsar Alexander I reinstated the previous protection of European bison and BPF, and prohibited all hunts in the forest (except hunts for predators). The Russian imperial administration aimed at turning BPF into the manufacture of timber, but it encountered several problems in realization of this plan. These were mostly the absence of surveys and maps of the forest, the lack of trained personnel, the weakness of the provincial administration and a myriad of

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_4

disobediences and misconducts by forest personnel. This meant that for a prolonged period (until the 1840s) no significant changes were made in the forest exploitation. The administrative division of the forest into twelve districts, and the structure and make-up of the forest personnel remained unaltered from the Polish royal period. Also, most traditional uses of the forest based on access rights remained in place, constituting a burning issue for the new forest administration.

Skirmishes with bands of Prussian poachers until 1807, war with Napoleonic France in 1812, and the Polish Uprising of 1830–1831, combined to influence the management of BPF in this period. The latter event especially, left forest personnel depleted, as eleven of the twelve guards and 109 of the 123 riflemen joined the fight. Many forest officers were forced into exile or prosecuted for their involvement in the insurrection. The owner of the adjacent Świsłocz Forest bordering BPF from the north-eastern side, Count Tadeusz Tyszkiewicz, was punished for his active participation in the uprising by the confiscation of his estates in 1832. Acts of repression after the failure of the November Uprising were also directed towards the Vilnius University and the education system in Grodno Province. For the next half-century, there were almost no scientific or research-based works on BPF. At the end of this period, in 1837, the Ministry of State Domains was created and with new energy set about introducing the new "rational" or "scientific" forest management in BPF.

The cultural heritage of this period is reflected in the local knowledge and traditional names of forest habitats—naïve interpretation of forest habitat diversity that was later included in the terminology of a new discipline of science, phytosociology. This was also the period of the first written mentions of local legends, that had probably been building up much earlier, and connected with remnants of past human activity like ground barrows and stone circles left by ancient and mediaeval dwellers of the forest.

By the beginning of the nineteenth century, the state of preservation of BPF acted as a magnet in drawing naturalists and travellers. A series of works published in the early nineteenth century by Brincken (1826), Górski (1829), Jarocki (1830a, b), Eichwald (1830) entered the scientific discussion and became the basis of the knowledge on BPF and natural forests until the end of the nineteenth century.

4.2 Former Royal Forest Under the Management of the Ministry of Finance

In 1795, BPF was incorporated into the Russian Empire, becoming the property of the state treasury—as was the case with most of the former royal properties of the Commonwealth of Both Nations. The new additions to the imperial state treasury were partially donated, partially rented, and forests came under the administration of the Forestry Department. This was a governmental agency established in 1798 within the structure of the Admiralty Board, and in the period 1802–1811 functioning within the Ministry of Finance (Forestry Department 1898; Lupanova 2017). One of the thirteen forest districts of BPF along with several villages was given to count Pyotr Rumyantsev (1725–1796), one of the long-time favourites of Catherine II. Over 1100 male serfs lived in these villages. In 1803, after Rumyantsev's death, his son attempted to sell the forest and the estate back to the treasury (NHARBG 1803). Unsuccessful in this bid, he then sold it to private hands, who soon managed to clear-cut the entire forest district (see Sect. 4.2).

In the first years after the 1795 partition, management of new state property "in the provinces newly annexed from Poland" was carried out under the supervision of the Expedition of the State Economy of the Senate. Governor-general of Lithuania, Nikolai Repnin, left the Polish administrators in their posts, retaining both staff and salary. In those years, Brest District was run by Krzysztof Engelhardt, former regiment-colonel of the Lithuanian Horse Guard, whose office staff consisted of a clerk, an accountant and three forestry officers (RSHA 1799–1801). Further changes in the management of state forests, shifts in the administration (in 1796, the Province of Lithuania was created, and in 1802 it was divided into Vilnius Province and Grodno Province. The reshaping of the external borders of this part of the Empire (on the basis of the Treaty of Tilsit in 1807) brought further confusion. Some of the documents from that early after-transition period ended up in the Grodno provincial administration (e.g. NHARBG 1802, 1816B), and some in the Forest Department of the Ministry of Finance in St. Petersburg (RSHA 1805–1811, 1808A, B, 1814–1828). Engelhardt and his subordinates retained their positions at least until 1801. After the creation of the Forest Department, Engelgard was appointed as an ober-forstmeister (according to the Russian nomenclature drawn from the German tradition; RSHA 1799–1801).

Under the rule of the most unpopular of the Russians Tsars, Paul I (ruled in 1796–1801), regional forestry management, and in particular the management of BPF, witnessed some strange decisions. In 1797, the Tsar ordered the killing of moose in all provinces of the empire, including BPF, to provide leather for trousers of his guard regiment. Białowieża's foresters engaged in a long correspondence, postponing the mass-killing of moose (and, in the meantime, estimating the number of moose at 125 animals). In 1798, the order to kill 40 moose reached Białowieża; yet the eventual fate of these animals is unknown (Hedemann 1935d). In January 1801, Paul I ordered the reorganization of the forest administration in the Lithuanian province following the model of the forest management in the Courland province, i.e. according to the German model. The task was entrusted to the Courland ober-forstmeister von Dershau, yet he was given no additional resources to implement it. Von Dershau's actions were limited to his visit to Vilnius, his correspondence with the Lithuanian ober-forstmeister Beckmann and a very brief note sent to the Forest Department. In this, von Dershau discussed the possibility of increasing the state forest income of the Lithuanian province from the current three thousand roubles a year to 100 thousand roubles a year. The Lithuanian ober-forstmeister Beckmann, on the other hand, responded to the Forest Department that the province lacked trained personnel, foresters' salaries were too small, and the forest administration was not provided with sufficient information by the local province administration. In 1802, the Forest Department was transferred to the Ministry of Finance and the Lithuanian

Province was divided into two smaller provinces, so that the order of now-deceased Tsar Paul was forgotten (RSHA 1801–1803).

Amidst this administrative turmoil, some local foresters started to report to the imperial administrators about the dangers looming over the fate of BPF and, especially, the European bison. In March 1802, the newly appointed Governor-general of Lithuania, General Levin August von Bennigsen, asked for information on the official status of European bison, what was being done to protect them from extinction, and why the peasants that used to tend and feed bison in the Polish times had been moved away. The head-forester of the Grodno Province answered that the bison population was in a poor condition, as the entire system of previous protection had collapsed. Some of the villages of beaters were donated to different people, and beaters themself were registered as state peasants. Further correspondence from BPF was even more alarming as ober-forstmeister Krzysztof Engelhard wrote that the number of bison, left without supplementary fodder and reliant on themselves, had fallen to around 200 animals and would continue to decrease if the former way of their support was not be reinstated (Hedemann 1931; see also NHARBG 1802). The eventual outcome of this correspondence was the decree issued in September of 1802 by Tsar Alexander I partially reinstating the previous bison protection:

> Cavalry General, War Governor of Lithuania Baron Beningsen!
>
> Considering the rarity of the species called European bison which dwells in Lithuania in the forest of Białowieża I deemed it important to protect them from killing and scaring away, and for the purposes of their feeding, to ascribe, as previously, villages left in Brześć economy: Ćwirki, Panasiuki, Kamieniki i Myzpnary Pauckie. In connection with that I order you, considering the local circumstances, to issue for acceptance of the Senate of the Empire all necessary concessions that those peasants should be granted. With benevolence, signed by His Majesty Tsar Alexander. St. Petersburg, 10th September 1802 (Karcov 1903).

In 1803, another decree followed stating that peasants from those villages, in total 489 adult men, were obliged to protect the European bison and the forest (Hedemann 1931). The Tsar's decrees did not any give clear instructions on how European bison should be protected or what actual duties the dwellers of four mentioned villages should undertake. These were left in the hands of the local authorities. It seems most plausible that the local administration restored the duties and privileges of the forest guards of BPF and other large state forests of the province (maybe there was also a need to explain what the animal that caused all the commotion looked like, as one of the archival files dated to 1802 and concerning peasants obliged to help in bison feeding was accompanied by a depiction of European bison, see Fig. 4.1). A few years later, the government agencies in St. Petersburg attempted to obtain a clear definition of what exact duties and privileges the forest guards of the Grodno province used to have. However, the Grodno Treasury Chamber was able to provide information about the privileges and land used, not the guards' duties (RSHA 1808A, 1814–1828).

According to data provided by the Grodno Treasury Chamber in 1807–1808, the size of the allotment for every 12 guards of the BPF varied from 38.5 to 114 desyatinas (1 desyatina = 1.09 ha) of arable land and meadows. (The size of the allotment was probably determined by the quality of the soil and the size of the part of the forest ascribed to the guard's responsibility). To cultivate this land, from 2 to 5 families

Fig. 4.1 Depiction of an European bison from an archival file on the number of peasants obliged to help in supplementary feeding of bison in BPF (RSHA 1802B)

of peasants (ogrodniks–crofters) were assigned to every guard's estate. Each of 45 families of ogrodniks had its own allotment from 3.25 to 13.5 desyatinas per family. The allotments of 107 riflemen varied from 6.75 to 29.5 desyatinas. In addition, most of them had the right to mow hay in forest glades: from 8 to 69 hay-carts for a guard; from 2 to 10 hay-carts for a rifleman; from 2 to 6 cart for an ogrodnik (RSHA 1808A).

At the beginning of the nineteenth century, following the German example, the Russian imperial administration made some attempts to "improve" its own state forestry in general, and not only in relation to harvesting timber for the navy. The Naval Ministry was interested in harvesting specific timber suitable for shipbuilding, and reports prepared by naval officers included only information on the type of the forest in different forest districts of BPF, sizes reached by the trees, and the distance to floatable rivers (RSHA 1808B, 1809; RSAN 1813–1829). Rational forestry aimed at turning all forests into having a focus on the manufacture of timber based on monocultures of the most valuable tree species planted in places evaluated as most suitable for them. This German model was imported directly by Russian forestry administration and so it is understandable how Russian forestry education was dominated by German forestry model (Kern 1903; Fedotova and Loskutova 2015; Loskutova 2016). However, there were still problems in terms of its full realization: the lack of trained personnel and the weakness of the provincial administration. An additional difficulty was the intricate subordination of forstmeisters to the provincial and imperial authorities. By the early 1812, the duties and officials of the Forestry Department were incorporated by the Department of State Domains of the same Ministry of Finance. Both in the form of a separate department, and within the Department of State Domains, forestry administration was mainly occupied with two tasks. First, it was obliged to ensure the supply of timber and firewood for state and public institutions of all ranks and kinds: from schools and hospitals to military fortresses and navy. The second urgent task was to provide cadastre of the state forests, including the provinces "recently joined from Poland" to find out exactly what forest areas had become state property and how their protection would be organized. It was necessary to survey the area, to draw maps and to mark the boundaries in the field. They also had to evaluate the stock of timber and make the plans for its "rational" use.

For this reason, a significant number of graduates of the Forest Institute in the 1810s were appointed as land surveyors at the forestry offices of the provincial State Chambers to "measure and describe state forests". Among them, there were the de Ronke (also spelt Ronka, Ronko, Ronca or Rongo) brothers, who graduated from the Forest Institute in St. Petersburg and began their careers in the Grodno province (RSHA 1810–1821). One of the brothers, Eugeniusz, became a forstmeister in BPF in early 1820s, and played an important role in the history of the Forest (see Box 4.1).

The forest administration, occupied with tasks of surveying and evaluating resources, had little chance to introduce new model of "proper" forestry. This meant that for a prolonged period (until 1840s) no significant changes were made in the forest exploitation. The administrative division of the forest into 12 districts, and the structure and make-up of the forest personnel, remained untouched from the Polish royal period. An example of how slow the changes were is the fact that it was only in 1808 that the authorities in St. Petersburg began to demand reports with Russian, not Polish units of measure, and it was then 13 years after the acquisition of BPF by the Russian Empire, when Polish *morga*, *włóka* and *pręt* were used along with Russian *desyatina* and *square sazhen* (RSHA 1808A). In 1810 report, ober-forstmeister Galisiewicz explained that guards were traditionally recruited from Polish nobility with a patch of land given by the king for their livelihood. Yet in answer to the Forestry Department's requests for detailed description of the exact size of land, duties paid by guards and the procedure of their appointment and removal from post, the Grodno administration could present only the data on the land-plots ascribed to guards, riflemen and beaters with the description of their duties in the most general terms (RSHA 1805–1811, 1808A, 1814–1828). In the 18th century, the guards' posts were usually hereditary, which resulted in the majority of guard families keeping their position until the first third of the nineteenth century. Comparison of guards' names in inventories of 1792 (LSHA 1792) and 1812 (RSHA 1810–1821) reveals that in two forest districts there was no change between 1792 and 1808, and in further five the post of guard remained in the same family (surname of the guard remained the same between 1792 and 1812). One of the files describes in detail the transfer of these posts from father to son and from older to younger brother (RSHA 1806–1847).

There was also a myriad of practical issues that the officials (forstmeisters and ober-forstmeisters) of the special management of state forests, Forestry Department of the Ministry of Finance, had to deal with. Since hunting for bison was now prohibited by law, it was the foresters' responsibility to investigate with the police all cases of bison death. In 1802 for example, a bison pelt was seen in the house in Kamieniec (20 km south from the border of BPF), spearheading an investigation that eventually led to identifying a rifleman from Świsłocz Forest, adjacent to BPF, who shot a bison supposedly confusing it with a wild boar (RSHA 1802A)—such an excuse is used by poachers even today. While BPF was expected to bring income to the treasury, neither the forest officials, nor the provincial administration understood how to organize the sale of the forest resources "by tickets". Each attempt to conclude a deal to buy timber or firewood, or lease haymaking or other type of use, provoked a cascade of letters exchanged among the BPF forstmeister, ober-forstmeister of Grodno Province, Forestry Inspector of Western Provinces, the Forestry Department and the Ministry of Finance (NHARBG 1816B, 1818, 1825; RSHA 1806–1807A). Although both the Forestry Department and the Ministry approved the sales of timber, some restrictions were made. It was possible to sell trees suitable for "civil" construction, but not for the navy, secondly, private buyers could buy wood only for their own needs but not for resale, the sale was intended for the local market only, not abroad, and lastly both forstmeister and ober-forstmeister should agree for the same (RSHA 1806–1807A, 1809, 1817–1829).

In addition to bureaucratic difficulties, the treasury set high prices and demanded the observation of a number of conditions that required additional labour costs, e.g. to utilize the tops and branches of trees. As a result, the majority of entrepreneurs preferred to buy wood from private owners, abundant in the province (Tsvetkov 1957; Anonymous 1849; NHARBG 1843).

Surviving documents show a disproportionately high (compared with later periods) frequency of dismissals in connection with court cases of misconduct. In the autumn of 1805, BPF forstmeister Karl Burger was dismissed and brought to trial, in 1806 ober-forstmeister of Grodno Province Peter von Mueller met the same fate. Again, in 1809, his successor ober-forstmeister Peter Trayden was brought down from his post and indicted. The next court case against BPF forstmeister Vityalis Gorbatovsky and ober-forstmeister Karl Gagman started in 1819 and lasted several years, revealing serious misconduct by those officials: cutting timber, including trees suitable for shipbuilding in 40 backwoods of BPF, allowing wood tar manufacture and potash burning, and forcing peasants assigned to the forestry to work for forest officials under the threat of beating, sending to army, or eviction from their houses (NHARBG 1805–1806, 1809–1824, 1816A, 1817, 1820–1825). Additional court documents filed in the archive in the nineteenth century (only a list survived to the present day) suggests that even more investigations against forest officials were led in this period (RSHA 1806–1815). To some extent, it was connected with an ambiguous hierarchy of forest officials within the province in the period 1802–1826: ober-forstmeister and forstmeisters were subordinate to the governor (subject of the Ministry of Internal Affairs) and the Forestry Department (subject of the Ministry of Finance), but not to the Treasury Chambers (local office of the Ministry of Finance).

Often, it resulted in Treasury Chambers not providing forstmeisters with necessary information and in fact not allowing them to fulfil their duties. Forstmeisters, in turns, were reluctant to fulfil Treasury Chamber's requests.

Although the "Plan of the organisation of Brest District Forests" written in 1796 by Jan Szczepanowski, administrator of state properties in Brest District (including BPF), saw "*burning wood-tar and birch-tar* (…) *abolished and charcoal burning prohibited, as the income to treasury is meagre and more loss by the destruction of the forest is incurred*" (NHARBG 1796), those court cases evidence that burning forest products was not abandoned completely. Despite Brincken's (1826) statement: "*since the exploitation of the forest was restricted not to disturb European bison, the income fell almost to zero. Chemical exploitation has met the same fate*", even the official financial accounts record income (albeit very small) from kilns until the late 1820s (RSHA 1828, 1829).

An important factor influencing the management of BPF in the early nineteenth century was the fact that the Russian Empire actively participated in the Napoleonic wars. Furthermore, until 1807, the northern border of BPF was also the boundary between the Russian Empire and Prussia, and was frequently trespassed by bands of Prussian poachers: "*Prussians did not care at all for Białowieża Forest, and bands of armed poachers were formed in villages on the forest border belonging to Prussia, which then intruded the forest abundant in game, and escaped back abroad. This state lasted until Napoleon's defeat of the Prussians, the creation of the Duchy of Warsaw (in 1807) and the dissolution of New East Prussia. However, in 12 years of this border existence, the number of bison fell from 700 to 300*" (Dyakowski 1908). It is however not known what sources Dyakowski used in estimating the number of bison—it is much more likely that the correct number was closer to the one from the end of the Polish period, oscillating around 300 individuals (see Chap. 3).

Regiments passing through BPF also brought inconvenience to local residents, demanding transportation, supplies and fodder. They occasionally felled trees and hunted and did not compensate the civilians for goods provided (NHARBG 1805–1806, 1809–1810). The same story was repeated during the war of 1812 and the Polish Uprising of 1830–1831 (NHARBG 1829, 1831, 1834–1835).

In 1802, all hunts in BPF (except for predators) were prohibited (Karcov 1903; Hedemann 1939), motivated by the need to protect the last living population of European bison. Around the same period the annual counts of the bison population, known from the royal period (Samojlik and Jędrzejewska 2010), were reinstated (Hedemann 1931), although the first known reports on bison numbers in the Russian period date to 1809 (Karcov 1903; RSHA 1813–1829). The diligence of forest personnel in ensuring that bison numbers did not drop was sufficient to trace and drive back the bison that left BPF to the adjacent, private Świsłocz Forest during the war of 1812 (NHARBG 1816B). Correspondence about the few bison that fled to Świsłocz Forest continued for 11 years, with eight bison returned to BPF in 1823 (RSHA 1816–1825; Karcov 1903).

Similar to the system of additional winter feeding dating back to 1700s, beaters had the duty to scythe forest meadows and prepare hay for winter feeding of bison (Samojlik and Jędrzejewska 2010). A common notion was that bison survived in

BPF and nowhere else on the continent because of a single or several plants characteristic only of Białowieża's woods. Although naturalists tried to disprove this belief, it survived long into the nineteenth century. Ludwig Heinrich Bojanus, professor at Vilnius University, explained that European bison disappeared from vast areas due to overhunting, pressure of human population, and destruction of bison habitat, not because of the lack of any specific plant (Bojanus 1825). Polish botanist, Stanisław Batys-Górski established already in 1829 that among grasses eaten by bison none was endemic to BPF (Górski 1829). Furthermore, experts of the scientific committee of the Ministry of State Domains gathered information from BPF's forest personnel indicating that bison fed on common grasses and the bark of trees (RSHA 1843–1846).

In 1803, Vilnius General-Governor Beningsen (the same person that was given the task of protecting bison by Tsar Alexander I) asked for permission to hunt an old bison in BPF and to send it for research and show purposes to the Vilnius University, but was only promised a bison that died of natural causes (Hedemann 1931). A series of reports on dead bison found in the forest dated to the first quarter of the nineteenth century were most probably a reaction to Tsar Alexander's order (RSHA 1813–1829). In 1819, after lively correspondence in which Professor Bojanus wrote that for the zoological cabinet of Vilnius University it is essential to acquire specimens of European bison, a species that sooner or later will be eradicated. If traces of this rare animal were not preserved for the sake of natural sciences, then foreign naturalists would blame the Vilnius University for "unforgivable levity in this case" (RSHA 1819–1821). Two bison were hunted for the University in 1821, and in 1829—two more, since the two killed in 1821 were poorly conserved (Sławiński 1931; LSHA 1829–1830). From the 1820s, the requests of universities and natural science museums for European bison skeletons, pelts, or entire animals were satisfied by the Russian emperors quite willingly (Samojlik et al. 2017; Fedotova 2018; Fedotova et al. 2018).

By the end of 1820s, the forestry administration discussed a practice to decrease the mortality rate of the bison, and this was the extermination of wolves (RSHA 1827–1833). The case started with forstmeister's Eugeniusz de Ronke's reports on European bison killed by wolves in March 1827. De Ronke explained that due to the complete ban on hunting in 1820, BPF has become a very favourable place for predators. De Ronke's first application to renew hunts for predators in BPF was not supported by the Grodno administration. Instead, the Grodno Governor and Ministry of Finance offered a clever way to hunt wolves using snare traps, but without any rifles, hunting dogs, and expense from the state treasury, to which forestry administration responded with a series of concerns. New reports on bison attacked by wolves in 1828 provoked a further series of letters. This time the Grodno Governor provided the Ministry of Finance with information on number of livestock killed by wolves in recent years in the province and offered to reward for killing a wolf in BPF, following Prussian model. In 1829, the Ministry of Finance allowed hunting in BPF of wolf (and other predators) but refused to pay any rewards for dead predators (Hedemann 1935b; RSHA 1827–1833). In the following decades, Grodno foresters discussed exterminating wolves with traps and poisons, but most of these methods were ineffective (RSHA 1839–1842, 1845–1847) . In the late 1840s, the practice of paying rewards for killed wolves was introduced in several Western and Southwestern provinces, including Grodno, and forestry administration together with the local police began to organize hunting raids on wolves (NHARBG 1846–1850, 1850–1851, 1867).

Another burning issue that the new forest management inherited from the royal times and had to resolve quickly was that of access rights. Already in 1799, all holders of such rights to BPF were obliged to present them before the administrators of the forest, but several volumes of collected copies of documents did not mean the problem was easily resolved (RSHA 1806–1807B). On the contrary, the scope of claimed rights and demands, and the complication of extensive documentation prolonged the case until 1812. During the Franco-Russian war, part of the documentation was destroyed, making the cases even harder to assess. In 1828, another call for documents confirming access rights was announced. Since the case involved state interests, the Ministry of Finance took part in the discussion and reminded participants that such rights should never include the prerogative to cut down any living trees (RSHA 1827–1829). The traditional uses of the forest encompassed by access right continued and there are documents covering the income from BPF from payments for beekeeping and haymaking on forest meadows. The latter constituted the major part of the forestry accounts in 1820, when logging in BPF was banned (RSHA 1827, 1828, 1829).

In the late 1810s, the imperial administration requested the introduction of a more transparent system of renting access rights for entire sections of the forest for three, six or twelve years on the basis of auction organized by the forstmeister. Also, tenants now had to pay for six months in advance. Probably St. Petersburg authorities saw the old system with local peasants renting small patches of forest and paying with money or in kind in early autumn, as inconvenient for management and unprofitable too. However, the new system was troublesome since neither forest officials, nor tenants understood how tenders should be organized, and misunderstandings occurred. Since a significant amount of money was needed to rent large areas, speculators took advantage and began to resell access rights to local peasants. At the same time, they were delaying their payment due to the forest administration and accumulating huge arrears (NHARBG 1818; RSHA 1817, 1837A, B).

Similarly to the access rights, the new administration of BPF inherited also a problem of wood-tar and charcoal makers (*budnicy*) settled in the second half of the eighteenth century in newly-erected villages (Budy, Teremiski, Pogorzelce and Masiewo) inside the forest. After the decline of the industries, the workers, excluded from serfdom or other duties, were an unruly group within society. This situation was observed by the authorities, but no actions were taken until 1824, when the Grodno Treasury Chamber asked forstmeister de Ronke to prepare report on the *budnicy* (RSHA 1837A). Apparently, this did not lead to any consequences, as in 1833, Grodno Province forester Anatieyenko informed the Grodno Treasury Chamber that the *budnicy* were occupied with agriculture and were free of any duties or taxes. Anatieyenko proposed to incorporate the *budnicy* into the forest personnel and confer to them the obligations of beaters. Nevertheless, this situation was not solved until the end of 1830s (RSHA 1837A).

After the acquisition of the Forest by the Russian Empire, fire prevention measures and the suppression of fires (already present in the late eighteenth century) were strengthened (Hedemann 1939). Modern studies on fire history show a substantial change in fire regime starting in the early nineteenth century. Instead of frequent but small, light fires as first occurred here from the sixteenth through to the eighteenth centuries, there were now occasional but intensive and disastrous fires during the nineteenth century (Niklasson et al. 2010). Historical sources from BPF repeatedly mention only three large fires at the beginning of the nineteenth century: in 1811, 1819, and 1834 (Brincken 1826; de Ronke 1830; Jarocki 1830a, b; Bobrovskii 1863; Genko 1903). Especially after the disastrous fire of 1811, fire protection measures became more strict (Genko 1903).

Before the forest had a chance to regenerate, war broke out in 1812. BPF was only partially affected by the conflict when it was utilized by both sides as a hiding place and transportation route. All of the main battles and skirmishes took place outside the forest (Karcov 1903). Some of the forest personnel were entangled in the warfare and some of the Grodno forestry officers were captured by the French army; some of the Polish forest guards of BPF even tried to join the fight on Napoleon's side, however, most of them returned to the service in 1813 (NHARBG 1813; RSHA 1811–1815; Szretter 1893). Some of the estates of forest guards and riflemen, especially those located near main roads, were looted or destroyed (NHARBG 1813, 1819–1820). Local residents tried to adapt to the dangerous times by setting up barns in the wilderness and hiding their property and food there (NHARBG 1813).

Behind the lines of war, an important cultural process took place. Both the army and administration of Napoleonic France were interested in Lithuania and Poland not only as a theatre of warfare, but also because of their resources and economic opportunities. Along with the Napoleon's Great Army, many educated naturalists, doctors, pharmacists and engineers came to the former Commonwealth of Both Nations. Among them were such eminent naturalists as Jean-Baptiste Bory de Saint-Vincent (1778–1846), Marcel de Serres (1780–1862), Jean-Louis Graffenauer (1775–1838) or Pierre François Dejean (1780–1845) (Daszkiewicz 2013). Napoleon greatly appreciated the natural sciences and supported their development to the point that some historians call the Napoleonic France the "empire of science" (Sartori 2003). Archives of the Historical Office of the Ministry of National Defence [Service Historique de la Défense] in Vincennes contain intelligence documents from the Napoleonic period. These include material on the territories of the former Polish-Lithuanian Commonwealth, usually contain a chapter on forests, and mainly concern the period immediately preceding the war of 1812 (Daszkiewicz 2010). However, whereas all information of military nature (roads, rivers, crossings) is very detailed, the description of forests is rather vague, for instance, "*….. the forests are full of bears, wolves, elks, bison* [in French *boeux sauvage*]*, lynx, beavers, wolverines, wildcats and all kinds of game and birds. Among the birds of prey, eagle and vulture are very often found here. Fish of numerous species live in rivers, lakes and ponds, visited by a large number of water birds*" (DHS 1812). And similarly, "*…..These forests are very old,* *very extensive and to some extent inaccessible, but almost everywhere are to some extent destroyed. They are often destroyed by fires caused by burning tree trunks by peasants who produce turpentine and tar, which they are obliged to deliver to their masters, by deliberate burning to change forest into cultivated fields, and fires after thunder strikes that no one ever tries to extinguish*" (DHS 1812; see also Daszkiewicz 2010).

Napoleonic naturalist-officers were very interested in studying European bison, and continued to look for the species in Prussia. Their attempts were based on local tradition, which proves that the memory of the occurrence of this species had survived for several decades after it went extinct (Lauzun 1908–1912). Eventually, Napoleonic officers contributed to developing the knowledge on bison biology, though it was not based on observation of free-living animals but rather on the description of captive European bison in Vienna (see Box 4.3).

According to the report on the state of the forest after the war, BPF did not sustain any significant damage through warfare, and the most notable impact it has left on the forest was wooden huts erected by the armies (RSHA 1813–1814). However, the royal hunting manor, a remnant of the royal times in the centre of Białowieża village, was burnt down. During the fighting, all the hunting equipment, nets and weapons from the manor were confiscated (Karcov 1903; Sztolcman 1924). The events of the Napoleonic war survived in local memory and were the subject of many tales, as reported by Hans von Auer, forester in BPF from the 1870s, "*the most important stories were those about the French crossing Narew river in 1812 (…) usually told when we camped near the French graves in the Narew valley. These were thousands of mounds, still visible in the forest. In the autumn, a lot of beautiful porcini and entoloma mushrooms grew between them*" (von Auer 1998). Despite the fact that the mounds described are, according to the current thinking, ancient, mediaeval burial sites (Krasnodębski et al. 2011), this shows how the 1812 war became a part of local tradition.

In the years following Napoleon's invasion of Russia, the need to protect European bison was one of the main factors shaping the forest management. As it was believed that felling trees and other work that required the presence of larger number of people in the forest scared the bison away, this was the official reason given for the ban on any kind of tree felling in BPF set in 1820. This was apart from dead trees usually cut along forest roads and firewood collected by local peasants for an annual pay (Brincken 1826; Hedemann 1935a; Karcov 1903; RSHA 1813–1829). Nevertheless, given the number of breeches and court proceedings as noted, the need to protect the forest itself from damaging use may have also played a role in the decision (RSHA 1822). Surviving document evidence that in the 1820s and 1830s St. Petersburg's forestry administration often issued limitation or prohibitions on tree felling in overexploited state forests in different provices (RSHA 1822, 1838–1842).

The November Uprising of the years 1830–1831, the Polish armed insurgence against the Russian Empire, also left an imprint on the history of BPF. News of rebellion of young officers in Warsaw reached the Forest by the end of 1830, but local conspirators decided to postpone any conflict until spring. In the meantime, all the rifles of forest service were commandeered by BPF forstmeister Eugeniusz de Ronke (see Box 4.1) and stored in his quarters in Królowy Most village on the southern border of the forest. In May 1831, the same de Ronke issued an order to all guards and riflemen under his supervision to report for duty and form an insurgent troop. One of the guards, Jakub Szretter, veteran of Kościuszko Uprising in 1794, and of Napoleon's invasion of 1812, joined the November Uprising with his five sons, wife and two daughters and played an especially important role in mobilizing the rest of forest service to join. Eventually, 11 of the 12 guards and 109 of the 123 riflemen joined the troop, armed by the arsenal that de Ronke had under his supervision (Hedemann 1935c; Śliwiński 1932; NHARBG 1832). The warfare lasted a relatively short time, and after several skirmishes inside the forest and on its outskirts, (the biggest battle took place in Hajnowszczyzna village), the uprising died out in BPF in June-July 1831. All of the forest guards and forstmeister de Ronke fled from the forest, and arrest warrants were sent after them (Pomarański 1925). Documents detailed the participation of forest personnel in the uprising, assessing also their previous record and accountability (NHARBG 1832). In the end, at least five of the guards were arrested (Hedemann 1935c) and majority of the rest were forced into exile. The Szretter family went to France, and de Ronke wandered between Prussia, France, and Switzerland (see Box. 4.1). Grodno Governor Mikhail Muravyov (later known as Muravyov-Vilensky and *Wieszatiel* [Hangman] for his forceful suppression of the 1863 uprising) requested to have all guards and lower-level forest personnel removed from their posts in the aftermath of the uprising. This was done by 1832 (Hedemann 1935c). The final note of the uprising in BPF was the rumour spread in the winter of 1833 that the rebels had returned to the forest. The rumour itself originated from the wife of one of the former forest guards, Anna Simson. She allegedly saw rebels, including de Ronke, organize food and weapon stores in the forest to start a new insurgence. The authorities reacted swiftly and sent the senior forester of the Grodno Province with a military unit to BPF. During several weeks of investigation, no traces of insurgents were discovered (NHARBG 1833–1835).

The failure of the November Uprising was an important pause in Polish history, but also in the history of scientific recognition of Białowieża's forest. Among numerous acts of repression after 1831, closing the Vilnius University, was one seemingly not the most devastating, but in fact with profound and long-lasting effects. An entire group of naturalists, who had started to discover BPF as a perfect place to study both European bison and also the biological diversity of the forest itself, was dispersed. The library and natural history collection were moved to Kiev. The whole education system in Grodno Province was crippled by a decreasing the number of secondary schools, which could have negatively affected not only the level of education in the region, but also reduced the employment opportunities for a naturalist. It can hardly be considered a coincidence that during the next few decades scientific or research-based publications on BPF appeared only sporadically, and the forest was the subject of only popular writings or official, statistical or administrative documents.

Although the personnel tied directly to the Polish traditional management of the forest were mostly removed from their posts in the aftermath of the events of 1831, at least some lower-level personnel were allowed to return to their previous duties (NHARBG 1832), and the management of BPF did not change much. New forest personnel were quickly recruited, mainly from the retired military from the Russian or Baltic provinces, and, "…*among them people completely unprepared*" (Karcov 1903). Sources from the 1830s document the prolongation of the period of chaotic administrative efforts to finally introduce effective forest management, but with little result—even the task of measuring and mapping the forest was not achieved (RSHA 1837–1838). The end of this period can be drawn in the year 1837, when the Ministry of State Domains was created. Following that, in 1839–1840, state domain management was separated from the Treasury Chambers into new provincial structures, Chambers of State Domains. The new Ministry of State Domains had ambitions to transform the management of state property (including state forests with BPF among them) on a new "rational" or "scientific" basis (Forestry Department 1898; Ministry of State Domains 1888).

4.3 Material Imprints: Reshaping the Map of the Forest

The period 1795–1837 brought two essential changes to the general area and to the border of BPF: (1) removal of one-thirteenth of its districts in the very beginning of Russian rule over the area, and (2) by the end of the period the incorporation of a forest that was never historically a part of BPF.

The chaotic first years of Russian reign over BPF exerted a long-lasting impact on the Forest. In particular, the thirteenth forest district (*Kraśniczańska*), a gift to Count Rumyantsev, was deforested, leaving a characteristic triangle on the eastern border of the forest—one of the most visible changes in the outline of the forest in the last two centuries (Mikusińska et al. 2013; see Fig. 1.1). Comparison of the two earliest known Russian maps of BPF shows how this section of the forest was separated from state properties. The first map, most probably based on a Polish map and prepared soon after the third partition, shows the thirteenth forest district as a part of BPF. The second map, dated 1799, shows an empty space in the place of previous Kraśniczańska district (Fig. 4.2), as it was no longer under state administration. When this part was deforested is not completely clear, yet the attempt at selling the Kraśniczańska district back to the state properties by Mikhail Rumyantsev in 1803. This was accompanied by the information that the district still had "wild cows" (NHARBG 1803) suggesting that by this time the area was at least partially forested. Hedemann (1939), based on unlisted archival sources, wrote that Mikhail Rumyantsev "…*promptly got rid of it, selling parts to different heirs, even Jews, and they from their side took care of the tree stands: the clank of axes and reports of bison flushed out from there meander in the archival sources from the beginning of the nineteenth century*".

The map of 1799 was prepared by an officer of the Naval Ministry (RSHA 1808B), in itself being evidence of the imperial military's interest in Białowieża's shipbuilding resources of Scots pine and oak. (This matter returned several years later—see below).

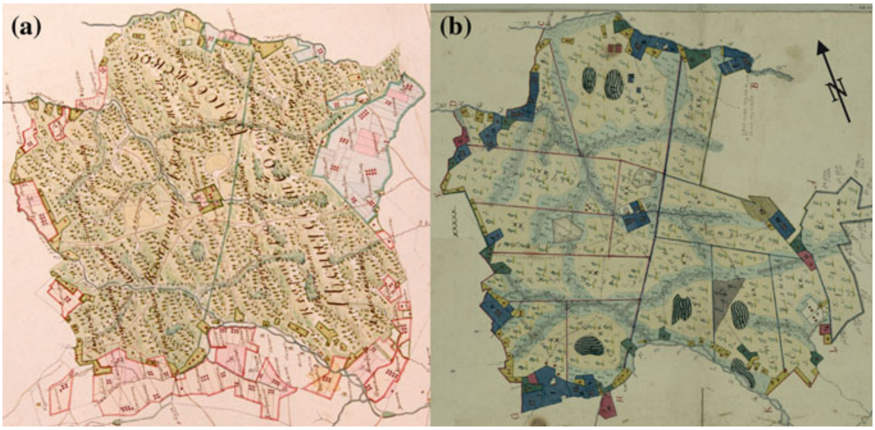

Fig. 4.2 Comparison of the earliest known maps of BPF drawn after the third partition of Poland: **a** Not dated, most probably prepared in 1795–1796, based on an older map from the Polish period and a part of a larger "Map of Brześć District currently lying within the boundaries of the Russian Empire" (CAHR undated). **b** Dated 1799 and entitled "Plan of Białowieża Forest showing which districts are within" (RSHA 1808B). Both maps were cropped and rotated for the purposes of fitting them together and clear presentation of BPF's area and outline

After the fall of November Uprising, Count Tadeusz Tyszkiewicz, a retired Brigadier General of the Duchy of Warsaw Army, settled in Paris and was punished for his active participation in the uprising by the confiscation of his estates. Since Tyszkiewicz was the owner of the Świsłocz estate and Świsłocz Forest bordering BPF from the north-eastern side, the Forest was also confiscated and put under the state administration (RSHA 1836–1847).

Other family members who did not join the uprising tried to challenge the confiscation, arguing that Tadeusz Tyszkiewicz was not the sole owner, but only the manager of the family property. Although the case lasted a long time, the Tyszkiewicz family were not successful in their attempt and lost the entire property (RSHA 1836–1847). In the villages of the Świsłocz estate there were about five thousand serfs, a few small factories, a brewery, a school, and even a hospital with a pharmacy. Having lost a reliable manager, the estate gradually fell into decay (RSHA 1838–1858).

4.4 Cultural Heritage of the Period: Local Knowledge on Forest Habitats and Białowieża's Legends

The memoirs of Piotr Szretter, a participant of the November Uprising and a member of a family of forest guards with traditions deeply rooted in the Polish period in BPF's history, contain a description of different forest habitats with their traditional names: "*bór, ostęp, bagno, ols, hrud czyli grąd, dolina, chruśniaki, łozy, gaje*" (Szretter 1893). Although Szretter brings back his memories from the beginning of the nineteenth century in his book, those names and their association with different forest types are much older. Names like *hrud*, *bór* or *gaj* already appear in 1559 in the inventory of royal forests (Revizya 1559). This centuries-old forest typology is also connected with traditional explanations of those environments, usually naïve and anthropomorphic. For example, describing coniferous forest (*bór*), Szretter writes: "*These splendid pine trees do not accept company of any skimpy tree, only those that bring fruit or greater use as: oaks, apple trees, birches, rowan,* etc." (Szretter 1893). Yet it was this naïve interpretation of forest habitat diversity and folk tradition of determining plant communities, it made its way to the new discipline of science—phytosociology. Its co-creator, Józef Paczoski, fully recognized this legacy: "*A certain notion about the types of plant cover existed for ages. Even the wild hunter tribes had to be aware of this as game, the basis of their existence, is connected with certain types of vegetation. The notion of bór, grud and oles dates back at least three hundred years in Białowieża*" (Paczoski 1927b).

In the turn of the nineteenth and twentieth centuries, the vegetation studies which Paczoski proposed to be named phytosociology (Paczoski 1896; Maycock 1967; Daszkiewicz 2004) attracted the attention of a growing number of naturalists. Already in the 1890s, Józef Paczoski, involved in cooperation with Polish and Russian botanists, speculated on the patterns shaping plant communities and discussed topics of the new scientific discipline (Paczoski 1896; Fedotova 2012). Many theorists of forest typology were also interested in traditional folk terms used to describe different forest habitats in BPF—for this reason terms like "*bór, grud and oles*" made their way both to the academic discipline—phytosociology, as well as to the applied one—forestry science (Genko 1903; Kründer 1909; Paczoski 1927a, b, 1928, 1930).

In this period of BPF's history (1796–1837), another interesting cultural phenomenon was noted for the first time in written sources, that of local legends. The process of building up a catalogue of folk tales, myths, and fables connected with the forest, its inhabitants, and royal visits, etc., most probably started much earlier. Centuries of royal protection had saved not only the forest canopy but also traces of past human presence. In BPF, remnants of human activity that would normally be ploughed a thousand times and over time levelled back into the earth, survived virtually untouched and this holds true to the present day (Krasnodębski et al. 2011). Preserved features include ground barrows and stone circles, left by ancient and mediaeval dwellers of the forest. In later centuries these were ascribed to different historical periods (for example, barrows treated as graves from the Napoleonic period) and as the basis for weaving local legends.

Already in 1826, Juliusz Brincken described such a legend, connected with Zamczysko Range: "*It is more than probable that in the past in this place there was a hunting palace of Polish kings and that the castle—having white towers [białe wieże]—gave the name to the village and the entire forest. Many wars that rumbled through this country have destroyed the ruins of this palace. In spite of this, it is still observed that badgers throw fragments of walls and even dishes from their dens sometimes*" (Brincken 1826). This legend of an old castle in the depths of the forest was rooted enough for the occasional visitor like Brincken to hear about it and also for the locals to firmly believe it. In 1827, the landowner of Tuszemla village and local judge, Antoni Skrodzki, applied for an official permission to excavate a part of the forest called Obołonie (adjacent to Zamczysko) in search for treasure (NHARBG 1827). In 1824, he learned about the treasure hidden in an underground dungeon from one of Tuszemla's inhabitants, peasant Maksym Niedźwiedzki, who accidentally discovered it while collecting mushrooms. Niedźwiedzki, intimidated by Skrodzki, revealed the place, but the landowner did not find the treasure there. Then Niedźwiedzki showed another place, and a third one, but none of them held the trove. Until that time, the news of exploration spread so widely that Skrodzki, so as not to be accused of wilful intrusion to the forest, sent a letter informing about the hidden treasure to the Grand Duke Konstanty in Warsaw. At the same time, Niedźwiedzki went to Grodno and described his findings to the Governor of Grodno Province, Mikhail Bobiatyński, adding that he barricaded the entrance to the dungeon with boulders (NHARBG 1827). The Governor found Niedźwiedzki's testimony convincing, went to the forest himself and organized excavations in the place pointed out by the peasant, in the presence of forstmeister Eugeniusz de Ronke. The dig revealed that it was impossible that a deep stone cellar was ever there, as trees have been growing in the spot for at least 200 years. No bricks or fragments of them were found, only stones and human bones among them. Disappointed, Bobiatyński became interested in an informer who caused all this confusion and learned from his wife that Maksym Niedźwiedzki started to neglect the household and avoid any work several years ago. He had devoted all his time to the church and walks in the wood, thus condemning his wife and six children to poverty and hunger. The Governor concluded that the treasure was a legend passionately shared by all the locals, including Skrodzki, but which had no basis in reality (NHARBG 1827).

The eyewitness, Eugeniusz de Ronke, added more details to this story, referring to Brinckens earlier description that a, "*Place called Zamczysko among common people is a graveyard, possibly pagan one, which we found out while digging the hill in 1825* [most probably a mistake, should be 1827] *in the presence of J. W. Bobiatyński, Governor of Grodno Provice. We discovered a large number of bones of people regularly buried, and under the head of each dead a small pot. However, despite all efforts, we failed to find any traces of masonry or crushed stone, which the badgers digging the burrows are supposed to throw out*" (de Ronke 1830). Similarly, Jarocki (1830a) described this place: "*Instead of the treasure or at least foundations of the castle walls, only the graves of regularly buried people were found there, of which there had been no mention among local tales. They date back perhaps to pagan times, because under the head of each skeleton there was a small pot, or so-called tear-box, which at that time, as history tells, was used to gather tears of relatives going to the funeral after and placed in the grave as the last proof of their attachment and respect*".

The process of building local lore and inventing legends has continued ever since, and is still in effect nowadays, with new places like "Place of power" ("*Miejsce mocy*", believed to hold a mystical energy) appearing on the map of tourist attractions of BPF. Nevertheless, some of the legends are verified by modern research—as it happened in the case of Zamczysko Range. Archaeological excavations conducted in 2003 revealed an early mediaeval Slavonic cemetery with inhumation burials under flat graves paved with stones, which were disturbed and removed by nineteenth-century and modern treasure hunters (Krasnodębski et al. 2005).

4.5 The Interest of Naturalists, Foresters and Travellers of the First Third of the Nineteenth Century in BPF

By the beginning of the nineteenth century, the state of preservation of BPF with its richness of forest environments, wildlife and relatively unaltered natural landscape, acted as a magnet in drawing naturalists and travellers. A series of works published in the early nineteenth century, was based on personal observations, information shared by forest personnel (guards and riflemen remembering times of Polish kings still served in BPF), on specimens at least obtained from Białowieża, entered the scientific discussion. Mainly due to political circumstances, this became the basis of the knowledge on BPF and natural forests until the end of the nineteenth century. Nevertheless, the first monograph of the Forest was written not by a scholar, but by a forester, Juliusz Karol Holte von den Brincken (1790–1846). He was born in the Duchy of Brunswick, studied forestry at the University of Göttingen, and became the head of the forestry in the Kingdom of Poland in 1818. His duty was to be superintendent all forestry administration on the kingdom, as well as lecturing on forest management and game management in the Special Forest School at the University of Warsaw. Additionally, Brincken served as an editor-in-chief of forestry journal “Sylwan” which was founded in 1820 (Daszkiewicz et al. 2004).

His most important work was “Mémoire Descriptif sur la Forêt Impériale de Białowieża, en Lithuanie”, description of the imperial forest of Białowieża published in 1826 and dedicated to Tsar Nicholas I (Brincken 1826). The book was written after two visits to BPF in September 1821 and February 1823. Brincken hunted during both visits with a special permission from the Tsar; each time killing one bison. His hosts were forstmeisters of BPF—in 1821 one who remained anonymous (most probably it was Karl Brenner, BPF forstmeister in the late 1810s—early 1820s), and in 1823 Eugeniusz de Ronke. They shared their knowledge of the forest with Brincken. In the book, Brincken presented the nature and history, physiographic conditions, and administration of BPF, sketching also a plan for future administration which focused on heavy felling of timber and substantial plantation forestry. He recognized the unique character of BPF as the last primeval forest in Europe and the last refuge of European bison. However, in his eyes, this was an obvious sign of the backwardness of the civilization of the former Polish-Lithuanian Commonwealth. Brincken’s work is full of inaccuracies, including zoological, botanical and even geographical mistakes (e.g. the hand-drawn map of the forest published by Brincken has distorted geometry, Fig. 4.3), which became the catalyst for many corrections in the works of other authors. Apart from the natural content, Brincken’s book included social observations showing inhabitants of the forest as bearish, raw and uncivilized, which was then widely repeated in popular works touching upon Białowieża. This point of view was

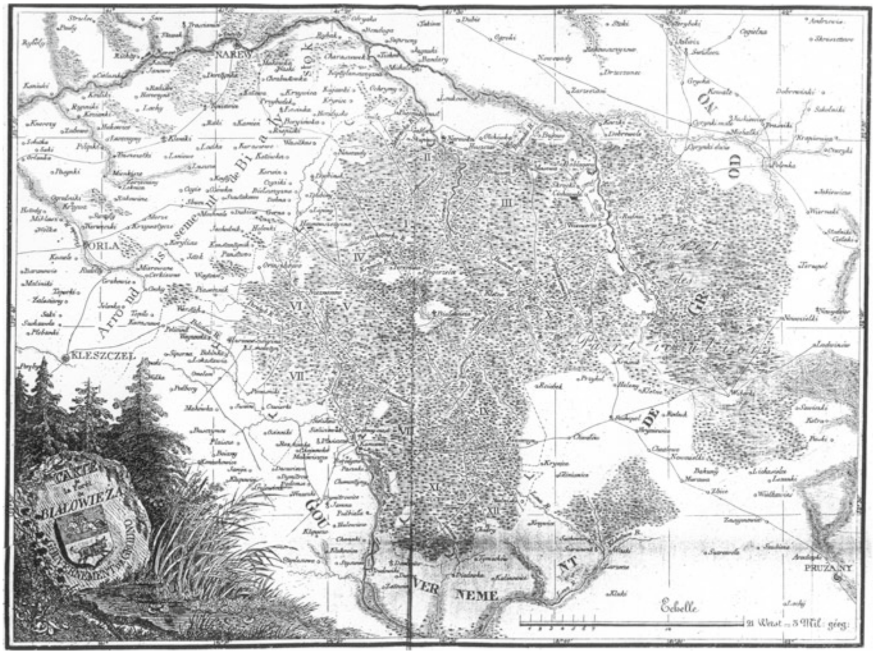

Fig. 4.3 Jakub Sokołowski's hand-drawn map of BPF which played only an illustrative role in Brincken's book (Brincken 1826)

also criticized and corrected by subsequent publications of naturalists and travellers visiting BPF after Brincken. Despite its weaknesses, Brincken's "Mémoire Descriptif sur la Forêt Impériale…" was the first monograph of BPF of such scale, triggering series of publications relating to its mistakes, and in result significantly developing the knowledge on the Forest.

One of such works was Stanisław Batys Górski's (1802–1864) "*O roślinach Żubrom upodobanych, jakoteż innych w Puszczy Białowiezkey*" ("On grasses preferred by European bison and others in Białowieża Primeval Forest"), published in 1829 (Górski 1829). Górski, a graduate of the Vilnius University, in 1829–1832, directed the Botanical Garden of Vilnius University. Instructed by his scientific supervisor, botanist Friedrich Wolfgang (1775–1859), Górski undertook an expedition to BPF, being the first botanist to study the forest. "On grasses preferred by European bison…" corrected Brincken's errors, and additionally demonstrated the diversity of Białowieża's flora as he discovered during his two visits in 1822 and 1826. He also rejected the then popular theory of how European bison fed on a specific grass endemic to BPF.

In the same year as Górski, a professor of zoology from Warsaw University, Feliks Paweł Jarocki (1790–1865), also visited BPF. Holding a special permission from Tsar Alexander I, he came to the forest to hunt two bison for the Zoological Cabinet of his university. After his visit, he published a work entitled "*O Puszczy Białowieskiej* *i o celniejszych w niej zwierzętach*…" ("On Białowieża Forest and its more accurate animals…") (Jarocki 1830a), incorporating descriptions of the forest and its animals, and also local traditions like forest beekeeping. While Jarocki corrected many of Brincken's mistakes, he himself relied heavily on a small article written by forstmeister Eugeniusz de Ronke (de Ronke 1830). This article was published in a popular daily newspaper by de Ronke as a direct answer to Brincken's book, rectifying information that contradicted forstmeister's own experience and knowledge. Brincken answered with his own article in the same daily publication, calling de Ronke ignoramus and displaying a sense of superiority (Brincken 1830). The third article in this series, written by Jarocki, fully supported de Ronke's observations (Jarocki 1830b).

In this busy period, another author addressed the issue of Białowieża's unique character. Karl Edward von Eichwald (1795–1876), a naturalist trained in Berlin, Switzerland, England, and France, who was a head of the zoology department at Vilnius University from 1827. In the years 1828–1829, he travelled through the Volyn, Podolia, Kherson, and Lithuania. After his journey, he published "Naturhistorische Skizze von Lithuanien, Volhynien und Podolien", with a small part devoted to BPF (Eichwald 1830). Information provided here is mostly taken from the work of other authors, supplemented with small amount of Eichwald's own observations or Eugeniusz de Ronke's tales. Heavy reliance on other authors' work brought a wave of criticism of Eichwald's monograph, to the point of accusations of plagiarism. This did not affect the fact that "Skizze" was a modern monograph which, published in German, had a wide uptake among the scientific circles of the era. Additionally, the map published by Eichwald (Fig. 4.4) was the first professional map of the forest, obviously based on forestry maps (see Fig. 4.2) and with superior quality when compared to the map published just 4 years prior by Brincken (Fig. 4.3).

Perhaps the most important identifiable role of BPF in the history of science at this time was through the work on European bison of Ludwig Bojanus (1776–1827). A professor at Vilnius University from 1806, he was a pioneer of comparative anatomy, parasitology and veterinary science in Poland and Lithuania. Bojanus authored approximately 70 scientific papers including the anatomy of the *European pond* turtle *Anatome testudinis europae*. This was widely recognized as a scientific masterpiece and one of the best monographs of comparative anatomy of the nineteenth century. However, he was also interested in European bison and in 1825 (until recently the date of publication was cited as 1827, corrected to 1825 by Daszkiewicz and Samojlik 2019) published a work that played a key role in developing an understanding of bison anatomy: "*De uro nostrate eiusque sceleto commentatio scripsit et bovis primigenii sceleto auxit*" (Bojanus 1825). This work, in addition to a detailed description of the bison's skeleton and skull, contains the first descriptions of two species (1) aurochs *Bos primigenius* and (2) steppe bison *Bison priscus* in accordance with the systematics of Linnaeus and the principles of the modern zoological nomenclature. Bojanus delivered a detailed description of European bison with special attention paid to the skeleton and reflected on the debate as to whether European bison and aurochs were separate species, stating that two different *Bovidae* species existed in Europe in historical times. Bojanus analysed historical sources and took

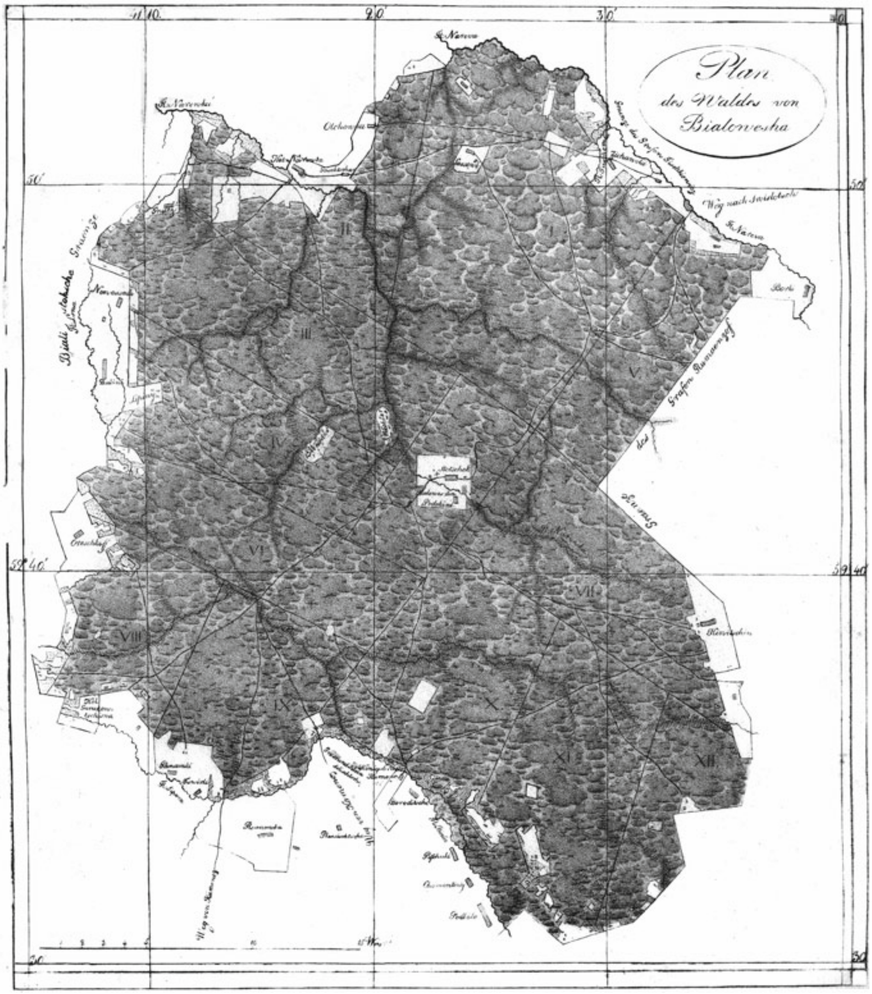

Fig. 4.4 Map published in 1830 by Karl Eichwald in "Naturhistorische Skizze von Lithuanien, Volhynien und Podolien" (Eichwald 1830)

into account more modern works, e.g. observations of Jean Emmanuel Gilibert. Apart from that, he used two bison obtained with permission and of Tsar Alexander I as a primary source of his observations and measurements. Bojanus was undoubtedly the first scholar who had such a rich research material on the bison at his disposal. Being rather a "cabinet scholar" than a field naturalist, it is doubtful whether he ever visited BPF himself. However, it is Bojanus's work that popularized BPF as European bison habitat in the scientific circles of Europe.

Bojanus not only delivered a detailed anatomical description of European bison but also discussed the matter of species' survival only in BPF, ascribing this to the process of gradual destruction of bison habitat: "*Because of the growing agriculture, these animals were surrounded by cultivated fields from everywhere, and exterminated by constant hunting. After felling forests near rivers in temperate zones of Europe, they were deprived of suitable places where they could survive.* As *they could not escape to the north due to severe winter climate not providing them with sufficient food, they sought the last refuge in the vast forests of Poland beyond the Vistula river. They would probably not have chosen such harsh conditions if they were not forced to do so. They would also gradually die out due to lack of food without great care of people, who, after banning any bison hunts, prevent this shortage of food by accumulating haystacks as winter fodder on forest clearings*" (Bojanus 1825).

Written in Latin, a language then widely known and used in the world of science, published in a prestigious and widely read "*Nova Acta Physico-Medica Academiae Cesareae Leopoldino-Carolinae Naturae Curiosum*", Bojanus' work was soon a part of the accepted scientific literature. This undoubtedly contributed to the dissemination of knowledge about European bison and their habitat in BPF.

Box 4.1 BPF forstmeister and insurrectionist Eugeniusz de Ronke

Eugeniusz de Ronke (other spelling: Ronca, Ronka, Ronko) (circa 1798–1875), descendant of a Swiss family settled in Russia, received training in forestry in the Forestry Institute in St. Petersburg. Together with his brother Vasily de Ronke (born around 1800) and cousin Ludwig de Ronke (born around 1796), also trained as foresters in Forestry Institute, he was assigned to the state forests of Grodno Province (RSHA 1810–1821). In 1822, he was appointed as the forstmeister of BPF, with his quarters in Królowy Most. He married Aniela Wesołowska, daughter of one of the forest guards, and apparently influenced by his new family, became Polonized. He was actively involved in the management of BPF and the conservation of European bison; and acquired a vast knowledge of the forest itself, its species diversity, and the behaviour of the associated wildlife. His name appears on a series of documents connected with the organization of Julius Brincken's second visit to BPF in 1823 (the first one took place in 1821, when the forstmeister of BPF was Karl Brenner, RSHA 1810–1821). Brincken was a prominent figure in the Russian administration, serving as the head of forests of the Kingdom of Poland, and as such has received a special permission to capture two bison in BPF. In 1821, Brincken failed to capture bison and in 1823, de Ronke opposed the idea of capturing live bison, arguing that "*methods of catching two young bison* (…) *proposed by baron Brincken are in both versions, i.e. by hand and using nets, inconvenient and dangerous for people and bison*" (NHARBG 1821–1824). Brincken eventually hunted one bison in 1823 (Brincken 1826).

De Ronke hosted Brincken in 1823 in his quarters in Królowy Most, and later served as host also to other travellers and naturalists visiting the forest: Feliks Jarocki, professor of Warsaw University in 1826 (Jarocki 1830a), Edward Eichwald, professor of Vilnius University in 1829 (Eichwald 1830), and most probably botanist Stanisław Górski in 1823 and 1826. De Ronke shared his knowledge with all his guests, helping them especially in finding, observing, and in some cases also hunting bison. After Brincken published his book on BPF (Brincken 1826), de Ronke corrected his mistakes in a newspaper article (de Ronke 1830). This demonstrated his vast knowledge of the forest and he also defended the forest communities and their traditions, which were neglected and demeaned in Brincken's book. De Ronke's explanations show him as having a lively interest in the daily life and new discoveries in the forest. He was the first to inform wider public about the unprofessional, yet very impetuous archaeological excavations in BPF in 1825 when a band of soldiers dug through a hill in Zamczysko forest range in search for a treasure hidden there according to local gossip. They simply found old burial sites with no treasure at all (de Ronke 1830; NHARBG 1827). Zamczysko was later professionally excavated and identified as and eleventh-century Slavonic cemetery (Krasnodębski et al. 2005).

De Ronke actively participated in the November Uprising "*sacrificing willingly a perfect post, estate, children, bright future, even beloved wife*" (Chałupka 2008). After the fall of the uprising he emigrated to France and then to Switzerland, trying to make his living in forestry. He attempted to return to Poland by foot, through Bavarian and Czech lands, and was captured in the Grand Duchy of Posen by the Prussian police and deported to France. His second attempt to get to the Grand Duchy of Posen ended with deportation again, but the third time turned out to be lucky. He settled in Miłosław in 1840, and took over the management of state and private forests there. He also shared his knowledge by writing forestry handbook and articles. He never returned to BPF.

Box 4.2 Jakub Sokołowski's depictions of the forest and its dwellers

Despite the forest of Białowieża being perceived as an interesting destination for naturalists' travels by the end of the eighteenth century, depictions of the Forest from this period are scarce. After Muntz's drawings (see Chap. 3), there are no known artistic works of Białowieża until 1821. In this year, Juliusz Brincken visited BPF for the first time, accompanied by Jakub Sokołowski (1784–1837). The latter was an illustrator and painter, with whom Brincken collaborated in his forestry journal "Sylwan" (published since 1820). Sokołowski was hired to document Brincken's visit and to prepare illustrations for Brincken's book, "Mémoire descriptif sur la forêt impériale de Białowieża en Lithuanie" (1826). Sokołowski prepared five figures for the book depicting a bison, a moose, an overview of Białowieża village, a scene with hunting party departing to the forest (Fig. 4.5), and a map of BPF (Fig. 4.3). During his stay

Fig. 4.5 Jakub Sokołowski's "Train de chasse—departure for the hunt in Białowieża", a hand-coloured aquatint from the collection of National Museum in Warsaw (NMW 1826). Black-and-white version of this drawing was printed in Brincken (1826). The drawing shows a scene described by Brincken himself: "*Hunting in the forest is very interesting for a foreigner hunting here for the first time. Imagine a group of over a hundred riflemen in grey uniforms with green ornaments. Some are on horseback, others on foot. Some of them keep packs of dogs. Others blow horns when approaching the forest. A group of several hundred peasants—beaters—can be seen from the side of the settlement. They form a strange and very gloomy gathering—all in brown clothes, bast shoes, wild looks. The guards are in charge of the gathering. The hunting retinue, which we present in detail in the illustration, usually consists of 40–50 small horse carts, sleighs in winter, pedestrians and horse riflemen drive in a fixed order after receiving a signal*" (Brincken 1826)

in Białowieża in Autumn 1821, Sokołowski made many more drawings, as sketches for his book illustrations and also original, small ones, showing various scenes e.g. a drunken forester, a young beater before the hunt or a small hut inside the forest (Fig. 4.6). Known mainly by his caricatures of higher society circles in Warsaw, he employed his talent in showing the dwellers

Fig. 4.6 Jakub Sokołowski's gouache "Hut in Białowieża Primeval Forest" from the collection of National Museum in Kraków (NMK undated)

of the forest, both in crowd scenes with hunters and beaters, and in smaller drawings of different personalities. His nature-based works have issues like distorted proportions and anatomical incorrectness of the depicted animals. But most informative is most probably the gouache painting with a small forest hut (Fig. 4.6). The drawing shows a glade with a small patch of pasture with cows, a vegetable garden and a thatched-roof house surrounded by dense forest. There is a pine tree in the foreground, with an interesting detail—a log with a beehive is attached to the tree, probably showing another form of traditional forest beekeeping.

Box 4.3 Miska—mysterious European bison of the Napoleonic period

European bison achieved a status of legendary beast by the end of the eighteenth century, and nineteenth-century naturalists were ready to invest considerable effort in attempts to observe and describe specimens of the species. Especially during the conquest of Europe by Napoleon's army there was an additional troop of naturalists for whom observing and obtaining an European bison became a matter of honour. Among numerous natural history collections confiscated in the occupied countries, there was a skeleton of a European bison from Vienna, called "Miska". Kept in Schönbrunn since 1788, it was officially of Transylvanian origin—yet the population in Hungary went extinct as early as the sixteenth century (Sztolcman 1924). Alive, Miska was observed by Marcel de Serres (1783–1862), French naturalist and an inspector of science and industry in Austria under French occupation. In his "Description of crafts and manufactures in the Empire of Austria", he included a description of Miska: "*many rare animals lived in the menagerie in Schönbrunn, to quote only the wild buffalo of Transylvania, known to naturalists under the name of bison* [in original, mistakenly: *aurochs*] (*Bos Urus*) (…). *What makes this description even more interesting is the fact that remains of the same individual I have seen alive have been deposited in the museum in Paris*" (de Serres 1815). On the following pages, de Serres detailed his observations of European bison. He came back to the subject of Miska many years later, in his 1845 work on the causes of migration of animals, but this time completely changing the origin of the animal from Vienna: "*this animal lived in Poland only in the great Białowieża Forest, which, thanks to the authorities' care, still had some individuals. In Schönbrunn near Vienna in Austria (in 1809), I saw a European bison captured few years earlier in Białowieża Forest*" (de Serres 1842).

Remains of Miska became famous when Georges Cuvier (1739–1862), one of the most important nineteenth-century figures in natural sciences and the founding father of comparative anatomy and palaeontology, posted its anatomical description in his "Recherches sur les ossements fossils" (Cuvier 1812). Cuvier compared the skeleton from Vienna with the American bison one, obtained from the United States. The French naturalist noted European bison's origins; its remains were kept in the Natural History Museum in Paris, where it was brought from the menagerie at Schönbrunn. In the nineteenth-century encyclopaedias and dictionaries, Cuvier's description was cited usually with an additional information, taken from British physician's Robert Townson "Travels in Hungary with a short account of Vienna in the year 1793", in which the author described bison kept in Schönbrunn: "*It was caught in Poland at a very young age and today it is completely tame*" (Townson 1803).

Another mysterious link between the Napoleonic expansion and Białowieża's bison is hidden in the archives of the Natural History Museum in Paris. In 1945, zoologist and specialist in the field of comparative anatomy Jacques Millot (1897–1980) published a report on his search for the oldest European bison specimens in the museum's collection (Millot 1945). After the search, Millot concluded that the bison from Vienna is absent both in the osteological collection and in laboratory records, which was odd due to the importance of this particular specimen in the history of science. But Millot's search yielded another surprising result: in the museum's collection of European bison specimens one is listed as "female from Lithuania. Emperor Napoleon". Careful investigation revealed that this female is indeed a male, but Millot's hypothesis that this is in fact Miska was proved wrong when he compared the Napoleonic bison with the one described by Cuvier—they were different. This bison was most probably hunted and send to Paris during French offensive of 1812.

Current research in the archives of the MNHN and the French National Archives brought no further information about the fate of Vienna bison, nor the location of Miska's remains. Although the origin of this bison remains a mystery, several traces connecting it with Lithuanian forests seem more plausible than the version that it was the last specimen of a species from Transylvania.

Box 4.4 Goutweed, tobacco and snuff in BPF

The sixteenth-century document describing the duties and prerogatives of royal foresters (Hedemann 1936) listed the right to collect various mushrooms, nuts and plants, among them *śnitka*, goutweed (*Aegopodium podagraria*). This plant is very common and easy to harvest, and it is also among the preferred herb-layer plants in red and roe deer diet (Gębczyńska 1980). In past, it had a medicinal use in podagra (gout) treatment and it was used as an edible plant since Roman period. Goutweed salads were prepared from the young leaves, and the older ones were cooked. Similarly to hogweed (see Box. 3.3), soups were also prepared using goutweed leaves. Consumption of this plant was especially important during periods of famine, and it was a valuable source of vitamins and microelements. Modern research also confirms the antifungal and antibiotic effect of substances contained in its leaves (Prior et al. 2007).

In his description of a type of the forest—*hruśniak*—Piotr Szretter noted that it is "*there where a poor peasant finds forest tobacco, and snuff-takers find a famous fragrant grass called tomka*" (Szretter 1893). Most probably Szretter referred here to an old practice that he knew from his experience, which today is one of the forgotten ethnobotanical traditions. Stanisław Bonifacy Jundziłł described tobacco as "*initially foreign but so assimilated that it reproduces itself from seeds without any effort. Peasants use the duly cultivated leaves to produce strong and unpleasant snuff*" (Jundziłł 1791). If we combine this description with the information shared by Franciszek Błoński, Karol Drymmer and Antoni Ejsmond after their botanical excursion to BPF in 1887 that tobacco is grown in rural gardens in Łozice and Teremiski villages on the border or inside the Forest, it is reasonable to assume the "forest tobacco" was wild-grown tobacco harvested by locals (Błoński et al. 1888).

Information about the snuff-takers collecting the famous *tomka* (hornwort or vernal grass, *Anthoxanthum*) is no less interesting. Jundziłł noted that its "*ear of grain, leaves, roots smell nice, and grinded with simple tobacco eliminates the unpleasant odour of snuff*" (Jundziłł 1791). The author also named other plants used as a substitute for snuff or as a snuff flavouring factor in the Grand Duchy of Lithuania: pasqueflower (*Pulsatilla*) is described as "*an acute plant*

(…) powdered and applied causes sneezing". Common hedge-nettle or wood betony (*Stachys officinalis*), "*strengthens the stomach*", and also, "grinded to dust and applied like snuff causes sneezing " (Jundziłł 1791). The latter, according to folk beliefs, had a variety of alternative applications: eaten on an empty stomach was supposed to protect against alcohol, but also protected against black magic and kept bad charms away.

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[NHARBG] 1809–1810. O prichinenii ubytkov Belovezhskim krest'janam mushketerskimi polkami (On causing damage to the Białowieża peasants by musketeer regiments). F. 1, O. 1, No 160 (in Russian)
[NHARBG] 1809–1824. Ob osvobozhdenii ot dolzhnosti Trejdena i vzyskanii s nego deneg za vyrublennyj kazennyj les (On dismissal of Trayden and collecting money from him for cutting down state woods). F. 2, O. 2, No 243 (in Russian)
[NHARBG] 1813. Po zhalobe forshtmejstera Belovezhskoj pushhi Gorbatovskogo na strazhnika Prushaka v podstrekatel'stve drugih strazhnikov na nepodchinenii emu (On the complaint of forstmeister of BPF Gorbatovsky on the forest guard Prushak in instigation of other guards to disobey him). F. 1, O. 1, No 321 (in Russian)
[NHARBG] 1816A. Po zhalobe strel'cov Seduna i Zhaka na lishenie ih imushhestva i vyselenii iz ih domov forshtmejsterom Gorbatovskim (On the complaint of riflemen Sedun 4 and Zhak about the deprivation of their property and eviction from their homes by the forstmeister Gorbatovsky). F. 1, O. 1, No 683 (in Russian)
[NHARBG] 1816B. Vedomosti o kolichestve pahotnoj zemli i sennyh pokosov, nahodjashhihsja v kazennyh pushhah i aukcionnye listy na prodazhu zubrovoj kozhi (Information on the area of arable and haymaking land in the state forests and the sale of bison pelt). F. 1, O. 1, No 736 (in Russian and Polish)
[NHARBG] 1817. Po zhalobe strel'cov Seduna i Zhaka na lishenie ih imushhestva i vyselenii iz ih domov forshtmejsterom Gorbatovskim (On the complaint of riflemen Sedun and Zhak about the deprivation of their property and eviction from their homes by the forstmeister Gorbatovsky). F. 1, O. 1, No 889 (in Russian)
[NHARBG] 1818. Ob otdache v arendnoe soderzhanie rybnoj lovli po rekam Belovezhskoj pushhi (On renting the right to fish in rivers of BPF). F. 1, O. 1, No 1178 (in Russian)
[NHARBG] 1819–1820. Po zhalobe strel'cov Belovezhskoj pushhi na forshtmejstera Brennera, pereselivshego ih v drugie mesta (On the complaint of riflemen from BPF on forstmeister Gagman for resettling them to other places). F. 1, O. 1, No 1869 (in Russian)
[NHARBG] 1820–1825. Ob uvol'nenii Grodnenskogo ober-forshtmejstera Gagmana za porubku kazennogo lesa v Belovezhskoj pushhe (On dismissal of Grodno ober-forstmeister Gagman from office for felling of state forest in BPF). F. 1, O. 1, No 2443 (in Russian)
[NHARBG] 1821–1824. O poimke zubrov v Belovezhskoj pushhe dlja otpravki ih chuchel v Varshavskij muzej (Of the capturing of European bison in BPF to send their pelts to the Warsaw Museum). F. 1, O. 1, No 2555 (in Russian and Polish)
[NHARBG] 1825. O porubke raznymi licami kazennyh lesov, o prodazhe lesomaterialov i po drugim voprosam (On felling of state forests by various persons, on the timber sale and other issues). F. 1, O. 2, No 874 (in Russian)
[NHARBG] 1827. Po prosheniju pomeshhika Skrodskogo proizvesti raskopki v predelah Belovezhskoj pushhe v urochishhe Obolon' dlja obnaruzhenija sokrovishh (At the request of the landowner Skrodzki to excavate Obolon Range in the Białowieża Forest to discover the treasure). F. 6, O. 1, No 262 (in Russian)

[NHARBG] 1829. Pretenzii zhitelej Grodnenskoj gubernii za produkty, vzjatye vojskami v 1812 g. (Claims of residents of the Grodno Province for the products confiscated by the troops in 1812). F. 1, O. 1, No 484 (in Russian)

[NHARBG] 1831. Zhaloby zhitelej na vojska, otobravshih proviant i imushhestvo (Complaints of residents for troops who confiscated food and property). F. 1, O. 3, No 1043 (in Russian)

[NHARBG] 1832. Spiski uchastnikov pol'skogo vosstanija 1831 g. v rajone Belovezhskoj pushhi i spiski chinovnikov lesnogo vedomstva gubernii (Lists of participants of the Polish uprising of 1831 in BPF vicinities and lists of forest officials of the Grodno province). F. 1, O. 27, No 63 (in Russian)

[NHARBG] 1833–1835. O rozyske uchastnikov pol'skogo vosstanija 1830–1831 gg. v Belovezhskoj pushhe (On the wanted list of participants of the Polish uprising 1830-1831 in BPF). F. 1, O. 27, No 350 (in Russian)

[NHARBG] 1834–1835. Po prosheniju krest'jan Krukovskovskj lesnoj dachi ob osvobozhdenii ih ot uplaty nedoimki, vvidu ih razorenija i perepiska o postavkah prodovol'stvija prohodjashhim vojskam v 1831 g. (On the request of the Krukowska Forestry district peasants for exemption from the payment of arrears, due to their ruin and correspondence about the supply of food to passing troops in 1831). F. 1, O. 4, No 290 (in Russian)

[NHARBG] 1843. O predostavlenii svedenij Lesnomu departamentu o kolichestve i sostojanii lesov v Grodnenskoj gub, prinadlezhashhih kazennym vedomstvam i chastnym licam (On providing the information from Forest Department on the number and condition of forests in Grodno Province, owned by government agencies and individuals). F. 1, O. 20, No 963 (in Russian)

[NHARBG] 1846–1850 Ob istreblenii volkov v Grodnenskoj gubernii (On wolves extermination in the Grodno province). F. 1, O. 21. D. 260 (in Russian)

[NHARBG] 1850–1851. Raporty iz zemskih sudov o chisle ubityh volkov po uezdam Grodnenskoj gubernii (Reports from the provincial courts on the number of wolves killed in the districts of Grodno province). F. 1, O. 29, No 687 (in Russian)

[NHARBG] 1867. O reguljarnom otpravlenii krest'jan Grodnenskoj gubernii v Belovezhskuju pushchu dlja istreblenija hishhnyh zverej (On regular delegation of peasants of Grodno Province to Białowieża Forest for extermination of predators). F. 10, O. 14, No 230 (in Russian)

[NMK] undated. Józef Sokołowski, Chata w Puszczy Białowieskiej (Hut in Białowieża Primeval Forest), National Museum in Kraków, NI 10 431, III- r.a. 2092

[NMW] 1826. Jakub Sokołowski, Train de chasse - wyjazd na polowanie w Białowieży (Train de chasse – departure for the hunt in Białowieża), National Museum in Warsaw, Gr. Pol. 2608

[RSAN] 1813–1829. O dubovyh i listvennichnyh lesah, proizrastajushhih v kazennyh dachah Vjatskoj, Kostromskoj, Kurljandskoj, Lifljandskoj, Grodnenskoj, Vilenskoj i Permskoj (On oak and larch forests in state domains of Vyatka, Kostroma, Kurland, Livonia, Grodno, Vilnius and Perm). F. 159, O. 1, No 981 (in Russian)

[RSHA] 1799–1801. O zameshhenii vakantnyh dolzhnostej forshtmejsterov (On vacant forstmeister posts). F. 1594, O. 1, No 38 (in Russian)

[RSHA] 1801–1803. Ob uchrezhdenii v Litovskoj gubernii lesoupravlenija na takom polozhenii, kak sushhestvuet v Kurljandii (On establishment of a forest administration in the Lithuanian Province based on Courland model). F. 1594, O. 2, No 20 (in Russian)

[RSHA] 1802A. Po raportu Lesnogo Departamenta s prilozheniem vedomostej o lesnyh dohodah, o nereshennyh delah, spiskami chinovnikov (Report of the Forest Department with the information on forest income, unsolved cases, lists of personnel). F. 1595, O. 2, No 33 (in Russian)

[RSHA] 1802B. O naznachenii izvestnoho chisla kazennykh krestian dla ukhoda i kormlenii zubrov v Belovezhskoj pushhi (On setting the numer of state peasants for care and feeding of European bison in Białowieża Primeval Forest). F. 1374, O. 5, No 25 (in Russian)

[RSHA] 1805–1811. O pereselenii malozemel'nyh krest'jan iz Belovezhskoj pushhi (On resettling smallholder peasants from Białowieża Forest). F. 1594, O. 1, No 1571 (in Russian)

[RSHA] 1806–1807A. Ob otpuske stroevogo lesa iz kazennoj Belovezhskoj pushhi poljaku Shambeljanu Savanevskomu (On the sale of the timber from state Białowieża Forest to the Pole Chambelain Sawaniewski) F. 1594, O. 1, No 1564 (in Russian)

[RSHA] 1806–1807B. Po rassmotreniu del o pravah raznyh lic v'ezzhat' v kazennye lesa Grodnenskoj gubernii (On access rights to state forests in Grodno province for different persons). F. 1594, O. 1, No 1569 (in Russian and Polish)
[RSHA] 1806–1815. Ob uchrezhdenii lesnoj inspekcii v vos'mi gubernijah, prisoedinennyh ot Pol'shi (On the establishment of a forest inspection in eight provinces incorporated from Poland). F. 1594, O. 1, No 1640 (in Russian)
[RSHA] 1806–1847. O strazhnikah Strizhevskih i Gutovskih (On guards Strzyzewskis and Gutowskis). F. 387, O. 1, No 635 (in Russian)
[RSHA] 1808A. O predostavlenii kvartiry Belovezhskomu fortshmejsteru, s vedomostjami o zemel'nyh nadelah lesnoj strazhi, senokosnyh lugah i bortevyh derevah (On provoning a house for BPF forstmeister with information on allotments for forest guard, as well as meadows and beehive trees). F. 1594, O. 1, No 1587 (in Russian and Polish)
[RSHA] 1808B. O prisylke k inspektoru po Lesnoj chasti planov Belovezhskoj Pushhi (On sending plans of Białowieża Forest to the forestry auditor). F. 1594, O. 1, No 1641 (in Russian)
[RSHA] 1809. O dachah, imejushhih korabel'nye lesa, po Grodnenskoj gubernii (On forestry districts with trees for shipbuilding in the Grodno Province). F. 1594, O. 1, No 1636 (in Russian)
[RSHA] 1810–1821. Formuljarnye spiski chinovnikov po lesnoj chasti Grodnenskoj gubernii (Formal lists of service of forestry officers in the Grodno Province). F. 379, O. 8, No 93 (in Russian)
[RSHA] 1811–1815. O komandirovanii v gubernii Vilenskuju, Grodnenskuju, Volynskuju, Podol'skuju, Kievskuju i Minskuju dlja opisanija i izmerenija vseh kazennyh 12 imenij 50 zemlemerov (On delegating to Vilnius, Grodno, Volhyn, Podole, Kiev and Minsk provinces fifty surveyors to measure all state estates). F. 1594, O. 1, No 1649 (in Russian)
[RSHA] 1813–1814. O sostavlenii opisanija lesov v gubernijah, gde prohodil neprijatel' (On descriptions of forests in the provinces where the enemy passed). F. 379, O. 4, No 202 (in Russian)
[RSHA] 1813–1829. O zubrah v Belovezhskoj pushhe Grodnenskoj gubernii nahodjashhihsja (About bison in Białowieża Forest of Grodno Province). F. 379, O. 4, No 7 (in Russian)
[RSHA] 1814–1828. O trebuemyh svedenijah po sluchaju opredelenija po Grodnenskoj gubernii strel'cov na mesto umershih, iz kakogo sostojanija, kakie platjat i kuda podati kak strel'cy, ogorodniki i kutniki (On the required information for appointing in Grodno Province new riflemen in the place of the deceased, of what state, and what taxes do they pay and where both riflemen, tenant farmers and crofters). F. 379, O. 4, No 312 (in Russian and Polish)
[RSHA] 1816–1825. O zubrah pereshedshih iz Belovezhskoj pushhe v lesa grafa Tyshkevicha (On European bison that went from Białowieża Forest to count Tyszkiewicz's forests). F. 379, O. 4, No 561 (in Russian)
[RSHA] 1817. Ob otdache v obrok senokosnyh i pahotnyh poljan po Grodnenskoj gubernii (On renting haymaking and arable lands in the state forests of Grodno province). F. 379, O. 4, No 691 (in Russian)
[RSHA] 1817–1829. O pravilah na vydachu biletov dlja vyrubki lesov i zavedenija shnurovyh knig (On the rules for issuing tickets for forest felling and establishing records). F. 379 O. 4, No 662 (in Russian)
[RSHA] 1819–1821. O dostavlenii zubrov v Zootomicheskij kabinet Vilenskogo universiteta (On the acquisition of European bison to the zoological cabinet of the Vilnius University). F. 379, O. 4, No 940 (in Russian)
[RSHA] 1822. O vozvyshenii lesnyh dohodov i o nabljudenii za vyrubkoj lesov (On the rise of forest revenues and on monitoring deforestation). F. 379. O. 4. No 1263 (in Russian)
[RSHA] 1827. O lesnyh dohodah i nedoimkah za 1826 g. po Grodnenskoj gubernii (On forest income and arrears for 1826 in Grodno Province). F. 379, O. 7, No 302 (in Russian)
[RSHA] 1828. O lesnyh dohodah i nedoimkah za 1827 g. po Grodnenskoj gubernii (On forest income and arrears for 1827 in Grodno Province). F. 379, O. 7, No 303 (in Russian)

[RSHA] 1827–1829. Po ukazu pravitel'stvujushhego Senata o pravah chastnovladel'cev na vezd v Belovezhskie lesa (On the decree of the Governing Senate on the rights of private owners to access Białowieża Forests). F. 379, O. 4, No 1662 (in Russian)
[RSHA] 1827–1833. Po predmetu streljanija volkov v Bialowiezskoi Pusche (On wolves shooting in BPF). F. 1589, O. 3, No 107 (in Russian)
[RSHA] 1829. O lesnyh dohodah i nedoimkah za 1828 g. po Grodnenskoj gubernii (On forest incomes and arrears for 1828 in Grodno Province). F. 379, O. 7, No 304 (in Russian)
[RSHA] 1836–1847. O likvidacii dolgov gr. Faddeja Tyshkevicha v svjazi s konfiskaciej u nego imenija Svisloch' (On the elimination of debt of count Tadeusz Tyszkiewicz in connection with the confiscation of his estate Świsłocz). F. 384, O. 1, No 1636 (in Russian)
[RSHA] 1837A. O revizii del khoziaistvennogo i lesnogo otdelenia Grodnenskoi kazennoi palaty (On the audit of the cases of the Economic and Forest Department of the Grodno State Chamber) Fond 379, O. 8, No 355 (in Russian)
[RSHA] 1837B. O lesnyh dohodah i nedoimkah po Grodnenskoj gubernii za 1837 (Information on forest income and arrears for 1837). F. 379, O. 7, No 314 (in Russian)
[RSHA] 1837–1838. Otchet po lesnoj chasti Grodnenskoj gubernii za 1836–37 gg. (Report on the forest management of the Grodno province for 1836-1837). F. 384, O. 1, No 1111 (in Russian)
[RSHA] 1838–1842. Po predlozheniyu Ministra Gosudarstvennykh Imuchestv o priostanovlenii prodazhi lesa v nekotorykh guberniyakh (On the proposal of the Minister of State Properties on the suspension of the sale of forest in some provinces). F. 387, O. 1, No 171 (in Russian)
[RSHA] 1838–1858 O ljustracii konfiskovannogo imenija Svisloch' (On lustration of the confiscated Świsłocz estate). F. 384, O. 1, No 1920 (in Russian)
[RSHA] 1839–1842. Po zapiske Grodnenskogo lesnichego Brinkmana o sposobah otravlenija volkov i lisic (The note of the Grodno forester Brinkman on poisoning of wolves and foxes). F. 1589, O. 3, No 38 (in Russian)
[RSHA] 1843–1846. Ob obrazcah trav, kotorymi pitajutsja zubry (On the grass species that are preferred by Eurbison). F. 398, O. 7, No. 7, No. 2083 (in Russian)
[RSHA] 1845–1847. Ob udah dlja lovli zverej (On traps for hunting animals). F. 387, O. 1, No 1043 (in Russian)

Chapter 5
Mixed Management Goals (1838–1860)

Abstract This chapter takes us though the period of mid-nineteenth century when the goals of management of Białowieża Primeval Forest (BPF) shifted from economic exploitation to preservation of European bison and creation of hunting reserve. The new Ministry of State Domains (created in 1837) with the new Forestry Department (created in 1843) aimed at combining the protection and exploitation of BPF. By 1849, the first forest inventory plan of BPF was finalized, and the forest was divided into rectangular compartments. However, most of the forest modernization projects, including creating plants, canals and a forestry school in BPF, were rejected by the Forestry Department. Furthermore, instead of the planned clear cuttings, the Forestry Department decided to first remove several thousands of culturally modified trees scattered throughout the forest, described by foresters as "damaged trees". Problems with external contractors selected to extract large quantities of wood from BPF delayed the introduction of clear cuts till the end of the period. The first Tsar's hunt (in October 1860) again made the preservation of European bison population a main goal of BPF management. Few outstanding naturalists visited BPF in this period, but it did not bring any major scientific publication on the Forest or European bison. The important cultural heritage of the mid-nineteenth century was the successful hybridization of European bison and cattle made by Leopold Walicki.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_5

5.1 Summary

The creation of the Ministry of State Domains in 1837 was followed by a change in attitudes towards BPF. At the request of the Department of Ship Forests of the Naval Ministry in 1839, the ban on timber felling was lifted and 2,834 large oaks in the northern part of the forest were selected for cutting. The operation turned out to be so unprofitable that the Naval Ministry abandoned the idea of obtaining trees from BPF. What followed was a period of mixed management goals and different points of view on use of forest's resources. Head forester of the Grodno Province, Dmitry Dolmatov proposed a number of projects to increase revenues from BPF. These involved straightening and connecting rivers with canals, establishing a forestry school in Białowieża, the erection of wood-tar, glass- and other workshops—but all these plans remained only on paper and not in reality. The first management plan for BPF was developed between 1843 and 1848. The initial step towards exploitation was division of the Forest into compartments of two square *versts*. The division meant not only virtual demarcation of borders on the map, but also actual cross-cutting of compartment lines in the forest itself. However, further introduction of rational forestry was hindered by the presence of over 83 thousand (or, based on later estimates, over 300 thousand) of culturally modified pine trees—chopped into in the lower part of the trunk by local community for resinous, kindling material. Scarred, modified, or "worked" trees carrying the memory of past traditional use and human-nature interactions were seen as an obstacle in introducing the modern, "scientific" forestry. In 1849, it was decided that these "damaged trees" needed to be exploited first before the actual management plan could be realized. However, problems with external contractors selected to extract large quantities of wood from different, often remote parts of BPF, prolonged the delays until the end of the period.

The first Tsar's hunt after the acquisition of BPF by the Russian Empire took place in October 1860, marking an important milestone. From this point the management effort gradually shifted from turning BPF into a model of rational forestry towards making it a hunting reserve of Russian imperial family and based on German hunting practice.

The most prominent cultural heritage of the period was the world's first successful experiment with creating bison-cattle hybrids (including a fertile male) conducted by Polish landowner Leopold Walicki using European bison obtained from BPF.

Although in the years 1838 to 1860, BPF was visited by few outstanding naturalists, they published remarks on their journeys rather than scientific descriptions of the Forest: Konstantin Arseniev (1845), Franz Müller (1859), and Jacques de Crévecoeur de Perthes (1859). The first Tsar's hunt in 1860 also became a subject of both official (Fuchs and Zichy 1862) and unofficial publications (Anonymous 1860). Following the spirit of the latter, articles criticizing the fraudulent exploitation of BPF were published, e.g. in "Kolokol" ("Bell", newspaper of the Russian opposition emigration in London, Anonymous 1861a, b, 1862).

5.2 The Historical Background: Continuous Attempts at Introducing Proper Forest Management and the First Tsar's Hunt in BPF

The creation of the Ministry of State Domains changed attitudes towards BPF as the old type of management, deeply rooted in the royal period, was fading into oblivion. The new, "rational" or "German" school of forestry management was to be introduced. But before the newly created ministry was able to realize its plan, officials of the Department of Ship Forests of the Naval Ministry arrived in BPF to check the availability of trees suitable for shipbuilding ready by the end of 1830s. For this reason in 1839, the ban on timber felling was lifted and 2,834 large oaks in the northern part of the forest were selected for cutting. However, finding an entrepreneur who would undertake such a challenge was complicated and time-consuming, with accusations of corruption and plotting (RSAN 1839–1840). Since a third of those oaks showed signs of rot, only 1,983 cut trees were eventually floated. The rest were sold to a private buyer, but eventually rotted on the spot, since there was no way to transport large trunks outside the forest. The Department also ordered 215 mast trees, but only six suitable pines were found and not a single one was cut. Financial calculations showed that the funds invested in acquiring oak trees were much larger than the actual worth of the material and "*the Department of Ship Forests eventually resigned from any acquisition of trees in Białowieża Forest*" (Genko 1902–1903).

It should be noted that in the 1830s, the Grodno Chamber of State Properties had repeatedly received reports from foresters informing them that BPF had many dead, dry or insect-damaged trees. In 1833, the head forester of the Grodno Province observed considerable damage caused by insects and suggested quick sale of such trees, as any fight with pests was deemed aimless (RSHA 1833–1844). The administration tried to organize the extraction and sale of such trees under the control of a "trustworthy" official, but the operation was unprofitable. Neither the Grodno nor St. Petersburg administrations wanted to allow private timber merchants to enter the forest, since they feared that this could harm bison (NHARBG 1836–1839; RSHA 1837A).

The first management plan for BPF was developed by the forestry inventory team working in the forest in the period between 1843 and 1848 (RSHA 1840–1849). It was the first inventory team that worked in the Grodno Province and one of the very first in the empire. BPF was measured and its timber stock was evaluated. In official accounts, BPF had 87,969 *dziesiatinas* (96,108 ha) of forest, 16,138 *dziesiatinas* (17,631 ha) of land under various uses (arable land, haymaking meadows, settlements and pastures), 7,972 *dziesiatinas* (8,710 ha) of wasteland with no use (peats or sands), and 5,512 (6,022 ha) *dziesiatinas* excluded from the forest administration made up of land leased by the state (RSHA 1840–1849, ; Genko 1903; Karcov 1903). The first forest management plan also assessed the composition of the forest: two-thirds of the area was covered by forests dominated by pine, one-fifth by spruce, one-thirtieth by oak, and the rest by various deciduous species. As a rarity, small populations of yew and fir were noted by forestry inventory team.

In accordance with the German model of "rational" forestry, BPF was divided into compartments of 2 square *versts* (~2.3 km^2) (Fig. 5.1). Over a period of five years after the end of the forest inventory, compartments were separated by cross-cut lines (see Sect. 5.2).

From the administrative point of view, in the 1830s, twelve forest sections were incorporated into three Pruzhana forestry districts (name derived from Pruzhana uyezd, i.e. municipality district): 1st Pruzhana Municipality incorporating all smaller areas of forest and groves in the Pruzhana district outside BPF, 2nd Pruzhana Forest

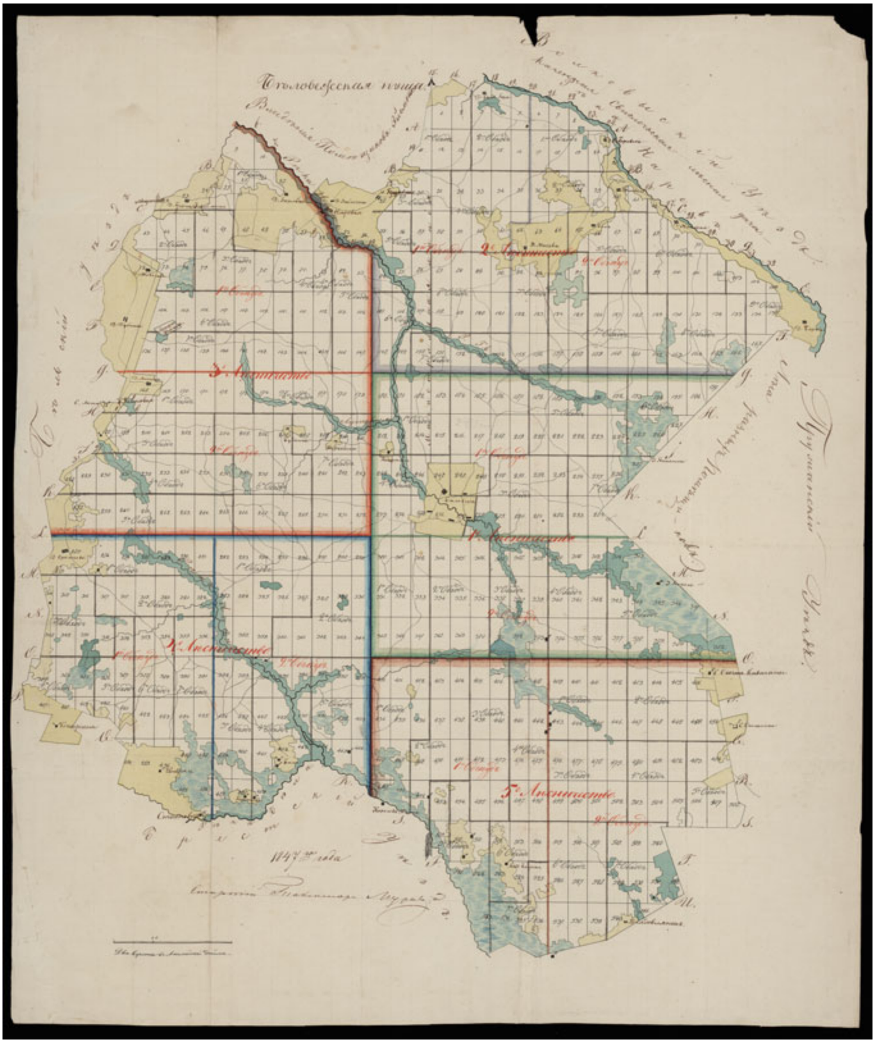

Fig. 5.1 Map of BPF dated to 1847—the first map showing the projected division into rectangular compartments (1 × 2 *versta* ~ 200 ha), with five forest districts marked (RSHA 1840–1849)

District (incorporating old Hajnowska, Leśnianska, Starzyńska and Stołpowiska sections), 3rd Pruzhana Forest District (including Augustowska, Narewska, Browska i Świetliczyńska sections), and 4th Pruzhana Forest District (embracing Okolnicka, Dziadowlańska, Podbielska and Krukowska sections). Two were managed by a forestry officer (*lesnichy*—forester), and the other two by deputy forestry officers (*podlesnichy*—deputy forester), subordinate to the forester. Deputy forestry officers occupied two of former guard estates (Hajnowska and Browska), and for this reason the number of guards was reduced to ten (RSHA 1840–1849).

Already in 1841, the Hajnowska section was separated and proclaimed a special "representational" forestry district, where "proper" forestry was to be introduced. A graduate of the Forestry Institute, Fyodor Cheredeyev, compiled a description and a plan for "proper" forest management. However, he quickly discovered that the Hajnowska section, due to its remote location from floatable rivers, roads and trade markets, was not a good choice for this task (NHARBG 1842; RSHA 1840–1849, 1841; Genko 1903). Consequently, the plan drawn up by Cheredeyev was put aside, and a forest inventory team sent to BPF (RSHA 1840–1849). In the course of the first forest management of BPF, the old division into guard's sections was finally abandoned and soon the BPF itself was divided into four forestry districts instead of three, although other projects were proposed, e.g. dividing BPF into five forestry districts (RSHA 1840–1846, 1840–1849, 1845–1856).

Each forestry district was supervised by one forester and divided into two to three subdistricts, which were in turn divided into several forest ranges. Some of the guards were offered the posts of "riding guards" with the prerogatives of deputy foresters. Riding guards had the duty to control how the permanent forest guard fulfilled their responsibilities. The permanent forest guard consisted of 77 families of the most "trustworthy" riflemen (later their number varied from 72 to 79). Additionally, 73 families of beaters also retained their previous roles such as haymaking on forest meadows for bison feeding, protecting the forest from fire and unauthorized logging, poaching, etc. (RSHA 1845–1856). Most former guards did not continue their service but were given the opportunity to stay on their estates for life without tax, and their heirs were promised only a small land tax would be levied (RSHA 1845–1856). Karcov (1903) summarized the life in the Forest in this period in an idyllic manner: "*With the new organization, everyone's life flowed freely. Excess land and spare time allowed for the establishment of orchards, gardens, cattle breeding, beekeeping, hence for each member of the forest service deportation from the forest would be the greatest punishment. Each rifleman had a uniform, good riding horse, saddle and state-owned rifle, and each had also own weapon. They were all excellent shooters, secretly engaged in merciless poaching*".

A careful study of the archival documents allows for a more comprehensive and complex understanding. The foresters of BPF, the forestry officers of the Grodno Chamber of State Properties, the forestry inventory teams, and auditors sent from the Forestry Department all had different points of view. They each offered their perceptions of forestry management and use of its resources. In this context, new projects were developed before the previous ones were implemented, and even before they were approved (RSHA 1840–1849, 1845–1856).

Poaching was a problem observed by forestry administration, and the decrease of penalty for killing bison from maximum of 150 roubles to only 25 roubles did not help. At this time, a bison's head was worth up to 500 roubles abroad (Karcov 1903). One of the groups most often accused of poaching were *budniks*, former makers of wood-tar, charcoal, and potash who settled inside the Forest in the eighteenth century. They seemed to be forgotten by the state administration. In particular, a group settled in the Forest in the 1790s to work in a potash factory but this soon stopped working and they were left to their own devices. Some of *budniks* engaged in agriculture and kept livestock; others, taking advantage of the remoteness of the forest guards' estates, were illegally producing small amounts of wood tar and birch tar and, of course, poaching. BPF's forestry officers, starting from 1820s, several times demanded eviction of *budniks*' villages of Budy, Teremiski and Pogorzelce. However, the Grodno Chamber of State Properties confined itself to assigning personal tax to *budniks* and stating that a land inventory was needed (NHARBG 1838; RSHA 1837A). Only after ten years was the land inventory was effective, and then the provincial administration did not find it necessary to evict the *budniks* but rather obliged them to pay standard tax and fulfil duties common for state peasants of that period. This was in fact drastic change after many years untaxed. The poorest of *budniks*, of which foresters wrote that "*it is impossible to understand what they live by*", were assigned to work in bigger farms (NHARBG 1838). Demands for their eviction were repeated in subsequent years but the Grodno Chamber and the Ministry of State Domains responded that *budniks* did not only not harm the forest, and even could be useful e.g. in extinguishing forest fires, and their relocation would mean "significant costs from the treasury" (RSHA 1857–1867, 1859–1867).

The specific character of *budniks* was romanticized by writer Sieroszewski: "*We—budniks! We have come here from all over the country, people of choice* (…). *They had their rights, their village leaders and their uniform: bast shoes and rugged clothing, and a golden belt on it. When such a budnik came to the inn on Sunday, he could beat everyone to no-ones opposition, as they would kill him immediately… They were such sharp-tempered!*" (Sieroszewski 1900).

Several archival documents concern the protection of the forest against fire. In 1840, Danilo Neuman, the head forester of the Grodno Province, proposed new regulations on this matter. He saw mainly herders and beekeepers as the main source of fire risk, therefore proposed that any burning of pastures or arable land in the vicinity or inside the forest should require the approval of the administration. This in turn, should ensure that enough people with shovels and buckets were present to keep any fire under control (RSHA 1839–1842B). In his description of the Hajnowska section, Fyodor Cheredeyev pointed out that if forest guards patrolled places where pastures and beehive trees were located sufficiently frequently, the peasants scrupulously observed precautions against fires. Only peasants in villages located in remote parts of the forest that could not be controlled by guards allowed themselves to be less careful with fire (NHARBG 1842). Furthermore, the foresters sometimes admitted that ground fires caused by herders and beekeepers did little harm to large trees but improved the vegetation for grazing and honey collection. At the beginning of the twentieth century, the administration even proposed to resume this practice in order to clean up some areas of spruce undergrowth and improve the feeding conditions for game (RSHA 1905–1911). However, officials from the provincial chambers regarded any forest fires as a threat and from the early 1840s proposed to ban beekeeping in state forests (Loskutova and Fedotova 2019). They also wanted to introduce a special fire-protection clause in the rules for grazing livestock in forests (RSHA 1839–1842B).

BPF was among the forests to which the inventory team was sent in 1843, i.e. in the first operative year of the Forestry Department. A year earlier, in 1842, one of the most prominent authors who wrote about BPF and European bison in the middle of the nineteenth century, Dmitry Dolmatov, arrived in the Grodno Province. He was invited to take up the post of learned forester (and then served as a head forester) of the province. His task was to reorganize forestry on a "rational" basis, mainly with an increase of income in mind. In addition to supervising the work of the forest inventory team, Dolmatov proposed a number of ambitious projects to increase revenues from state-owned forests. These included straightening and connecting BPF rivers with canals, the establishment of a school to educate foresters in Białowieża, and the erection of wood tar, glass and other workshops (Genko 1903; RSHA 1840–1849; SARF 1860). Those plans, however, remained only as ideas on paper since during the evaluation of his projects it turned out that Dolmatov had underestimated the costs of implementation and overestimated future revenues. However, others also observed the need to make forest rivers floatable to improve sales and this was noted in Cheredeyev's reports as early as in 1842 (NHARBG 1842; RSHA 1840–1849).

The first Tsar's hunt after the acquisition of BPF by the Russian Empire took place in October 1860. According to the official description of the hunt, preparations lasted just weeks (Fuchs and Zichy 1862), yet an anonymous article on the hunt draws a different picture of an enormous and longer lasting effort to gather animals, erect new buildings, and to mend both roads and bridges (Anonymous 1860). Tsar Alexander II was accompanied by several German princes to whom the Tsar was related, along with high-ranking officers from the Russian army and the court. Two hunts on 6th and 7th October took place in a fenced area located on the remnants of an old hunting garden of the Polish kings at Teremiska. Animals were driven in front of the Tsar and other hunters and shot from a safe distance. To commemorate the hunt, several larch trees were planted (Fig. 5.2), and later a bison monument was erected in front of the hunting enclosure (Fig. 5.3). During the two days, 28 bison, 2 moose, 10 fallow deer, 11 wild boar, 16 wolves, 16 roe deer, 7 foxes, 4 badgers and 2 hares were killed (Fuchs and Zichy 1862). According to an anonymous article detailing the preparations for the hunt, the majority of the game was brought to BPF from other forests and hunting enclosures (Anonymous 1860). The hunt marked an important milestone—from this point, the management effort gradually shifted from turning BPF into a model of rational forestry towards making it a hunting reserve of Russian imperial family. This was to be organized in accordance with German hunting practice.

The management plan from 1843 to 1848, incorporating the locations of clear-cuts and the amounts of timber to be produced, was accepted by the Special Forestry Committee in St. Petersburg (RSHA 1849). However, the Grodno Chamber of the

Fig. 5.2 Tsar Alexander II directing the planting of commemorative trees after the hunt of 1860, surrounded by foreign princes and the imperial entourage. In the foreground, there is imperial court painter Michaly Zichy. Title illustration from the book “Hunt in Białowieża Forest” (Fuchs and Zichy 1862), drawn by Michaly Zichy

Fig. 5.3 Drawing of a projected monument commemorating the first Tsar's hunt in BPF in 1860, which was later erected around 1862. Illustration from the book "Hunt in Białowieża Forest" (Fuchs and Zichy 1862), drawn by Michaly Zichy

Ministry of State Domains informed the authorities that there were present in the forest 83,704 large pine trees damaged in the lower part of the trunk by chopping pieces of wood (For more on the origin and significance of those culturally modified trees see Sect. 5.2). The Grodno Chamber recognized such trees as still valuable in timber market, but susceptible to quick deterioration. It, therefore, proposed to exploit them first before the actual management plan was realized (RSHA 1840–1849, 1850). The Special Forestry Committee reconsidered the decision on introduction of the new management plan elaborated in 1843–1848, and in 1849 suspended this until the damaged pine trees were removed from BPF (RSHA , 1849). In effect, thousands of culturally modified pine trees were stamped and selected for felling. However, a new problem arose: with narrow and meandering rivers incapable of floating big trees, and poor land transportation routes (as reported by the learned forester Funk's note from 1841: "*during the Polish period these rivers were convenient for rafting, but now they are so overgrown that they can no longer serve as floating routes*"—RSHA 1840–1841). Moving large quantities of wood from different, often remote parts of BPF posed an insurmountable and unprofitable task. This was evidenced by the 1845–1848 attempts at selling trees cut in BPF by the local administration to local merchants, Zawadzki and Zabłudowski, and to foreign companies, Mallan, and

Thomson & Bonar (Genko 1902–1903). To bypass this problem, the forestry administration decided to invite a contractor from outside, who would be responsible for the extraction and transportation of damaged trees. In 1854, Berlin-based merchant C.F.L. Buggenhagen, represented in Russia by S.A. Simund was contracted to extract 60,000 pine trees "damaged in the bottom part of the trunk" in the course of the next 5 years (Hedemann 1936). Instead, Buggenhagen and his representatives selected mainly large-sized trees of high quality for felling. Despite the fact that the contract directly prohibited the cutting of trees "suitable for shipbuilding", especially shipmast trees, Buggenhagen's workers felled them after first damaging bottom parts of their trunks. To make matters worse, Buggenhagen claimed that his men were able to find only 17,200 damaged pine trees instead of promised 60,000, and put forward a motion to select the remaining number from old but undamaged trees. Adverse comments about Buggenhagen's and Siemundt's misdemeanours reached the ministry, and a commission was sent to the Forest to inspect the case. The commission estimated the number of damaged trees to be at least 300,000 and accused Buggenhagen of deliberately overlooking them (RSAN 1856–1862, 1857, 1861–1862). The Naval Ministry pressed charges against Buggenhagen, but his prestige and influence among the higher authorities in the Russian Empire resulted not only in freeing him of any charges, but also in granting him an additional 10,000 trees from BPF as compensation for his losses and expenses incurred to improve floatability of rivers (Hedemann 1936). Also, before the case against Buggenhagen was concluded, the demand for mast trees and wood for shipbuilding in general diminished; the age of iron warships was beginning and shipbuilding timber rapidly began to lose the status of a strategic resource. Probably, for this reason, the Naval Ministry did not continue the dispute with Buggenhagen (RSAN 1856–1862, 1861–1862; RSHA 1857–1860).

After the end of Buggenhagen's contract in 1860, forest management was still aiming to remove damaged pine trees from the forest, and prevent any new scarring of trees by traditional use. In 1861, the head forester of the Grodno Province, Adolf Eichwald, proposed to supply to the local peasants to satisfy their demand for kindling material, the pine trees cut during preparation of compartment lines set by the first forest management plan. The plan included also removal of the network of traditional pathways cutting the forest and instead introduction of straight-line roads based on compartment lines. This helped prevent quick and discrete access to the forest by unauthorized people (RSHA 1861B). Despite all efforts, meandering forest pathways, as well as the practice of the chopping of resinous wood from pine trunks survived until the twentieth century. This activity was listed among other undesired behaviours of herders pasturing cattle in the forest, along with poaching and carving indecent writings on forest compartment poles (RSHA 1912–1913).

5.3 Culturally Modified Trees and Division of BPF into Compartments as Material Imprints of the Period

One of the most evident material imprints of this period in BPF was the division of the forest into rectangular compartments based on the first forest management plan elaborated in 1843–1848. The division meant not only virtual demarcation of borders on the map, but also actual crosscutting of compartment lines in the forest itself. A total of 541 compartments in BPF and 125 in Świsłocz forestry district were delimited by drawing straight vertical lines in a distance of 1 *wiorst* (1067 m) from each other, and straight horizontal lines in the distance of 2 *wiorst* (RSHA 1840–1849). According to the management plan, each compartment was a rectangle covering 2 square *wiorst* (227.6 ha, but the actual sizes oscillated between 200 and 250 ha). Cutting of compartment lines was a complicated and time-consuming task, as the gap between the compartments measured 2 (occasionally more) sazhen (1 sazhen = 2.16 m). In some cases, this task was not completed even after five years.

In the period 1838–1860, another type of material imprint was documented and became one of the most hotly debated issues in the management of BFP. Scarred, modified or "worked" trees carrying the memory of past traditional use and human nature interactions were seen as an obstacle in introducing the modern, "scientific" forestry. This was similar to the problems over access rights (see the previous chapter), cattle pasturing in the forest (Samojlik et al. 2016) and traditional beekeeping (Karpiński 1948; Loskutova and Fedotova 2019).

The first type of culturally modified trees were pines scorched and chopped in the bottom part of the trunk. Pine was traditionally used for acquisition of resinous wood for kindling, produced by scorching the lower part of trunk. The resinous parts of the injured trunk were then chopped off and collected, and scorched again, to repeat the process. This type of utilization usually did not kill the tree, nor heavily hazard its stability. This practice, dating back to the royal period, was not seen as desirable by timber-oriented, "scientific" forestry based on German forestry school. In this context, pine trees used for the collection of resinous wood were recorded as wasted timber, verbatim trees "*spoiled in the lower part of the trunk by chopping pieces of wood*" (RSHA 1850). Such damage was recorded on numerous pine trees in the forest, and an official inventory of those was carried out in 1844. In total, 83,704 damaged trees were recorded in twelve forest sections of BPF (Świsłocz Forest was a part of Volkovysk forestry district and was inventoried later) (RSHA 1850). According to this inventory, "wasting pine timber" could be ascribed to different groups of people present in the forest. This ranged from people casually passing through the forest, herders driving their cattle to forest pastures, and craftsmen producing wood and birch tar, to beekeepers and haymakers scything meadows inside the forest. The majority of trees were damaged in the past, yet the head forester of the Grodno Province Danilo Neuman observed that this activity was still present in the forest and local people smuggled chopped pieces of resinous wood under their clothes to bypass the guard posts (RSHA 1840–1849). Furthermore, he accused forest guards of being involved in the process: whenever new damage happened, guards would soot the trunk themselves to make it look like a remnant of old activity.

As described in Sect. 5.1, the priority given to the removal of damaged pine trees in forest management, along with technical problems with transportation of large amounts of wood, significantly delayed the introduction of the modern forest management to BPF with its planned clear-cuts. Three years of effort by forest inventory specialists preparing the first forest management plan were partly wasted and merchants and companies contracted to remove damaged trees were not following the prescribed sequence of cutting. Denunciations, prosecutions and court proceedings resulted in prolonged periods of no cutting, and the situation led to delay in the introduction of modern forest management. Before the problem of damaged trees was solved, a shift towards hunting occurred in the forest management in connection with the first Tsars' hunt in the Forest in 1860. Forest management then began to shift from timber- to game-oriented practices which culminated in the turning of BPF into a hunting reserve as part of the Tsar's private properties in 1888 (Genko 1903; Karcov 1903).

The second type of culturally modified trees were pines with carved beehives, resulting from the forest beekeeping. This was one of the most widespread and long-lasting forms of traditional forest use (see Chap. 3). Beekeepers carved artificial hollows for beehives mainly in Scots pine (rarely using other tree species for this purpose), with one tree hosting usually one, occasionally two beehives located on average at 7 m above ground (Blank-Weissberg 1937; Karpiński 1948). Not only the artificial hollows carved in the trunk made beehive trees stand out among other trees in the forest: they were also equipped with constructed elements like a wooden cover over the beehive or wooden hooks hammered into the trunk for the beekeepers to hang the cover while they were working. There were also other characteristic modifications such as ownership marks carved in the lower part of the trunk and protective measures against bears and other animals considered beehive-pests by beekeepers. Such measures incorporated sharpened poles placed beneath the tree, wooden platforms around the trunk preventing animals from climbing the tree, and heavy logs hung on a rope in front of the beehive. The latter acted as a swinging heavy pendulum when bears pushed them away to get to the honey (Blank-Weissberg 1937; Karpiński 1948).

After peaking with more than seven thousand beehives, including over 900 active ones inventoried at the end of the royal period (Hedemann 1939), the number of recorded active beehives dropped by the late 1830s to around 300–350 (RSHA 1837B). Forest beekeeping, side-by-side with other traditional uses, was considered harmful or at least undesirable by the forest administration. Thus in 1838, the Ministry of State Domains ordered a survey of this type of activity in state forests in the entire empire. Arguments against forest beekeeping were numerous. They ranged from insufficient income from beekeeping for forestry administration, to damaged trees, beekeepers' unauthorized access to the forest, frequent fires, and to the general remark of the backwardness of the old way of beekeeping. In the 1840s, a decision of gradual transition from forest beekeeping to house apiaries was made (RSHA 1843), thus prolonging the presence of beekeeping, and beehives, in BPF. In 1862, the

Forestry Department requested from beekeepers to move their tree-beehives from the forest to apiaries and homesteads (Forestry Department 1862; RSHA 1862–1865). However, thanks to petitions and direct sabotage, the removal of beekeeping from state forests was postponed until the 1870–1880s (Loskutova and Fedotova 2019; RSHA 1861–1872, 1871–1900). In BPF, when the Forest was incorporated into the Tsar's private properties in 1888 "*forest beekeeping underwent complete destruction: beekeepers were gradually removed from the forest and existing beehives were taken over by forest administration. From that time, beehives devoid of their owners and caretakers were overexploited and in an express pace met their doom*" (Karpiński 1948).

Both kinds of culturally modified trees are still present in contemporary BPF. Scorched pines with traces of chopping wood for kindling and hollows carved by past beekeepers are located mainly in the strictly protected area of Białowieża National Park (for more details, see Samojlik et al. 2019).

There was also probably another type of culturally modified tree present in BPF in the period being discussed. These were lime trees with strips of bark peeled. Among various uses of small-leaved lime (see Sect. 3.3), bast tearing was probably the one most widespread and considered an important right of local dwellers. Lime bast was used across the entire Russian Empire and the activity was considered as essential for the entire class of peasants by the experts of the Ministry of State Domains (Köppen 1841). It was even argued that bast tearing for the production of lime bast shoes had lesser impact on Russian forests than a tanning industry would if all peasants' shoes were to be replaced by leather ones (Teplouhov 1862). Alongside the official view that there was no commercial use of lime wood in BPF, other sources confirm that lime bast was being used (only for local purposes, not commercially) until the late nineteenth century. In the Grodno Province in 1880s, tearing bast from young lime trees led to visible absence of lime in private forests, and it was found only in protected state forests, among them BPF. Peasants of the province rarely wore shoes other than bast ones, as every herder pasturing cattle in the forest "*returning home with the herd in the evening, brings back a stock of bast*" (RSHA 1880). This type of culturally modified tree, the peeled lime tree, was not mentioned in inventories or reports. It is possible that such trees were not observed because increased anthropogenic pressure resulted in the removal of the entire utilized tree. Bast peeling was only one among many uses lime has experienced as bark, wood, leaves, and flowers were also traditionally used (Samojlik et al. 2019).

5.4 Cultural Heritage of the Period: Successful Experiments with Creating Bison Cattle Hybrids Conducted by Leopold Walicki

European scientists were already interested in European bison by the eighteenth century. Experiments such as crossing European bison with different species of *Bovidae* and domestic cattle were planned by Georges-Louis Buffon to check the status of the species and answer several other questions of origins of domestication, the concept of degeneration of the species, etc. (Buffon 1764). It is probable that some attempts towards realizing Buffon's plan were made in the Polish Lithuanian Commonwealth, as French professor of natural history Jean-Baptiste Dubois de Jancigny wrote in his 1776 book: "*When it comes to European bison, it was due to* [Buffon's] *genius, great in the observation equally to the Nature itself, to put it in the Bos family. I must honestly admit that my doubts were not fully dispelled by his argument, as according to information I obtained in Poland attempts at crossing bison with domestic cow were numerous, yet all failed*" (Dubois de Jancigny 1778). However, the first known description of an actual experiment was published by Jean-Emmanuel Gilibert, who unsuccessfully attempted to cross-breed a bison female with a domestic bull (Gilibert 1781). Along with Gilibert's observation that bison calves refused to suckle on cows (Gilibert 1781), this fuelled the common eighteenth-century belief in a biological barrier between species. Such barrier, that would not allow European bison calves to be fed by cows and also prevented any hybridization, was described as "hatred" (Dubois de Jancigny 1778) or "natural antipathy" (Gilibert 1781) between wild and domesticated animals. This common assumption, along with the fact by the beginning of the nineteenth century those animals were rare in zoological gardens, and the only known free-living population of lowland bison was located in the remote area of Europe, hindered any hybridization experiments for around seventy years. Such prejudices, preventing the advance of science, took many years to unpick and refute.

First, a success in feeding European bison calves with cows' milk was achieved when bison calves were captured for London Zoo and St. Petersburg imperial court in 1846. The head forest manager of Grodno Province Dmitri Dolmatov described the life of captured bison calves held in the estate of one of BPF's forestry officers, correcting the eighteenth-century misconceptions repeated continuously by authors of his era. "*I have turned my attention particularly to refute by experience the erroneous opinion, accredited by all the writers who have treated on this subject, namely that the calf of the Bison cannot be suckled by our domestic cow. This fable has been repeated even in the work of an esteemed writer of our times, Baron de Brinvers* (Brincken), *who relying upon the recital of another writer, the learned Gilibert, asserts that two female Bison calves, caught in the forest of Bialowieza, seven weeks old, constantly refused the teats of a domestic cow; that they consented, indeed, to suck a goat, but as soon as they had had enough, they repelled their nurse with disdain, and grew furious whenever they were put to a domestic cow. M. de Brinvers had not himself the possibility of verifying this fact; and he cites traditions, communicated to him by the old inhabitants of the environs; for if any one of the forest guards, or the peasants who inhabit the forest, had even met a Bison calf, parted by any accident from its mother, he would rather have left it, than seized and nursed it, in contravention of the severe law, which prohibits the capture or killing of a Bison. It was therefore only the supreme order of His Majesty the Emperor, emanating from the desire expressed by Her Majesty Queen Victoria to possess in her Zoological Garden two living Bisons, which has enabled me to rectify the error above mentioned*" (Dolmatov 1848).

Dolmatov's observations of European bison calf playing with domestic cattle and being fed by cows (Dolmatov 1848, 1849a, b) were published and widely commented throughout Europe (Brehm 1877; Gervais 1855; Viennot 1862). Having confounded the myth of the impossibility of young bison being suckled by domestic cows, forestry administration aimed also at learning experimentally if hybridization between species was possible. When a wealthy landowner and agronomist Leopold Walicki offered to test this hypothesis in 1847, the Ministry of State Domains allowed to transport two young bison to Walicki's estate Wilanów near Grodno. Walicki, irrespective of the eighteenth-century prejudice, managed to obtain fifteen hybrids of European bison and cattle *Bos taurus* in the years 1847–1859 (Fig. 5.4), among them one fertile male hybrid from the first generation (Karcov 1903; RSHA 1854–1862). This achievement was never repeated, as even contemporary experiments conducted at Mammal Research Institute, Polish Academy of Sciences failed to obtain a fertile male in the first generation (Krasińska 1988).

Walicki did not publish results of his experiments himself, and his motivations are known only through mention in other authors' works and his own reports to officials of the Ministry of State Domains (RSHA 1854–1862). Combining the size and strength of the bison with docility of cattle should have created a perfect working animal: "*about four years ago under the act of his highness* [Tsar], *a number of juveniles*

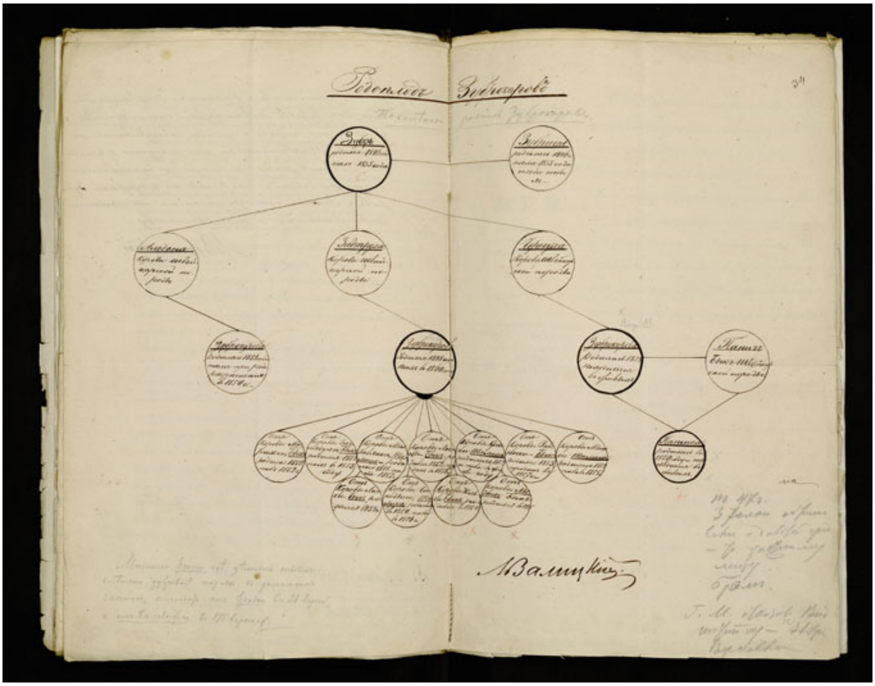

Fig. 5.4 Walicki's graph listing hybrids obtained during his experiments with cross-breeding European bison and domestic cows in years 1847–1859 (RSHA 1854–1862)

were transferred to surrounding landowners. An attempt to create a new breed by crossing them with cattle was made. The new breed was to be bigger, stronger and thus more useful, as in this area cattle, similar to horses, are small and weak" (Müller 1859). Similarly, the goals of Walicki's endeavour were described as to check "*(1) the possibility to breed and multiply bison in farm conditions, maintaining the natural beauty, health and size of the animal, (2) the possibility to cross them with domestic cattle, and if the strength, size, beauty and wildness is not lost during the process*" (Bobrovskii 1863).

The issue of hybridization was in the centre of one of the most heated scientific debates in the mid-nineteenth century, as it cast doubt on the definition of species based on the criterion of inability to obtain fertile interspecific hybrids. Surprisingly, Walicki's successes in obtaining hybrids between different genera did not enter scientific circulation of that time and were absent from the nineteenth-century discussion on the definition of species and hybridization. Despite being mentioned both by Eichwald (1853) and Müller (1859), prominent naturalists of the half of the nineteenth century, Walicki's experiments were not recognized by European scientific circles. Some authors of the era considered European bison belonging to the same genus as cattle, so it was easy for them to overlook Walicki's experiment and count it towards hybridization between species of the same genus. Naturalists, on the other hand, have paid much closer attention to Dolmatov's feeding of bison calves by cows.

Two other heated scientific debates of the era that could have benefited from Walicki's experience: a dispute on the degeneration of species and a dispute whether European bison and aurochs belonged to the same species. Yet, both have almost completely neglected Walicki's experiment.

The idea of degeneration of species puzzled many naturalists of the second half of the nineteenth century, and cattle were often used as an example of this process. Walicki was never mentioned in this context, but Dolmatov's observation of friendly behaviour between bison and cows was used as a proof of bison slowly degenerating: "*Gilibert resided in Poland for a long time and had an opportunity to closely study four of these animals kept in captivity. They had to be fed by goats as a result of their stubborn refusal to suckle on cow which was first brought to them. They maintained this hostility to domestic cattle, and whenever cows were driven to the same enclosure, bison chased them away. Despite similar statements by various authors, Mr. Dimitri de Dolmatof, Grodno province forest administrator, in a memo from 1847 stated that events he repeatedly witnessed contradicted this opinion and that young bison were well fed by the domestic cow. Maybe you can reconcile these statements admitting the existence of some kind of degeneration of modern bison compared to their great ancestors*" (Viennot 1862).

The lack of a reproductive barrier between bison and domestic cattle could have been used as an argument towards treating bison and aurochs as the same species and, furthermore, assuming that the bison is in fact the ancestor of modern cattle. The dispute itself was resolved by August Wrześniowski's "Studien zur Geschichte des polnischen Tur" which finally concluded that bison and aurochs were separate species (Wrześniowski 1878).

5.5 Travellers and Naturalists Recognition of BPF and the Unofficial View on the First Tsar's Hunt in the Forest

After the failure of the November Uprising, Vilnius University was closed and up until 1842 only two faculties, medical and theological, were allowed to function and these were deprived of university status. Zoological collections of the Vilnius University were transferred to the St. Vladimir Imperial University in Kiev. In effect, higher education and research into the natural sciences were effectively removed from Vilnius. This, along with other repressive measures of the imperial government against intelligentsia of Lithuanian provinces (Dolbilov 2010) hindered the scientific study of BPF and the region for several decades.

In the years 1837–1860, BPF was visited not only by professional naturalists, but also by several foreign and Russian travellers, officers, and amateurs. Even during the shortest visits to the Forest, these often left written evidence of their experiences of Białowieża's nature.

Konstantin Ivanovich Arseniev (1789–1865) was a Russian historian, statistician and geographer, and member of the St. Petersburg Academy of Science. He visited the Forest in 1845 and his very general and brief remarks on BPF are reminiscent of knowledge passed on in school books. They were made up of essential bullet points with occasional brief anecdotes (Arseniev 1845). This suggestion may be close to the truth as in 1830s Arseniev served in the role of a history and statistics teacher of heir to the throne, later Tsar Alexander II.

In 1851, Franz Müller, professor of animal anatomy and veterinary art from the Vienna Veterinary Institute visited BPF to acquire two bison. He was to supervise their transport to Vienna. Despite Müller being a valued author of works on animal anatomy and physiology of domestic animals, his visit to the Forest did not have any prominent scientific results. He left an account of the hunt organized for him, supplemented with a description of European bison biology compiled from previous works and published in 1859 (Müller 1859). However, one aspect of interest is that his note from BPF contains an original observation of the bisoncattle hybrid. Müller visited Leopold Walicki (see Sect. 5.3) and was the first to report the success of his hybridization experiments. Published in German, in a prestigious journal edited in Vienna, Müller's description entered European scientific discourse on such matters.

Jacques Boucher de Crévecoeur de Perthes (1788–1868), was a founder of prehistorical archaeology and a naturalist who visited BPF in 1859. His interest in the Forest was different from any previous foreign visitor since as a dedicated investigator of animal remains in archaeological excavations, he now observed live populations of animals known previously from excavated bones (Perthes 1859).

Białowieża's bison were noted by the French natural science community, as evidenced by R.T. Viennot's presentation at the meeting of the French Acclimatization Society in Paris (Viennot 1862). Founded in 1854, the Acclimatization Society was one of the most important French scientific societies with its goals to acclimatize economically-useful plants and animals. Therefore the society was interested in various views on enriching local fauna and flora with new species. However, the Acclimatization Society also played a pioneering role in French nature conservation and both these reasons probably explained the society's interest in BPF as the last refuge of European bison. Viennot's text was again, as in previous cases, a compilation of information from various sources and containing errors. Nevertheless, it played its role as a French popular science publication that reached wide audiences and informed them about Europe's last wild bison population.

An event that had long-lasting cultural impact on the history of BPF and the future role of the Forest itself was the first Tsar's hunt in 1860. As noted, management of the forest was shifting from the idea of introducing modern, timber-oriented forestry and towards a game-oriented hunting reserve. This was in the aftermath of the imperial hunt, but the event also brought more attention to BPF. Audiences belonging to different partitions, parts of former Polish-Lithuanian Commonwealth, were interested not only in the official view on the successful and trouble-free process of the hunt (Fuchs and Zichy 1862), but also in the unofficial descriptions. The latter were published in an anonymous article in "Czas", the Cracow-based journal, in November 1860, i.e. weeks after the actual event (Anonymous 1860). This work offered a behind-the-scenes peek at preparations for the hunt (that reportedly had lasted several months), revealing several problems the administration encountered. First, the cost of organizing such hunt was enormous, from erecting new buildings in a matter of weeks and equipping them with foreign furniture, to employing thousands of people to fence large area of the forest and to mend the forest roads. Secondly, gathering animals in the hunting enclosure posed a real problem, therefore most of the game was bought outside the forest and transported (with problems, e.g. one of the bears escaped during travel, approached a farmstead and was killed on spot, Anonymous 1860). Lastly, the author mentions continuous problems with fraudulent contractors and periods of chaotic tree-felling that changed the look of the forest to the extent that during hunt, the Tsar should be "*driven along the same road several times and shown only what they want, and not shown that he should not see*" (Anonymous 1860).

Information by the anonymous author was partially confirmed (at least on the scale of preparations for the hunt) by letters from the assistant to the preparator of the Zoological Museum of the Imperial Academy of Sciences, Pamphil Ivanov, who was sent to BPF to handle trophies. Ivanov arrived to BPF two weeks before the start of the hunt, and met a large group of high-ranking officials, including the minister of state domains, Alexandr Alexeev Zelenoi and jegermaister Pavel Karlovich Ferzen (SPB ARAS 1860). It was believed Zelenoi, as the organizer of the hunt, wanted to surpass the hunt organized for the Tsar by the Tyszkiewicz family in the Vilnius Province in 1858, and to draw upon the model of the royal hunt of King August III in BPF in 1752 (Egorov 1996; Mina 1859).

There was great interest in this pompous event and its background of hidden fraudulent exploitation of BPF. For example, there were two articles published in "Kolokol" ("Bell", newspaper of Russian opposition emigration) in October 1861 and January 1862. Since the described events occurred in 1850s, we include remarks on those articles in this chapter. "Kolokol" was published by Alexander Herzen, one

Fig. 5.5 Cover of Dolmatov's manuscript "History of aurochs or bison and Białowieża Primeval Forest"—author of the drawing unknown (NMW 1851)

of the loudest critics of imperial politics and a well-known writer of the period. Both articles by an unknown author centre on the chaotic and fraudulent exploitation of forest resources of BPF and fit well into "Kolokol's" line of heavy criticism towards Russian agrarian policy (Anonymous 1861a, b, 1862).

Box 5.1 Dmitry Dolmatov's futuristic plans of BPF's management

Dmitry Dolmatov (Dalmatov) was born to the family of solicitor of the Saransk district court, Yakov Maksimovich Dolmatov, around 1812. Dmitry studied in the St. Petersburg Forestry Institute at the state's expense in 1826–1830, and immediately after graduating, began his service in the Department of State Domains of the Ministry of Finance. In 1835, he was appointed as a district forester in the Nizhny Novgorod Province and in 1842 was promoted to the post of "learned forester" of the Grodno Province. It was the second most important position in state forestry management within the province and after successful service, Dolmatov was promoted to provincial forester in 1845 (RSHA 1841–1856).

Dolmatov is known as the author of modernistic, though not entirely realistic, projects for the "rational" exploitation of the state forests in the Grodno province. One of his ambitions was to improve the floating routes for rafting timber and firewood from BPF. Dolmatov proposed to clear and straighten

Fig. 5.6 Drawing "Białowieża's aurochs or bison" from Dolmatov's manuscript "History of aurochs or bison and Białowieża Primeval Forest"—author unknown (SARF 1860)

smaller rivers and build new channels that would allow delivery of Białowieża's wood to Grodno, Brest, and other cities. However, inspection of his projects, revealed that Dolmatov's assessment of the situation in BPF was biased, and he even changed the reported data in levels surveys. Yet he still managed to have several forest inventory parties operating in the province and consequently grew the income from state forests in the Grodno province (RSHA 1840–1849).

In his projects and published articles, Dolmatov compared Białowieża Primeval Forest with "*an abandoned factory in which the expensive machineries and stocks rot and rust, and everything leads only to increase of losses*" (Dolmatov 1846). He insisted that intensive, rational forestry management would be beneficial not only for the economy of the region, but also the forest itself. In addition to the canals, he proposed to build new factories and to restore the old ones to produce iron, tar, glass, and potash, both in BPF and in its vicinity.

Dolmatov's duties included the control of hunts and live-capture of European bison (each time with the Tsars' permission), and usually requested by European universities, natural collections and menageries (see Samojlik et al. 2017). In 1846, Dolmatov supervised the capture of European bison for the London Zoological Society; and a year later the hunt for the Zoological

Museum of the Imperial Academy of Sciences in St. Petersburg. The first event led to Dolmatov's correspondence with the head of the natural history department of the British Museum Richard Owen in late 1840s to early 1850s. Resulting from this, Dolmatov published a note on European bison in the Proceedings of the London Zoological Society and received the medal of the Society. At the same time, more bison were captured. Of these, two were sent to St. Petersburg and two others stayed in Białowieża to be used for domestication attempts. These two bison were subsequently presented to Leopold Walicki for his experiments on bison domestication and hybridization (see Sect. 5.3).

All these successes, connected with the fact that Dolmatov published articles on BPF and European bison in Russian forestry and hunting journals (e.g. Dolmatov 1849a), resulted in his election to the Russian Geographical Society. In the 1850s, he worked on his *magnum opus* the "Natural History of the aurochs or bison and Białowieża Primeval Forest". But neither the Forestry Department nor the Russian Geographical Society found it feasible to publish this book (A RGS 1853–1854). The manuscript was most probably used as a source of data on European bison and BPF by several authors, including G. Karcov. The manuscript's revision of 1860 (Fig. 5.5) is still kept in the State Archive of the Russian Federation (SARF 1860). The volume also includes a watercolour scene of wolf pack chasing European bison in the forest (Fig. 5.6).

By the end of 1848, Dolmatov was appointed as the provincial forester for Perm Province, and in 1853, for Novgorod. However, this period of service was less successful and in 1855 he left the state forestry management (RSHA 1841–1856). He served as the provincial postmaster in Ufa, and then in Vyatka. Out of twelve of Dolmatov's children (he was married three times), the youngest one, Alexander (1873–1937), deserves special attention since his photographs were used to illustrate Karcov's monograph on BPF (Karcov 1903). Dolmatov died in Vytka in the late 1876 or early 1877 (Anonymous 1877).

Box 5.2 Mihály Zichy in the imperial forest

Mihály Zichy (in Russian Michail Alexandrovich Zichy) (1827–1906) was a Hungarian painter and graphic artist, who worked mostly in the Russian Empire. Educated in Budapest, he continued to study drawing and painting at the Vienna Academy of Arts. In 1847, the Grand Duchess Elena Pavlovna (Princess Charlotte of Württemberg, wife of Grand Duke Michael Pavlovich) invited him to St. Petersburg to teach her daughter drawing and painting. After few difficult years in St. Petersburg, Zichy earned a reputation among Russian artists, art lovers and aristocratic patrons. His painting of the coronation of Emperor Alexander II (1856) earned him election to the Imperial Academy of

Arts as an academician of watercolour. In 1859, he was appointed as a court painter (RSHA 1859–1874). He kept this post until 1873 and after a break for a journey through Europe, he returned to this position in 1880. Although he was famous as a painter throughout Europe, in Russia he was mostly known as a draftsman-chronicler of imperial court ceremonies, entertainment, and family events, caricatures of people close to the court and scenes from imperial hunts (Aleshina 1975).

Zichy accompanied Tsar Alexander II during his hunt in BPF in 1860, and by the early 1861 he presented a series of watercolours depicting the scene of the Tsar's arrival at the Białowieża village, hunters in the forest, hunted animals and the hunters planting commemorative trees after the event (Fig. 5.2). His works were published in 1862 in the book "The Hunt in Białowieża Forest" (Fuchs and Zichy 1862). The text of the book was written by Viktor Yakovlevich Fuchs (1829–1891), an official of special assignments at the Ministry of State Domains, and then at the Ministry of Internal Affairs (Egorov 1996). It is likely that Fuchs used Dolmatov's manuscript about BPF and European bison to compile the text. "Hunt in Białowieża Forest" was intended as a gift for participants of the 1860 hunt and a total of 210 copies were printed in Russian and 60 in French. A smaller share of the edition made with particular luxurious, golden edged, calfskin binding. Due to the small number of books printed, this edition immediately became a bibliographic rarity (Egorov 1996).

Zichy returned to Białowieża in 1894 to depict the hunt of Alexander III, but in this case his drawings remained either scattered throughout different publications or unpublished (Aleshina 1975; Kantor-Gukovskata and Printceva 2005).

Box 5.3 Bison grass

Bison grass (*Hierochloe odorata*) is the plant most often associated with European bison and the forest of Białowieża. There are two species of this grass known under the colloquial name of "bison grass": *Hierochloe odorata* (L.) P. Beauv, called *żubrówka* in Polish, *zhubrovka* in Russian and sweetgrass or holy grass in English, and *Hierochloe australis* (Schrad.) Roem. & Schult. Due to the high concentration of coumarins, bison grass is characterized by an intensely pleasant smell of freshly cut grass. Although harvested worldwide, bison grass from Białowieża seems to have a more intense smell than that sold for example in the Parisian Jardin des Plantes. It is difficult to say whether this is related to the geographical variability of the species, habitat conditions, or the result of selecting individuals with a higher concentration of coumarins for the purpose of flavouring alcohol.

The plant is not endemic for BPF since its geographical range includes Eurasia and North America. In the latter, it was and still is used and highly valued by the native people as a medicinal and aromatic plant (Hart 1979; Shebitz and Kimmerer 2004). Similarly, as bison grass from BPF was associated with European bison, the North American herb is connected with American bison (Szalay 1919).

Eastern-European traditional medicine prescribed the external use of bison grass in the form of infusions, and internally for ailments of the digestive system. Vodka with bison grass (*żubrówka–zhubrovka*) is recognized around the world, with a European bison on the label and the shard of grass dipped in alcohol. According to the current producer, it was manufactured already in the seventeenth century, and in the eighteenth century became one of the favourite drinks of Polish gentry. The fact that Białowieża was in the middle of an important communication route between two major cities of the Commonwealth of Both Nations, Kraków and Vilnius, supported the popularity of this particular brand. In the nineteenth century, Stanisław Batys Górski, based on information from local forest personnel, described the grass as especially attractive for bison (Górski 1829). Other authors noted its attractive smell, not only in the vegetative period, but also dried and scenting haystacks offered to bison in winter (Bobrovskii 1863; Przybylski 1863). The first mention of its organized extraction from BPF in the Tsars' period was dated 1912: "*When describing the Forest, one should mention bison grass—it is a marshy grass with a pleasant aroma. In 1912, collection of bison grass was organized for the first time and 100 lb* (40.38 kg) *of dry grass was sent to Moscow to a manufacturer producing vodka, as well as perfume and soap. Perfume "Fresh hay" was based on bison grass*" (Golenko 1935a, b; see also Daszkiewicz et al. 2008).

Recently, Białowieża's bison grass, (along with cucumbers traditionally pickled in Narew River), has been the subject of an international programme aiming at retracing traditional crafts and local products, to support sustainable use and promote biodiversity (UNEP-WCMC 2003).

Box 5.4 Manna grass

The region of BPF, and the entire former Commonwealth of Both Nations, has a long tradition of harvesting and consuming manna grass (*Glyceria fluitans*): "*It had an important function as a component of tribute paid to local landowners by villagers, which led to the persistence of manna gathering even when this was not sustainable for peasants themselves. Manna grass was always an expensive food due to its time consuming gathering, but appreciated for its sweet taste and often served as dessert*" (Łuczaj et al. 2012).

Manna, its collection, cultivation and application were described by French naturalists visiting the Commonwealth. Jean-Étienne Guettard (1715–1786) observed that in Poland and Prussia manna is used to prepare groats and soup, and recommended cultivation of this grass in France (Guettard 1768). It was also recommended as a fodder plant growing in wetlands, which were usually treated as wastelands unsuitable for agriculture in the eighteenth and nineteenth centuries. Its cultivation and use was also described by the Napoleonic officer and naturalist Jean Philippe Graffenauer (1775–1838), who also noted an interesting dessert made from manna cooked in wine (Graffenauer 1809). The usual way to cook manna was to treat it as groats, boil it in milk, or add it to various soups (Buc'hoz 1787).

Manna was (and is) a quite common plant in BPF. The tradition of its collection and consumption was particularly widespread in Podlasie and Augustów regions: "*Especially peasants in Augustów and Podlasie are engaged in collecting manna. In the morning, before the dew dries, they collect grains with wet sieves, shake it lightly and clean from husks. Then it can be put to domestic use or sold*" (Waga 1847–1848). It is also confirmed in the reports of the governor of Grodno Province in the second half of the 1830s, where manna is mentioned as a crop used by local residents of Kobrin, Brest and Pruzhany districts: "[there] *grows a manna grass, used to collect grains, from which nourishing, tasty and healthy porridge is made*", as well as by Bobrovskii (1863) and Krestovskii (1876). Łuczaj et al. (2012), quoting Ciszewski (1925), writes: "*Village women from the Białowieża Forest usually sell the collected kasza to Jews, who pay them 50–70 kopekas per garniec [pot], depending on the year's crops*". This might be a trace of an organized collecting of manna in BPF not only for personal use but also for sale.

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[NHARBG] 1838. Perepiska s Grodnenskoj kazennoj palatoj o tjazhelom polozhenii krest'jan dereven' Budy, Teremiski i Pogorel'cy i ob oblozhenii ih nalogom (Correspondence with the Grodno State Chamber on the plight of the peasants of the villages of Budy, Teremisky and Pogorel'tsy and on their taxation). F. 1. O. 4. No 755 (in Russian)
[NMW] 1851. Historia tura czyli żubra w Puszczy Białowieskiej (History of aurochs or European bison in Białowieża Primeval Forest), National Museum in Warsaw, Gr.Pol.97
[RSAN] 1839–1840. O zloupotreblenijah po operacii rubki i dostavki korabel'nyh lesov iz Belovezhskoj pushhi (On abuses in the operation of tree cutting and delivery of ship timber from BPF). F. 410. O. 1. No 800 (in Russian)
[RSAN] 1856–1862. O nezakonnoj porubke v Belovezhskoj pushhe lesov pri zagotovlenii sosnovyh breven dlja inostranca Bugengagena (On illegal logging in 10 Belovezhskaya Pushcha when harvesting pine logs for a foreigner Bugenhagen). F. 410. O. 2–1. No 1272 (in Russian and German)

[RSAN] 1857. O naznachenii podpolkovnika Fedorova dlja osmotra Belovezhskoj pushhi (On appointment of Lieutenant Colonel Fedorov to inspect Belovezhskaya Pushcha). F. 860. O. 1. No 1301 (in Russian)

[RSAN] 1861–1862. O dostavke iz Danciga v Kronshtadt konfiskovannogo machtovogo lesa, otobrannogo u inostrannogo gostja Bugengagena (On the delivery of confiscated mast trees from Danzig to Kronstadt, taken from a foreigner Bugenhagen). F. 153. O. 1. No 55 (in Russian and German)

[RSHA] 1833–1844. O pojavivshihsja v Belovezhskoj pushhe vrednyh dlja sosnovyh lesov nasekomyh (About insects harmful to pine forests appearing in Białowieża Primeval Forest), part 1–3. F. 387, O. 1, No 5030, 5033, 5034 (in Russian)

[RSHA] 1837A. O revizii del khoziaistvennogo i lesnogo otdelenia Grodnenskoi kazennoi palaty (On the audit of the cases of the Economic and Forest Department of the Grodno State Chamber). Fond 379, O. 8, No 355 (in Russian)

[RSHA] 1837B. O lesnyh dohodah i nedoimkah po Grodnenskoj gubernii za 1837 (Information on forest income and arrears for 1837). F. 379, O. 7, No 314 (in Russian)

[RSHA] 1838. Po otchetu o sostojaniju Grodnenskoj gubernii za 1837 g. (On the report on the state of the Grodno Province for 1837). F. 1281, O. 3–1838, No 40 (in Russian)

[RSHA] 1839–1842B. Zapiska ob ohranenii lesov ot pozharov i o vospreshhenii zavodit' v lesah borti (Note on the conservation of forests from fires and on the prohibition of carving beehives in forests). F. 398, O. 3, No 551 (in Russian)

[RSHA] 1840–1846. O samovol'nyh porubkah lesov v 1839 g. proizoshedshih i ob uchrezhdenii postojannoj lesnoj strazhi v Grodnenskoj gubernii (On the unauthorized forest cutting in 1839 and the creating of a permanent forest guard in the Grodno province). F. 387, O. 27. No 7 (in Russian)

[RSHA] 1840–1849. O vvedenii v Belovezhskoj pushhe pravil'nogo lesnogo hozjajstva (On introducing proper forest management in Białowieża Forest). F. 387, O. 1, No 7057–7059 (in Russian)

[RSHA] 1840–1841. O sushhestvujushhih v gubernijah sposobah vydelyvanija lesnyh proizvedenij (On the methods of forestry production existing in the provinces). F. 387, O. 1, No 384 (in Russian)

[RSHA] 1841. Gajnovskaja dacha, chernovoj plan (Hajnowska forestry disctricts, a draft map). F. 380, O. 1, No 1635 (in Russian)

[RSHA] 1841–1856. O gunernskom sekretare Dolmatove (On forestry officer Dolmatov). F 387, O. 1, No. 1739 (in Russian)

[RSHA] 1843. Otnositel'no oblozhenija pchelovodstva obrokom i o perehode ot bortevogo k pasechnomu (Regarding the taxation of beekeeping dues and the transition from beehive to apiary). F. 398, O. 7, No 1937 (in Russian)

[RSHA] 1845–1856. O vyselenii iz dach Grodnenskoj gubernii semejstv zashtatnoj strazhi (On resetelment the fired forest personnel from the state forests of the Grodno province, Part 1. F. 387, O. 2, No 22855 (in Russian)

[RSHA] 1847. Po rassmotreniju taksacionnyh rabot v Belovezhskoi Pushche za 1843–1846 (On reviewing of the forestry inventory work in PBF in 1843–1846). F. 387, O. 25, No 5 (in Russian)

[RSHA] 1849. Po rassmotreniju taksacionnyh rabot v Belovezhskoi Pushche za 1848 g. (On reviewing of the forestry inventory work in BPF, 1848). F. 387, O. 25. No 6 (in Russian)

[RSHA] 1850. O vyrubke v Belovezhskoi Pushche vmesto naznachennoj lesoseki povrezhdennyh stvolov vyborochno (On selective logging of damaged trees instead of the designated cutting in BPF). F. 387, O. 25. No 6 (in Russian)

[RSHA] 1854–1862. O zubre, dannomu pomeshhiku Valickomu dlja opyta v smeshenii krovi s domashnim skotom (On the bison given to landowner Walicki for experiments with mixing blood with experience in mixing blood with cattle). F. 387, O. 27, No 34 (in Russian)

[RSHA] 1857–1860. O komandirovanii v Belovezhskuju pushhu vmesto Bul'meringa polkovnika Dlatovskogo (On sending to BPF Colonel Dlatovsky instead of Bulmering). F. 387, O 2. No 22327 (in Russian)

[RSHA] 1857–1867. O vyselenii krest'jan iz Belovezhskoj pushhi (On peasants resetlement from BPF) F. 384, O. 6. No 1308 (in Russian)
[RSHA] 1859–1867. Ob ustrojstve lesnoj strazhi v Belovezhskoy Pushche i prochih dachah Grodnenskoj gubernii (On the reorganization of the forest guard in the BPF and other state forests of the Grodno province). F. 387, O. 2, No 22086 (in Russian)
[RSHA] 1859–1874. O pozhalovanii akademiku Zichi zvanija zhivopisca Ego imperatorskogo velichestva (On the award to academician Zichy the title of painter of His Imperial Majesty). F. 472, O. 18, 112/949, No 35 (in Russian)
[RSHA] 1861B. O nastojashhem sostojanii Belovezhskoy Pushchi i o budushhem ee ustrojstve (On current condition of BPF and on projects to improve its management). F. 387, O. 25. No 27 (in Russian)
[RSHA] 1861–1872. Ob ustanovlenii obrochnoj platy za ustroenie pasek v lesah kazennogo vedomstva (On the establishment of fee for the renting land plots for of apiaries in the state forests). F. 387, O. 3, No 25121 (in Russian)
[RSHA] 1862–1865. Ob iskljuchenii iz chisla bortevyh uhozhij iz chisla obrochnyh statej (On exclusion of beekeeping trees from the lists of rented articles). F. 387, O. 3, No 24412 (in Russian)
[RSHA] 1871–1900. Ob ustrojstve pasek v kazennyh lesah (On making apiaries in state forests). F. 387, O. 3, No 26942 (in Russian)
[RSHA] 1880. Statisticheskie svedenija o lesnoj torgovle v Grodnenskoj gubernii, I otdel. Les i ego obrabotka (Statistical information on forest trade in Grodno Province, 1st branch. Forest and its products). F. 387, O. 28, No 1707 (in Russian)
[RSHA] 1905–1911. O rashodah po Belovezhskomu zverincu v 1906 (On expenses for game reserve in BPF in 1906). F. 515, O. 80, No 406 (in Russian)
[RSHA] 1912–1913. O razreshenii raznym litsam prava past'by skota (On allowing different persons the right to graze livestock). F. 515, O. 80, No 1058 (in Russian)
[SARF] 1860. Dolmatov D. Istorija tura ili zubra i Belovezhskoj pushchi (The history of aurochs or bison and Białowieża Primeval Forest). F. 728, O. 1, No 2620 (in Russian)
[SPB ARAS] 1860. Perepiska po delam Zoologicheskogo Muzeja za 1860 g. (Correspondence on the matters of Zoological Museum for 1860). F. 51, O. 3, No 30 (in Russian, German and French)

Chapter 6
The Restoration Period (1861–1888)

Abstract The history of Białowieża Primeval Forest (BPF) experienced a pivotal moment with the royal hunt by Alexander II in 1860. The impact of the hunt was to trigger a gradual change in BPF management (creating a game reserve, purchasing deer, exterminating predators, restricting logging) that ultimately transformed it into the private hunting ground of the Russian Tsars; a process completed in 1888. There was a major shift from the incremental moves towards commercial forestry driven by nineteenth-century scientific forestry towards being a hunting ground based on principals of German "rational" game management. The changes that the era of Great Reforms brought in the 1860s, sparked conflicts between former state peasants and BPF administration over traditional forest use. At the same time, as this change was sweeping through the forest interest in the natural world and in new ideas such as evolution and natural selection were emerging throughout Europe and North America. With this new enlightenment, the forest became subject to increasing interest and scrutiny from a more educated society. A consequence of this change was that the forest appeared regularly as a topic in publications devoted to nature and forestry. Furthermore, the region was visited by prominent artists and writers and perceptions and ideas relating to the forest and its European bison emerged from that time and some remain with us today.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

6.1 Summary

The hunt by Alexander II in 1860 began the gradual change in the management of BPF that eventually led to its transformation into the Tsar's private hunting ground in 1888. At the same time, the Forest became the subject of interest to an educated society.

After the new forest inventory in 1861–1862, all felling of living trees was stopped and all hunts were halted for the purposes of bison protection and the organization of further possible Tsar's hunts in BPF. (Royal hunts took place in 1865, 1875, 1879,

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_6

1880, 1881 and 1885). A *zverinets* (a menagerie or game reserve) with bison, fallow deer, roe deer, and wild boar was created in the hunting enclosure fenced for the purposes of 1860 hunt. The enclosure eventually covered 874 ha and incorporated different types of forest habitats, including a stream and two artificial canals, and by the end of the period hosted eight bison, 200 red deer (reintroduced species originally present in BPF), up to thirty fallow deer (an alien species introduced here for the first time), a few roe deer, and wild boar.

Year 1863 saw the Polish January Uprising spreading to the former Grand Duchy of Lithuania and BPF and the Forest became the centre of concentration for insurgents from the entire Grodno Province. The insurgent forces passed through BPF several times and skirmished with Russian forces, but the uprising received much smaller support from local population than the 1831 one.

Great reforms in the Russian Empire in 1860s saw peasants on state-owned lands receiving allotments of hayfields located inside or on the border of the Forest, which reduced grazing areas for European bison and restricted their access to water, increasing the number of people present in BPF. Forestry administrators saw especially the latter as a threat, together with herders' and beekeepers' activity in the forest, often connected with ground fires in the forest. In an attempt to remove beekeeping from state forests, the Forestry Department ordered the speeding up of transition from forest beekeeping to apiaries. Beekeepers filed numerous complaints and protest letters, and received support from the head of BPF, which postponed the decision to remove traditional beekeeping from BPF until the 1880s. After 1888, forest beekeeping was discontinued.

In the period 1861–1888, no well-established researcher visited BPF, but the Forest appeared regularly in publications devoted to nature and forestry. For example, Piotr Bobrovskii compiled information on Grodno Province, forestry specialists Nikolai Kholshevnikov and Victor Tutsevich wrote on BPF management for the sake of bison, Władysław Zaleski and Vsevolod Krestovskii published travel reports, Adam Honory Kirkor posted an encyclopaedic entry on BPF, Antoni Wałecki reviewed the information on European bison and beaver, and Józef Siemiradzki documented geological information on the Forest. In 1882, later becoming a Nobel Prize winner, Henryk Sienkiewicz visited BPF.

6.2 Forestry Department's Attempt at Creating a Game Reserve in BPF

The beginning of the period was marked by the outcome of the hunt by Alexander II in the BPF in 1860. This event shaped the approach to the forest protection and management for the remainder of the nineteenth century and more than a decade of the twentieth.

The hunt attracted the attention of many members of the Romanov family to European bison and BPF itself. Thus began the gradual change in the management of the forest that eventually led to the transformation of this are into the Tsars' private hunting ground in 1888. Additionally, stoked up by popular articles and a luxurious book describing the Forest and the Tsar's visit in 1860 (Fuchs and Zichy 1862), the Forest became the subject of interest of educated society.

The Ministry of State Domains ordered a revision of BPF in 1861 by an inventory team under the leadership of the forest specialist Adolf Eichwald. One of the members of the forest inventory team was Nestor Genko, who three decades later led an inventory of BPF and published a monograph on forest communities and forestry work in BPF (Genko 1903).

In 1861–1862, this forestry inventory team worked in BPF, measured the Forest, estimated the timber and firewood volume and made new plan of forest use. Compared to previous measurements from 1843–1847, the Forest area had decreased by 3,741 ha, and at the same time land under different uses increased by 3,082 ha, the unused land increased by 1,350 ha but the total area of BPF remained more-or-less the same (112,713 dziesiatinas or 123,141 ha; Genko 1903).

Eichwald's team returned to Dolmatov's ideas of digging river canals to enable wood rafting and to create wood tar manufacture in the northern part of the forest, yet these plans were again rejected by the Ministry of State Domains. In 1863, similarly as after the forest inventory in 1840s, for the purposes of bison protection all felling of living trees was halted (RSHA 1865), and the plans to intensify forest management again remained on paper alone. In his work, Genko suggested it was rather the fear of misconduct in the wood trade that motivated the ban imposed by the ministry (Genko 1903).

Recognizing the uniqueness of BPF, the Ministry of State Domains has determined that one of the four foresters should be considered the head of BPF and submit directly to the head of the Grodno Administration of State Domains (bypassing the regular chain of command of the forestry administration). Adolf Eichwald became the first head forester of BPF (RSHA 1861).

Given the difficulties with gathering animals for royal hunts (see Sect. 5.3) and foreseeing further possible royal hunts in BPF, Eichwald as the head forester of BPF outlined a plan to preserve and promote the main game species of the forest. The plan consisted of: (1) continuing the prohibition of all hunts, (2) creating the *zverinets* (a menagerie or game reserve—an enclosure fencing few hundreds ha of the forest) with bison, fallow deer, roe deer and wild boar in the hunting enclosure fenced for the purposes of 1860 hunt, (3) creating similar small enclosures for bison in all forest districts of BPF and (4) eradicating all wolves (Karcov 1903; NHARBG 1861). Except for creating small game reserves, all Eichwald's propositions were accepted by the ministry and executed in 1861–1866.

Once again it was stated that hunting inside the forest was prohibited, and the administration should and would intensify the fight against poachers. In the ironic turn of events, the only cases when poaching was reported in the following years, turned out to be raids against wolves organized by forestry administration (NHARBG 1868–1870).

Eichwald's successor in the post of head forester, Karl Strahlborn, in 1866 proposed to establish stable herds of European bison, wild boar, red deer, fallow deer, and hares in the enclosure to provide game for royal hunts and supply the population in the forest with "fresh blood". Work on the *zverinets* continued, and the enclosure eventually covered 800 diesiatinas (874 ha) and incorporated different types of forest habitats, a stream and two artificial canals. Almost a half of the enclosure was reserved for bison, the rest divided into yards for red deer, fallow deer and roe deer, and wild boar (the idea of keeping hares there was dismissed) . In 1865, in exchange for four bison (1 bull and 3 cows) captured alive and sent to Prince Pless's estate in Silesia, BPF received 18 deer (5 males and 13 females) (Karcov 1903). By the end of this period, in 1888, the menagerie held from eight to twenty-four bison and 200 red deer (with several deer escaping to the forest through holes in the fence). In the period from the 1860s to the 1880s, the enclosure also held 20–30 fallow deer. These animals were descendants of four individuals that survived the 1860 hunt and fifteen others that were purchased in 1864. There were also a few roe deer and wild boar (Karcov 1903). The Białowieża's wild boar were considered the largest in Europe, and taken to St. Petersburg several times for royal hunting menageries. In 1894, couple of dozens of wild boar were donated to the German Kaiser (RSHA 1857–1861, 1866–1867, 1892–1896). Although the forest administration had a very limited budget to maintain the enclosure, the number of animals increased or remained stable, with occasional incidents like the one in 1875, when around 10 fallow deer were killed by lynx (Karcov 1903).

Among the Great Reforms in the Russian Empire in this period, the most important for BPF administration was the reform of the Ministry of State Domains in 1866 that guaranteed the state peasants the right to use their land allotments in return for set payments (Ministry of State Domains 1888; Druzhinin 1958). In the case of BPF, the change resulted in many peasant communities receiving additional areas of hayfields located inside or on the border of the forest. This meant reduced grazing areas available for European bison, restricted access to water (since most of the haymaking areas in the riverine meadows were given to peasants), and increasing numbers of people present in the Forest. Foresters protested against the decisions of the Ministry of State Domains to endow peasants with these meadows pointing out that remaining bison meadows were of poorer quality, marshy, or overgrown with shrubs, resulting in low quality of the collected hay. These arguments were not taken into account (NHARBG 1869).

Another problem for the BPF administration was the general standardization of taxes and natural duties conducted by the government during these years. The beaters, 102 peasant families from four villages located on the southern and eastern border, were previously obliged to prepare hay for bison on forest meadows. However, they now started to pay regular taxes, and the forestry administration was forced to find another way to provide hay for bison, for example by hiring the same peasants for this purpose (RSHA 1874). The new system required a better way of monitoring. For example, in 1869, it was reported that preparation of hay for winter feeding of bison was not satisfactory. Another option for organizing hay harvesting guaranteed the effectiveness of haymaking, but halved the amount of bison hay. In this case, haymakers preparing hay for the bison took half of the hay for their own purposes, with the forest administration supervising the hay distribution. Peasants had the right to take their half of hay only after the bison hay was delivered to feeding points (RSHA 1889–1894).

The supplementary feeding of European bison was slightly modified. Once again it was stated that places where bison were feeding should be protected in summer against cattle pasturing and haymaking by local people. Part of hay was traditionally left on meadows, but most was transported to shacks built in various compartments of the forest in 1870. Gradually, other feeding constructions were also erected and by the end of the period, there were 4 barns for fodder, 8 wooden roofs, and 10 feeding racks (Karcov 1903).

Data on hay provided for bison in this period is available only for years 1868–1886 (Karcov 1903; NHARBG 1875, 1876, 1877, 1878, 1879, 1880, 1881, 1882, 1883). An average yearly supplement of hay amounted to over 306 tonnes (0.6 tonne of hay per bison), ranging from 75 to 470 tonnes in different years (Samojlik et al. 2019).

In 1863, the January Uprising Polish broke out in the Kingdom of Poland, quickly spreading to the lands of the former Grand Duchy of Lithuania. Although the January Uprising received less support among the local residents of BPF than the previous one in 1831, the Forest became the centre of concentration for insurgents from the entire Grodno Province. The insurgent forces, commanded by Onufry Duchyński and Walery Wróblewski (graduate of the Imperial St. Petersburg Forestry Institute) (Grzywacz 2012) passed through BPF several times. They skirmished with Russian forces and marked their trail with graves. According to the memoire of Wróblewski's adjutant, Ignacy Aramowicz, insurgents passed through BPF with difficulty, manoeuvring through fallen trees and dead branches—in these circumstances, Wróblewski called BPF "gardens of Gedymin", referring to the twelfth-century Lithuanian Grand Duke (Kozłowski 1981).

Royal hunts by Grand Duke Nicholas Nikolaevich the elder, his son Grand Duke Nicholas Nikolaevich the younger, and Grand Duke Vladimir Alexandrovich took place in BPF in 1865, 1875, 1879, 1880, 1881 and 1885. Russian princes were often accompanied by German dukes and counts (SARF 1860–1915). Many of trophies of those hunts ended up in national natural history collections (Fedotova 2018; Samojlik et al. 2017).

In the years 1870–1871, BPF was once again inventoried and measured which this time suggested 82,638 dziesiatinas (90,282 ha) of forest, 2,586 dziesiatinas (2,825 ha) under different uses, and 8,429 dziesiatinas (9,209 ha) of wastelands, in total 93,655 dziesiatinas (102,316 ha) (Genko 1903). The decrease of the total area under forest administration compared to measurements from 1861 was explained by the amount of land offered to peasants in the light of the reforms of 1860s. As a result of the inventory, the administrative division of BPF was changed: instead of the previous five, four new forest districts were created, with the idea of using the Forest's rivers and streams as natural borders of the districts (Genko 1903).

In 1875, for the purposes of bison protection, a fence was built to divide the outer parts of the Forest not frequented by bison (this part of the Forest was to be exploited) and the inner part, where bison were present. Reportedly, the fence survived only until mid-1880s , when it fell apart (Genko 1903). In the new management plan of 1877, timber production was renewed but only in the outer part of the forest which lacked bison. The rotation age for Scots pine, oak and ash was set again as very long (to 180 years, and changed to 200 years in 1885) (Genko 1903). Reports from 1886 prepared for the purposes of transferring the forest to Tsars' domains (RSHA 1886), again stressed timber felling as one of the main factors threatening the bison population. Exploitation lasted until 1888, and was evaluated by forestry specialists as very limited (RSHA 1886–1895).

The policy of predator eradication was continued (NHARBG 1850–1851, 1867), and was sufficiently effective to drastically limit the populations of wolf and lynx, and completely exterminate brown bears from BPF (Karcov 1903; Samojlik et al. 2018; see Box 6.3).

Nevertheless, in 1869 it was still reported that many bison calves were killed by predators, especially wolves, and the forest services were accused of ineffective eradication of predators (NHABG 1869). This information contradicted data published by Karcov (1903), which indicated that only 19 bison were killed by wolves, and 4 by bears, in the period between 1861 and 1888.

In 1869, the head forester of BPF, Alexey Zhohovskii, prepared a report on bison conservation measures to explain the losses in bison population and to propose measures to enhance bison breeding in the Forest (NHABG 1869). Means of enhancing bison breeding by Zhohovskii included employment of ten bison wardens that would be responsible for more precise bison counting, keeping track of bison whereabouts and current numbers, and observing hay produced for them. Following this recommendation ten bison warden posts were introduced to BPF in the 1870s. In 1884, the number of bison wardens was increased to 22 (Karcov 1903).

In the management rules, it was also proposed that old solitary bulls, believed to join herds during rut and scare away or kill young adults and so preventing them from breeding, should be killed off in the late autumn or at the end of winter (NHABG 1861). This plan was not executed, and the reason behind that was explained in 1869. It was felt that local people would not understand this means of management and would treat it as an incentive to poach bison (NHABG 1869).

The next revision of the forest management planning took place in 1884–1885, with very limited new measurements . From 1885, the clear cuts were renewed , but the results of tenders for forest clearing were disappointing, with less than a half of plots offered for clear cuts by the forest administration eventually sold (Genko 1903; Karcov 1903).

In September 1888, BPF with Świsłocz Forest were transferred to the Ministry of Imperial Court (Appanage Department, *Departament Udielov*). This was in exchange for the equivalent appanage forested area in the Oryol and Simbirsk provinces. Thereby, the task of turning BPF into the Tsars' private hunting reserve started in 1860 finally came into fruition (RSHA 1886, 1886–1895).

6.3 European Bison from BPF in the Collections of Museums, Universities and Curiosity Chambers Around the World

From the beginning of the nineteenth century, obtaining a European bison specimen (in form of a skeleton, or stuffed animal) became an objective for many natural history collections, museums, and universities across Europe. However, the rarity of the species and the fact that since the decree of 1802 the main goal of forest management in BPF was to preserve bison, made the administrative procedures for obtaining bison for scientific purposes complicated and time-consuming. Already in the beginning of the nineteenth century, such requests were only very reluctantly granted. In 1803, the General Governor of Vilnius Province, Levin August von Bennigsen asked for permission to hunt an old bison in BPF in order to present a stuffed specimen to the Imperial Vilnius University. In response, Tsar Alexander I ordered: "*Wait before one of these animals dies, and then pass the skeleton and mannequin*" (Hedemann 1931).

A new application from Vilnius University came in 1819, initiated by Ludwig Bojanus (1776–1827), who explained the need to obtain a specimen of a species which "*would sooner or later be completely exterminated*". Alexander I agreed for the hunt and in January 1821 a male and a female European bison were killed (RSHA 1819–1821). Acquired specimens, as well as osteological material from museums in German lands and Baltic provinces of the Russian Empire allowed Bojanus to describe two separate species of extinct bovines: the aurochs, *Bos primigenius* and the steppe bison, *Bison priscus* (Bojanus 1825). Nevertheless, the dispute on bison and aurochs identity was prolonged until publication by August Wrześniowski resolved their taxonomic distinction (Wrześniowski 1878).

Three bison were killed in 1823, two for the museum of Warsaw's Forestry School (Brincken 1826) and one for the St. Petersburg Kunstkamera (NHARBG 1821–1824), with the latter used in 1836 by Karl von Baer to compare it with the pelt of Caucasian bison (Baer 1836). Alexander I's successor Nicholas I (reign 1825–1855) softened the policy against bison hunts. Yet obtaining the Tsar's permission to shoot or live-capture such animal was still easier to obtain if it was justified by scientific motivations. Data on European bison hunts with the Tsar's permission in the nineteenth century and the early twentieth century show that there were more "scientific" than purely "recreational" hunts in BPF. In the period before the first imperial hunt of 1860, 8 bison were shot for recreation, and 33 for museums or natural history collections (Samojlik et al. 2017; Fedotova 2018).

To obtain such a precious gift, museums and scientific institutions had to employ all bureaucratic, diplomatic, and even personal means. The bureaucratic procedure of obtaining the Emperor's permission was formalized in 1847 (RSHA 1847). For foreign museums and collections, the first step was to contact the Ministry of Foreign Affairs (through the Ambassador in St. Petersburg), which in turn contacted the Minister of State Domains, which then applied directly to the Tsar. Russian institutions applied to the Ministry of Education which had to forward the request to the Minister of State Domains, with the rest of the procedure being the same

(Fedotova 2018; Fedotova et al. 2018). After the Tsar's approval, the entire bureaucratic machinery of organizing a hunt in BPF (which involved all forest personnel and several hundred beaters recruited from local residents) began. Museums had to delegate a naturalist and taxidermist to hunt the animal and conserve the specimen, and also take care of transportation of the large animal (an extremely difficult task before the railway to Białowieża was opened in 1897). A good example of the bureaucratic and logistical struggle that these hunts posed is the case of European bison obtained for the Zoological Museum of the Imperial Academy of Science in 1847–1848. In 1847, the head of the museum Johann Brandt (1802–1879), following his plan to expand the collection as a base of state-of-the-art zoological studies (Brandt 1865), applied for the Tsar's permission to shoot three bison in BPF (SPB ARAS 1847–1858). In the subsequent exchange of letters, Brandt explained that the European bison specimen the Academy of Science received in 1823 was in a bad condition, destroyed by insects. The occasion to kill the requested bison arose in autumn of 1847, with one of the most prominent naturalists of the Academy, Alexander von Middendorff (1815–1894) agreeing to travel to the Grodno Province. The Ministry of State Domains and the Ministry of Education interfered in the plan, ordering Augustus Schuster, the taxidermist of the St Vladimir University in Kiev to travel to BPF to stuff animals killed by Middendorff, but also to kill an additional two bison, stuff them and store in the Grodno Chamber of the State Domains for the purposes of future requests (RSHA 1847–1849). As it turned out, none of the parties involved in the plan were happy with this plan: the Academy was supposed to cover all the expenses, Shuster had to leave his post in Kiev for a prolonged period, and St Vladimir University's zoological cabinet was to be left unattended for this period. However, this was not the end of problems. When Middendorff received the Tsar's permission to shoot three bison and arrived to Grodno in December 1847, he learned that a hunt was already scheduled, but at the end of the month. Not being able to stay in Grodno that long, Middendorff killed one bison (a solitary bull wandering outside BPF), processed the pelt and skeleton and preserved some internal organs in alcohol (SPB ARAS 1847–1858). Schuster arrived at BPF after Middendorf had already left and participated in the hunt scheduled for late December. He then spent almost six months preparing the two stuffed animals ordered by the Ministry of State Domains—most likely one of them was sent to the St Vladimir University in Kiev in return for Schuster's prolonged absence (Bajko 2004).

The first Tsar's hunt organized in BPF on 6th–8th October 1860 for Alexander II and his entourage resulted in the killing of 28 bison. Apart from recreational purposes, the hunt served as an occasion to reinforce the diplomatic connections (based on family ties) with German princes. The idea of the hunt could come from similar event organized by Tyszkiewicz in 1859 (Mina 1859), but also was also used as an opportunity to fulfil pending requests for European bison exhibits. Since 1858, Western European museums had filed several requests for bison (Jena, Giessen, Stockholm, Dresden, Waldau, Hannover and Strasbourg; RSHA 1858, 1858–1860). Alexander II most probably wanted to not only capitalize politically on the Białowieża hunt, but also to present himself as a patron of science, offering the outcome of the hunt to Russian (and foreign) universities. A taxidermist from the Zoological Museum of the Russian Academy of Science, Pamphil Ivanov, was sent to BPF two weeks

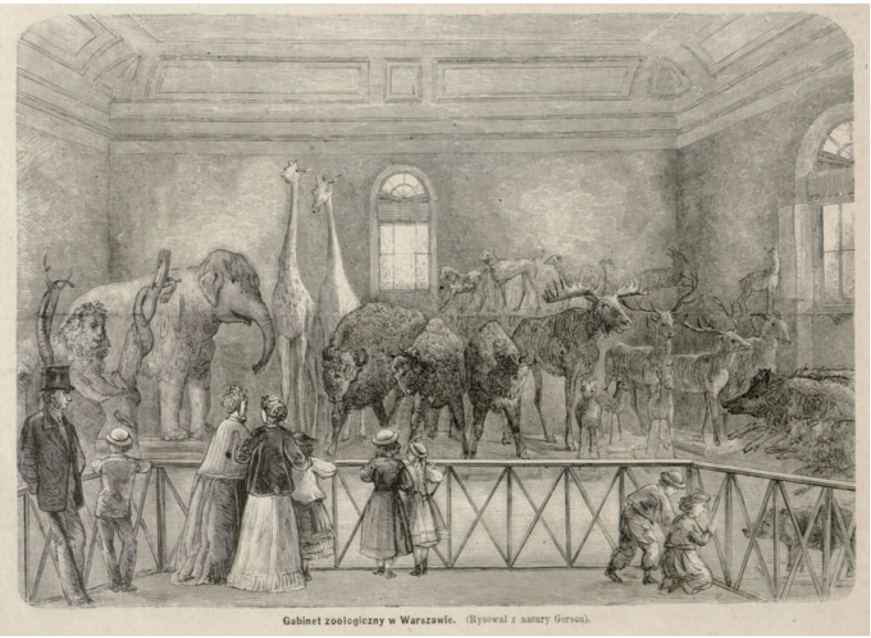

Fig. 6.1 Three stuffed European bison in the Zoological Cabinet of the Warsaw Main School, drawing by Wojciech Gerson from "Tygodnik Ilustrowany" (Taczanowski 1869)

before the hunt to make sure all conditions necessary for his work would be observed (SPB ARAS 1860). However, despite all promises of help, he was left alone with his enormous task. He managed to skin all the animals within ten days, but some of the carcasses deteriorated in the meantime (as no proper cold storage was prepared). In February 1861, after finishing his work with the pelts and skeletons, Ivanov packed them in several packages, separate for the Zoological Museum of the Imperial Academy of Sciences, Russian universities (Moscow, Kharkov, Kiev, and Dorpat), the Russian Imperial Court, German princes, and additionally for Freiburg University Heavy parcels were sent from Białowieża in May to July of that year (SPB ARAS 1860, 1861; RSHA 1861–1862). Seven packages were mistakenly sent to Freiburg University (five of them were intended for German princes and the Russian Imperial Court. In 1862, Freiburg University had returned some of the exhibits it had mistakenly received and the Russian Emperor and German princes eventually got their trophies (RSHA 1861–1862; Fedotova 2018; Fedotova et al. 2018).

Despite these hunts being a bureaucratic and logistic nightmare, many prominent museums, collections and universities received European bison specimens throughout the second half of the nineteenth century (Samojlik et al 2017; Fedotova 2018; Fedotova et al. 2018). In many cases, these exhibits became main attractions, presented in the centre of the hall, and overshadowing even such exotic species as giraffes, elephants, or lions (as it was the case in Warsaw's Zoological Cabinet: Fig. 6.1).

Shipping European bison specimens from BPF to numerous European natural collections had a long-lasting impact reaching beyond the simple fact of obtaining a prestigious object by interested organization. It meant also popularizing the species and especially the place where it found its refuge, i.e. the forest of Białowieża. The importance of this "familiarization" of European societies with bison cannot be overestimated in the light of Europe's concerted effort to reinstate to population of bison after their extinction in 1919 (de Bont 2017).

Thanks to animals captured alive in BPF, several zoological gardens and animal reserves e.g. Berlin, Moscow, Constantinopole, Gatchina, Pszczyna, Schönbrunn, and Dresden, were able to breed their own bison throughout the nineteenth century (Karcov 1903; Sztolcman 1927). Gatchina and Pszczyna especially, became important as breeding centres, with Pszczyna being a source of bison for zoos throughout Europe (for more details on Pszczyna's bison—see Sect. 8.3).

6.4 Traditional Beekeeping and Beekeepers in BPF

Throughout the most of the nineteenth century, the forestry administration considered the traditional ways of forest utilization by local people harmful to the forest and adversely affecting the introduction of "proper", "rational" forest management. But to remove centuries-old types of use was not an easy task, especially in the case of traditional beekeeping. In the royal period, this activity was the second most widespread type of forest utilization based on access rights to BPF (Hedemann 1939). Beekeepers' knowledge was passed from generation to generation, especially the craft of carving artificial hollows on average at 7 m above ground (from 3.3 to 15 m) in Scots pine trees (occasionally oak and Norway spruce), attracting wild bees to settle in them, tending them, protecting from weather impact and against brown bear, pine marten, or forest dormouse and collecting honey and wax once or twice per year (Blank-Weissberg 1937; Karpiński 1948). The beekeepers' activities created a specific type of landscape with trees with artificial hollows carved in the trunk and other man-made elements visible: wooden covers of beehives, hangers hammered into the trunk, ownership marks carved in the bark. There were also other forms of beehives in BPF: wooden platforms with logs with beehives placed in tree crowns and similar single logs attached to tree branches (see Fig. 4.5). Some of the beehive trees were also equipped with protective measures against animals, such as sharpened poles dug at the base of the tree, wooden "collars" embracing the trunk to prevent animals from climbing the tree, and logs hung on a rope acting as a pendulum when bears pushed them while trying to get to the beehive (Blank-Weissberg 1937; Karpiński 1948).

Hereditary rights to beehive trees held by beekeepers, a code of law specific to their guild, and especially the deeply-rooted tradition of this type of use made it a hard row to hoe for the forestry administration. In the first third of the nineteenth century, when forest management was still under the supervision of the Ministry of Finance, forest beekeeping was tolerated as one of the traditional crafts of peasants. However, it was observed that in BPF the annual rent paid for using beehives gradually decreased, especially compared to the royal period. In the 1790s for example, rent from around 1,000 beehives was collected (Hedemann 1939), but by the 1820s, this amount had reduced threefold (RSHA 1828, 1829)—which was consistent with the general trend of forest beekeeping being gradually replaced by household apiaries (Loskutova and Fedotova 2019). Julius Brincken visiting the Forest in 1823 noted that the evolution towards house apiaries was visible: "Gradually, beehives in trees are replaced with hives. Such change is simple and does not bring any loss to the owners. They are allowed to cut trunks 6–7-feet long for hives and to place them in the garden where, in spring, bees settle" (Brincken 1826).

After the transfer of responsibilities over BPF to the Ministry of State Domains, a more heated argument began over the presence of beekeeping in state forests. In 1838, the ministry ordered an inventory of all forest beekeeping in the state forests, along with an overview of its positive and negative aspects. The inventory revealed that traditional beekeeping was still exercised in at least eighteen provinces of the Russian Empire (excluding former Kingdom of Poland) in the late 1830s to the early 1840s (RSHA 1843; Loskutova and Fedotova 2019).

The arguments against forest beekeeping were numerous: from the low income from this activity, through unauthorized access and the continuous presence of beekeepers in state forests, damage to trees with carved beehives, and the low productivity of traditional forest beekeeping. Probably most important from the point of view of forest management, was the argument that beekeepers needed fire to handle bees and often intentionally started ground fires in the forest. The latter argument was confirmed by research into the fire history of BPF, which showed that in the period when beekeeping activity reached its peak, in the seventeenth to eighteenth centuries, low-intensity, small-scale ground fires happened very often, with mean stand scale fire intervals of 6 years (Niklasson et al. 2010).

The Ministry of State Domains ordered an analysis of actual impact of traditional beekeeping by an expert, Mikołaj Witwicki (1764–1853), the precursor of the modern beekeeping and author of beekeeping manuals (published in Polish and Russian, e.g. Witwicki 1829). Witwicki in his notes addressed to the ministry, defended forest beekeeping and pointed out its various benefits—from financial income, through advantages of wild bees over domesticated ones, and to the fact that experienced beekeepers could become the most reliable assistants of the forest guard in protecting forests (RSHA 1838–1841, 1839–1842, 1842). Not convinced by his arguments, in 1840 the ministry asked two other experts for their opinion. These were Piotr Prokopovich (1775–1850) and Alexander Pokorsky-Zhoravko (1813–1874). Their opinions contradicted Witwicki as they saw forest beekeeping as a primitive stage of development of modern beekeeping, much less profitable compared with house apiaries, and also risky from the point of view of beekeepers climbing high trees to get to their beehives (RSHA 1838–1849). Also there was the lack of control over beekeepers, accusations of idleness, stealing honey and timber, and even hiding fugitives in the Forest (RSHA 1839–1842). Overall, both recommended transforming forest beekeeping into apiaries by the means of increasing rent for forest beekeeping, and strengthening the supervision by forest guards. They suggested prohibiting the carving of new beehives, and at the same time creating favourable conditions for renting land for apiaries and training peasants in "improved" beekeeping (RSHA 1812–1877, 1838–1841, 1838–1849; Loskutova and Fedotova 2019).

Negative arguments almost tipped the balance, but the Minister of State Domains Count Pavel Kiselyov stated that thetransition from forest beekeeping to house apiaries had to be "not in a form of strict order but as gradual, voluntary act resulting from peasants' conviction of apiary superiority over forest beekeeping" (RSHA 1838–1849). Therefore the state forestry administration forgot about beekeepers for another two decades. During these years, a significant share of beekeepers moved their bees to house apiaries.

During the inventory of forest beekeeping, only a few provinces provided arguments supporting this activity, and Grodno Province was among them. The Grodno Chamber of State Domains explained that in the three-state forests of the province (including BPF) there were almost eleven thousand beehives (both with bees and empty) from which annually a rent of about a thousand roubles was collected. Furthermore, this was observed to do little harm to forestry, as only old trees were used for beehives (RSHA 1838–1849).

A change in attitude towards forest beekeeping was brought by reforms of the Ministry of State Domains in the 1860sand termination of ministry's care over state peasants (1866). In the result of the reform, peasants living on state lands received arable land, haymaking meadows and pastures in communal property; however, access rights to traditional forest resources were limited and under tighter cotrol (Gins and Shafranov 1914). In 1862, the Forestry Department of the Ministry of State Domains ordered the discontinuation of all low-yielding rents, including beekeeping to reduce the administrative efforts connected with them (Forestry Department 1862; RSHA 1862–1865).

This regulation did not go unopposed as beekeepers filed numerous complaints and protest letters to the chambers of state domains and to the ministry, which also received similar letters from forest guards, in many cases renting beehives themselves (RSHA 1862–1865). In BPF, the head forester Karl Stralborn defended the beekeepers, most probably due to the fact that since the times of Polish-Lithuanian Commonwealth a significant part of beehives was rented by Białowieża's forest guards. Stralborn's intervention helped postpone the decision to remove traditional beekeeping from BPF until the 1870s (RSHA 1861–1872). At that time however, the attitude towards traditional beekeeping started to change, with more and more foresters seeing them in more positive light. For example, Forestry Department received a note from Mogilev Province in 1884, and from Vilnius Province two years later—in both forestry officers described the negative impact of banning beekeeping in the forests and proposed the return of this use. By the late 1880s, the Forestry Department allowed to develop local temporary rules of beekeeping in state forestry districts, and by 1903 created general rules for state forests in the entire Empire (RSHA 1871–1900; Loskutova and Fedotova 2019). BPF, however, was already at that time incorporated into the Tsars' private properties, and there was to be no return for forest beekeeping (Karcov 1903). At this moment "*forest beekeeping underwent complete* *destruction: beekeepers were gradually removed from the forest and existing beehives were taken over by forest administration. From that time, beehives devoid of their owners and caretakers were overexploited and in an express pace met their doom*" (Karpiński 1948).

6.5 BPF in in the Works of Naturalists, Artists and Travellers Until 1888

In the period 1861–1888, no well-established researcher visited BPF, which did not mean that it fell out of the scientific and cultural circulation. The Forest appeared regularly in publications devoted to nature and forestry, and in more popular writings, usually accompanied with drawings depicting the forest or European bison, which remained the main attraction for public (Fig. 6.2). Often, these publications repeated the same information, so here we will focus only on a few more original approaches to the description of BPF in this period. To this end, we consider the works of two forestry specialists Nikolai Kholshevnikov and Victor Tutsevich on BPF management for the sake of bison, and travel reports by two popular authors—Władysław Zaleski and Vsevolod Krestovskii. We also note Adam Honory Kirkor's encyclopaedic entry on BPF, Antoni Wałecki's review of the information on European bison and beaver, and Józef Siemiradzki's account of geological information on the Forest. Among these works, a compilation of information on Grodno Province by Piotr Bobrovskii should be mentioned in the first place.

The description of the Grodno Province (Bobrovskii 1863) was created as a volume in the series "Materials for Geography and Statistics of Russia collected by officers of the General Staff". The main value of Bobrovskii's work is the collection of statistical information on the province available in the early 1860s: on its topography and hydrology, the use of natural resources, about trade, agriculture and forestry, about the dynamics and composition of the population, about roads, local administration, religion and education. A separate article on BPF included herein, was compiled using both already published books and new materials provided by Adolf Eichwald directly from the Forest. The book also contains a map of BPF that was probably the preliminary result produced by the forest inventory team working in BPF in the early 1860s (Fig. 6.3).

In the 1870s, the "Forestry Journal" presented two big articles and several small notes on BPF. The journal was published by St. Petersburg's Forestry Society and addressed to forestry specialists of the Russian Empire, having a significant impact on them. The first paper was published in 1873 (Kholshevnikov 1873), and the second one in 1878 (Tutsevich 1878). Both were written by members of forest inventory teams working in BPF (Kholshevnikov in 1871, Tutsevich in 1877), and borrowed a lot of data from published sources (especially in the description of bison). Kholshevnikov, however, objected to the majority of naturalists who agreed that bison was doomed to extinction. He saw no signs of degeneration in the species and explained

Fig. 6.2 European bison on a drawing accompanying a popular description of the species (Anczyc 1876)

the decrease in numbers only with the lack of food, as a significant part of the riverine meadows was given to peasants based on 1866 reforms. As a remedy, Tutsevich and Kholshevnikov proposed to remove peasants from BPF and introduce controlled burning to improve the feeding conditions for bison and to save the pine from undergrowth of less valuable species that hinder it—primarily spruce. What was unique in Tutsevich's approach was the fact that his paper basically defended the abandonment of modern forest management in BPF for the sake of European bison.

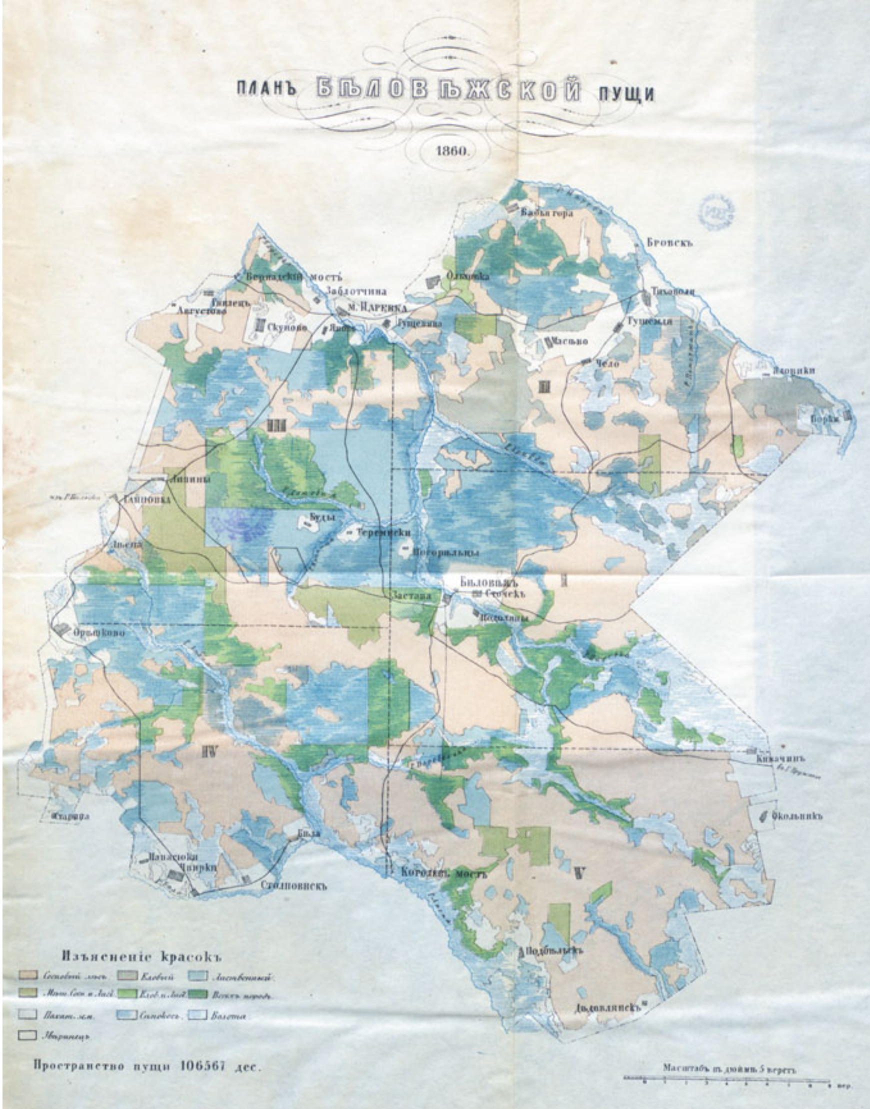

Fig. 6.3 Map of BPF prepared by the forestry specialists in 1861–1862, published in 1863, showing the forest composition (coniferous, deciduous and wet forest distinguished) and division of the Forest into five forest districts (Bobrovskii 1863)

Tutsevich aimed to answer the question if "*proper forest exploitation of Białowieża Primeval Forest may have a negative impact on the population of European bison and what actions could favour the preservation and spreading of this species?*" (Tutsevich 1878).

Observing that bison protection was so important to the administration that wood exploitation was drastically limited (no sale of raw material from 1859 to 1877), Tutsevich argued that the controlled and limited exploitation of the forest was not incompatible with the policy of bison protection: "*It should be noted that Białowieża's bison do not belong to cruel or extremely wild animals, which could be particularly disturbed by the close proximity of people. It is often observed that bison approach a herd of domestic cattle and graze with it. Therefore, it seems to me that the preservation of this species in the forest depends more on providing them with the basic needs, that is water and food, than on completely stopping the exploitation of forests solely to ensure peace of these animals*" (Tutsevich 1878). Furthermore, he noted that bison had ten established resting places in BPF and further two in Świsłocz Forest where they return after wandering through woodlands. Therefore, it would be possible to plan the exploitation of the forest in a way that ensures that only one of those resting places is disturbed, leaving the rest at the species' disposal. However, "*only selective cutting should be employed, not to modify the character of tree stands to which the bison are accustomed to and which, most probably, are a necessary condition for their survival*" (Tutsevich 1878).

It was not only articles for specialized audiences that were published in this period. Two popular descriptions of travels to BPF appeared in the Russian literary magazine "Russian Messenger" and popular Polish weekly "Wędrowiec" (Wanderer). The former, written by Russian bestselling author Vsevolod Krestovskii (1840–1895), was published in 1876, the latter, by journalist Władysław Zaleski, in 1881. Both texts can be treated as some of the first purely "touristic" accounts of BPF (the authors had no education in natural sciences for example), and published in the period when the tourism in the Forest was not yet developed. Both articles attach great importance to patriotism and national identity, although from completely opposite points of view.

Both Krestovskii's and Zaleski's motivations to write the article were also educational. Zaleski noted that the young generation of Poles would know more on the virgin forests of Brazil and would more quickly point out the capitals of middle Africa than know about the forest of Białowieża "*about which they rarely even heard. And definitely nobody knows where exactly this forest lies*" (Zaleski 1881). In contrast to almost every other text on BPF addressed to specialists, which point out how well-known the Forest was across Europe, Zaleski saw a complete lack of knowledge on the Forest in the younger generation. Krestovskii's introductory remarks are very similar as he complains that "*educated Russian people*" know Europe better than "*cities and regions that are part of our fatherland*" (Krestovskii 1876).

Both texts are full of remarks on travel routes by train and horse-cart or accommodation options that were almost ready to include in a travel guide. In addition, both authors exchange malicious comments on the Russian administration (Zaleski 1881) and Polish landowners, focusing on the anti-Polish sentiments of peasants (Krestovskii 1876). Zaleski described his fear that he might not see the bison at all in BPF with a specific sense of humour, stating that bison are so rare in the Forest that some of the local dwellers live their entire life without seeing one. When he finally saw the animal, he noted that the locals observed it with no lesser pleasure than he, a visiting resident from Warsaw. Zaleski was also the first to postulate the development of Białowieża tourism and stress the necessity to develop needed infrastructure: "*There are never any tourists in Białowieża, as if the forest lay somewhere deep in Siberia, unless there is a hunt, then a lot of locals and officials will come, rather to watch visiting dignitaries than to admire the Lithuanian nature. Under such conditions, no clever entrepreneur has yet been smart enough to set up a hotel or inn; there is only the most ordinary Jewish tavern, which I do not recommend to anybody; much better and easier to find a place for couple nights at some wealthy peasant or forester*" (Zaleski 1881).

Zaleski has left very naive descriptions of Białowieża's nature, representing a typical approach of "townspeople"—he complained about biting insects, mentioned only poisonous and dangerous mushrooms and plants, and frankly admitted that he knew nothing about botany, nor wished to learn. Nevertheless, Zaleski strongly advised everyone to visit BPF: "*I advise each fellow countryman to go and see the last place in our country which has retained its primevalness of old ages, along with the last* [herd of] *bison in the whole world. Let no one be deterred by the cost, because you can easily* [illegible] *with thirty roubles, which is not too much for an interested tourist, and let everyone remember that they will see far more interesting things than anywhere else*" (Zaleski 1881).

In 1882, in the multi-volume encyclopaedic series "Zhivopisnaya Rossiya" ("Picturesque Russia") , a description of BPF by Adam Honory Kirkor (1818–1886) was published. The author was the co-founder and one of the most active members of the Temporary Archaeology Committee and Curator of the Museum of Antiquities in Vilnius. He was known not only from his archaeological works, but also as a publisher of several journals in Vilnius, St. Petersburg and Kraków. His book "A walk around Vilnius and its surroundings", published in 1856 received much attention and served as an inspiration for numerous similar historical studies (Bernstein 1930). In the period between 1849 and 1854, Kirkor was appointed as a member of the Vilnius provincial Statistical Committee—his tenure was cut short by authorities when "harmful content" was discovered in his article about Prince Witold, supposedly provoking "regret over the lost independence of Lithuania". However, it is possible that in his encyclopaedic entry he used information from the period of work in the Statistical Committee (Antoniewicz 1913).

Kirkor's text was important as one of the first works of esteemed archaeologists of the era concerning BPF. He wrote: "*Białowieża Primeval Forest has its history, its legends and the tombs of forefathers protected as sanctuaries. Only the forests of Białowieża can rightly be called virgin, and their inaccessible backwoods and impassable thickets resemble the times of the primitive man. This forest is a valuable monument of the distant past: being there, one travels in thoughts to the times when most of the area currently occupied by Lithuania was covered with such inaccessible, dense forests*" (Kirkor 1882). Kirkor's text is a compilation of information from various earlier publications, with virtually no new data.

The article "*Bison and beaver (according to the latest news about them)*" published in "Pamiętnik Fizjograficzny" (Physiographic diary) in 1885 attempted to present both endangered species in a popular article intended for wide audiences. This was by zoologist Antoni Wałecki (1815–1897), a curator of the Mineralogical Cabinet of the Main School and the Imperial University of Warsaw. Wałecki (1885) emphasized that the presence of these species raised the scientific significance of fauna of the forests of the former Polish-Lithuanian Commonwealth with regard to the rest of Europe. The core of the article was composed of information compiled from other works, paying attention to the historical reduction of the range of occurrence of the species, quoting ancient works, medieval chronicles, the hunts of great Lithuanian princes and Polish kings, travellers' descriptions, and finally concluding that the survival of the species required the transfer of some animals to other forests, This was so that that bison was not limited to only one location.

In September 1885, a popular science weekly "Wszechświat" ("Universe") based in Warsaw published an article "Puszcza Białowieska". Its author, educated in Warsaw and Dorpat, was Józef Wacław Siemiradzki (1858–1933). He was known mainly for geological and paleontological work, as well as natural exploration of Central and Southern America. He worked closely with the Warsaw Zoological Cabinet. His article on BPF was prepared in the period when he worked on the geological monograph of Poland and his trip to Białowieża was most probably undertaken to conduct a geological and paleontological survey to include this area in his main work. In the monograph, he later wrote about layers of chalk discovered BPF, dilution layers and their stratigraphy in Narewka River, as well as fossils found in the Forest and surrounding area. The article in "Wszechświat" is important for the history of research in BPF, as Siemiradzki was the second geologist and palaeontologist conducting his study here. The first was Antoni Giedroyć (1848–1909), who surveyed the Grodno Province commissioned by the St. Petersburg Mineralogical Society in 1878. Yet his research was published a year later than Siemiradzki's (Giedroyć 1886).

Siemiradzki described the geological composition of BPF, listing several connections between the type of soil and trees, mosses and even snails living on them. As a palaeontologist studying ancient fauna, Siemiradzki was interested in the topic of species extinction, especially in the context of the nineteenth-century theories of degeneration and inevitable extinction of European bison. As he wrote with a sense of humour "To be in Białowieża and not see a bison is the same as to be in Rome and do not see the pope" (Siemiradzki 1885). He gave his opinion on the causes of decline in bison population blaming the poorly-managed forestry, not any natural causes or tendencies in the biology of the species. He also pointed out the inconsistency, in his eyes, in bison protection, with severe punishments for killing a bison but only on paper not in practice, and the guards having too many ties to local poachers (Siemiradzki 1885).

Box 6.1 Nobel prize winner Henryk Sienkiewicz in BPF

Henryk Sienkiewicz (1846–1916) was one of the most famous Polish writers and literary Nobel Prize laureate of 1905. His most important works were historical novels, e.g. Quo Vadis, set in the ancient Rome or trilogy of novels set in the seventeenth-century Polish–Lithuanian Commonwealth. His essays and serialized novels published in press won him enormous popularity with Polish readers, which later translated to his international success. Descriptions of nature figured prominently in his writings, regardless of the geographical location of his books. In 1882, Henryk Sienkiewicz visited BPF together with Zygmunt Gloger (Fig. 6.3), Polish historian and ethnographer. Both recounted the visit in different forms—Gloger published a photograph album "Białowieża in album" in 1903 (Gloger 1903), dedicating it to Sienkiewicz. The latter himself published a memoir "From Białowieża Forest" (Sienkiewicz 1899).

Sienkiewicz was captivated by the vastness, darkness and stillness of the dense forest and compared it with his experience with wilderness in North America: "*How silent! But how silent. Spacious, serious, good! The eye, despite only by moonlight, sees pine trees somehow eternal, more prodigious than elsewhere*" (Sienkiewicz 1899). He was also struck by the diversity of tree species, and the height of single trees—accurately observing the process of "race to light", in which trees grow tall, branchless trunks with crowns high above the ground to gain advantage in the competition for sunlight. After observing a herd of bison and red deer driven by beaters in front of him, Sienkiewicz noted "*unwittingly, the hand searched for a shotgun, which is prohibited in this wilderness. The whole procession was moving no more than steps away from us. Not a single shot would miss. Bison, scared by screams of beaters blocking their way, circled for some time in front of us. Small calves, black as beetles, hornless and ball-like, funny, with head and front of the body disproportionately developed compared to the rest, jumped over all obstacles swift like roe deer*" (Sienkiewicz 1899).

Sienkiewicz's account was not free of stereotypes and cliché, characteristic for popular writings of the nineteenth century: he described lynx hunting red deer by jumping off trees, wolves besieging forest villages in winter, and finally—unmanaged forest doomed to its demise: "*Thinking that the forest should be left untouched for its benefit is wrong. First of all, it should be kept clean, and what is this forest like in this respect? In vast spaces, piles of fallen trees are here almost everywhere. These heaps, rotted and decaying, sometimes pile up several feet high. Roots of enormous fallen trees tower above them like houses. A falling old tree plucks the ground that covered its roots, creating a hollow, filled then with rainwater. Such backwoods present a strange and frightening view: torn ground, full of traps, trunks, roots, twisted branches covered by moss of filthy dampness fill the entire space; amidst this wood chaos shines the swamp—creating a picture of all mixed together, broken, destroyed,* *wild, decayed and rotting. Even the air is heavy there, saturated with the smell of rot and decay*" (Sienkiewicz 1899).

Sienkiewicz also repeated the legend connected with the Zamczysko Range about an old castle, which was supposedly located there: "*who erected this castle? Maybe the Yotvingians? Maybe some of the Lithuanian dukes? Those are only assumptions. Ages passed over this ruin and the tenth generation of trees grows on it. Only the folk tale, like spider's web, is still attached to the debris*" (Sienkiewicz 1899). Interestingly, the same place inspired another popular writer of the time, Eliza Orzeszkowa. In her words "*once there was a castle of proud, powerful and brave princes*" in Zamczysko, and the magical forests of Białowieża are still "*full of terror and wonder*" (Orzeszkowa and Romski 1904). Such remarks were supported by Gloger, who also saw ruins of an old building in the stones covering the hill in Zamczysko, explaining the depression at the top of the hill as remains of fallen cellar (Gloger 1903)—when in fact it was probably the evidence of chaotic treasure hunt in 1820s (see Box 4.1).

Sienkiewicz's remarks on dwellers of forest villages are especially interesting, as they present quite different picture to the one sketched by Juliusz Brincken's book (Brincken 1826), showing the forest folk as uncivilized and unmannerly. In Sienkiewicz's eyes, "forest peasant *is prudent and by no means reckless. These people deserve closer attention in every respect. Closed in the wilderness, disconnected from the rest of the world, they experienced less external influence ad retained more old customs, traditions and characteristics of the old than inhabitants of the adjacent lands. It is also the folk that has never known no lord—it was the royal forest—so it only fulfilled its duty to the king, and besides that, was free* (…). *Most of forest peasants are tall and very slim, almost skinny. Their attire consists of a grey homespun coat reaching the knees, linen salwar and clogs made of lime bast, attached to the leg with a twine, similar as in their neighbours from outside of the forest. Their unusually delicate complexion I did not see anywhere else struck me the most. They probably owe it to the life in shadows and darkness of the forest. Less exposed to sunlight, they tan less. I saw few blond people among them. Almost everyone has dark blond hair and big, thoughtful blue eyes. Those are pretty and amiable faces, with delicate features and spiritualized expression. It is a sober, hospitable and exceptionally devout folk*". Sienkiewicz even goes as far as to propose a complex investigation of the forest dwellers' livelihood, traditions and lore—an undertaking that was not fully realized until our times. But in the writer's eyes, the same pertains to the forest itself: "*the entire forest is equally insufficiently known, and it assuredly contains true animal and plant treasures, in the most part concealed so far*" (Sienkiewicz 1899).

It must be noted that Sienkiewicz was not the only one to contradict the stereotypical descriptions of Białowieża's residents as uncivilized. The head forester of the Grodno province in the 1840s, D. Dolmatov, directly protested against negative assessments of forest dwellers: "*unavailingly some of travellers attribute these quiet and hardworking villagers stupidity and wild nature similar to the forest around them. Quite contrary, most of them are literate, of sober behaviour, sharp-witted, healthy-built, beautiful—especially the women*" (SARF 1860). Although officials of the Białowieża's forest administration often noted that local residents demanded exorbitant fees for any work and services (because of this, workers had to be invited from other counties and even provinces), they also noted their utmost good nature. One of the examples, taken from the letter of Evegenii Alekseev (BPF head forester, see Sect. 7.3) to his friend zoologist, is especially colourful and anecdotal: "*wife of [the clerk] Piasecki ran off with another clerk, a bachelor, but after spending two weeks away, she returned to her husband, who accepted her with joy. From this you can see how forgiving the people of Białowieża can be, and do not forget this fact when you write about their biology*" (SPB ARAS 1909–1913).

Fig. 6.4 Henryk Sienkiewicz and Zygmunt Gloger in Białowieża Primeval Forest in 1882 (Gloger 1903)

Box 6.2 Artistic trip of the painter Bolesław Łaszczyński to Białowieża

Bolesław Łaszczyński (1842–1909), was a painter educated in Poznan, Belgium, Heidelberg (where he studied archaeology) and in Munich. He visited BPF in 1883 and was already a recognized artist, with exhibitions in Warsaw and Kraków, when he came to the Forest. Despite the fact he was neither the first, nor the most famous painter who came to BPF, he was probably the only one who left a description of his artistic visit. It was published by the Polish National History Society in the years 1928–1929 in “Ziemia” ("Earth") magazine (20 years after the painter’s death).

In the second half of the nineteenth century, nature became one of the most important sources of inspiration for artists in Eastern Europe, but it was not the only motivation to visit places of natural interest. During the period of the emerging concept of nature protection, artists’ interest in wilderness was often motivated by the need of landscape protection. Artistic works shaped the social perception of wild nature. In some cases, picturesque depictions of nature played a large role in convincing wider audiences to conservation arguments, as it was the case of artists from the Barbizon School and the Fontainebleau Forest in France (Fig. 6.4).

Łaszczyński also went to the forest with a specific artistic programme. Fascinated by the works of Adam Mickiewicz, and wishing to illustrate these, he went on a trip to the poet’s native land in Grodno and Vilnius provinces. BPF was supposed to serve as a model for illustrations of the Lithuanian forests prepared by him for the fourth book of Mickiewicz’s “*Pan Tadeusz*”. The painter’s account of the visit provides a completely different approach to the Forest than the one in works of foresters and naturalists, containing also several botanical errors (corrected at the end of published article in “Ziemia” by Bolesław Hryniewiecki).

Organization of such “artistic tours” encountered several administrative obstacles, which Łaszczyński overcame with a good dose of humour: “*Painters had a difficult time trying to collect landscape studies due to the new regulations of the Russian authorities. Fearing the possible harassment along the way, I first made efforts with police chef and later with the general-governor to obtain an iron letter protecting me from the gaggles of the gendarmes and firefighters; I also attempted to get a permission to carry a revolver: both requests, obviously too dangerous for Tsar, were firmly refused. I packed the revolver in my pocket and in place of the iron letter I had more convincing papers—roubles*” (Łaszczyński 1928–1929). In the light of anti-poaching regulations and strict policy against firearms in private hands in BPF, it is not surprising that Łaszczyński’s attempt at legalizing his pistol failed. But apart from joking remarks, Łaszczyński’s account provides much interesting information, e.g. about spring heather-burning to facilitate the growth of grasses and the first known description of the use of bison grass (*żubrówka—zhubrovka*) for to give fragrance to alcohol (see Box 3.1, Chap. 5). As a painter, he noted the visual characteristics of dwellers of forest villages: dressed for holidays in blue-checked pants and jacket with blue, yellow and red squares, they were perfect models for Łaszczyński's drawings (unfortunately, no pictures by this artist are known from Białowieża). His remarks about propriety, hospitality and amiability of locals are similar to Henryk Sienkiewicz's observations: "*Their hospitability resembles the one from past centuries. When I entered the house of the eldest of the family, old Pawluk, he greeted me shortly and hosted with demijohn of mead and loaf of whole-wheat bread, aroma of which corresponded well with the aroma of butter prepared in May. Words would not suffice to tell about the butter with the smell of forest grasses*" (Łaszczyński 1928–1929).

Fig. 6.5 Brown bear fighting European bison in BPF in 1844, drawing by Nikolai Shyshkin (Karcov 1903)

Box 6.3 Brown bear in BPF

In the royal period, the brown bear was one of the most valuable game species in BPF, part of *animalia superiora*, i.e. big game exclusively reserved for the purposes of royal hunts. It was subject to strict legal protection (similar to European bison, moose, red deer or lynx) in the three subsequent collections of laws—the Statutes of Lithuania (1529, 1566 and 1588). According to the first Statute (from 1529), big game poaching was punishable by death, later changed to prison and draconian fines. The fine for brown bear poaching was 480 grosz, equal to four horses (Samojlik and Jędrzejewska 2010).

Most historical mentions of Białowieża's bears were connected with incidents during royal hunts: from the first in 1426, when the King Władysław Jagiełło broke his leg while chasing a bear, to the last in 1784, when a bear lightly injured a member of King Stanisław August Poniatowski's chase (Samojlik 2005). Apart from being hunted by monarchs, wild bears were captured in the entire Grand Duchy of Lithuania and transported to royal or magnate castles, where they were used as a source of entertainment. In villages and towns of the Grand Duchy of Lithuania, as well as in Balkans or in Russia the lower society classes also enjoyed travelling bear shows, during which cruelly trained bears danced or performed tricks. Until the end of the eighteenth or even the mid-nineteenth century, bears were treated as valuable game or folk attractions, and were rarely regarded as a threat. Numerous traditions and customs were built around beneficial aspects of bear trophies: a pelt was almost an obligatory interior element for castles or mansions, medicinal and magical power was ascribed to different body parts of bears (bear fat was supposed to heal wounds and cutaneous diseases, bear eyes worn in a sack served as a good charm (Kluk 1779).

Bears were rarely mentioned as pests in this period, and if they were, it was usually in the context of traditional beekeeping. Several methods of mitigating negative impact of this species on forest beekeeping were reported from BPF, e.g. a log hung in front of beehive and acting as a pendulum, beating a bear that would push it away to get to honey and bees (Karpiński 1948; Samojlik et al. 2003).

A change was brought with the rise of "rational" game management originating from the German model of forest management in the nineteenth century. According to its theory, all predators should be treated as pests, harmful to "proper" game management. Already in 1820s, this policy was proposed by forstmeister Eugeniusz de Ronke, who saw predators as detrimental for the development of European bison population. Since the end of 1840s, hunting all predators (brown bears, wolves, lynx and smaller carnivores) became an official obligation of state forest officials, including the forest guard in BPF (RSHA 1827–1833, 1840–1849). Białowieża's bears became the main target of extermination plans after several reported sightings of bear attacks on European bison. The most widely commented event happened in 1844, when the struggle between an old bison and a bear reportedly destroyed around 400 m^2 of the forest (SARF 1860; Karcov 1903). This event, recalled on an illustration from the beginning of the twentieth century (Fig. 6.5), was not the most tragic or damaging one. In 1846 for example, a bear killed five bison and several cattle and severely injured two riflemen before being shot down by a warden (RSHA 1840–1849). After these events, bear extermination intensified, also by poachers, who desired the highly valued pelt of the animal. By 1860s, there were almost no bears in BPF (Karcov 1903), which was confirmed by the fact that during the organization of the first Tsar's hunt in 1860, bears had to be bought from Russian trainers to be transported to Białowieża. One of them escaped and was shot in a nearby garden, it is not known what happened to the other, but eventually no bears were present at the Tsar's hunt (Anonymous 1860; Daszkiewicz et al. 2012).

In the period from 1870 to the 1880s, the forest personnel were obliged to file reports to the administration of the Grodno Province on the number of "pests" killed (Karcov 1903). Single bears were shot in 1870, 1871, 1874, 1877 and 1878. Afterwards, bears were seen only sporadically. One bear wandering in the forest in 1879 was immediately killed (Genko 1903). In 1889, the list of rewards issued to BPF's forest personnel for predators killed, mentioned the reward of 25 rubls for a bear (RSHA 1889A). However, there is no evidence that such a reward was ever paid. The catalogue of the Russian State Historical Archive in St. Petersburg contains the title of a document "On brown bear appearing in Białowieża Primeval Forest". The document did not survive, but it suggests perhaps that in July 1908 a bear, or at least its tracks, were observed in the Forest (Samojlik et al. 2018).

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[NHARBG] 1878. Svedenija o sostojanii Grodnenskoj gubernii za 1877 g. (Information about the state of Grodno Province for 1877) F. 1, O. 15, No 1343 (in Russian)
[NHARBG] 1879. Svedenija o sostojanii Grodnenskoj gubernii za 1878 g. (Information about the state of Grodno Province for 1878) F. 1, O. 16, No 184 (in Russian)
[NHARBG] 1880. Svedenija o sostojanii Grodnenskoj gubernii za 1879 g. (Information about the state of Grodno Province for 1879) F. 1, O. 16, No 591 (in Russian)
[NHARBG] 1881. Svedenija o sostojanii Grodnenskoj gubernii za 1880 g. (Information about the state of Grodno Province for 1880) F. 1, O. 16, No 953 (in Russian)
[NHARBG] 1882. Svedenija o sostojanii Grodnenskoj gubernii za 1881 g. (Information about the state of Grodno Province for 1881) F. 1, O. 16, No 1243 (in Russian)
[NHARBG] 1883. Svedenija o sostojanii Grodnenskoj gubernii za 1882 g. (Information about the state of Grodno Province for 1882). F. 1, O. 17, No 204 (in Russian)
[RSHA] 1812–1877. Perepiska Vol'nogo jekonomicheskogo obshhestva po voprosam pchelovodstva Correspondence of the Free Economic Society on beekeeping) F. 91, O. 1, No 352 (in Russian)
[RSHA] 1819–1821. O dostavlenii zubrov v Zootomicheskij kabinet Vilenskogo universiteta (On the acquisition of European bison to the zoological cabinet of the Vilnius University) F. 379, O. 4, No 940 (in Russian)
[RSHA] 1827–1833. Po predmetu streljanija volkov v Bialowiezskoi Pusche (On wolves shooting in BPF) F. 1589, O. 3, No 107 (in Russian)
[RSHA] 1828. O lesnyh dohodah i nedoimkah za 1827 g. po Grodnenskoj gubernii (On forest income and arrears for 1827 in Grodno Province) F. 379, O. 7, No 303 (in Russian)
[RSHA] 1829. O lesnyh dohodah i nedoimkah za 1828 g. po Grodnenskoj gubernii (On forest incomes and arrears for 1828 in Grodno Province) F. 379, O. 7, No 304 (in Russian)
[RSHA] 1838–1841. Pchelinyj promysel v kazennyh lesah Mogilevskoj gubernii (Beekeeping in state forests of Mogylev Province) F. 398, O. 2, No 376 (in Russian)
[RSHA] 1838–1849. O pchelinom promysle v kazennyh lesah (About beekeeping in state forests) F. 398, O. 2, No 172 (in Russian)
[RSHA] 1839–1842. Po zapiske Grodnenskogo lesnichego Brinkmana o sposobah otravlenija volkov i lisic (The note of the Grodno forester Brinkman on poisoning of wolves and foxes) F. 1589, O. 3, No 38 (in Russian)
[RSHA] 1840–1849 O vvedenii v Belovezhskoj pushhe pravil'nogo lesnogo hozjajstva (On introducing proper forest management in Białowieża Forest) F. 387, O. 1, No 7057–7059 (in Russian)
[RSHA] 1842 O bortevom pchelovodstve v kazennyh lesah (On beekeeping in state forests) F. 381, O. 2, No 366 (in Russian)
[RSHA] 1843 Otnositel'no oblozhenija pchelovodstva obrokom i o perehode ot bortevogo k pasechnomu (Regarding the taxation of beekeeping dues and the transition from beehive to apiary) F. 398, O. 7, No 1937 (in Russian)
[RSHA] 1847. O zubrah v Belovezhskoj pushhi Grodnenskoj gubernii (On European bison in Białowieża Primeval Forest of the Grodno Province). F. 733, O. 90, No 113 (in Russian)
[RSHA] 1847–1849. O dostavlenii dlja Zoologicheskogo muzeja Akademii Nauk chuchela zubra (About delivery of stuffed bison for the Zoological Museum of the Academy of Sciences) F. 735, O. 2, No 653 (in Russian)
[RSHA] 1857–1861. O priobretenii dlja zverinca Lisinskogo uchebnogo lesnichestva dikih kabanov (On the acquisition of wild boars for the menagerie of the Lisinsky training forestry district) F. 387, O. 2, No 23051 (in Russian)
[RSHA] 1858. Ob ubitii odnoho zubra v Belovezhskoy pushchi dlja Agronomicheskogo instituta Waldau (On killing one European bison in Białowieża Primeval Forest for Waldau Agronomic Institute) F. 387, O. 2, No 21767 (in Russian and German)
[RSHA] 1858–1860. O snabzhenii zubrovymi shkurami Jenskogo, Gessenskogo, Drezdenskogo i Stokgol'mskogo muzeev (On supplying Jena, Hessen, Dresden and Stockholm museums with Euripean bison pelts) F. 387, O. 2, No 21761 (in Russian and German)

[SARF] 1860–1915. Kniga rospisej pochetnyh posetitelej Belovezhskoj pushchi (The book of signatures of honorary visitors of BPF) F. 601. Оп. 1. Д. 1554 (in Russian)

[RSHA] 1861. O nastojashhem sostojanii Belovezhskoy Pushchi i o budushhem ee ustrojstve (On current condition of BPF and on projects to improve its management) F. 387, O. 25. No 27 (in Russian)

[RSHA] 1861–1862. O vydache zubrovyh skeletov i shkur 5 inostrannym princam i dlja Muzeumov i universitetov (About issuing bison skeletons and pelts to 5 foreign princes and for museums and universities) F. 387, O. 2, No 23602 (in Russian and German)

[RSHA] 1861–1872. Ob ustanovlenii obrochnoj platy za ustroenie pasek v lesah kazennogo vedomstva (On the establishment of fee for the renting land plots for of apiaries in the state forests) F. 387, O. 3, No 25121 (in Russian)

[RSHA] 1862–1865. Ob iskljuchenii iz chisla bortevyh uhozhij iz chisla obrochnyh statej (On exclusion of beekeeping trees from the lists of rented articles) F. 387, O. 3, No 24412 (in Russian)

[RSHA] 1865. Ob usilenii ohrany i regulirovke otpuska lesa iz Belovezhskoi Pushchi (On strengthening the protection and regulation of timber sales from Białowieża Forest) F. 387, O. 25, No 41 (in Russian)

[RSHA] 1866–1867. O priobretenii zverej dlja vysochajshih ohot (On the acquisition of animals for the imperial hunts) F. 387, O. 3, No 25602 (in Russian)

[RSHA] 1871–1900. Ob ustrojstve pasek v lesah kazennogo vedomstva (On making apiaries in state forests) F. 387, O. 3, 26942 (in Russian)

[RSHA] 1874. O perechislenii summ po smete Lesnogo departamenta na soderzhanie zubrov v Belovezhskoi Pushche (On transfer of the sums according to the estimate of Forestry Department for the bison addition feeding in the BPF) F. 1152, O. 8, No 282 (in Russian)

[RSHA] 1886 Otchet chinovnika osobyh poruchenij Departamenta udelov Rahmaninova i uchenogo lesnichego Svetlova po osmotru imi Belovezhskoj pushchi i Svislochskoj dachi (The report of the special assignments officer of the Appanage Department Rachmaninov and learned forester Svetlov on their inspection of BPF and Świsłocz forestry district) F. 515, O. 42, No 3416 (in Russian)

[RSHA] 1886–1895 Ob obmene kazennyh lesnyh dach Belovezhskoj Pushhi i Svislochskoj dachi na udel'nye zemli (On the exchange of state Białowieża Forest and Świsłocz forestry district for appanage land) F. 515, O. 42, No 3415 (in Russian)

[RSHA] 1889–1894. Ob ustrojstve v zverince Belovezhskoj pushchi kormovyh saraev, pogreba s ambarom i zaborov, a takzhe po soderzhaniju zverinca voobshhe (On the construction of feeding racks, cellars with a barn and fences, as well as on the expenses on the game reserve in general) F. 515, O. 42, No 2941 (in Russian)

[RSHA] 1889A O prinjatii mer k istrebleniju hishhnyh zverej i ptic v Belovezhskoj pushche (On taking measures towards extermination of predators and birds of prey in BPF) F. 515, O. 43, No 37 (in Russian)

[RSHA] 1892–1896. O rashodah po soderzhaniju Belovezhskogo zverinca v 1893-1894 gg. (On expenses for Białowieża game reserve in 1893-1894). F. 515, O. 42, No 3429 (in Russian)

[SARF] 1860. Dolmatov D. Istorija tura ili zubra i Belovezhskoj pushchi (The history of aurochs or bison and Białowieża Primeval Forest) F. 728, O. 1, No 2620 (in Russian)

[SPB ARAS] 1847–1858. O priobretenii dlja Zoologicheskogo muzeja chuchela zubra iz Belovezhskoj pushchi (On the acquisition of stuffed bison from BPF for the Zoological Museum) F. 2, O. 1-1847, No 9 (in Russian and German)

[SPB ARAS] 1860 Perepiska po delam Zoologicheskogo Muzeja za 1860 g. (Correspondence on the matters of Zoological Museum for 1860) F. 51, O. 3, No 30 (in Russian, German and French)

[SPB ARAS] 1861 Perepiska po delam Zoologicheskogo Muzeja za 1861 g. (Correspondence on the matters of Zoological Museum for 1861) F. 51, O. 3, No 31 (in Russian, German and French)

[SPB ARAS] 1909–1913. E Alekseev's leters to A Mordvilko. F. 348, O. 2, No 4 (in Russian)

Chapter 7
The Tsars' Private Hunting Ground (1888–1915)

Abstract In September 1888, Białowieża Primeval Forest (BPF) together with the adjacent Świsłocz Forest was transferred to the Tsar's family private property managed by the Ministry of Imperial Court. This was in order to secure better protection for European bison and turn the Forest into the Tsar's private hunting ground. In the years following the transfer, the imperial palace designed by Nicolas de Rochefort was erected, along with accompanying infrastructure. Game management was given priority in the overall administration of the Forest, and royal hunts were organized in 1894, 1897, 1900, 1903 and 1912. Forest inventories carried out in 1889–1890 and in 1909–1911 assigned low volumes of oler timber to be extracted. This extraction was further limited according to Tsar Nikolai II's wish to sustain the original, primeval appearance of the Forest. In the period from the late nineteenth to the early twentieth century, a wave of botanists, zoologists, forestry scientists, and veterinarians, passed through BPF and produced several landmark works on various key aspects of its natural values. However, more rigorous control of the new administration over forest use led to increased number of conflicts with local dwellers. This chapter considers the management and administration of the Forest during this critical period in its history.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

7.1 Summary

In September 1888, BPF together with the adjacent Świsłocz Forest was transferred to the Tsar's family private property managed by the Ministry of Imperial Court, to secure better protection for European bison and turn the Forest into the Tsar's private hunting ground. In the first years after the transfer, the imperial palace, the palace park, roads, railway, and other accompanying infrastructure, along with administrative buildings and apartments for forest administration personnel were constructed. The palace, designed by Nicolas de Rochefort, was especially the symbol of imperial

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_7

splendour. An English-style park was designed around the imperial palace by Walerian Kronenberg. The railway connecting Hajnówka and Białowieża was opened in 1897, and in 1903 a hardened road connecting Bielsk, Hajnówka, Białowieża and Pružany was built.

Game management, now separated from the forestry administration, was given the priority in the overall administration of the Forest. Supplementary feeding developed with the construction of new feeding points By 1902, there were 321 feeding racks, 64 barns for fodder, and 27 wooden roofs scattered throughout the Forest. The herd of bison had grown significantly: from 403 in 1890 to 727 in 1900. This increase bison numbers stopped at the very beginning of the twentieth century, and the population oscillated around 600–700 animals. To make the annual bison count results more reliable, reported numbers were based on three complementary methods.

Game management was involved in organization of royal hunts in 1894, 1897, 1900, 1903 and 1912, while fighting with poachers was also a major activity of the forest administration. The BPF administration had little means of exerting pressure on local dwellers suspected of poaching except from limiting their access to forest pastures, employment for forest works, or access to firewood for favourable price.

In the late nineteenth to early twentieth century, botanists, a wave of zoologists, forestry scientists, and veterinarians, passed through BPF and produced some milestone works on different aspects of its natural values. These included botanists Błoński et al. (1888), Błoński and Drymmer (1889), Paczoski (1897–1900, 1930). In 1906–1908, a scientific expedition to BPF was organized by Nikolai Kulagin, resulting in series of publications covering different aspects of European bison biology.

Two forest inventories of BPF were carried out: in 1889–1890 and in 1909–1911. The forest inventory team of 1889–1890 was aware that BPF was a historical monument, and assigned low volumes of timber to extract with high felling age. Tsar Nikolai II wished to sustain the original, primeval appearance of the Forest and ordered to further limit even those small planned cuts. The inventory team of 1909–1911 divided the Forest into "economic" and "protected" plots, but even in the "economic" plots the main emphasis was put not on the revenues from the forest, but on the forage needs of the game and the aesthetic values of BPF.

In 1914, the first natural history museum of BPF was opened, representing the unique values and diversity of life of the Forest (Daszkiewicz et al. 2006). In August 1915, the majority of museum exhibits, along with other belongings of the Appanage Administration and the imperial palace, were evacuated to Moscow.

7.2 BPF Under the Management of Appanage Administration

In 1888, BPF and the adjacent Świsłocz Forest were transferred by the Ministry of State Domains to the Tsar's family private property (Russian *Udely*, meaning appanages, hereditary property of tsar's family), managed by the Appanage Department (from 1892—the Main Directorate of Appanages), as a structure of the Ministry of Imperial Court. This was in exchange for the equivalent forests in Oryol and Simbirsk Provinces. Preparation of this transfer took two years and the process itself was over four months, from September to December 1888. The aim of this exchange was to secure better protection for European bison and turn the Forest into the Tsar's private hunting ground (Genko 1903; RSHA 1886, 1886–1895, 1888–1890).

In the 1880s, the lands in the vicinity of BPF were still economically underdeveloped, and BPF itself brought little income due to restrictions in timber production and the Forest's remote location from wood markets. Low level of profitability of BPF's forestry districts meant, in turn, that the forestry administration did not have the financial opportunities to reorganize the management of the forest and the bison protection system. Financial restrictions were in the larger part removed when Tsar Alexander III expressed the wish to change Białowieża into a game reserve, and this allowed the management system to change (RSHA 1886, 1886–1895). In the first years after the Forest was transferred to Romanov family's private domains, the administration was engaged mainly in major construction work. This included the imperial palace, palace park, roads, railway and other accompanying infrastructure, along with administrative buildings and apartments erected for administration and personnel. The latter was an especially burning issue, since due to lack of appropriate offices, BPF's administration had to be located in Vilnius in the first years after transition (RSHA 1890). At the same time, the administration had to familiarize itself with the state of BPF and develop a plan for its management. The new forest inventory was completed under the supervision of Nestor Genko in 1889–1890 (Genko 1903; see Sect 7.3).

Instead of the previous five forestry districts (four in Białowieża and one in Świsłocz Forest), three new ones were formed—West, East and South. However, after a few years the administration returned to the old division into five parts, called "estates", in accordance with the nomenclature of the appanage administration. The administrator of BPF became directly subject to Department of Appanages in St. Petersburg.

All the previous foresters were still officials of the Forestry Department, and as such were transferred to other state forestry districts. In their place, new specialists were employed. Many forest guards retained their posts, and for posts of jegers (hunters) the appanage administration invited retired low-ranking military officials from other provinces. These were primarily from St. Petersburg Province because of fears that employing local dwellers would mean inviting poachers to wear the uniform. However, employing jegers from remote areas turned out to be expensive and complicated, and already by 1888–1889 local residents were employed for such posts (RSHA 1889B, 1894–1900, 1901, 1902A, 1914–1915B; Karcov 1903).

Game management, now separated from the forestry administration, was given the priority in the overall administration of the forest (RSHA 1890–1891). Furthermore, European bison management underwent a fundamental reorganization. In the first years, the game reserve was managed by Hans von Auer, who had served as a head of one of the BPF forestry districts since 1875 (von Auer 1998). In this period, the administration's resources were almost fully depleted by construction works, and therefore expenses for additional feeding and conservation of bison increased only slightly (RSHA 1889–1894, 1890–1893, 1891–1893). In 1894, the newly created post of Game Manager was given to Josif Newerly, previously working in the Tsars' hunting ground in Spała. Josif Newerly held this post until his death in 1914, and then was replaced by his former subordinate Ewald Bark. The Game Manager supervised the work of initially three and later five oberjagdmeisters, fifteen jagdmeisters (or rangers), eleven hunters, one (later three) warden of the game enclosure, one jagdmeister-secretary, and twenty-two bison guards, each with specific uniform and a metal badge confirming their position (Figs. 7.1 and 7.2). According to Newerly's and Karcov's opinions, these uniforms, although stylish, were too theatrical and impractical during the hunt due to the excessive amount of metal ornaments and hunting knives (Karcov 1903).

Since the 1890s, a new model of game management was gradually introduced to BPF. It was based on promoting ungulate species by additional feeding, reintroduction of extinct species (red deer), and the introduction of alien species (fallow deer), as well as augmentations (roe deer), and attempts to "refresh blood" of existing populations of bison. A number of construction works started, incorporating the erection of a new game management building near the animal enclosure, new fences

Fig. 7.1 Uniforms of forest and game personnel in BPF, along with their equipment and arms in 1899 (RSHA 1899B). (1) Hunter, (2) and (3) hunting and bison guards, (4) *Jäger* (ranger), (5) forest keeper and (6) forest guard. See Fig. 7.2 for photographs of badges worn by the forest and game personnel

Fig. 7.2 Metal badges worn by forest and game administration personnel in BPF: bison warden (left) and forest keeper (right). Photograph by Tomasz Samojlik

for the menagerie itself, and the construction of feeding points. By 1902, a final number of 321 feeding racks, 64 barns for fodder and 27 wooden roofs were achieved and scattered throughout the forest. Some of these points were also supplemented with artificial licks, large wooden troughs buried halfway into the ground and filled with clay saturated with a strong salt solution (Karcov 1903; RSHA 1889–1894, 1890–1893, 1891–1893). Thanks to a more thorough additional feeding in winter, the herd of bison had grown significantly: from 403 in 1890 to 727 in 1900.

Several attempts to "refresh" the bloodline of the European bison were made but all failed and animals brought from Gatchina or Caucasus died soon after transfer. The same happened with American bison received from the Duke of Bedford in exchange for European bison (Karcov 1903; RSHA 1896–1902, 1898–1908, 1907A). Game management had its major successes, though. Siberian roe deer, brought to BPF several times in the 1890s, crossed with the local population and stimulated its growth. The reintroduction of red deer that started in the mid-1860s found its fruition under the appanage administration. In 1890, the first hundred red deer were released from the enclosure, and by 1895 about 400 lived in the forest. In 1910s, their number reached over five thousand. Also, the introduction of fallow deer was successful and by 1900, there were 600 in the forest (RSHA 1889–1894, 1896–1902, 1913–1915A). In the beginning of the twentieth century, red, roe, and fallow deer were reproducing to such an extent that many experts expressed their concern over bison feeding conditions and forest regeneration under the pressure of such numbers of deer (Wróblewski 1912b). The hunting administration tried to control the deer populations by shooting and by selling or donating them to other parks but only with limited success (RSHA 1909–1913). Translocation of red and roe deer has left a long-lasting effect with complicated genetic structures of the contemporary populations of both species in BPF (Niedziałkowska et al. 2012; Olano-Marin et al. 2014).

In an effort to keep large game confined to BPF, following the example of other hunting parks, Newerly proposed several times to build a strong high fence . The construction started in 1910s, yet it was not completed by the beginning of WWI (RSHA 1911–1913, 1914–1915B).

Although it was European bison that attracted most of the attention of game management of BPF, supplementary feeding affected all ungulates. Bison were additionally fed for a prolonged period of time, from 166 days in 1895, to 202 days in 1905. Each early autumn, the Game Manager was to specify the amount of additional fodder needed for bison and other ungulates (Karcov 1903). In addition to hay, traditionally provided to bison, to enrich the diet of bison the game management started from 1895 to buy clover and tubers, and occasionally oat from areas beyond the forest. In addition, meadows inside the forest were kept open for the purposes of bison feeding. The hunting administration also knew that bison debark trees and occasionally feed on stems of trees, so each winter, near feeding points they chopped down several hundred aspens (a species of low forestry value) to feed the ungulates. The forestry administration strictly monitored the situation so that valuable tree species would not suffer; RSHA (1904–1907). However, the increase in the number of bison, which was significant in the early years of intensive feeding, stopped at the very beginning of the twentieth century. In the next decade, the population oscillated around 600–700 animals. The increase in the number of deer lasted the entire appanage period (RSHA 1896–1898, 1896–1902, 1897–1898, 1898–1908, 1899–1903, 1901–1903, 1901–1904, 1902–1903, 1903–1907, 1904–1907, 1913–1915A, 1914–1915A; Samojlik et al. 2019).

Newerly also reorganized the system of the annual counting of bison (and other ungulates) in BPF. To make the count results more reliable, three complementary methods were introduced. The first was a straight line continuation of traditional winter counts after fresh snowfall, engaging a large number of people in a complicated logistical task. On the day of count, no fodder was put in the feeding racks, which aimed at forcing animals to disperse in search of food in the entire forest. Tracking started in all sections of the forest simultaneously, with guards noting all animals entering and exiting their range. The second method was based on counting of bison concentrating around feeding racks during feeding. The third method was the least formalized and was based on the guards' own observations and calculations in their ranges. In effect, three different values of bison abundance were produced annually (starting from 1900) and it was the duty of the head of game management to unify them for the purposes of official reporting (RSHA 1899–1903).

Another duty of the head of game management was to directly manage all matters related to the organization of royal hunting (up to the smallest details set up in the hunting plan). The hunts were planned as events lasting from several days up to a couple of weeks, involving around 500–700 people (up to 500 soldiers of regiments quartered near BPF and from 250 to 300 local peasants as battue). There were special stands for shooters, and usually a tent for serving breakfast in the forest. The hunts of Nicholas II were especially luxurious (RSHA 1894–1903, 1912; Karcov 1903).

In August and September 1894, Alexander III hunted in the Forest (81 animals were killed, including 7 bison). The Tsar died a few months later, and in 1897 a hunt was organized for his successor, Nicholas II. During over a week of hunting, 209 animals were killed, including 36 bison. This hunt was filmed by the Tsar's cinematographer Bolesław Matuszewski (the outcome being a document entitled "Travels and hunting in Białowieża and Spała" is unfortunately considered lost; Skaff 2008). Nicholas II returned to BPF three more times: in 1900 (680 animals killed, including 45 bison), 1903 (467 animals, 12 bison), and 1912 (472 animals, 11 bison) (Karcov 1903, RSHA 1894–1903, 1911–1913). Additional hunts in 1910 and 1914 were planned but both were cancelled. The first was due to epizootic disease among the European bison and the second one due to international circumstances (RSHA 1910,).

Display of trophies after each hunt was organized with pomp and splendour, and frequented not only by local dwellers, but also visitors from Grodno, Białystok, Brest and Warsaw (Karcov 1903; Semakov 2002). A large weighing scale was bought to weigh the hunted animals; and the Tsar's or Grand Dukes' entourage often included an artist, a photographer, and occasionally a taxidermist (RSHA 1896–1898, 1912A).

Fighting with poachers was another major activity of the forest administration. Hans von Auer, head of the game reserve until 1894, wrote that a wedding in forest villages would rarely do without "roasted European bison or at least wild boar" (von Auer 1998). The first administrator of the forest in the appanage period, Mikhail Andreevskii, noted that poachers would enter the forest "like the market for meat" (RSHA 1889B). Due to the fact that the new administration had more resources than in previous periods, the conflict with poachers intensified. In 1889, possession of any firearms was prohibited for people living in BPF and in the area up to 15 km around it, but the implementation of this law was not immediate. Even five years later, Josif Newerly alerted the administration in St. Petersburg that there is "no village, in which there would not be several firearms hidden" and called for a more strict fight with poaching (RSHA 1894–1895). The forest administration appealed several times to the Ministry of Internal Affairs not to leave poachers unpunished (RSHA 1894–1895, 1896–1902; NHARBG 1896–1899). Since local policemen were burdened with their own tasks, and the forest personnel did not have legal rights to search houses of people suspected of poaching, the appanage administration funded five additional policemen posts, starting from 1895. In 1901, the number of policemen was increased to eight (RSHA 1895–1902). Each year, policemen together with forest guards confiscated dozens of firearms from local dwellers (Karcov 1903; RSHA 1896–1902; NHARBG 1896–1899).

Although the BPF administration had a list of names of "the most delinquent poachers" and kept tabs on them (Karcov 1903; RSHA 1895–1897, 1895–1902; NHARBG 1899–1915), the BPF administration had little means of exerting pressure on local dwellers except from limiting their access to forest pastures (see Box 7.4), employment for forest works, or access to firewood for favourable price (RSHA 1902B). In the period 1893–1896, there were 132 court cases with 240 suspects connected with poaching, (including not only poachers caught red-handed, but also finding traps and snares in peasants' households, insults towards policemen during inspection, and even attacks), out of which 57 ended with conviction, 27 with exculpation, and the rest were not yet solved in 1896 (NHARBG 1896–1899; RSHA 1895–1904). Sometimes the fight between game administration of BPF and poachers was quite literal: there were occasional shootings in the forest, with people wounded and killed (RSHA 1901, 1903A, 1906A, B). One of the forms of revenge from poachers was probably the theft or slaughter of horses belonging to forest guards, as evidenced by several archival cases of requests for funds to purchase new horses for forest personnel to replace the killed or stolen ones (RSHA 1905B, 1906A, 1909).


Для покупателя.

Билетъ на побочное пользованіе въ удѣльныхъ лѣсахъ

Выданъ Управляющимъ имѣніемъ Бѣловѣжскимъ Удѣльнымъ ... дня ... 1911 года.

№ 3

Округъ Бѣловѣжскій.

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Объѣздъ № 11		
Обходъ № 8		
Кварталы №№ 563		
Сданное въ побочное пользованіе пространство 20 дес. — кв. саж.		
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Управляющій имѣніемъ

Fig. 7.3 Payment ticket for minor forest uses, e.g. collection of berries, mushrooms and firewood in BPF (RSHA 1912–1913)

The game management considered it necessary to limit the access of local residents to the forest to avoid "unnecessary disturbance" to animals. Some measures to restrict access to mushrooms, berries, and other forest fruits were taken in the previous period, when local dwellers were granted access to these resources for free but based on special ticket (Fig. 7.3). In 1900s, the procedure was more stringent—tickets were issued for free or for a small payment and only for people known to the administration as "trustworthy" (Karcov 1903; NHARBG 1899–1915, 1913A). Peasants were also allowed to collect dead wood to build fences around their meadows and arable land to protect them from wild ungulates but only on selected days, and after receiving special, 'free from pay' tickets. In 1889, 1546 such tickets were issued (Więcko 1984). Pressure of local dwellers on forest increased during years of poor harvest: in such circumstances, the administration of many provinces allowed free access to firewood, wood pastures, haymaking, collecting mushrooms, berries and other forest fruits (NHARBG 1891–1896; RSHA 1891–1895, 1898–1905).

In 1913, BPF's administration signed a contract with the Moscow pharmaceutical company for the supply of bison grass (see Box 5.3). For this reason, the collection of *zhubrovka* was placed under strict control: it was collected only in certain compartments (to avoid disturbing animals) by families of the forest guards, then stored by the administration, and the money for collected grass was paid only in autumn. House of anyone suspected of illegal collecting of *zhubrovka* was subject to inspection by guards and confiscation of collected grass (NHARB 1913B).

7.3 Imperial Palace and Other Material Imprints of the Period in BPF

The Tsar's private hunting ground demanded an adequate headquarters to host the monarch and his guests. In 1889, the decision was made to erect the palace in the place formerly occupied by hunting manors of the Polish kings—the first was built there in the late sixteenth century and was destroyed in the 1670s (Samojlik et al. 2014), the second was first mentioned by written sources in the beginning of the eighteenth century and survived until the 1830s (Samojlik 2005a). The design of the palace was entrusted to architect and engineer Nicolas de Rochefort (1846–1905). The palace was built in the period (1889–1894) and most of the materials used for the erection of the palace were obtained locally; bricks were manufactured in a brickyard located nearby, stones for hardening the hill were brought from the forest, and the interior was furnished using ten species of trees found in BPF (Karcov 1903). The palace had 120 rooms, with the cellar and part of the first floor designated for service (the rest was occupied by the dining, billiard, and coffee rooms), the second floor taken for the imperial family, and the top floor for guest rooms (Anonymous 1894). The construction of the palace was completed by 1894 and in August of that year the Romanov family visited it for the first time (Karcov 1903). Shortly after, Alexander III died and the new emperor, Nicholas II, became the owner of the palace. Alexander III severely limited personal and family expenditure, especially compared to his son. Nicholas II, famous for his love of luxury, demanded a complete overhaul of palace interior decoration. Guest rooms and halls were designed and furnished with splendour: each room had its own theme (e.g. room decorated with postage stamps or playing cards), walls were decorated with paintings, among them ones made by Polish painter Zdzisław Jasiński, depicting centaurs and nymphs (Chestnykh and Kettering 2010). Once finished, the palace became one of the most often painted and photographed objects in BPF (Fig. 7.4). Despite the enormous expenses spent on finishing the palace, Nicholas II visited Białowieża only four times—in 1897, 1900, 1903 and 1912, never staying there for more than two weeks.

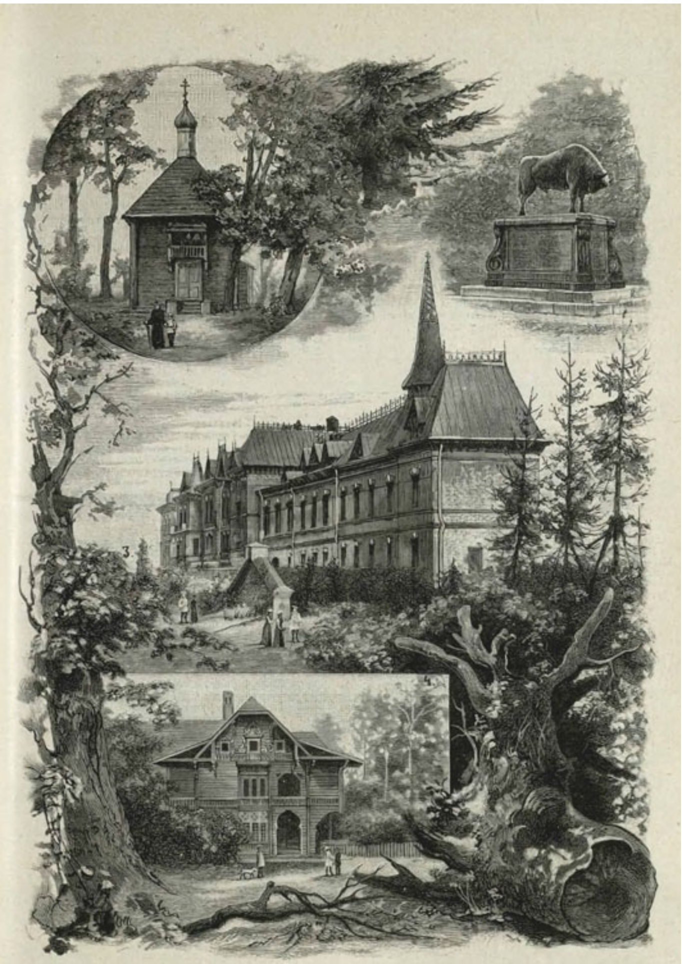


Fig. 7.4 Imperial palace, orthodox church in Białowieża, oberjegermaister's house and the bison monument commemorating the 1860 hunt on a drawing from "Niva" weekly magazine (Anonymous 1894)

Several other buildings were erected in the similar architectonical style in the vicinity of the palace: Hunters' House (with billiard hall and large bathroom), Hofmarschall's Lodge, houses for court officials, stables for forty horses, kitchen, electric station, telephone station, laundry, electric mill, icehouse, garage, bakery, etc. (Bajko 2001; Karcov 1903). Close to the palace, an Orthodox church was erected and was consecrated in 1895 (Bajko 1994). Decoration was finalized in 1897, with the porcelain iconostasis delivered from St. Petersburg. At a distance of about 2 km from the palace, the office building of BPF administration was erected, along with apartments for its employees. In the spring of 1912, the new head manager of BPF, Mitrofan Golenko (1863–1943), concerned with water quality in Białowieża, had a water treatment and ozonation station (in this period, infusing water with ozone was used to disinfect drinking water) erected in the vicinity of the palace (RSHA 1912A, B).

An English-style park was designed around the imperial palace. The designer of the park, Walerian Kronenberg (1859–1934), was one of the most popular gardeners and park planners at the turn of the nineteenth and twentieth centuries. Around 200 species of trees and other plants were planted in the park. Kronenberg also designed a smaller and more modest park at the other end of Białowieża village, surrounding buildings of the Appanage Administration (Chestnykh and Kettering 2010).

In 1901, a pheasantry was arranged in the park, and several species and subspecies of pheasants were kept there (four of them were later added to the collection of the museum—see Sect. 7.3). During the Tsar's visits, the enclosure was opened and pheasants walked freely in the park (RSHA 1901–1904). Probably some other animals were kept in the park or in the game reserve, for example arctic foxes, which were animals of the polar region that were subject of several attempts at fur farming in the Russian Empire in the beginning of the twentieth century. The only trace of this attempt in BPF are catalogues of the Zoological Institute of the Russian Academy of Sciences in which skulls and skins of arctic foxes sent from BP in 1908, 1911 and 1912 are noted (SA ZIN 1906–1912).

The old ponds, dating back to the era of the first royal manor in Białowieża and that by this time were almost completely overgrown with vegetation and turned into swamps, were deepened and enlarged. In the spring of 1912, during preparation for the Tsar's visit, small carp were released to the pond and two boats were bought for the entertainment of younger Romanovs. Also a playground and a lawn for tennis and other games were prepared for them (RSHA 1912A).

In 1897, a railway connecting Hajnówka and Białowieża was opened, with two stations erected in Białowieża: the palace station near the imperial palace and the freight station 2 km away (Bajko 2001). Trains to Hajnówka and Bielsk ran twice per day (NHARBG 1913B).

A hardened road connecting Bielsk, Hajnówka, Białowieża and Pružany was built in 1903. The road itself was very modern, with roadside ditches and iron bridges with ornamented handrails (a number of the latter still survive, especially in the contemporary Belarussian part of the Forest). Building of a hardened road made BPF accessible to cars. In 1903, the first automobile was bought, and during the 1912 hunt there were already two of them in Białowieża: one (made by Prunel Phaeton) for urgent issues of the appanage administration, the second (Benz) for theroyal family. The request for a "motor carriage" from February 1903 lists several reasons why a car is necessary in BPF These were to realize the Tsar's urgent orders and quick travel during imperial hunts and, on a daily basis, to travel to Pruzhany with errands and to Bielsk for receiving and depositing money. The first car, a Prunel Phaeton with four-valve Piper engine was bought in St. Petersburg in spring 1903 (RSHA 1912A). In summer 1915, cars along with other appanage properties were transported to Moscow. In October 1918, one of them, the Benz, was repossessed by the All-Russian Central Executive Committee of Soviet Russia (SARF 1918–1919).

Inside BPF, a network of roads was created; some reserved solely for hunting purposes, and heavy transportation was prohibited on them. These roads required constant attention, e.g. clearing of dead wood, digging drainage ditches, sowing grasses and creating bridges (RSHA 1892–1896A, 1896–1902). An agronomist and botanist Ivan Klingen (1851–1922) was consulted on what grass should be sown on these roads (RSHA 1897–1898). Some roads, especially in Białowieża village, were turned into avenues with the use of horse-chestnut, apple, pear, oak, etc. Tree saplings (up to 300 annually) were also given to local peasants by their requests to plant near their houses (RSHA 1902C, 1904; NHARBG 1913B).

Many transformations under the appanage administration required land exchanges and led to protracted conflicts between peasant communities (or individual peasants) and the administration. For example, some peasants were not satisfied with the compensation for the plot of land in front of the palace, where the old ponds were restorated. The administration stated that this plot was a swamp with bad grass, suitable only for cattle resting on a hot day; still, some peasants have claimed that this plot was good mowing and arable land (RSHA 1892–1902). The construction of roads and straightening of rivers were particular actions that led to a number of conflicts (RSHA 1897–1902, 1899A).

Although the organization of hunting was a priority, attempts at modernizing the forest management were still undertaken. The officials studied the sales markets, developed projects for organizing factories in BPF and its vicinity, considered the proposals of large western trade companies to engage in logging in the forest (RSHA 1891–1892, 1892–1896B, 1894–1898). However, the actual implementation of those plans was limited. In 1895, approximately 40 km in a straight line from BPF, a modern sawmill with steam engines and other latest technological advances was built near the railway station Strabla on the River Narew, with the aim to increase the profitability of BPF. A few years later, the appanage administration considered that management of the sawmill was too troublesome and rented it to private entrepreneurs (RSHA 1890–1891, 1908–1912). Small-scale production continued, e.g. wood tar and turpentine kilns which were present inside the Forest already in the previous period (RSHA 1894, 1912C); in the 1900s sixteen wood tar kilns were an important source of employment of local people (Wróblewski 1907, 1909). The administration of BPF surveyed also possibilities of introducing other types of production. An idea to erect a glass factory (which would use the accumulated dead wood, not suitable for sale) appeared and a specialist was sent to BPF to gather data on the quality of sand and limestone in the Forest. The project was eventually discarded, both for economic and nature protection reasons (RSHA 1891–1892). Private entrepreneurs also attempted at opening glass factory and tannery in BPF: the first project turned out to be completely unprofitable (RSHA 1894–1898), the second was rejected, as it posed too much of a risk of epizooty and pollution (RSHA 1902C).

The abundance of ungulates in BPF in the early 1900 s led to their increased impact on the forest and raised the concerns of the administration. To keep ungulates away from regenerating forest, the cutting area was enclosed with a fence for several years. In case the administration found it necessary, pine, oak, and less often ash were planted. In 1896, six tree nurseries of 0.5 ha were established but by 1904 the administration reduced their number to three. Additionally, shrubs and trees valuable to game were planted: Chinese wolfberry, different types of willow, apple, pear, chestnut. Due to the fact that the logging in BPF was limited (see below), the area of plantations was also small and rarely exceeded two hundred hectares per year in the 1900s (RSHA 1902C, 1904).

Villages located in the middle of the Forest were a constant headache for the appanage administration. Similarly to the previous period, the management attempted to remove peasants from the Forest, which often turned out unfeasible (RSHA 1913). However, by 1910, the administration managed to exchange more than 2500 ha of peasant lands in the forest for areas outside BPF (although in 1910 half of those areas were still used by the previous owners). Plots with poorer soil were left under natural succession and more fertile areas were used for haymaking. Both the Białowieża and St. Petersburg administrations believed that local production of feed would be more profitable and reliable than buying it outside BPF (RSHA 1909–1913). The first experiments were conducted in 1910, and the actual plantations started under the new head manager of the forest, Mitrofan Golenko. In 1912, an 8-year crop rotation plan including clover, oats, alfalfa, vetch and potatoes was developed, seeds, equipment and mineral fertilizers were purchased. In the fall of 1913, oats and clovers were harvested from 65 ha (RSHA 1912–1914, 1912–1915). In the summer of 1914, fodder was grown on 260 ha, and on three hectares of earth apple (*Helianthus tuberosus*) was cultivated. Additionally, in 1914, work was carried out to improve haymaking conditions on 110 ha of the Nikor bogs. All this provided more than half of fodder requirements for ungulates of BPF.

7.4 Botanists, Zoologists, Forestry Scientists and Veterinarians Interest in BPF and Opening of the First Natural History Museum in Białowieża

The Forest of Białowieża and its biological diversity were a subject of increased interest from naturalists in the early nineteenth century. The rapid development of different branches of knowledge about BPF and European bison was then cut short for a half of century and it was only the late nineteenth-early twentieth century when botanists, zoologists, forestry scientists and veterinarian returned to BPF and produced some milestone works on different aspects of its natural values.

From the 1822 and 1826 travels of Stanisław Batys Górski, BPF was not visited by professional botanists until 1880s, when three botanists, Karol Drymmer, Franciszek Ksawery Błoński and Antoni Ejsmond, toured the Forest. All three graduated from the Imperial University of Warsaw, but were only able to engage in their botanical research in their free time (Drymmer worked as an official at excise tax board, Błoński was a factory doctor in a sugar factory, and Ejsmond taught natural history in secondary schools). The results of their travels to BPF in 1887 and 1888 were published in "Pamiętnik Fizjograficzny" (Błoński et al. 1888, Błoński and Drymmer 1889), an annual publication containing "pure" scientific materials from widely understood natural science, or more precisely—materials for the description of Poland falling within the scope of botany, zoology, geology and anthropology (Majewski 2005).

Błoński, Drymmer, and Ejsmond were specialists in the fields of various taxonomic groups and wanted to prepare as complete a list as possible of the plants of the forest flora. This was in accordance with the standards of botanical research of that period. As a result, 1337 species of plants and fungi were collected. The works of Błoński, Drymmer and Ejsmond are still cited by botanists today. Additionally, Błoński was a specialist on "cryptogams" and his remarks on fungi, ferns and mosses based on analysis of specimens collected during expeditions became the basis of the first flora of spore plants of BPF (Błoński and Drymmer 1889). Among the specimens found in BPF, several species were previously unknown to science (see Box 8.3).

Józef Paczoski (1864–1942) visited BPF several times in the 1890s. He was a specialist in the flora of the southern and western provinces of the Russian Empire, the Balkans and Poland. He was the first to coin the term phytosociology and outlined tasks of this scientific discipline (Maycock 1967). In 1897–1900, he published the results of his research in Polesian flora with exhaustive geographical and environmental commentaries (Paczoski 1897–1900). Although Paczoski never completed his university education, as a researcher he was in more favourable conditions compared to Błoński, Drymmer and Ejsmond. His trips were sponsored partly by few scientific societies and partly by the Department of Agriculture. Thanks to the well-established relations with both Russian and Polish botanists, Paczoski had the opportunity to work with herbaria of dozens of naturalists, including those of Błoński, Drymmer and Ejsmond, and even Stanisław Górski and Willibald von Besser (1784–1842). Paczoski then returned to BPF in the 1920s to work as a scientific director of Reserve Forest District (later transformed into the Białowieża National Park), and continued to publish works on the Forest, crowned by the 1930 book "Lasy Białowieży" ("Forests of Białowieża"). In this book, based on botanical, geographical and ecological arguments, Paczoski argued that the communities of BPF should be considered the least anthropogenically disturbed lowland forests of Europe (Paczoski 1930). Paczoski's phytosociological ideas helped to confirm that BPF was an exceptional forest not only because of the European bison population but also because of the preservation of natural, "primeval" forest communities along with the natural dynamics of their development and succession. It was also Paczoski who defended natural processes against human intervention, including the impacts of the bark beetle *Ips typographus*, e.g. protesting against the removal of bark beetle-infected spruce from BPF (Daszkiewicz and Samojlik 2016).

Although there was a plethora of works devoted to European bison in the mid-nineteenth century, most were written either by non-professionals or by professional naturalists that never visited BPF. It might seem strange that one of the most important zoologists and geographers of the Russian Empire of the mid-nineteenth century, Alexander Middendorf, who visited Pushcha in 1847 and hunted for a bison for the Zoological Museum of the Imperial Academy of Science (Fedotova 2018), did not use this opportunity to publish any paper on this topic. Most probably he was too busy working on materials collected during his Siberian expedition from years 1842–1845 (Sukhova and Tammniksaar 2015).

It was only in 1905, when Nikolai Mikhailovich Kulagin (1860–1941), a professor at the Moscow Agricultural Institute, whose field of research interests can be characterized by the broad term “applied zoology”, suggested organizing a scientific expedition to BPF for a comprehensive study of European bison. Kulagin who was esteemed by his colleagues as an agile scientific organizer, managed to obtain funding from the Appanage Administration and put together a good team of researchers (RSHA 1905–1914). The expedition involved a wide range of zoologists of different specialties. Two researchers, Alexander Mordvilko (1867–1938), an entomologist and parasitologist) and Konrad Wróblewski (1864–1945), a veterinarian with ambitions in “pure” zoology), worked in BPF permanently for two and a half years (1906–1908) (see Box 7.2). Professors from St. Petersburg and Moscow made several shorter trips. Zoologists collected more than a hundred specimens of European bison (found dead in the forest or selected for shooting as old or sick) for various anatomical, histological, and pathological studies. Skulls, pelts and skeletons of 86 bison went to Zoological Museum of the Imperial Academy of Sciences, and more than a dozen went to the Zoological Museum of Moscow University (The Catalogue of the Mammalogical Department of Zoological Museum of the Moscow State University; The Catalogue of Osteological Department of ZIN). Reports on the progress of the expedition and its scientific achievements were widely publicized, both in scientific journals and in more popular periodicals (see for example Wróblewski 1907–1909). Members of the expedition studied bison in all possible aspects of contemporary “modern zoology”. This included their feeding and influence on vegetation as well as competition with other ungulates; their reproduction and the sex and age structure of the population; causes of death, and their parasites and diseases; morphological and histological structure of their organs, etc. (Kulagin 1919, 1928; Ognev 1926; Wróblewski 1908a, b, 1912a, b, 1927). A detailed analysis of this extensive material obtained by zoological museums in St. Petersburg and Moscow made the naturalists conclude that the bison’s internal constitution did not bear any signs of degeneration and that the European bison population was limited only by ecological reasons and anthropogenic pressure. This conclusion disproved the prevailing notion of the nineteenth century that European bison was doomed to extinction due to “degeneration” of the ancient species. Data collected by members of Kulagin’s expedition played an important role in the European bison restitution (Wróblewski 1927; Zablockij 1939).

Zoologists collected not only bison, but also representatives of almost all higher animal taxa from BPF. Additionally, Mordvilko and Wróblewski collected anatomical, histological, and parasitological samples from bison and other ungulates (SA ZIN 1906–1912; ARAS 1906–1909, 1906–1913).

In 1906, Dmitry Rusanov, a veterinarian in the service of Grand Duke Nikolai Nikolayevich, visited BPF. He prepared a report on the state of health, nutritional conditions and the capacity for the continued existence of bison and other game species in the Forest. This indicated that many bison suffer from parasites and diseases (RSHA 1905–1914) but his recommendations basically repeated those made several times by the head of game administration of BPF, Newerly. These were to restrict the access of domestic livestock to BPF, to provide clean existing water holes and create new ones, to drain marshy meadows to improve the quality of summer pasture, to plant deciduous trees, to control spruce succession, to pay special attention to additional feeding of young bison, and to regularly cull old and ill bison. Rusanov (as most other specialists) believed that marshy meadows and contaminated watering places were the main cause of high parasite infestation of BPF's ungulates.

In June 1910, an outbreak of an epizootic of livestock began in the vicinity of BPF, and lot of dead wild ungulates were found in the Forest itself. The BPF administration ordered that grazing of domestic livestock in BPF be limited and requested veterinary support from St. Petersburg. Samples were sent to the St. Petersburg's Bacteriological Laboratory of veterinary administration for analysis, as the district veterinarian suspected anthrax. Reports on the progress of the epizootic and the number of corpses of wild ungulates found in the forest were sent every 5–10 days. In two weeks, three veterinarians and several veterinary students arrived, yet the weather has already changed from heat and drought to cool weather and rain. New carcasses of bison were not found, and bacteriological analysis showed that deaths of animals were caused by Pasteurellosis or Bollinger's disease rather than anthrax. Later microbiologists evidenced that although *Pasteurella multocida* can cause bovine hemorrhagic septicemia in cattle, it is, in fact, a widespread bacteria. Over a two-month period, the team of veterinarians, veterinary assistants and students, surveyed livestock in the surrounding villages, supervised the destruction of corpses, oversaw the sanitary improvements in cowsheds, treatment of cow-houses, vaccinated against anthrax, and also dealt with the beginning foot-and-mouth disease (RSHA 1910–1912). A detailed report on the sanitary condition of the forest and local livestock, the progress of the epizootic, and the fight against it was published two years later (Eckert and Fedders 1912).

The epizootic outbreak of the first half of summer probably led to cancellation of the imperial hunt planned for August-September 1910 and forced the BPF administration to request a veterinarian for the permanent BPF staff. In 1913, such a post was created with former district veterinarian Vyacheslav Zaustsinsky serving as the first BPF veterinarian. In addition, BPF's administration managed to open a small bacteriological laboratory (RSHA 1912–1914).

Zaustsinsky collected an interesting array of veterinary exhibits for the museum (RSHA 1913–1915B, see below). In early 1914, he recorded the loss of 16 bison from a liver parasite, the common liver fluke, *Fasciola hepatica* (RSHA 1913–1915A).

In June 1915, with the front line of war approaching, Zaustsinsky recorded anthrax epizootics in livestock and wild ungulates (RSHA 1915A).

In the appanage period, two forest inventories of BPF were carried out: in 1889–1890 and in 1909–1911. Both are represented by archival and published data. In case of the former, the results of the inventory were published by Genko (1903) and forest management maps survived in the collection of RSHA (1889–1890B). From the latter, several archival files survived and few papers were published by Alekseev (1915, 1916a, b, c). In addition, forest specialists came to BPF on various occasions resulting in several published and manuscript documents, e.g. work of a group of forest specialists and entomologists in connection with the outbreak of nun moth *Lymantria monacha* in 1907–1908 that resulted in a publication by Kründer (1909).

For the preparation of the new forest management plan, one of the most highly qualified forestry officials in the Appanages, the author of forestry handbooks, Nestor Genko (1839–1904) was invited to BPF (RSHA 1889–1890A). In his youth, in 1861–1862, Genko had already participated in preparing the forest management plan of BPF under the leadership of Eichwald. In a few years, Genko published a monograph on BPF, which to this day remains an important source of knowledge on the past forest environment and its management (Genko 1903). The map compiled by the Genko taxation team was discovered in the collection of RSHA (1889–1890B) (see part of the map in Fig. 7.5). This is the oldest detailed forest management map of BPF in existence and is especially valuable because the forest management maps of the first half of the twentieth century are considered lost.

During the inventory works in BPF, Genko consulted with BPF senior forester Eduard Walleburger and with specialists in St. Petersburg, e.g. professor of the Forestry Institute, Piotr Vereha. The forest inventory of BPF was carried out in accordance with the state-of-the-art forestry science, including the forest types concept. The names of the types were borrowed from the traditional names used by forest dwellers to describe different parts of BPF, recorded already in the sixteenth century (Inspection 1559), and in detail described by Piotr Szretter (see Box 4.3).

Genko and specialists working in BPF after him (Alekseev 1916a; Kründer 1909) indicated that objectives of the forest inventory often varied from preserving the habitat of European bison to increasing income from forest exploitation, with the former prevailing. Forest inventory teams were well aware that BPF was a historical monument—in cases when timber was extracted from BPF, the volumes were low with high felling age assigned (160, 180 and even 200 years for pine and oak), and clear-cuts were rarely introduced. According to Genko's plan, the first step in the management was the removal of over-mature and dead trees using selective cutting. In 1894, professor of St. Petersburg Forestry Institute Mitrofan Turski proposed to reduce the size of planned cuts, and in 1897 Tsar Nikolai II decreased it even more, noting that removal of old and decaying trees takes away the original, primeval appearance of the forest (Genko 1903; Karcov 1903).

Clear-cuts were carried out only in the Świsłocz Estate, in which forest exploitation differed both for historical reasons (as Świsłocz Forest, until the Interwar period, was never considered a part of BPF) and due to the fact that imperial hunts were not planned there. Nevertheless, the felling age was as high as in BPF itself, and the

Fig. 7.5 Part of the forest management map prepared under the supervision of Nestor Genko, dated 1889, showing the inner part of Białowieża Primeval Forest with the variety of different forest habitats (RSHA 1889–1890B)

area of cuts was limited to an average of 125 ha per year for the last 5 years of the nineteenth century (RSHA 1891–1902, 1905–1908).

In June 1907, the management of BPF reported to the officials in St. Petersburg about the nun moth outbreak that damaged mostly spruce. As Norway spruce was considered increasingly valuable for timber markets, specialists were sent to the Forest. First, was the forestry specialist of the Appanage Administration, Leonid Yashnov, then two entomologists and a team of forestry specialists led by A.A. Kründer. The latter, after his stay in Białowieża, presented a report on the state of BPF during a meeting of the St. Petersburg Forest Society "with the demonstration of shadow figures on the big screen" (Kründer 1909). Kründer described the types of forest in BPF in accordance with the typology proposed by Genko (1903), based not only on the dominating tree species but also soil and moisture conditions. He also described the possible dynamics of some of the forest types as a result of certain disturbances, such as fires. He believed that in some cases fires might contribute to paludification of the soil. Kründer's description of forest types was accompanied by information about the grassy layer of each of them and photographs of these types. This report also contained an overview of the fight with nun moth outbreak (Kründer 1909).

The first entomologist that arrived to BPF was Ivan Shevyrev from St. Petersburg Forestry Institute. He developed a plan to combat nun moth, but considering how costly it would be, he proposed to leave the development of the outbreak to the "natural course of things". However, BPF administration did not agree with this proposal and requested for opinion of another entomologist. The entomologist of the Main Directorate of Appanage, Konstantin Demokidov, arrived to BPF and introduced several methods of outbreak control for the summer of 1908. These methods incorporated protecting especially valuable trees with caterpillar glue, cutting down heavily damaged spruce, and setting "trap trees" to prevent bark beetles settling on trees weakened by nun moth. The most interesting, however, were in this case not the measures applied against insects themselves, but the possibility that Demokidov had to carry out some experiments and observations to monitor the effectiveness of these methods. The studies have shown that the measures taken did not stop the reproduction of insects. Most moth eggs overwintered successfully and Demokidov predicted that the following season would bring another outbreak (RSHA 1908–1909).

In 1909, preparation of the next forest management plan was entrusted to the Appanage Administration forestry specialist Leonid Yashnov (1860–1936), but the same year it passed to Evgeny Alekseev (1869–1930). Alekseev worked in the forest inventory teams of the Forestry Department, and in 1904 was appointed as an assistant to the head manager of BPF. He remained in this post until the eve of WWI, and in the early 1920s he became a professor of forestry in Kiev. Alekseev's work in BPF reflects the discussions of that time about the theoretical grounds for forest inventory. Alekseev was interested in soil science, forest ecology, and the influence of "natural" factors on the economic value of the forest. It was visible, for example, in his paper devoted to supposedly unprecedented level of resin in Białowieża's pines (Alekseev 1916a, b). Although several documents from Alekseev's forest inventory survived, the most interesting materials have not been discovered, e.g. documents illustrating thoroughness of Alekseev's study, like several hundred soil pits dug to study soil conditions across the entire BPF (Alekseev 1915).

Alekseev noted that in 1909–1911 the main emphasis was put not on the revenues from the forest, but on the forage needs of the game and the aesthetic values of BPF. In relation to this, he noted the fears of forest specialists that the overpopulation on the forest by game would result in destruction of the undergrowth. Forest management plans prepared under supervision of Alekseev focused mainly on selective logging, with priority given to over-matured trees of the most valuable species (pine, oak and ash). Less valuable ones (birch, aspen and spruce) were planned for selective logging to give more space to young pines, oaks and ashes. Alekseev also noted the increasing market value of spruce, including it in the forest management plan (Alekseev 1915).

All authors responsible for preparation of forest management plans confirmed that the Appanage period was the first time that BPF received full attention from specialists in forestry sciences. In accordance with the priorities of their time (Russian forestry science of the late nineteenth-early twentieth century), they paid great attention to the dynamics of forest communities and their soils. They indicated that the forest of Białowieża is and should be treated primarily as a "historical monument". Their remarks on the aesthetic values of BPF include also reflections on its primeval character. Alekseev, describing the goals of forest management in 1909–1911, noted that the hunt for the "antediluvian beast" amidst "the overmatured tree stands give the illusion of a pristine forest untouched by the hand of man". He, as well as many of his colleagues understood that such "untouched" forests are just a concept, nevertheless worth preserving (Alekseev 1915).

In the second half of 1912, the BPF administration started preparations to create the first natural history museum (RSHA 1913–1915B). Organization of the museum was entrusted to the new administrator of the forest, appointed in 1912, Mitrofan Golenko . During planning of the museum, Golenko asked the Appanage Administration for permission to appeal to the population through newspapers and local message boards with a call to donate historical objects that "*characterize the historical and economic life of the forest*" to the museum. However, the authorities rejected Golenko's request stating that the museum should focus on the natural life of the forest with sections devoted to imperial hunts and forest management (RSHA 1913–1915A, B).

Golenko ordered all his subordinates to collect "everything that is characteristic for the forest's nature and life". This was to include samples of soils, plants, animal and mushrooms, hunting equipment, including poachers' weapons, as well as photographs, prints and paintings presenting the life of the Forest (RSHA 1913–1915A, B).

Between the spring and autumn of 1913, the erection of the additional wing of the Appanage Administration building (originally constructed in 1890s) was completed. On the first floor of this wing, there were additional rooms for officials (e.g. forester's office and a drawing room), and the museum was to be located in the second floor. In November 1913, taxidermist Ivan Hudyma from St. Petersburg started working on stuffed animals and birds for the exhibition. Exhibits devoted to soil science and forest management were prepared most probably by or under the supervision of Evgeny Alekseev. In November 1914, BPF administration reported to St. Petersburg that the museum was almost completed: "*in general, the museum is very interesting and beautiful. Undoubtedly, hundreds of tourists visiting the forest will have a full impression, and students of the educational institutions will get a lot of useful information after visiting the museum*" (RSHA 1913–1915B).

By the end of 1914, the museum had over 500 exhibits, among them over 300 stuffed animals. Apart from some omissions (e.g. almost no representation of animal groups other than mammals and birds), the museum represented the unique values and diversity of life of BPF very well. Alekseev, who was a graduate of the St. Petersburg Forestry Institute and an active member of St. Petersburg Forestry Society, collaborating with Golenko, was able to implement many ideas taken from the Forestry Museum at the St. Petersburg Forestry Institute to Białowieża's first museum.

Despite the outbreak of World War I in 1914, the new head of BPF administration—Yurii Lwow—cared about the museum. He applied to St. Petersburg for money to employ a museum warden and complained about the lack of space to properly display all collected exhibits (both requests rejected by the Appanage Administration as "untimely"). The museum was opened to the public, and Lwow wrote in his letter: "*the Białowieża Museum is eagerly frequented by the local population and people visiting the forest, and has gained a fame of an educational institution. Most importantly, with its gradual replenishment, it will be an informative and pictorial monument of BPF for future generations*". Visitors were recorded during the period between 25th April and 5th July with a total of 350 people attending the exhibition (RSHA 1913–1915A, B). In August 1915, the majority of museum exhibits along with other belongings of the Appanage Administration and the Imperial Palace were evacuated to Moscow. Most probably for some time, all property was stored in Neskuchny Palace, but from 1918 various items were given to different institutions and today the fate of natural exhibits remains unknown. The only traces of the first natural history museum in Białowieża remain in archival files (especially the full catalogue of the museum, RSHA 1913–1915B) and Mitrofan Golenko's paper devoted to the museum (Golenko). There is also a small chance that the stuffed animals and birds from BPF museum could still be held in State Darwin Museum (Moscow), as recently antlers of deer and horns of bison from the Białowieża imperial palace were discovered there (Miloserdov 2016).

7.5 View from Outside: The British Recognition of the Forest and European Bison

The fame of BPF as the last refuge of European bison spread through Western Europe in the course of the nineteenth century. This resulted in increasing numbers of scientific publications, popular articles, travel memoirs, encyclopaedic entries, etc. Even a brief and cursory overview of all writings published in German, French, Spanish, Italian and English would take up a disproportionate part of this book, therefore we decided to focus on just one point of view in this subchapter, the British recognition of BPF and European bison.

By the mid-century, the Forest and its bison became the subject of interest for English naturalists and entered the mainstream of public interest through popular articles and travel memoirs, e.g. connected with bison hunting. The most prominent scientists such as Lyell, Murchison, Owen and Darwin, who played key roles in the development of natural sciences were interested in Białowieża and European bison. Charles Lyell wrote about European bison and their extinction outside of Lithuania in historical times in "The geological evidences of antiquity of man" (Lyell 1863). The Darwin Correspondence Project (www.darwinproject.ac.uk) shows that Darwin knew Brincken's book through Weissenborn's article published in "The Magazine of Natural History" (Weissenborn 1838). Darwin mentioned Białowieża's bison in his "On the origin of species" as an example of the inbreeding risk in the shrinking populations of wild animals (Darwin 1859).

However, until around the mid-nineteenth century, English naturalists knew European bison and BPF mostly from published literature and data. Historical sources indicate that the King of Prussia, Friedrich Wilhelm I, who donated two bison to Anna of Russia in 1739, donated also one or two bison to London in 1730s (Hagen 1819). However, the oldest of the items in the British zoological collections connected with BPF that we currently know of is a male European bison skull, donated to the Museum of the Royal College of Surgeons of England in 1838, by Georges Cuvier's colleague Adolph Wilhelm Otto (1786–1845) (Flower and Garson 1879–1884).

This situation has changed thanks to geologists Roderick Impey Murchison (1792–1871), who undertook three expeditions in the European part of Russia and the Ural in the 1840s. Murchison was interested in bison in the context of geological changes and the extinction of large mammals. He mentioned bison both in the article on the extinction of mammoths (Murchison 1845) and in the geological inventory of Russia (Murchison et al. 1845). Murchison never visited BPF and in the first years of 1840s he had no possibility to see live European bison elsewhere. Nevertheless, he was probably able to see skeletons of European bison in St. Petersburg's zoological collections. Murchison saw geological changes as the main reason of the extinctions of many species, including European bison, paying less attention to anthropogenic pressure. The degeneration of species was another theory he supported, which was seemingly evidenced by ancient bison bones—some of which were larger than the contemporary living members of the *Bison* genus. Also descriptions of old royal hunts during which supposedly enormous bison were killed seemed to support the degeneration theory. Murchison was invited to Russia by Nikolai I, who hoped that the famous geologist would discover coal and other useful minerals in the empire. Murchison took advantage of his connections and the Tsar's favour and, as a member of the Royal Commission of the British Museum, asked for a bison pelt and skeleton for the Museum. His request was granted in 1845. Such replenishment of the collection was a treasure for Richard Owen (1804–1892), a palaeontologist famous for his reconstructions of prehistoric animals: "*The specimens have been presented by his Imperial Majesty to the British Museum, where I have had the opportunity of comparing the recent bones with those of the fossil Aurochs since the foregoing sheets went to press*" (Owen 1846). Owen used the European bison skeleton to determine the difference between the aurochs and the European bison and to examine the relationship between the skeleton of fossil and contemporary bison and aurochs (Owen 1846).

Serving also as a member of Zoological Society of London, Murchison asked for two live bison for the London Zoo, which was managed under the aegis of the Society. This request was also fulfilled (See Box 5.3). It was also Murchison who introduced the Dolmatov's description of European bison and BPF to the London Zoological Society (Dolmatov 1848). Thanks to this event, Dolmatov corresponded with Owen

(SARF 1860). Dolmatov's observations and data published in Proceedings of the Zoological Society were later used by several authors.

Bison sent to London Zoo received names Sarah and Billy and enjoyed a great popularity. A natural history sculptor and artist Benjamin Waterhouse Hawkins (1807-1894) prepaerd a project of a monument depicting the pair (Bramwell and Peck 2008). When one of the bison died at the age of two years and five months in 1847, Owen conducted the autopsy. He published its results and description, along with the measurements of this bison. He assumed that the cause of death was "*raw cold and heavy fogs*" (Owen 1848). In 1868, one more live bison was purchased for London's Zoological Garden from the Zoological Society of Amsterdam; however this one also did not live long.

It is worth noting that the long-lasting confusion between names of bison and aurochs in continental Europe in the English literature was further confused by the term buffalo, used for American bison. A reader's letter to "Nature" sums this situation well: "*In his excellent article on the extermination of the American bison, "R.L." remarks (Nature p. 11) on the transatlantic practice of miscalling that animal a "buffalo"; but on the next page he writes of "its European congener the Lithuanian aurochs". This is to perpetuate a common error at least as bad. The "aurochs" (= ox of yore), Latinized by Caesar in the form urus, is or was the Bos primigenius or B. urus of scientific nomenclature. It is wholly by mistake that in its extinction as a wild animal its ancient name was transferred to the bison or Zubr*" (Newton 1890).

By the 1890s, European bison caught the attention of Herbrand Arthur Russell, 11th Duke of Bedford. From 1899 to 1936 he was a President of the Zoological Society of London and directly responsible for saving two species of large mammals from extinction, the Himalayan tahr and Père David's deer. In 1899, the Duke of Bedford received three bison from BPF for his park at Woburn Abbey (RSHA 1898–1908). Since the majority of European bison held in captivity in the Central Europe did not survive World War I, the Woburn Abbey herd played an important role in post-war bison restitution (de Bont 2017; Zablockij 1939).

European bison fell into the scope of interest of palaeozoologist, author of the catalogues of the Natural History Museum and many books on natural history, Richard Lydekker. Bison appeared in the catalogue of the Museum and in Lydekker's book (Lydekker 1901), and author was able to visit BPF together with his daughter in August 1907 (RSHA 1907B).

At least two British travellers and hunters, Algernon Herber Percy and Edward North Buxton, contributed to the popularization and increase in knowledge on European bison and BPF. Sir Algernon Heber-Percy's (1845–1911) main career was in the army but he was mostly known as a traveller and hunter, especially fond of exotic travels to the Middle East. In 1894, he published a description of his travel to BPF and hunt organized there for him in 1879. Scrupulously described, both the hunt and circumstances surrounding it, were undoubtedly interesting for the British audience. Heber Percy's text was accompanied by two drawings made by Charles Whymper (1853–1941), known for his depiction of landscapes, game animals and hunts (Fig. 7.6). Heber Percy received the permission to hunt in BPF thanks to his

Fig. 7.6 Charles Whymper's drawing of two European bison heads—trophies of Algernon Heber-Percy's hunt in BPF in 1879, published in 1894 (Heber Percy 1894)

acquaintance with Frederick Temple Blackwood, commonly known as Lord Dufferin (1826–1902), one of the most influential figures in Victorian England. This, at least partially, explains the effort and engagement of higher officials of the Grodno Province into the organization of Heber Percy's stay (Heber Percy 1894). In the visitor's register of BPF, Algernon Heber-Percy's name is written next to that of his wife Alice and the name of the wife of Grodno governor Olga Ceumern who guided the English tourists in Białowieża (SARF 1860–1915). The trophies from the hunt in BPF, two bison heads, are still displayed in Heber Percy's family manorial home at Hodnet Hall in Shropshire England. Today the manor and its garden and park are open to the public, and the bison heads are the pride of the "tea room" at Hodnet Hall.

Edward North Buxton (1840–1924) was a British conservationist and a founding member of the Society for the Preservation of the Wild Fauna of the British Empire. He visited BPF "*in order to see the Bisons (Bison europaeus) in the Emperor of Russia's forest, where he was successful in approaching near enough to a part of the herd to obtain some photographs of these animals*" (Buxton 1899). Accompanied by Josif Newerly, Buxton had a chance to acquaint himself with the state of the forest by the very end of the nineteenth century and noted: "*With the exception of the* *meadows which border the latter, and a few clearances for cultivation round small villages, there are no open spaces: consequently although the timber, which consists mainly of oak, elm, birch, spruce, and fir, is very fine, the forest is tame and wanting in variety. This monotony is enhanced by the unfortunate practice of removing all wind falls, a most short-sighted policy, as I think, because nothing so assists the warmth, shelter, and sense of security of a forest, for wild animals, as fallen timber, through the branches of which a tangle of wild growth quickly penetrates and forms a natural screen*" (Buxton 1899). From the point of view of today's knowledge on forest ecology Buxton's remark on "removing all windfalls" being a short-sighted policy sounds very modern. Nevertheless, the fact that Buxton complains about the "monotony" of the forest and does not mention pine, one of the most distinctive tree species in BPF, could suggest he saw only a small portion of the forest. This was maybe just the hunting grounds prepared for imperial visit, thus cleared from all windfall wood.

At the end of the nineteenth and the beginning of the twentieth century, naturalists (European in general and British in particular) were interested in European bison mainly in connection with three scientific problems. (1) Firstly, could this large, prehistoric species be preserved from inevitable demise? (2) Secondly, what were the causes of bison extinction?—a topic directly related to discussions about the extinction of the Pleistocene megafauna as a whole. In this respect, the question on the role of man in extinctions of large animals (like mammoth or bison) often appeared. The scientific debate on this issue at the end of the nineteenth and the beginning of the twentieth century exerted its influence even on fiction, in which the theme of prehistoric people became fashionable (see the novels by Ernest d'Hervilly or J.-H. Rosny). These palaeozoological discussions are still ongoing as a part of a wider debate, e.g. the discussion of the extinction of the megafauna in North America (the overkill hypothesis), which often spreads to political debate (the Overkill hypothesis and its opponents, Meltzer 2015) and involves a number of political issues. (3) The third field of interest was the idea of species degeneration and notion that due to this gradual degeneration bison were doomed to extinction. Indirectly, this line of thought was connected with the eugenic movement started with ideas of Francis Galton and widespread in the late nineteenth and early twentieth century among the British educated circles (Bashford and Levine 2010).

Box 7.1 Konrad Wróblewski's veterinary research on European bison

Konrad Wróblewski (1865–after 1945) was born in the family of Lithuanian peasant Osip Žvirblis in the vicinity of Vilnius. In the nineteenth and early twentieth-century Lithuanian language and culture were subjected to Polonization or Russification, therefore the Lithuanian surname Žvirblis (sparrow) was "translated" into Polish as Wróblewski. The family was not able to financially support their son's education, but in 1889 Konrad managed to enter the Dorpat Imperial Veterinary Institute. In 1894, he received the title of Master of Veterinary and started to build his career in this field (ENA 1889–1895). In the early 1900s, Wróblewski worked in St. Petersburg as a veterinarian at the railway and in his free time—at the Zoological Museum of the Imperial Academy of Sciences, which resulted in craniological paper (Wróblewski 1905) and the invitation to Kulagin's expedition (RSHA 1905–1914).

Wróblewski arrived to BPF in the late 1906. He turned "the bison barn", where game were weighed and gutted after the royal hunts, into an anatomical laboratory with heating, lighting, tables covered with zinc, and running water. Wróblewski made a small bacteriological laboratory as well. Before Wróblewski's research, knowledge of bison's food preferences was based on accidental observations of eating certain plant species; and the causes of death when it occurred were described by vague and obscure words. Wróblewski opened the corpses of many dozens of dead animals, analysed the contents of their stomachs and the character of tooth erosion, collected parasites from their internal organs, counted the number of those parasites, and identified their species (ARAS 1906–1913, 1906–1909; Wróblewski 1907–1909, 1912b, 1927). He also made some bacteriological experiments trying to determine the causative agent of wild boar plague and also made probably the first attempt to quantify the nutritional value of BPF's meadows. From blood samples collected by him, a new species of the bison blood parasite was described and named *Trypanosoma wrublewskii* after him (Kingston et al. 1992; Wróblewski 1908a, b, 1909, 1912a). His research showed that the main food of bison were grasses, in contrast to deer, which prefer shoots of trees and shrubs (Wróblewski 1912b). Wróblewski's works are probably the best source on biology and ecology of bison living in the last natural population of lowland bison (Wróblewski 1927). His research gives possibilities to determine the level and causes of bison mortality in the early twentieth century and to compare them with the recent data (Krasińska and Krasiński 2013; Karbowiak et al. 2014).

Wróblewski stayed in BPF till the autumn of 1908, and summarized his observations in Russian and German veterinary journals. The report on his work was to be part of a large European bison monograph with parallel text in Russian and French, but the Appanages did not complete its publication (RSHA 1905–1914; ARAS 1915–1917). It was eventually published in Poland in 1927. Wróblewski came back to BPF in 1930–1931, when he worked in the Directorate of State Forests in BPF organizing a bacteriological laboratory. His breeding and veterinary tips for preserving the Białowieża population (Wróblewski 1912a, 1927, 1932) became instrumental in the restitution of European bison (Krasińska and Krasiński 2013).

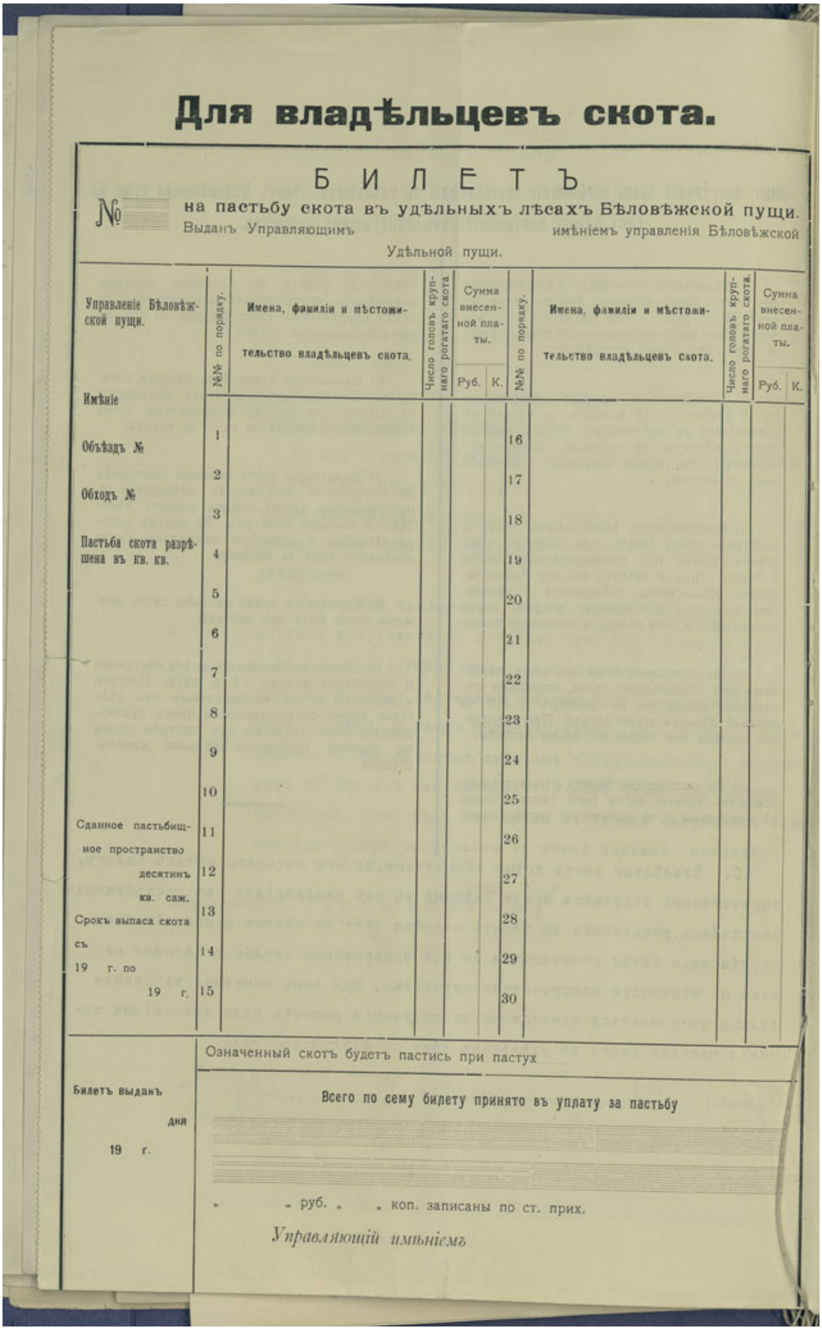
Fig. 7.7 Photograph of the road to Białowieża by Vishnyakov (1894)

Box 7.2 Evgeny Vishnyakov's photographical journey through BPF

Evgeny Petrovich Vishnyakov (1841–1916) was a Russian infantry general. Nowadays, he is mostly known for his landscape and nature photographs. In 1889, two volumes of his photographs were published under a common title "Photographs from nature" with a subtitle "Sketches by pen and photo"—a similar subtitle accompanied his album of photographs from BPF, published in 1894. He travelled to BPF in summer of 1892 together with Ivan Shishkin, painter famous for his landscapes of Russian forests. Shishkin himself was personally invited to BPF by Tsar Alexander III (Pikulev 1955). Photographs (see Fig. 7.7) are accompanied by a text, majority of which serves as a summary of knowledge on BPF and European bison collected from various sources, not without errors (number of bison in 1894 estimated at 1700), or simplifications (e.g. low increase in bison explained by old infertile bulls not allowing younger ones to mate with cows). But there are several passages in the text where author's artistic look on the Forest prevails: "*This diversity of tree species of wood plants gives the forest an individual form and colour. Like in a kaleidoscope, bright and dark greens linger on the background blue sky. All trees are rushing up towards the sunlight with peculiar energy, which allows for* *picturesque and beautiful development of their crowns. Powerful pine trunks stand like a slender columns of a mysterious temple with a dome weaved by treetops. Because of this, the interior of the forest acquires a special majesty, and if it were not for the motley young thickets, it would gain even greater dignity*" (Vishniakov 1894). What is interesting to note, especially in the light of contemporary conflict around the future of BPF management and Polish State Forests fight to defend Norway spruce stands against bark beetle, is the view on spruce presented by the photographer: "*Both the coniferous and deciduous habitats were penetrated by an uninvited guest – Norway spruce, which obscures the beauty of buxom pines and oaks and disturbs the scenic view sought with a lot of effort by a painter or nature lover*" (Vishniakov 1894). Some photographs were made from the same observation point as Shishkin's paintings, for example "Felled oak in BPF" (now in Art Museum in Yaroslavl). Vishniakov's remarks on the reasons behind the protection of the Forest sound strangely contemporary: "*The slow current of rivers as well as swampy banks do not allow for floating of wood. This, in turn, was one of the reasons why the forest was saved; the desire to protect European bison living in the forest and announcing laws to that end allowed for preservation of one of the oldest forest areas, remnant of primeval forests of Central Europe, which are scarce today, just like bison surviving in the Forest*". Also his opinion on the need to restore beaver population in the Forest was ahead of its time: "*Despite the fact that it was possible to preserve such rare species like European bison in BPF, it was unfortunately not possible to preserve equally rare beaver, once swarming Narew, Narewka, Leśna, Łutownia and other rivers.* (…) *Probably even now it would be possible to breed and introduce the species to BPF, especially considering the fact that the latter favours such endeavours*" (Vishniakov 1894).

Box 7.3 The impact of cattle pasturing inside BPF

Wood pastures for domestic stock, similar to other traditional types of forest resource use existed for centuries with very little formal governance (Rackham 2004; Bergmeier et al. 2010) and became a major problem in the nineteenth century, when modern forest management prioritizing timber production was introduced. Cattle pasturing for centuries was one of the main factors shaping European landscapes (Vera 2000) that created of so-called "wood pasture" (Rackham 2013; Rotherham 2013), by definition contradicting the goals of "proper" forest management. Livestock grazing can have a detrimental effect on tree regeneration and in combination with other traditional forest uses can lead to the development of heterogeneously structured forest stands (Mayer and Stöckli 2005; Mayer et al. 2005; Hjeljord et al. 2014).



Для владѣльцевъ скота.

БИЛЕТЪ

№ на пастьбу скота въ удѣльныхъ лѣсахъ Бѣловѣжской пущи.

Выданъ Управляющимъ имѣніемъ управленія Бѣловѣжской Удѣльной пущи.

Управленіе Бѣловѣжской пущи.

Имѣніе

Объѣздъ №

Обходъ №

Пастьба скота разрѣшена въ кв. кв.

№№ по порядку.	Имена, фамиліи и мѣстожительство владѣльцевъ скота.	Число головъ крупнаго рогатаго скота	Сумма внесенной платы. Руб.	К.	№№ по порядку.	Имена, фамиліи и мѣстожительство владѣльцевъ скота.	Число головъ крупнаго рогатаго скота.	Сумма внесенной платы. Руб.	К.
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Срокъ выпаса скота съ 19 г. по 19 г.

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Билетъ выданъ дня 19 г.

Всего по сему билету принято въ уплату за пастьбу

руб. коп. записаны по ст. прих.

Управляющій имѣніемъ

Fig. 7.8 Pasturing ticket—printed blank form for cattle owner specifying the area, number of cattle and annual pay for pasturing inside BPF (RSHA 1912–1913)

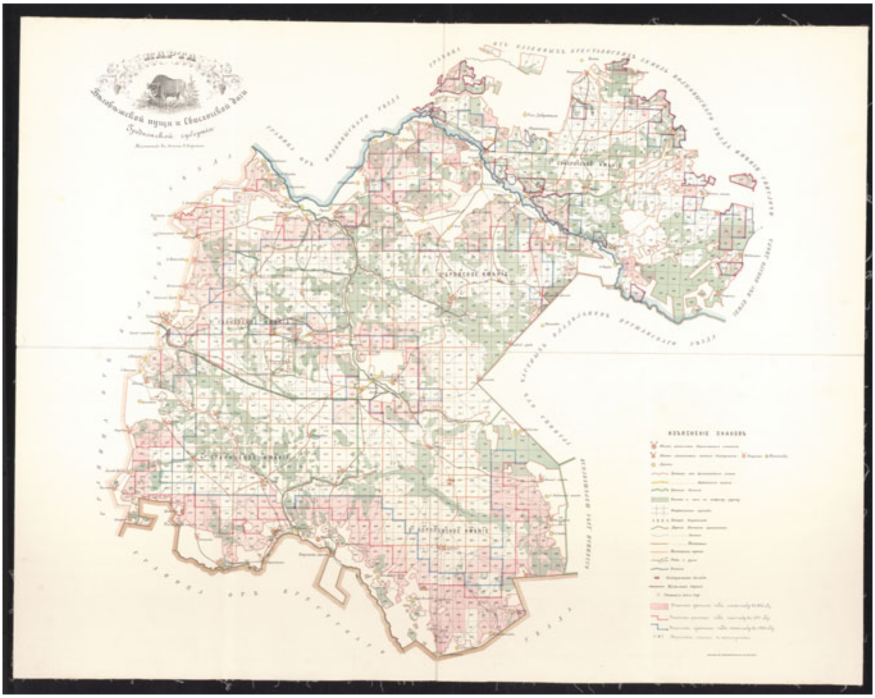

Fig. 7.9 Map of Białowieża Primeval Forest and Świsłocz Forestry in Grodno Province, showing the pasturing range in 1897, 1904 and 1905 (RSHA 1905A)

In BPF, cattle pasturing inside the forest was confirmed in written sources since at least the sixteenth century (Hedemann 1939; Samojlik 2007) and persisted after the forest was incorporated into the Russian Empire. Forestry and especially game administration fought with cattle pasturing inside BPF, still considered an undeniable privilege of the local residents. The activity itself, and the herders, accused of poaching, setting fire and other misconducts, were seen as a factor harmful for wildlife and BPF itself (Karcov 1903). Forest administration incorporated different measures in an attempt to control and limit the forest pasturing, from strict delimitation of the area of forest pastures ascribed to each of the villages, through control over the annual pay for area of pasture and cattle brought to the forest, to special pasturing contracts signed by each of the villages. The latter stated the area under pasture on one hand, and obligations of cattle owners on the other. These obligations incorporated some basic terms: no pasturing in the night, prohibition of bonfires, strict no timber felling or hunting policy, etc. Peasants were also obliged to pay compensation for any damages done in their forest pastures. Special tickets were printed and prepared for each of the villages from which cattle were brought to the forest (see Fig. 7.8). BPF administration attempted to tighten the control over pasturing but it met the resistance of local communities, documented in dozens of complaints sent to the authorities in Grodno and St. Petersburg (RSHA 1889A, 1902B; NHABG 1893, 1905, 1913C). As a result, the administraton had to lower the annual rates for cattle pasturing in BPF several times (1.40 roubles in the period 1894–1900, 1.20 roubles from 1901, and 1 rouble from 1905), although it was already low compared to similar rates in private forests (RSHA 1908).

Archival documents, especially the “Map of Białowieża Primeval Forest and Świsłocz Forestry in Grodno Province” showing the pasturing range in 1897, 1904 and 1905, are evidence of frequent changes of the area where cattle pasturing was allowed 1897 and 1905 (RSHA 1905A; see Fig. 7.9). It is easy to explain these changes by observed depletion of resources, especially taking into account more modern, Paczoski’s (1930) and Faliński’s (1966, 1986) remarks on the destruction of undergrowth by cattle in the twentieth century. But there is another explanation: changes of pasturing areas could have been used as a method of strengthening the policy regulating woodland use, and changes of pasturing areas could be seen as a penalty for poaching and other misconducts of local dwellers (RSHA 1889A, 1902B; see also Samojlik et al. 2016). Although the administration of BPF would rather see the forest devoid of cattle, as competitors for the same resources the game species use, it was practically impossible to remove this privilege from local peasants (in fact cattle pasturing survived in BPF until 1970s; see Samojlik et al. 2016).

Box 7.4 Antoni Kamieński’s graphical memoirs from BPF

Antoni Kamieński (1860–1933) was an artist born in Vilnius who studied in St. Petersburg and Paris. He was a keen illustrator of the most widely read Polish literature of his period, including Sienkiewicz, Sieroszewski, Orzeszkowa and Prus, and he also published in a popular weekly “Tygodnik Ilustrowany”. Valued for his allegoric scenes and portraits, he did not shun nature as a topic of his works.

In 1912, Warsaw’s publishing house “Świat” (“World”) published Kamieński’s “Szkice z Puszczy Białowieskiej” (“Sketches from Białowieża Primeval Forest”) (Kamieński 1912), a set of 14 charcoal drawings with foreword written by Polish writer and journalist Zygmunt Bartkiewicz (1867–1944). Kamiński’s portfolio contained drawings of BPF nature (“Białowieża giant”, “European bison”, “Windfall”), but more often of local dwellers and their daily life (“Budy village”, “Yarn spinners”, “Budy village (yard)”, “Lumbermen shack”, “Road from Bielsk to Białowieża”, “Cart”, “Old man”,

Fig. 7.10 Antoni Kamieński's drawing from BPF: "Meeting on a road" (Kamieński 1912)

"Lithuanian and Belarussian", "Białowieża girl"). One of the drawings, "Meeting on a road" (Fig. 7.10) is particularly interesting—it shows a two-horse cart that has been stopped on a road by a bison. Two men, apparently lacking any other resource to remove bison from their way, start a small fire in the middle of the road. The bison is covered in smoke, but there is no indication that this trick worked—the animal is calm and steady, not looking like it wants to ever move from the road.

Bartkiewicz's foreword focuses on Białowieża folklore and legendary status of the forest, yet does not keep away from commenting and criticizing the modern, commercial approach to BPF: "*The new man loves strong words and although his hands are weak, they are always eager to destroy what thousands minds, thousands caring hearts build through ages. How much delighted one would be, if only allowed, to take an axe and chop the ancient woodland trunk by trunk, tear and burn the rest, and then sew the fertile openness with fresh cereal. And then announce with pride to the world that it was his doing that in the place where ancient wild forest rustled, oat was born*" (Bartkiewicz 1912).

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[RSHA] 1895–1902. Ob otkrytii v Belovezhskoj Pushche dolzhnosti policejskih urjadnikov (On the opening of the posts of police officers in the BPF). F. 515, O. 42, No 3999 (in Russian)
[RSHA] 1895–1904. Po isku k krest'janam Fome Galke i Antonu Diku o vzyskaniju s nih ubytkov za ubitogo imi losja i zubra (On the claim of penalty from peasants Foma Galka and Anton Dik for a moose and a bison they killed). F. 515, O. 38, No 411 (in Russian)
[RSHA] 1896–1898. O rashodah po soderzhaniju Belovezhskogo zverinca v 1897 g. (On expenses for game reserve in BPF in 1897). F. 515, O. 42, No 3609 (in Russian)
[RSHA] 1896–1902. O rashodah po soderzhaniju Belovezhskogo zverinca v 1896 g. (On expenses for game reserve in BPF in 1896). F. 515, O. 42, No 4346 (in Russian)
[RSHA] 1897–1898. O rashodah po soderzhaniju Belovezhskogo zverinca v 1898 g. (On expenses for game reserve in BPF in 1898). F. 515, O. 42, No 4002 (in Russian)
[RSHA] 1897–1902. O pravilah splava lesa po r. Narev, Narevka, Gvozna, Lutovnja i Lesna i o vkljuchenii jetih rek v nomenklaturu splavnyh putej, a takzhe ob assignovanii sredstva na ispravlenie rusel rek (On the rules of wood floating on the river Narew, Narewka, Hwozna, Łutownia, Leśna and on the inclusion of these rivers in the nomenclature of floatation paths, as well as on the allocation of funds for the correction of river beds). F. 515, O. 42, No 4000 (in Russian)
[RSHA] 1898–1905. O dopushhenii krest'jan, postradavshih ot neurozhaja, k besplatnym pol'zovanijam v udel'nyh lesnyh dachah (On permission, to peasants affected by crop failure to free use in appanage forestry districts). F 515, O. 42, No 4483 (in Russian)
[RSHA] 1898–1908. O rashodah po Belovezhskomu zverincu v 1899 (On expenses for game reserve in BPF in 1899). F. 515, O. 80, No 172 (in Russian)
[RSHA] 1899–1903. O rashodah po Belovezhskomu zverincu v 1900 g. (On expenses for game reserve in BPF in 1900). F. 515, O. 42, No 4148 (in Russian)
[RSHA] 1899A. O voznagrazhdenii krest'jan za ubytki, prichinjaemye udel'nym lesom v Belovezhskoj Pushche ih rybnoj lovle (On the remuneration of peasants for the losses caused to their fisheries by appanage administration). F. 515, O. 43, No 311 (in Russian)
[RSHA] 1899B. Risunki formy obmundirovanija (Drawings of the uniform style). F. 515, O. 87, No 1436 (in Russian)
[RSHA] 1906C. O vospreshchenii postroyki kozhevennogo zavoda vblizi Pushchi (On the prohibition of the construction of a tannery near the Forest). F. 515, O. 42, No 4501 (in Russian)
[RSHA] 1901. Ob opredelenii na sluzhbu i uvol'nenii egerej i ober-egerej v Belovezhskoj Pushche (On appointment and firing of jegers and ober-jegers in BPF). F. 515, O. 42, No 3790 (in Russian)
[RSHA] 1901–1903. O rashodah po Belovezhskomu zverincu v 1902 g. (On expenses for game reserve in BPF in 1902). F. 515, O. 42, No 4150 (in Russian)
[RSHA] 1901–1904. O rashodah po Belovezhskomu zverincu v 1901g (On expenses for game reserve in BPF in 1901). F. 515, O. 42, No 4351 (in Russian)
[RSHA] 1902A. Ob utverzhdenii novogo polozhenija ob ohote v Belovezhskoj Pushche (On the new instructions for hunting personnel in BPF). F. 515, O. 42, No 4007 (in Russian)
[RSHA] 1902B. Po zhalobe krest'jan Pogorel'skogo obshhestva na pritesnenija so storony administracii Belovezhskoj pushchi (On complaints of Pogorzelce community peasants on oppression from BPF administration). F. 515, O. 43, No 442 (in Russian)
[RSHA] 1902C. O lesokul'turnyh rabotah v Belovezhskoj pushche (On silvicultural works in BPF). F. 515, O. 42, No 4006 (in Russian)
[RSHA] 1902–1903. O rashodah po Belovezhskomu zverincu v 1903 g. (On expenses for game reserve in BPF in 1903). F. 515, O. 42, No 4154 (in Russian)
[RSHA] 1903A. Ob opredelenii na sluzhbu i uvol'nenii egerej, ohotnikov, o vydache im posobij (On the appointment and release of rangers and hunters and on issuing funds to them). F. 515, O. 42, No 4158 (in Russian)
[RSHA] 1903B. O priobretenii koljaski avtomobil' dlja nadobnostej Upravlenija Belovezhskoj pushchi (On purchasing a car for the needs of BPF administration). F. 515, O. 42, No 4155 (in Russian)

[RSHA] 1903–1907. O rashodah po Belovezhskomu zverincu v 1904 (On expenses for game reserve in BPF in 1904). F. 515, O. 42, No 4583 (in Russian)
[RSHA] 1904. O lesokul'turnyh rabotah v Belovezhskoj pushche (On silvicultural works in BPF). F. 515, O. 42, No 4584 (in Russian)
[RSHA] 1904–1907. O rashodah po Belovezhskomu zverincu v 1905 (On expenses for game reserve in BPF in 1905). F. 515, O. 42, No 4585 (in Russian)
[RSHA] 1905A. O komandirovanii g. Rejharda v Belovezhskuyu pushchu dlja vyjasnenija nekotoryh voprosov otnositel'no nedorazumenij mezhdu krest'janami i udel'noj administraciej (On Mr. Reichard's trip to BPF, to solve some misunderstandings between peasants and the appanage administration). F. 515, O. 42, No. 4572 (in Russian)
[RSHA] 1905B. Ob opredelenii i uvol'nenii egerej, ober-egerej, ohotnikov i konno-policejskih urjadnikov Belovezhskoj Pushchi, o vydache im posobij (On appointment and dismissal of jegers, ober-jegers, hunters and police officers BPF, on issuing funds to them). F. 515, O. 42, No 4357 (in Russian)
[RSHA] 1905–1908. Po ustrojstvu Svislochskoj dachi (On the management plan of Świsłocz forestry disctict). F. 515, O. 80, No 405 (in Russian)
[RSHA] 1905–1914. Po zapiske prof. Kulagina o nauchnom issledovanii zubrov (On the note of prof. Kulagin about the scientific study of bison). F. 515, O. 80, No 409 (in Russian)
[RSHA] 1906A. Ob opredelenii i uvol'nenii egerej, ober-egerej, ohotnikov i konno-policejskih urjadnikov Belovezhskoj Pushche, o vydache im posobij (On appointment and dismissal of jegers, ober-jegers, hunters and police officers BPF, on issuing funds to them). F. 515, O. 42, No 4498 (in Russian)
[RSHA] 1906B. Ob ubijstve brakon'erov v Belovezhskoj pushhe (On killing poachers in BPF). F. 515, O. 42, No 4502 (in Russian)
[RSHA] 1907A. O prinjatii ot vel. kn. Sergeja Mihajlovicha kavkazskogo zubra dlja Belovezhskoj Pushchi (About receiving a Caucasian bison for BPF from the grand duke Sergei Mikhailovich). F. 515, O. 42, No 4591 (in Russian)
[RSHA] 1907B. O razreshenii raznym licam poseshhenija Belovezhskoj Pushche (On allowing different people to visit BPF). F. 515, O. 43, No 78 (in Russian)
[RSHA] 1908–1909. Perepiska entomologa Demokidova s upravljajushhim Belovezhskoj Pushchi o merah bor'by s shelkoprjadom monashenkoj (Correspondence of entomologist Demokidov with BPF manager on control methods of nun moth). F. 515, O. 80, No 2043, in Russian)
[RSHA] 1908–1912. Vedomosti lesoustrojstva Belovezhskoj pushhi (Information on the forest management of BPF). F. 515, O. 67, No 1163 (in Russian)
[RSHA] 1908. O razreshenii raznym litsam prava pastby v lesnyh dachah Belovezhskoj Pushchi za platu i bezplatno (On allowing different persons to pasture in Białowieża Forest for pay and free of charge). F. 515, Op. 43, No. 825 (in Russian)
[RSHA] 1909. Ob opredelenii i uvol'nenii egerej, ober-egerej, ohotnikov i konno-policejskih urjadnikov Belovezhskoj Pushchi, o vydache im posobij (On appointment and dismissal of jegers, ober-jegers, hunters and police officers BPF, on issuing funds to them). F. 515, O. 80, No 787 (in Russian)
[RSHA] 1909–1913. O rashodah po Belovezhskomu zverincu v 1910 g (On expenses for game reserve in BPF in 1910). F. 515, O. 80, No 662 (in Russian)
[RSHA] 1910. Rasporjazhenija otnositel'no predpolagaemogo Vysochajshego priezda v Belovezh (Orders regarding the upcoming tsar's arrival to Białowieża). F. 515, O. 70, No 113 (in Russian)
[RSHA] 1910–1912. O pojavlenii sibirskoj jazvy v Belovezhskoj pushche i o bor'be s nej (About the appearance of anthrax in BPF and the fight against it). F. 515, O. 80, No 775 (in Russian)
[RSHA] 1911–1913. O rashodah po Belovezhskomu zverincu v 1912 g (On expenses for game reserve in BPF in 1912). F. 515, O. 80, No 897 (in Russian)
[RSHA] 1912A. Rasporjazhenija po Vysochajshemu priezdu v Belovezhskuy Pushchu (Orders for the tsar's arrival to BPF). F. 515, O. 70, No 259 (in Russian)
[RSHA] 1912B. Proekt fil'troozonnoj stancii pri Belovezhskom dvorce (A project of ozonefilter station near Białowieża palace). F. 515, O. 88, No 2258 (in Russian)

[RSHA] 1912C. Po hodatajstvu evreev-smolokurov Belovezhskoj Pushchi ob ostavlenii ih na zhitel'stve v Pushhe (On the request of the Jewish wood-tar makers to let them stay at BPF). F. 515, O. 39, No 1305 (in Russian)
[RSHA] 1912–1913. O razreshenii raznym litsam prava past'by skota (On allowing different persons the right to graze livestock). F. 515, O. 80, No 1058 (in Russian)
[RSHA] 1912–1914. O rashodah po Belovezhskomu zverincu v 1913 g (On expenses for game reserve in BPF in 1913). F. 515, O. 80, No 1068 (in Russian)
[RSHA] 1912–1915. Po voprosu ob uluchshenii senokosov v Belovezhskoj pushhe (On improving hayfields in BPF). F. 515, O. 80, No 1064 (in Russian)
[RSHA] 1913. Po zhalobe krest'jan na administraciju Pushhi za chinimye raznogo roda pritesnenija (On the complaint of peasants on BPF administration for various kinds of oppression). F. 515, O. 80, No 1188 (in Russian)
[RSHA] 1913–1915A. O rashodah po Belovezhskomu zverincu v 1914 g (On expenses for game reserve in BPF in 1914). F. 515, O. 80, No 1194 (in Russian)
[RSHA] 1913–1915B. O ustrojstve muzeja pri Upravlenii Belovezhskoj pushhi (On the creation of museum by BPF Administration). F. 515, O. 70, No 300 (in Russian)
[RSHA] 1914. Rasporjazhenija po predpolagavshemusja Vysochajshemu priezdu v Belovezh (Orders regarding the tsar's arrival to Białowieża). F. 515, O. 70, No 346 (in Russian)
[RSHA] 1914–1915A. O rashodah po Belovezhskomu zverincu v 1915 g. (On expenses for game reserve in BPF in 1915). F. 515, O. 80, No 1317 (in Russian)
[RSHA] 1914–1915B. O reorganizacii ohotnich'ej administracii Belovezhskoj Pushchi (On the reorganization of BPF hunting administration). F. 515, O. 80, No 1316 (in Russian)
[RSHA] 1915A. O padezhe zverej v Belovezhskoj pushche ot zaraznyh boleznej (On deaths of animals in BPF from infectious diseases). F. 515, O. 80, No 1501 (in Russian)
[SARF] 1860. Dolmatov D. Istorija tura ili zubra i Belovezhskoj pushchi (The history of aurochs or bison and Białowieża Primeval Forest). F. 728, O. 1, No 2620 (in Russian)
[SARF] 1860–1915. Kniga rospisej pochetnyh posetitelej Belovezhskoj pushchi (The book of signatures of honorary visitors of BPF). F. 601, O. 1, No 1554 (in Russian)
[SARF] 1918–1919. Ob jevakuacii imushhestva Belovezhskogo dvorca i o ego hranenii (On the evacuation of the property of the Białowieża Palace and its storage). F. R-1056, O. 1, No 27 (in Russian)
[SA ZIN] 1906–1912. Scientific Archive of Zoological Institute, Russian Academy of Sciences. Knigi postuplenij kollekcij za 1906-1912 gg. (Collections registration books for 1906–1912). F. 1, O. 6, No 13–30 (in Russian and Latin)

Chapter 8
The End of the Imperial Epoch

Abstract The beginning of World War I did not significantly affect the daily life of Białowieża Primeval Forest (BPF) until the approaching battle-front forced the evacuation of forest and of the game management employees, along with the palace and appanage properties. The Germans entered Białowieża on August 17th, 1915, and left it on December 28th, 1918. However, the three years of the German occupation was a crucial episode in the history of BPF, with the effects felt even to this day. The rapid exploitation of the Forest's resources started with around 2.6 million cubic metres of timber harvested in the period 1915–1918. The German administration used mainly prisoners of war and local residens as forced labourers. The population of European bison and other ungulates fell dramatically in the first months of the occupation. A glimmer of hope, but with the backdrop of the overexploitation of forest resources during WWI, was Hugo Conwentz's effort to protect the Forest and European bison. The German occupation drastically changed the status of the Forest—from the Tsars' hunting reserve with large areas excluded from any exploitation to heavily logged source of timber for German economy. The latter demanded the forest resources that BPF could supply.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

8.1 Summary

The beginning of World War I did not significantly change the daily life of BPF until the approaching battle-front forced the evacuation of forest and game management employees, along with the palace and appanage properties. The Germans entered Białowieża on August 17th, 1915, and left it on December 28th, 1918. The three years of German occupation was a crucial episode in the history of BPF, with the effects felt to this day. The rapid exploitation of the forest's resources started with around 2.6 million cubic metres of timber harvested from a total area of 6500 ha of BPF in the period 1915–1918. A glimmer of hope against the backdrop of overexploitation

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_8

of forest resources during WWI, was Hugo Conwentz's effort for protection of the Forest and European bison. Conwentz visited BPF in 1916, and persuaded the army administration to exclude the heart of the forest from any exploitation (the area that later served as a basis of the Białowieża National Park). Nevertheless, the German occupation led to the decline and eventual extermination of Białowieża's European bison population and wasteful exploitation of the forest. The German occupation drastically changed the status of the forest—from the Tsars' hunting reserve with large areas excluded from any exploitation, to a heavily logged source of timber for a German economy, craving forest resources.

During the occupation period, German scientists carried out a number of scientific studies in BPF with a general goal of identifying the natural resources of the newly acquired Forest.

The nineteenth century was a period when the process of the modernization of forestry, was completed for most parts of Europe. Against this backdrop, BPF was an outstanding exception. From the fourteenth century, as a royal hunting ground it was excluded from commercial use. After seizure of control over BPF and adjacent areas by the Russian Empire, new administrators attempted to "civilize" the Forest but were unable to fully realize their plans due to the conservation requirements of European bison on the one hand, and a strong connection between the local population with traditional uses of forest resources on the other. In spite of repeated attempts, until the end of the nineteenth century, the plan to introduce rational, modern forest management in BPF was not achieved.

The international concern about European bison and BPF during WWI showed that the awareness of the species and the place of its existence rose prominently during the nineteenth century. Nevertheless, the alarming tone of different articles published during the war turned out to be accurate, as in 1919 bison's fate was sealed with the last specimen poached in BPF. But the events that followed, especially the creation of the International Society for the Protection of the European Bison, and the subsequent successful international initiative of bison restoration, proved that this awareness was not in vain.

8.2 State of the Forest During WWI and the Extermination of European Bison

The first months of World War I brought little changes to the daily life of BPF. The political tension was felt as early as mid-summer 1914, and the Tsar's visit planned for August 1914 was cancelled already in July (RSHA 1914A). In August and early autumn of the same year, some of the forest and game personnel were drafted to the army, and funds for game management were reduced. Despite that, the usual planned work continued: repair and construction of fodder sheds, roads, houses of forest personnel (RSHA 1910–1915, 1913–1915A, 1914–1915A, B). Work on the construction of a fence that was supposed to completely stop forest ungulates from entering peasants' fields and neighbouring forests slowed down, but did not stop. Similarly, work on the improvement of grasslands for haymaking and production of fodder for bison and other ungulates continued as planned, with minor adjustments because of the shortage of labour. The main manager of BPF, Golenko, informed the Appanage Administration that the harvest of 1914 from appanage fields and haymaking meadows covered more than half of the fodder planned for the winter of 1914–1915. Golenko pointed out that without an increase of fodder production, the situation with supplementary feeding of the game in February–March 1915 would be catastrophic. Eventually, the amount of fodder planned for the winter of 1914–1915 was about 10–15% lower compared with the previous winter (RSHA 1913–1915A, B, 1914–1915A, B).

The successor of Josif Newerly in the post of game manager of BPF, Ewald Bark, reported in August 1914 that there were more red deer in the Forest than Newerly had indicated. Bark suggested that Newerly deliberately underestimated numbers of red deer and had not undertaken even half of the planned shooting; and this was due to his personal fondness for the species. Bark then proposed to shoot 1,425 of 6,600 red deer, a little more than half of 1,380 fallow deer, as well as 725 wild boar out of more than 2,000 in the winter of 1914–1915, and to offer the venison for the army. After the annual count of game in February 1915, it turned out that the number of red deer exceeded 8,000 even after intensive shooting (RSHA 1913–1915).

In the spring of 1915, Professor Nikolai Kulagin resumed correspondence with the Main Directorate of Appanage on the publication of results of the scientific expedition to BPF. Both naturalists and Georgii Karcov (coordinating this publication in the appanage administration) persisted in the publishing of results according to scientific standards, in separate issues, which was opposed by the head of the Main Directorate of Appanage, Prince Kochubey, who wanted a single, lavishly illustrated volume with parallel text in French and Russian. The publication in the latter form was almost started in November 1915, but was halted due to errors discovered in the French translation (ARAS 1915–1917). The printing was then not resumed and the results of the expedition were published only in the post-war period, at the lowest cost, without illustrations or the French translation, and in separate issues; Wróblewski published his book in Poland (Kulagin 1919, 1928; Ognev 1926; Wróblewski 1927).

In May 1915, the forest authorities reported a solitary male European bison damaging crops fields, which coincided with the visit of a general of the Russian army and member of the Romanov family, Charles Michael, Duke of Mecklenburg, in BPF. The Duke requested permission to kill the bison, and as result, the last bison hunt of a member of imperial family in BPF took place on 30th May 1915 (RSHA 1914–1915A).

In March 1915, the veterinarian of BPF, Zaustsinsky, recorded *pasteurellosis* as a cause of death on a few red deer, but estimated the risk of an epizootic outbreak as low. With the war front approaching, and in prolonged drought conditions, anthrax was brought to the BPF in summer 1915. By the end of June, 135 dead wild ungulates were discovered, including four European bison. Forest guards and game wardens that were not drafted for military service were all engaged in the search for animal carcasses and their destruction. Anthrax was confirmed as the cause of death of game by Nikolai Ushinsky (1863–1934), professor of bacteriology (RSHA 1915A).

Professor Ushinsky arrived in Białowieża in July 1915 to arrange a Red Cross hospital in the buildings of the palace park (RSHA 1915B). Soon, however, the evacuation of employees, along with the palace and appanage properties began, with the latter transported to Neskuchny Palace in Moscow. Some of the personnel received new appointments in the Moscow appanage administration (RSHA 1915–1916, 1916).

The Germans entered Białowieża on August 17, 1915, and left it on December 28, 1918. The three years of German occupation was a crucial episode in the history of BPF. Its effects are felt to this day. As it was customary, German army produced bucolic postcards so that soldiers could send them to their families (Fig. 8.1). The cards present a very idyllic image of the ancient forest (with officers posing among standing and windfallen giant trees), forest villages and their inhabitants (Fig. 8.2). Also, many writings softened the picture of Białowieża under the German occupation; however the reality was completely different.

The head of the German forest administration was Georg Escherich, a hunter and a traveller, from 1913 employed at German forestry administration. In Cameroon, a German colony at the time, he organized three scientific expeditions to prepare forest exploitation of that part of Africa. As a result, he produced a project of colonial

Fig. 8.1 Photograph titled "One of the storms felled huge trees in the forest of Białowieża". First photographs from BPF were often used as postcards for German soldiers (Europeana 1914–1918A)

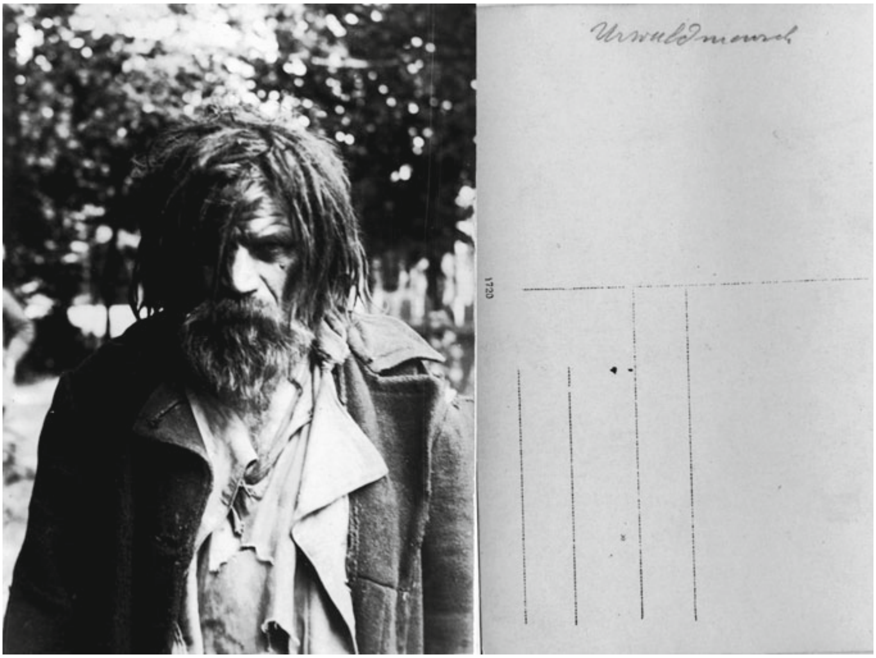

Fig. 8.2 Photograph of the forest dweller used as postcard for German soldiers. On the back, word "Urwaldmensch" [wood man] (personal collection of T. Samojlik)

forestry administration and forest exploitation in German Cameroon. After being wounded at the beginning of WWI, he was moved to BPF to plan the rapid exploitation of the Forest's resources (Schoenwald 1971). Escherich obviously succeeded, as in the period 1915–1918 around 2.6 million cubic metres of timber were harvested on a total area of 6500 ha of BPF (Więcko 1984). In his memoirs, Escherich was especially proud of this Białowieża period (Escherich 1934, 1935).

Historians draw attention to the colonial character of German management in BPF and its analogy with German colonial policy in the forests of Cameroon (Sunseri 2012). This comparison is quite accurate not only because of the Escherich's personal experience from Cameroon but also other aspects. The treatment of occupied forests and local dwellers was as primitive, there was plunder management of forest resources, exploitation of slave labour forced from local residents and prisoners of war, but this was also alongside the desire to protect large game for hunting or to undertake a research programme to assess the resources of the "new colony".

However, there was a glimmer of hope on the backdrop of pillaging and overexploitation of forest resources during WWI. German researchers observed the enormous biological diversity of BPF, its rich history, including archaeological artefacts preserved in the Forest, and ancient forest features, that hundreds of years of human presence did not erase: "*If we were now to assess the described above impact of* *human and game on the forest, we must, in my opinion, come to the conclusion that all human interventions, like excavated graves, prove to be very remote and not more than local changes to the forest*" (Rubner 1918). Even before the German army seized BPF, Hugo Conwentz (1855–1822), German botanist, director of the natural history museum in Gdańsk (originally a Polish city, in 1793 incorporated by Prussia) and then *Staatliche Stelle für Naturdenkmalpflege in Preußen* (State Organization for Natural Heritage Conservation in Prussia) asked for protection of the Forest and European bison. As an author of a "natural monument" idea, founder of institutional nature protection in Germany, and exponent of internationalization of nature protection (Hoffman 2003), he enjoyed international scientific reputation and charisma. Conwentz visited BPF in 1916, and persuaded the army administration to exclude the heart of the Forest from any exploitation and stop construction of railway designed to cross the centre of BPF. This area later served as a basis of Białowieża National Park. The event was commemorated in 2012 with a tablet devoted to Conwentz's achievements in saving BPF, and by naming one of Białowieża's grandest oaks after him (Wajrak 2013).

From the point of view of environmental history of BPF, the German occupation can be analysed in regards to three main points: (1) the decline and extermination of Białowieża's European bison population, (2) the scientific research carried out by Germans, and (3) the wasteful exploitation of the Forest and its consequences.

Jan Sztolcman, the deputy director of the natural history museum in Warsaw and a delegate of the State Council for Nature Conservation at the First International Congress of Nature Conservation, which took place in Paris on May 31st—June 2nd, 1923, initiated an international action of European bison restitution by summarizing the extinction of the species during WWI:

"*The beginning of the war did not change the situation. The local administration was able to take appropriate steps. The withdrawal of Russian troops, as evidenced by the counselor Escherich, the administrator of Białowieża during the German occupation, did not cause any significant losses. The critical period began only after his compatriots arrived in Białowieża. According to his own words, "Feldgrauen"* [the "nickname" of the German military from the colour of their uniforms] *shot several dozen bison. German authorities, however, managed to bring order. Despite that, the destruction of bison continued. On one hand, local poachers, and on the other, 800 to 1000 Russian soldiers left in the forest to harass the German army, hunted bison. Members of the Russian unit lived almost exclusively from hunting, eating meat and exchanging pelts and horns for bread, vegetables, tobacco, etc. with the local population.... In this way, the number of bison decreased in 1917 to 120 animals. In 1918, according to Mr. Escherich, thanks to the severity of the German administration, the number grew to 200. However, within few months between the withdrawal of the German troops and the establishment of the Polish authorities in Białowieża, almost all bison were killed*" (Sztolcman 1926).

At the beginning of WWI, there were 727 bison in BPF according to the estimations done for the additional winter feeding period of 1914–1915 (RSHA 1914–1915A). When the German army entered the Forest, both regular soldiers and officers began shooting animals. Due to the pressure of the scientific community, the German command banned the hunting of bison: "*As early as September 25, 1915, a ruling regarding hunting was issued by Lieutenant-General von Seckendorff, which declared: "We desire to preserve Wisent* [European bison in German] *herd as far as possible, although this is enemy territory, so as to convey to posterity a Natural Monument of peculiar value". Thus the best hopes for the future were entertained, but then came the collapse of the German power and the revolution of November, 1918. On December 16, 1918, shortly after the revolution, Major Escherich wrote to "Staatliche Stelle": "In consequence of the events of the past weeks the military forest administration can no longer exercise any control over the protection of game in the forest of Bialowies and consequently the Wisent herd of 170–180 head is seriously reduced. The imminent retreat of the German troops increases considerably the danger of extermination of the animals and thus extinction of the species is to be feared'. In fact it seems that all or nearly all the remaining wisents have been shot by inhabitants and the retiring German soldiers, among whom discipline had been undermined by the revolution*" (Ahrens 1921).

Jan Sztolcman was the key figure in saving the European bison from complete extinction and the first to declare the extirpation of bison in BPF in March 1919, when he found last bison recently killed by poachers. He initiated the international cooperation of researchers, activists and patrons that allowed European bison reintroduction (de Bont 2017), and until his death in 1928 was the most important person connected to the Polish policy of saving the species. However, both his and Theodor Ahrens' knowledge of bison during the occupation of BPF came from secondary sources. Furthermore, even if Sztolcman was aware of the extermination of bison by regular German army, for political reasons he did not want to openly blame the Germans—much of the funding for the restoration of bison was of international, including German origin (de Bont 2017). In addition, if Sztolcman blamed the Germans, he blamed undisciplined soldiers, but not officers.

It is rather symptomatic that both Russian zoologists and nature conservation activists, calling on the new Soviet government to protect nature, turned to the authority of their German colleagues. In particular, Franz Schillinger (1874–1934), in a 1922 note prepared for the State Committee for the Protection of Monuments of Nature about urgent measures to be taken to protect rare species, quoted Hugo Conwentz's words on German efforts to rescue Białowieża's bison during WWI and conserving a small bison herd in Pszczyna. He kept silent about the role of the German military in their extermination (ARAS 1922).

Therefore, texts of both these eminent scholars and the leading actors of bison rescue action require commentary. Jan Sztolcman's information about Russian guerrillas in BPF is not supported by other sources and seems very unlikely, given the fact of the strong German military presence in the Forest, as well as heavy exploitation and production infrastructure that reached almost every corner of BPF. For the same reasons, also massive poaching in the forest seems inconceivable—it is hard to imagine German army tolerating armed bands of poachers roaming the forest. Escherich himself wrote about fighting poaching with an "*iron hand*", at the same time criticizing the Russian management of the forest for breeding too many animals which destroyed the forest (Escherich 1934). Under his supervision and according to his game management plan, game was heavily reduced: 600 ungulates were shot in 1916 and 1,000 in 1918 (Escherich 1934).

Ahren's text, on the other hand, paints a picture in which bison died out in spite of the German protection of the species, and the local poachers and marauders from the German army (in the period when German troops retreating) were to blame. This is contradicted by the fact that the most drastic reduction in bison population (from 727 to 160) occurred in the first period of German occupation, in the interval of mere months before protective measures were taken in autumn of 1915 (Fig. 8.3). And even after bison hunts were officially banned, hunting permits for various dignitaries and high-ranked military officers were issued very often: among others German Emperor Wilhelm II, Paul von Hindenburg, Prince Leopold of Bavaria, commander in chief of the 9th army, Frederick Augustus II, Grand Duke of Oldenburg, Prince Friedrich Leopold of Prussia, and even the famous "Red baron", Manfred von Richthofen (Escherich 1935). Although there are no official data on how many permits were issued, Escherich's declaration that only seven bison were killed during these hunts is unreliable (Escherich 1934). Memoirs of Pierre Loevenbruck, a veterinarian, officer of the French army, and prisoner of war used by the Germans as

Fig. 8.3 Massive exploitation of forest resources in BPF during WWI (Europeana 1914–1918B)

a beater during hunts in BPF, incorporated a description of a hunt for a large bison by a German lieutenant, well after ban on bison hunts was issued (Loevenbruck 1955). The destruction of BPF's bison took place under the German occupation, despite efforts of Hugo Conwentz. When the Polish administration finally arrived at BPF, the fate of the species was already sealed, even if individual bison were still present in the Forest. Their number and the date of the last bison killing are different in various sources (Daszkiewicz 1999).

During the occupation period, German scientists carried out a number of scientific studies in BPF with a general goal of identifying the natural resources of the newly acquired forest. Firstly, an "*expedition to the ancient woods of Białowieża took place from the end of October 1915 until January 1916. The main part of scientific collection sent to Germany consists of numerous skeletons and animal specimens, especially European bison*" (Anonymous 1916; Escherich 1917; Genthe 1918; Rubner 1918; Rörig 1919). After the scientists left, BPF saw an unprecedented scale of exploitation, with research goals limited to general collection of information about virtually "everything": "*Next, from June 1916, the German army administration, besides devastation of Białowieża, carried out scientific research. The plan was to (1) study the geology and carry out meteorological observations, (2) study fauna and flora, (3) create a geological, zoological and botanical collection, (4) collect biological observations of all kinds, (5) photograph more interesting natural specimens*" (Anonymous 1918–1919). The results were published in the years 1917–1919 in the form of a series of four small volumes, printed by the publishing house Paul Parey in Berlin. The results obtained in Białowieża were also used by German scientists in publications published in the later period. These works, although of unequal value, have made a contribution to the knowledge of the Forest. BPF's flora and the main animal species, especially European bison, were studied and described throughout the nineteenth and twentieth centuries. However, vast areas of biological diversity were not investigated, and German works from this period were able to fill in some gaps. David Geyer was the first professional malacologist (mollusc specialist) who studied the malacofauna of BPF (the more important that the location of the molluscs collected by Mordvilko and Wróblewski in 1906–1908 is unknown), and his work listing 84 Mollusca from the Forest was a valuable contribution to the overall knowledge of the forest fauna (Geyer 1917A, B). Similarly important was the entomological research carried out by Karl Escherich, brother of Georg, comparing the entomofauna of the primeval and managed forest (Escherich 1917). At the same time, botanical research was less innovative and even criticized for large errors as to the occurrence of species cited in the work (Paczoski 1927). Six European bison skulls from BPF were added to the collection of Museum für Naturkunde in Berlin in 1915 and 1917 (Hilzheimer 1920).

The German occupation drastically changed the status of the Forest—from the Tsars' hunting reserve with large areas excluded from any exploitation to heavily logged source of timber for German economy, craving forest resources. For the implementation of the robbery economy, six sawmills, a wood wool factory, dry wood distillation factory with 128 retorts, and a factory of wood houses were erected in villages inside and on the border of the forest (Więcko 1984). Apart from enormous amounts of wood extracted (around 2.6 million cubic meters), one of the most important aspects of this exploitation was the construction of a narrow-gauge railway cutting through the forest and enabling transportation of material to sawmills and other factories. Part of colonial exploitation policy was also the use of forced labour of local residents and prisoners of war for all types of work. The narrow-gauge railway was intensively used to transport timber in the 1920s–1930s by Polish, and in 1939–1941 by Soviet wood industry (Bajko 2001; Więcko 1984).

German troops left Białowieża in December 1918, destroying buildings, disassembling part of industrial facilities, plundering and devastating the Tsars' palace. Fortunately, the Forest, despite robbery exploitation, was not entirely destroyed: the heart of BPF was excluded from exploitation and was in 1921 proclaimed as "reserve forestry", later, in 1932, transformed into Białowieża National Park. But even in the exploited part, "*cutting them, which should be regarded as simply plundering, destroying the current forest stand, usually did not interfere with the natural regeneration of the forest* (…) *and thus enabled relatively quick regeneration of the forest*" (Paczoski 1930), especially in the light of the fact that selective cutting was employed. How this natural regeneration and the irrepressibility of the BPF was later put to the test using technological means left here by retreating German army, and how the rest of the twentieth-century history of the Forest unfolded, is a separate story, not less complicated than the one presented in this book.

Seeing and assessing the destruction of the Forest during WWI, Józef Paczoski summed up a century of Russian rule in the forest: "*When I visited BPF for the first time in 1893, there was no railway, no narrow-gauge railway, nor all that is disfiguring the forest today. At that time, there was no excess of animals, which in the times just before the World War, almost completely ravaged the entire forest bottom, and finally began to die out due to overabundance. In the absence of communication routes that could encourage forest exploitation on a wider scale (the nearest the railway station—Bielsk was about 50 verst from the forest), few floatable forest rivers could carry only a very small amount of wood compared to the size of this enormous forest. As a result, felling embraced only as much forest, as it was possible to float in rivers. Most often oak and pine were sold abroad as the first-class building material. It is absolutely understandable that in such transportation conditions, not only the entirety, but the pristineness of this complex were largely guaranteed*" (Paczoski 1930).

8.3 Long-Lasting Impact of BPF Management, Protection and Exploitation in the Long Nineteenth Century

The nineteenth century was a period when the process of modernization of forestry, which began at least a century earlier with the introduction of a new rational, scientific forest management, was introduced in most parts of Europe. Against this backdrop, BPF was an outstanding exception. From the fourteenth century, it was excluded from commercial use as a royal hunting ground and only briefly, at the end of the eighteenth century, it entered the orbit of economic reforms aimed at making it a profitable enterprise. After seizing control over BPF and adjacent areas by the Russian Empire, new administrators of these territories faced the difficult task of reconciling several factors influencing the management of the forest, usually completely contradicting each other. On the one hand, there was a strong belief that the forest needs to be "civilized", expressed by both officials who managed state assets and forestry specialists who visited BPF, for example the head forester of the Kingdom of Poland, Juliusz Brincken (Brincken 1826). On the other hand, the local population was connected with the forest and various benefits derived from it through rights granted by Polish kings and Lithuanian grand dukes, and their rights needed to be respected (Hedemann 1939; Samojlik 2010). The third factor that the forest administration had to pay close attention to was the need to preserve European bison, a curiosity called the "wild cow", for which BPF has become the last lowland refugium in Europe. When assessing the overall impact of management on the state of BPF during and after the end of the long nineteenth century, it should be taken into account that these factors (and a whole lot of other minor ones, such as the shortage of trained forest personnel in the first decades of Russian rule over the Forest, conflicting interests of different ministries crossing in the forest, two national uprisings and an episode of the Napoleonic wars that took place in the forest and its outskirts) created a Gordian Knot, most probably unsolvable at all.

In spite of repeated attempts, until the end of the nineteenth century, the plan to introduce rational, modern forest management in BPF was not achieved. This can be explained by a complex of reasons, but probably the most important of them should be considered regular and frequent changes of policy towards BPF: shorter periods of more intensive forest exploitation were intertwined with longer periods of total prohibition or at least a significant restrictions in logging for the sake of European bison. This peculiarity was noted by forest specialists writing about the history of forest management in BPF (Genko 1903; Kründer 1909).

The first period (1796–1837), was dominated by administrative turmoil. Firstly, there were frequent changes of supervision of the forest between the Expedition of the State Economy of the Senate, Forestry Department of the Naval Ministry, Forestry Department of the Ministry of Finance and eventually Department of State Domains of the Ministry of Finance. Secondly, there were attempts at assessing and regulating access rights for traditional utilization of the forest of local communities. Lastly, national uprising in 1830–1831 resulted in a significant part of the local forest personnel being killed, expelled from the Forest, or removed from their posts.

After the period of more intensive logging in the 1810s, there was a total ban put on any timber extraction in the 1820s, accompanied with prohibition of hunting, even for predators. Main long-lasting impacts of this period incorporated the removal of one of thirteen forest districts of BPF in 1795, and incorporation of Świsłocz Forest, adjacent private forest owned by count Tadeusz Tyszkiewicz, which was confiscated as a punishment for his participation in the November Uprising (in 1831).

In the period 1838–1860, the new Ministry of State Domains was determined to introduce "proper" forest management yet the Naval Ministry was given priority in selecting material suitable for shipbuilding from BPF. However, acquiring the shipbuilding wood in BPF proved to be too expensive even for the Naval Ministry. In the period 1843–1848, the first management plan was prepared and the entire BPF was divided into rectangular compartments measuring 1 × 2 *verst* (~218 ha). But the actual planned exploitation with clear-cuts did not start, mainly due to a large number of "damaged" trees (being actually culturally-modified trees—remnants of traditional, centuries-old forest utilization (see Sect. 5.2)), which were designated to be first removed from the forest before planned cuts could take place. Extraction of those trees exposed plethora of problems, from fraudulent entrepreneurs and forestry personnel, through lack of technical means to transport large timber, to unfloatable forest rivers. Naval Ministry complicated the situation even further, opposing the sale of timber from BPF abroad (as this "strategic resource" could be used by potential adversaries), but not providing any transportation means to deliver timber from BPF to Russian shipyards.

The first imperial hunt of 1860 brought a substantial change of the overall policy and attitude towards BPF which was from that point on gradually turned into hunting reserve of the imperial family. The following period, 1861–1888 saw three new management plans produced (in 1861–1862, 1870–1871 and in 1877) but short periods of intensified timber production were again mixed with management focused on promotion of game species with European bison at the first place and eradication of predators. In the place where 1860 hunt occurred, a menagerie—enclosure with the main game species—was created. The head of one of four forest districts of BPF served at the same time as the manager of the menagerie. Forest compartments were divided into squares measuring 1 × 1 *verst*; a system that is still being used in BPF. In 1888, the Forest was transferred from the Ministry of State Domains to the Ministry of Emperor's Court, becoming an actual private hunting ground of the imperial family. In the last period, 1889–1915, the administration of BPF tried to simultaneously combine "modern" forestry and "aesthetic" aspects of game reserve of a primeval beast, European bison, with Tsar Nikolai II personally wishing for old and dead trees not to be removed to conserve the pristine look of the forest (1897). For this reason, for the most part of the last period logging was carried out only in the Świsłocz Forest whereas cutting of living trees was prohibited in the main part of BPF for almost half of this period. In the last forest management plan of 1909–1911, the Forest was divided into the "preserved" part (in which only dead or dying trees were to be felled) and the "managed" part, in which clear-cuts were to be introduced after removing all old trees. Game management dominated over forest management, and erection of imperial palace, opening of a railway and preparation of hunts were given priority over topics of forest exploitation.

Paradoxically, one of the grandest achievements and the subsequent heritage of the imperial period in the history of BPF consists of something that was never achieved. This was due to different circumstances, and the Forest was never turned into rationally managed plantation deprived of its pristine features. Other achievements incorporated the diligent care of European bison and other ungulates, the transfer of bison to game reserves and zoos, and scientific study of the biology of bison. All of these came together to allow the reinstatement of the species and to eventually facilitate the return of the bison to the wild after WWII.

The imperial palace in Białowieża, captured by German army, was one of the most frequently photographed places in the entire Forest (Fig. 8.4). Already devoid of the most valuable interior equipment that had been transported back to Russia in 1915, the palace deteriorated during the German retreat in December 1918 (Bajko 2001). Nevertheless, after Polish administration took over Białowieża in 1919, the palace was still in good condition and became one of the main tourist attractions of BPF, also serving various new purposes such as hosting the Polish president during official hunts, serving as the first location of the Polish natural history museum of BPF, a forestry school, and the Catholic chapel (Chestnykh and Kettering 2010; Kordiukiewicz 2006). Furthermore, the interior of the building was well preserved: "*interior of the palace is made almost exclusively of wood, walls covered with boards, wooden ceilings, sculptures and rich decorations also made of wood. Walls and ceilings are abundantly decked with painted plafonds and frescos. Also stoves and chimneys made with variety of colourful tiles of different shapes serve as a grand decoration of the palace*" (Szymański 1925).

The palace was badly damaged during WWII, and its remains were finally destroyed in 1958. Currently, the place is occupied by the new building of the administration of Białowieża National Park, including the new natural history museum of

Fig. 8.4 Photograph of dead bison from the beginning of German occupation (dated 1916) (personal collection of T. Samojlik)

the national park. Some of the original properties and furniture of the palace were distributed between different organizations in Moscow between 1918 and 1920, e.g. part of the imperial hunting trophies from the palace (European bison, red deer antlers) fell to the Darwin State Museum and a few stuffed animals (including couple of European bison) ended up at the Department of Applied Zoology of Politechnical Museum (Miloserdov 2016; SARF 1918–1919).

8.4 European Bison from BPF in European Zoos and Nature Parks

The earliest mention of live bison capturing in BPF in the long nineteenth century dates back to 1821, when bison were about to be caught for the Skierniewice game reserve (within the Kingdom of Poland, and a puppet state of the Russian Empire). There is no information if bison were actually delivered to the Tsar's menagerie, as only letters by forester Eugeniusz de Ronko criticizing the method of catching bison proposed to him were preserved (NHARBG 1821–1824).

In 1846, the famous British geologist Roderick Murchison asked Nicholas I for a pair of live European bison for the London Zoo (see Sect. 7.4). Perhaps the negative experience from the 1820s caused the rather serious approach of forest administration to this case. It was decided that bison would be caught and then kept in captivity for a year to make them acquainted with humans and to ensure they would survive. The plan was successfully implemented with the direct participation of Dolmatov (see Box 5.1). Apart from a pair of bison for London, another pair was sent to St. Petersburg, and another was left in BPF for "domestication experiments" (see Sect. 5.3). A pair of bison sent to St. Petersburg was also supposed to undergo domestication experiments though these were limited to keeping animals in one of the farms of the imperial court. Sustaining bison there brought a lot of inconvenience to personnel (RSHA 1848, 1852).

The imperial hunt in BPF in 1860 aroused interest in obtaining live bison: in 1861, two pairs of young bison were brought to the Gatchina imperial game reserve near St. Petersburg (RSHA 1861). However, the first attempts to keep bison in Gatchina were unsuccessful: although young were born, they quickly died, and the adult bison all died by 1866, too. A year later, in 1867, another four bison were brought to Gatchina, but they all fell within few months. In 1868–1869, two couples were brought there again (RSHA 1866–1870). Improvements in the maintenance of animals allowed to keep eight live bison in Gatchina by 1881 (RSHA 1881). After that, the Gatchina bison successfully reproduced (up to 35 specimens in the 1910 s) and were donated to game reserves and parks of aristocrats and zoos (RSHA 1912–1914, 1914B, 1917A).

In 1862, the Amsterdam Zoo received two bison from BPF and in 1864 the Moscow Zoo—three bison (Karcov 1903; NHARBG 1864). In 1873, bison was given to the Sultan of Constantinople (Karcov 1903). In the 1860s–1870s, European bison were able to live several years in zoos and sometimes even produce offspring, though most calves died shortly after birth (Usov 1865). In 1865, 4 bison (3 females and one male) were presented to Prince von Pless Jan Henryk XI Hochberg (Hans Heinrich XI) for his hunting park in Pszczyna. The Pless herd successfully multiplied and in 1893 the prince asked for 5 more females to refresh the blood and prevent degeneration. His wish was granted, and this entire introduction was officially treated as "*return of European bison to Germany*" after the last bison was killed in Prussia in 1755 (Langkavel 1895).

Like other hunting parks of the European aristocracy, the Pszczyna reserve represented a kind of "acclimatization park" for large game, especially deer. For decades, European red deer were subject to crossing attempts with imported deer species, including wapiti deer from Canada and Siberia (Lantz 1910). Unlike the imperial hunting reserve of Gatchina, which has very few documents about animals, the park of Prince Pless had accurate records of bison calves born there (a total of about 200). Although many of young bison died soon after, the Pszczyna herd became the largest population of European bison besides BPF and the North Caucasus (Böhme 2017).

Pszczyna bison were fed in the winter with 10 kg of good quality hay and 2 kg of oats per day for each individual. Attempts to feed the peels and barley were considered to be unsuccessful because the bison struggled to digest this food. For comparison, in BPF's enclosure in the 1910s bison were given 6.3 kg of good quality hay, 5.5 kg of clover and 3.5 kg of root crops per day, and in the forest itself—4 kg of good quality hay, 3.5 kg of clover and 2.25 kg of root crops per day (RSHA 1912–1914). The Pszczyna herd was big enough to start hunting. Emperor Wilhelm II often hunted in Pszczyna where he reportedly killed two bison with one shot (Hanswijck de Jonge 2007), which is rather doubtful. Pszczyna bison hunting was described by Red Baron, the famous German WWI pilot, Manfred von Richthofen, who also hunted in BPF during the German occupation (Richthofen 2005).

Animals captured alive in Pszczyna were also sold, e.g. to Berlin zoo in 1888 (Böhme 2017) or to the United States. A pair of bison was purchased for New York Zoological Park in 1903, and "*after a long and severe journey, the animals arrived on the April 15, very thin and weak, and much bruised on the hindquarters. They were immediately liberated in a very comfortable corral that had been specially prepared for them (…). Ever since their arrival, their condition has steadily improved, and by autumn they should be in fine condition. (…) So far as we know, the specimens now in the New York Zoological Park are the first to arrive in America; and the acquisition of the good examples of the species so rare and difficult to represent may fairly be accounted a stroke of good fortune*" (Anonymous 1903).

Pszczyna's population proved to be very important also from scientific point of view—it visibly denied the theory that bison would not be able to breed outside BPF, and also served as the subject of research (independent from Białowieża), e.g. on the behaviour of bison (Langkavel 1895). However, bison were believed to be especially difficult to sustain in captivity and breed: "*menagerie Jardin des Plantes owned bison many times but they all quickly died*" (Huet 1891).

Similarly as with the BFP population, Pszczyna's one also suffered a disastrous fate during and after WWI. In 1918, the herd consisted of 60 individuals, but it was depleted by poachers down to just three. Later, the bull from Plisch became the founding father of the reinstated lowland European bison population.

In the appanage period (1888–1915), the imperial family agreed to send European bison from Białowieża quite often, although some requests were rejected. The Duke of Bedford received three bison from BPF (two females and a male) in 1899 for his famous park at Woburn Abbey, and the Austrian emperor received bison from Białowieża several times for his menagerie in Shönbrunn, in particular—in 1903 and 1907. Friedrich von Falz-Fein who had an acclimatization park on his estate of Askania-Nova, (nowadays in Ukraine), was known for his conservation efforts regarding Przewalski's horse. He received European bison twice: in 1901 and 1910 (Karcov 1903, RSHA 1907–1908, 1909–1913). In Askania-Nova it was observed for the first time that European bison did well in the steppe and tolerate the diet of steppe grasses, and steppe hay in winter. It was also observed that they cross well with American bison there—this process was continued during the Soviet era producing a herd of hybrids. Nowadays they live in the North Caucasus and are considered by some breeders to be a separate subspecies—*Bison bonasus montanus* (which was refuted by the scientific community, see Pucek 2002).

At least two more representatives of the aristocracy declared their wish to breed bison in their game parks. Count Joseph Potocki arranged a park in Pilawin (currently in Ukraine) in his estate in the Volhyn Province, for which he received two female and one male European bison from BPF in 1905 (RSHA 1903–1907), and then a pair of bison from Gatchina. The Pilawin herd successfully reproduced for several years and Potocki planned to hybridize European and American bison. However, the old male European bison became aggressive and killed all other males, including this own offspring and the male American bison. Potocki then asked for another male from BPF and received it in 1913 (RSHA 1912–1914).

In 1914, Prince Vyazemsky asked for bison for a game park in his mother's estate near St. Petersburg. The Main Appanage Administration supported his requests arguing that experiments on "acclimatizing and reproducing bison in the conditions of the northern part of Russia" would bring "scientific merit". Nicholas II allowed to give him three bison from Gatchina (RSHA 1914B).

In early 1914, four bison (three females and a male) were brought to the Crimean hunting grounds of the Tsar. As in all cases described above, European bison were not the only representatives of rare species, the breeding of which was planned in the park. By 1917, the number of bison increased to eight (RSHA 1913–1914, 1917A, B). Some animals (but not European bison) survived the time of WWI and in 1923 a Crimean Nature Reserve was organized in this territory. In 1940, bison hybrids from Askania-Nova were brought to the Crimea, but this attempt was unsuccessful due to the further outbreak of war.

Not all requests for bison received a positive response from Nicholas II. Carl Hagenbeck (known wild animals merchant and supplier of animals for the Romanovs), many times asked for European bison, offering for them any money and animals in exchange. His requests were crowned with success only once. In 1899, an employee of Hagenbeck brought a batch of deer to Białowieża exactly at the time when an aggressive female bison was to be shot at the menagerie. Hagenbeck's subordinate pleaded the exchange of this animal for four deer from China and succeeded (RSHA 1898–1908). Apart from this one case, Hagenbeck received at least one bison from the Caucasus (RSHA 1912–1914; Zablockij 1939) and others, probably from Pszczyna.

The Volhyn hunting society, which created an acclimatization park of more than 10 thousand ha for breeding wood grouse, wild boar, moose and deer, was able to get deer from BPF, but not bison (RSHA 1904–1907). The owners of the St. Petersburg Zoo asked for a European bison several times but were able to get one only from Gatchina in the early 1900s. In 1914, the female from the BPF was obtained to complement the pair with the previously obtained male (RSHA 1912–1914).

In the summer of 1915, BPF was captured by German troops. In the first month, a significant part of the bison population was killed, and only in September the German administration began to take late measures to protect bison. It is difficult to say exactly how many bison were sent to zoos and hunting parks located in the territories controlled by Germany in those years. Majority of bison kept in game parks and zoos of Germany and the Russian Empire survived WWI, but were poached during the revolutions and the Civil War. In 1917, the Academy of Sciences in St. Petersburg unsuccessfully attempted to preserve European bison in the estate of princess Vyazemskaya (RSHA 1917C). The most of Gatchina bison were probably exterminated in the winter of 1917–1918 (RSHA 1917B). The same happened in the German territories (Bont 2017; Zablockij 1939). In Soviet Russia European bison and their hybrids were preserved in Askania Nova, and few bison survived in the zoos of large cities (Zablockij 1939).

8.5 The International Concern about European Bison and BPF during WWI

In the issue dated September 27th, 1915, the Spanish magazine "Alrededor del Mundo" published an anonymous article entitled "European bison and war: how the European conflict threatens a rare animal". The article is one of the few examples of interest in the fate of BPF and bison at the other end of Europe, nevertheless acting as an evidence that part of the Western European media kept track of the situation in BPF during the First World War under the German occupation: "*When the war reaches a terrible size, as now, it not only destroys works of art or cathedrals, but also threatens nature and can lead to the extinction of entire species. Probably this applies to one of the most interesting species of European fauna, bison. It survived only on the Russian Lithuania, not counting individual specimens living in private parks and zoological gardens, such as the Duke of Bedford in England or Count Potocki in Volhynia* [the western part of Russian Empire, nowadays Ukraine]. *In the last weeks, Lithuania is a place of fierce Russian-German fights that will lead to the most extreme situations. It is said that also Białowieża Forest, where the last wild bison live, is currently the theatre of war* [in fact active battles bypassed the area of BPF]" (Anonymous 1915A).

The article was published shortly before Hugo Conwentz's intervention that resulted in excluding a part of the Forest from plundering exploitation in an attempt to protect what was left of bison herd there, but perfectly captured the mood of the moment, portending the eventual doom of the species: "*with the highest probability* *it can be said that the European war will lead to the destruction of European bison, today a species as rare as related American bison, known for colourful stories from the wild West*" (Anonymous 1915A).

European bison became the subject of interest also in newspapers published in different countries, e.g. Russia or France (Anonymous 1915B). The "Bulletin of the Kharkov Society of Nature Admirers", published by naturalists but focused on the general public, had a special section called War and Nature, which reported on threats that nature experienced during the war. Two notes about bison appeared in one of the issues. The first was a reprint from the "Kharkov Provincial Gazette", which told a fantastic story about bison fending off "one German detachment": "*with blood-shot eyes, dishevelled hair, bison rushed swiftly at the bewildered Germans. German bayonets broke under the onslaught of the broad foreheads of bison, and indiscriminate shots aroused the frantic rage of the forest monsters even more, jumping and trampling on the Germans, they tore and tossed everything until it fell into pieces*" (Anonymous 1915C).

The second note was a reprint from "Yuzhny Kray", a newspaper in Kharkov. It was a report on European bison sold to the Stockholm zoo by Hagenbeck with the conclusion that the German administration has captured many bison from BPF and is now selling them around Europe (Anonymous 1915C).

The first of the above stories was used in the context of war propaganda also in France: on 18th November 1915, a French newspaper reported: "*Exceptionally unexpected allies fought for us in* (…) *the famous Białowieża Forest, the only forest that has free-living bison, a species almost extinct and currently counting only 653 animals* (…). *It is said that one of troops stood eye to eye with a herd of bison. Germans, who have never seen bison, stopped surprised and terrified. Bison seemed to be equally surprised. But a shot was fired and the herd burst into anger, throwing itself at soldiers. A terrible fight ensued, bayonets spattered against the skulls of angry animals, and many Germans were crushed under bison's hooves*" (Anonymous 1915B).

The revolutionary events in Central and Eastern Europe (including Russia) had put all bison herds in game parks and zoos at the brink of survival. Russian zoologists and environmental activists, just like their Western associates, tried to draw the attention of the new authorities to the need to protect the remaining bison, mostly unsuccessfully. Such attempts were especially visible when the Grodno Province and North Caucasus were under the control of the new Russian authorities (or there were hopes that they would soon be under their control). At the end of December 1918, professor Nikolay Kulagin wrote about European bison and BPF to the People's Commissariat of Education, demanding to take "the most energetic measures" to protect this "outstanding monument of nature". Next month, this issue was discussed at a meeting of the Central Forest Department of the People's Commissariat of Agriculture and led to the creation of a commission devoted to ascertaining the state of BPF (Khrustalev 1980; ARAS 1922).

Before the extinction of European bison was officially confirmed, an anonymous note (signed "G.G.") appeared in the bulletin of the French Geographical Society, expressing similar concern about the fate of the species during the war. The note presents a short and simplified history of the nineteenth-century protection of bison (including an erroneous explanation of bison population decline caused by the degeneration of the species), leading up to the outbreak of war: "*Unfortunately, war treated bison with extreme cruelty. Areas, where these animals live, passed between sides of the conflict many times. We have virtually no clue as for the fate of bison in Caucasus. Regrettably, all evidence indicates that the three herds living in Poland and Lithuania were destroyed, forever*" (Anonymous 1918).

These are just few evidences that international audiences were interested in the fate of European bison. The concerns raised in anonymous notes turned out to be accurate, as in 1919 bison's fate was sealed with the last specimen poached in BPF. But the events that followed, especially the creation of the International Society for the Protection of the European Bison in 1923 and successful international initiative of bison restoration (Krasińska and Krasiński 2013A, B; de Bont 2017) proved that awareness about the bison raised in the nineteenth and early twentieth centuries did not go in vain.

Box 8.1 BPF in Polish children's literature

At the turn of the nineteenth and twentieth centuries, a new phenomenon appeared in the Polish literature (similarly as in many other countries): books for young readers about nature and forests, combining interesting characters and adventure stories with popularization of science. Undoubtedly some of them were inspired by books published by Jules Verne and writers influenced

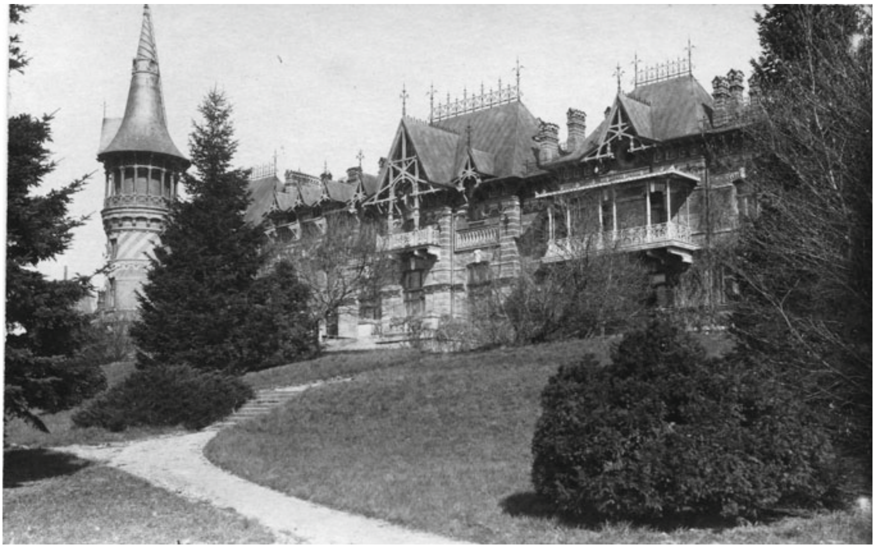

Fig. 8.5 Photograph of imperial palace taken by German soldiers, dated 1916 (personal collection of T. Samojlik)

Fig. 8.6 A postcard entitled "From Białowieża Primeval Forest. An excursion in the forest", dated 1908, published by Polish Tourist Society (personal collection of T. Samojlik)

by him, very popular among Polish readers in that period. Such literature grew on the basis of positivism, which paid special attention to natural sciences, education and popularization of knowledge. The best known among these popular natural science novels was "Doktor Muchołapski" ("Doctor Flycatcher") written by an archaeologist, historian and naturalist, Erazm Majewski (1858–1922). Similarly popular and also unrelated to BPF were books "Wilk, psy i ludzie. W Puszczy" ("Wolf, dogs and people. In the Forest") by Adolf Dygasiński (1839–1902) and "Synowie Puszczy" ("Sons of the Forest") by Władysław Umiński (1865–1954).

BPF first appeared in literature for young readers at the turn of the nineteenth and twentieth centuries in "Z opowiadania młodego chłopca. W puszczy" ("From the story of a young boy. In the Forest") by Zuzanna Morawska (1840–1920), published in Warsaw in 1898. The topic of this and similar novels usually orbits around children's holiday excursion to the forest of Białowieża (Fig. 8.5). Characters in the book have many adventures but there is one common experience: meeting with European bison. These kind of topics became very popular in connection with the development of tourism, especially in the beginning of the twentieth century, when BPF became accessible by train, almost to everybody. This was humorously summarized in the novel "Z Puszczy Białowieskiej" ("From Białowieża Primeval Forest") by Bohdan Dyakowski (1864–1940), who described the meeting of his characters, young boys, with the forest guard in Białowieża: "*Yes, yes, now these excursions of young people to the forest became very fashionable. Maybe it's good, but for us it is only trouble" they need to be followed. There is enough bandits and revolutionaries among these young masters, though seemingly look decently*" (Dyakowski 1908).

Dyakowski's novel, published in Warsaw in 1908, is one of the most interesting examples of children's literature connected with Białowieża from that period. The author, lecturer at the Pedagogical Studies of the Jagiellonian University, was a respected educator and populariser of natural sciences. Although "From Białowieża Primeval Forest" tells a story of fictional excursion of three 15-year old boys and their 20-year old guide to the Forest, it is in fact a true compendium of knowledge about BPF's natural values and history. The book contains a description of train travel, history of the Forest (from royal times to period of imperial forest), history of the extinction of the beavers, description of ancient graves and archaeological works, of largest trees and types of forest communities, information on food preferences of European bison, all situated on the background of actual places in Białowieża village and the forest itself. Dyakowski quotes Mickiewicz, Brincken, Jarocki, Błoński, Drymmer and Gloger, giving the reader also an overview of the nineteenth-century literature on BPF and supplementing the knowledge from actual excursions to the Forest (Fig. 8.6).

The motif of nature conservation lingers throughout the book, sometimes expressed in young heroes' thoughts and dialogue: "*European bison are very interesting example of the old European forests fauna and deserve protection not for the purpose of occasional hunting but because we should not let this interesting monument of nature go extinct. We organize various museums, zoological cabinets and others in cities, but no museum or even the most beautiful cabinet can replace such a natural menagerie, such a forest with animals inhabiting it almost freely. Learning the values of such natural museums, people try to protect and save from destruction various places that are the remains of an ancient nature*" (Dyakowski 1908).

In a novel "Duch puszczy podlaskiej" ("The ghost of Podlasie forest") published in 1924, Mikołaj Mazanowski (1861–1944) showed BPF as a promised land of a family who inherited an abandoned settlement inside the forest. The family arrived to BPF to find it rich, pristine and breathtaking, but also causing a lot of problems, usually solved by young protagonist of the book in a fairy-tale manner (Mazanowski 1924). Here, the topic of nature conservation played only a secondary role: the most important message was to live the life in concordance with nature, but also with other people. BPF was personified in a titular character of a ghost that possesses magical power to turn people into animals, and bestows this power upon good people living in the forest.

More realistic approach was taken by Antonina Żabińska (1908–1971), author of "Jak białowieskie rysice zostały warszawiankami" ("How lynxes from Białowieża became residents of Warsaw"), published in 1936. The author, wife of the director of Warsaw ZOO, described the life of a pregnant lynx female in BPF, showing the forest as a safe haven, full of hideouts and food. Idyllic life was then ruthlessly interrupted by a man, catching and taking away young lynx—by showing the emotional connection between mother and lynx kittens, Żabińska was able to transmit a strong conservation message.

BPF was usually idealized and shown as a place of meeting with true nature in the literature for children and young readers in the beginning of the twentieth century (Nosek 2016). Experience of being in the forest and meeting its animal dwellers served the purpose of shaping an attitude of respect and understanding for nature's needs—in this respect the century-old children's literature is very close to the contemporary one.

Box 8.2 Absent and mythical animals from BPF

Throughout the nineteenth century, there were several cases of species reported to occur in BPF that were in fact absent from the fauna of the Forest. Such errors were sometimes repeated and persisted in subsequent publications longer than a century. The basic mistakes resulted from incorrect taxonomic identification, most often based on assigning species or subspecies status to different colouristic or morphologic forms of the same species. The bear was one of such problematic species. Although there was only one species of brown bear *Ursus arctos* in BPF and the entire Europe, it was often described in division to three variations: "*almost black bears*", "*medium tawny bears*", "*small with silvery fur*" (Brincken 1826). The unification of all brown bears into one polymorphic species was finally done by Alexander Middendorf (Middendorff 1850–1851), but in the cases of other species confusion persisted.

Sometimes the mistake of listing various species instead of one existing in reality was associated with the use of different names for the description of the same animal. Piotr Szretter's description of three different snakes ("*forest, water and house snakes*") pertained to the same species: grass snake *Natrix natrix*. Another kind of error originated from the uncertainty between old and modern species names, e.g. the grouse, currently meaning *Lagopus lagopus was* previously used for *Tetrax tetrax* (Tomiałojć 2003).

The largest number of errors, however, was of biogeographic nature, i.e. species known from other areas of Poland and Lithuania (sometimes from accidental occurrences or even imports) were ascribed to BPF. Some of them were probably known from fur trade, like sable, others were kept as house pets for example a wolverine at the Lithuanian court (Gilibert 1802). According to Tomiałojć (2003), the list of species ascribed to BPF but historically absent from the Forest incorporated wolverine *Gulo gulo*, sable *Martes zibelina*, wildcat *Felix sylvestris* and the Old World flying squirrel *Pteromys volans*. This list should be supplemented by the steppe polecat *Mustela eversmanii* quoted by Brincken (1826), a mistake that probably originated from steppe polecat pelts sold at Lithuanian fairs.

The majority of such errors came from the first half of the nineteenth century but there were also much more recent misleading cases. In 1908–1912, oberjeger Bark sent several skulls from BPF, describing them as arctic fox *Alopex lagopus*, species never historically connected with the Forest (SPB ARAS 1906–1912). Most probably pelts were taken from animals kept in captivity, either by the imperial palace or in the menagerie.

Another category incorporates animals that probably were present in BPF but went extinct at an early but unknown period. This was very likely the case for Eurasian wild horse, tarpan (*Equus cabalus ferus*), present in Brincken's description of BPF: "*Less than a century ago the tarpan ("wild horse") used to live in the forest. Even 40 years ago, it was seen, though rarely, in the northern parts of Lithuania* (…). *The extermination of this species in Lithuania occurred due to the actions of people who caught young and hunted for old individuals. The last horses caught were transferred to the property of Count Zamoyski near Zamość, where they were kept for quite a long time along with other animals. Since there was no use of them, about 20 years ago* [i.e., around 1800] *they were all caught and distributed among peasants. Apparently, today you can recognize this breed among peasant ponies*" (Brincken 1826). No other sources confirm tarpan presence in BPF, but there are relatively many reports on the presence of this species in historical times in in the Lithuanian forests and in Eastern Poland (Antonius 1933). Information about the wild horses are also present in Napoleonic intelligence reports from the beginning of the nineteenth century, but it is well possible that they were simply a part of the collective memory of local population, repeated already after the extinction of the species. Nevertheless, Brincken visited BPF when the memory of tarpan was still fresh, therefore his information should be considered at least partially correct and not neglected as unfounded (Vuure 2015). This story became a basis of one of the most interesting breeding experiments of the twentieth century: in the 1920s, Polish biologist Tadeusz Vetulani became interested in primitive ponies from the vicinity of Biłgoraj which he considered to be descendants of those last tarpans released by Zamoyski. Vetulani assumed that these horses retained tarpan's traits and attempted to reinstate the wild horse by cross breeding individuals of a murky colour, with hair turning white in winter and with short standing mane (Vetulani 1928). In the interwar period, a reserve of such horses was created in BPF—they were captured by the German army during WWII and transported to Germany. After the war, Tadeusz Vetulani and his pupils continued their work, eventually creating the breed of Konik (Polish primitive horse) (Daszkiewicz 2001).

Dolmatov (SARF 1860) indicates the presence of European mink in BPF. The species was also present in the early editions of the lists of predators for which the forest administration paid a reward (RSHA 1889, 1891). In later versions, the mink disappeared from the list (RSHA 1896). The administration reduced the list removing valuable fur-bearing animals (since the pelt itself was considered a reward for the hunter), but perhaps also the animals for which rewards were not claimed. European mink was included in the list of stuffed animals ordered for the Białowieża natural history museum, but the species was not mentioned in the catalogue of the finished exhibition (RSHA 1913–1915B).

The catalogue of the museum itself could be considered a first list of bird species present in BPF in the beginning of the twentieth century, although it was compiled by a taxidermist and has gaps and inaccuracies. This is especially the case for other animal groups, e.g. in bats only "flying mouse" and "flying rat" are mentioned, and in insectivores only a shrew, hedgehog and mole are listed (RSHA 1913–1915B).

Box 8.3 Mushrooms of BPF

Collecting mushrooms was one of the most long-lasting traditional uses present in BPF, which is also today considered a fundamental prerogative of local communities. Mushrooms have always played an important role in the cuisine of the inhabitants of BPF (as well as in most of areas of Poland, Lithuania, Belarus and Russia), to the point of local cuisine being often associated with a special, delicious "opieńkowa" (honey fungus soup). It is prepared with honey fungus, potatoes, onion, meat and cream.

In the eighteenth century, even the poorest were obliged to pay a "tax" for collecting mushrooms in BPF (Hedemann 1939). Local peasants were quite knowledgeable when it came to distinguishing many species of mushrooms and evaluating their usefulness in kitchen. Piotr Szretter wrote that there are so many species of mushrooms recognized in the Forest that "it would be hard to describe types and names of those", and that they are edible "fresh, salted, dried or sour". And those that were not edible, also found their use in the local household: flybane was used to kill flies, and common stinkhorn *Phallus impudicus* was collected and sold as medicine (Szretter 1893). The latter use was a part of the wider tradition of the former Polish-Lithuanian Commonwealth (Pranskuniene et al. 2018).

Dimitri Dolmatov described the common and widespread use of one other mushroom—"plastinochnik orukh" (most probably *Sparassis crispa*, called cauliflower fungus), which had the appearance of a "*huge sponge, sometimes weighing up to a pood* [around 16 kg]". As Dolmatov described it: "*It is* *characteristic only for coniferous forests and is usually placed near the very root of the trunks of large-sized mature pines* (...). *It has a strong resinous smell but boiled in oil orukh has a quite pleasant taste and is very nutritious. In the autumn, when the peasants gather in groups of 12–20 people to stalk wolves or to assist in bison hunts, each of them puts a piece of butter in a pot with finely shredded orukh. When boiled, the pot is quickly filled with a mushroom decoction, which together with butter and salt makes a quite tolerable and nutritious food for indiscriminate local residents*" (SARF 1860).

In the nineteenth century, Białowieża's fungi became also a subject of interest for naturalists, making BPF an important research area for mycology for many years. The first attempt at cataloguing fungi species in the Forest was undertaken by Franciszek Błoński in cooperation with Józef Rostafiński. Błoński wrote about the mycological wealth of the Forest: "*Today it is difficult to even estimate the total number of species: most likely it ranges from 1200 to 1500 or from 1500 to 2000; which figure to accept as the most credible, I dare not to judge today*" (Błoński and Drymmer 1889). Modern mycological studies (Kujawa et al. 2018) show that there were almost 900 species of fungi collected for the purposes of exhibitions in Białowieża, and the total number of fungi species in BPF is now estimated at 5000 (Gierczyk et al. 2018).

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Chapter 9
Conclusions—Learning the Past to Understand the Future of BPF

Abstract Białowieża Primeval Forest (BPF) entered the long nineteenth century having recently lost its royal forest status in 1795. By the end of the period, it was once again deprived but this time of its status as an imperial hunting reserve. Nevertheless, in the meantime, the human-forest relationship evolved and countless scientific and literary publications and works of art devoted to BPF were produced. These were evidence of BPF's rise to prominence as a symbol of the diversity, longevity and persistence of nature. Together with the European bison (regarded as a curiosity and sought-after addition to any natural history museum, university collection, zoological garden or private hunting park), the Forest grew to become a reference point in many scientific debates. BPF survived sufficiently long for its pristine character to be appreciated by key naturalists and environmental activists. A scientific discussion on the "naturalness" of forests began in eighteenth-century France with the definition of terms "natural", "pristine" and "primeval" but as merely hypothetical states. Then the nineteenth century brought the gradual realization that in the European lowlands such a place actually survived. In this chapter, we show how knowledge of the nineteenth-century history of BPF is relevant for current conservation. Indeed, the debates about the future management of BPF, its protected status, and the limits of acceptable human intervention in forest ecosystems, demand better understanding of historic processes.

Keywords Environmental history · Forest history · Traditional use · Forestry · Ecological history · Forest management · Game management · Primeval forest · Conservation · Anthropogenic impact · History of science

Białowieża Primeval Forest entered the long nineteenth century freshly deprived of its royal forest status in 1795. It then ended it deprived again, but this time of its status as an imperial hunting reserve. In the meantime, there were numerous dramatic events, ambitious undertakings and their failures, far-reaching plans and their clashes with reality. But it is a story of the Forest, *pushcha*, a living entity with resilience far beyond the grasp of the nineteenth-century foresters, entrepreneurs and other actors. Thousands of documents from the period are preserved in Russian, Belarussian, Polish, Lithuanian, French, German and even Estonian archives. These remarkable documents depict the evolution of the human-forest relationship, and countless scientific and literary publications and works of art, evidence the rise to prominence of

T. Samojlik et al., *Białowieża Primeval Forest: Nature and Culture in the Nineteenth Century*, Environmental History 11,
https://doi.org/10.1007/978-3-030-33479-6_9

BPF as a symbol of the diversity, longevity and persistence of nature. Together with the European bison as a curiosity and sought-after addition to any natural history museum, university collection, zoological garden or private hunting park, the Forest became a reference point in many scientific debates. These discussions were orbiting around topics of naturalness, human impact on nature, and the chances of saving certain species or entire areas into the future.

The period of the history of BPF contained in this book was undoubtedly crucial in terms of the outcomes for the region. It is easy to envisage a scenario in which BPF became just another managed forest, focused on maximizing profits in foreign markets and turned into forestry plantation. In these conditions, European bison would have met its demise without any chance of species recovery and BPF would have been demoted to the status of substandard woodland with ill-advised game management leading to the collapse of key aspects of the ecosystem. Yet BPF survived long enough for its pristine character to be appreciated by a number of key naturalists and environmental activists, as predicted by Józef Paczoski (Paczoski 1896, 1927a, b), so as to become a unique area of scientific research in various fields of natural sciences.

A scientific discussion on the "naturalness" of forests began in eighteenth-century France with the definition of terms "natural", "pristine" and "primeval" but merely as hypothetical states (Daszkiewicz and Samojlik 2018). Then the nineteenth century brought the realization that European lowlands actually still had such a place. From the point of view of history of science it is interesting to observe how the view on the Forest has changed with the development of natural sciences. Originally considered impenetrable forest and the "mainstay of European bison", BPF became the subject of essential research. This work allowed understanding of the features a forest should present in order to be considered natural, pristine, or primeval; it also contributed to twentieth-century concepts of nature protection. The development of the latter emerged from a simplistic approach of the protection of royal or imperial property, to a complex system integrating conservation of forest habitats with natural cycles, disturbances, and ecological successions.

Although the story presented in our book ends more-or-less a century ago, the knowledge of the nineteenth-century history of BPF is relevant for current conservation. The on-going debate concerning the future management of BPF, its protected status, and the limits of acceptable human intervention in forest ecosystems demand better understanding of historic processes. There are often historical arguments to show how deep the anthropogenic impact on the Forest was and how much "naturalness" has been lost during the nineteenth and early twentieth centuries. We believe this book will help inform stakeholders in this dispute through awareness of objective, document-based history of human presence and intervention in BPF in this period. The story also illustrates how long and complicated was the process of recognizing BPF as the best-preserved forest of the European lowlands—the same process that current forestry management aims to scupper, reducing the role of BPF to a production forest (Wesołowski et al. 2016, 2018; Żmihorski et al. 2018).

Analysis of historical documents, publications, maps and works of art concerning BPF in the long nineteenth century has led to key conclusions. Some of these are obvious and even trivial, but others are unexpected and possibly beneficial for the contemporary conservation of the Forest:

(1) Initially (since the very early nineteenth century), decisions on conservation of BPF were motivated by the presence of European bison in the Forest and it became a leitmotif for the entire imperial period (1795–1915). This shaped the authorities' approach to any attempts at more intensive exploitation of forest resources.
(2) In the long nineteenth century, dense pristine woods were considered to be the indigenous habitat of European bison. This is in contrast to the current state of the knowledge that forests are the last refuge of the bison, and the most suitable ecosystems for bison were forest-steppe open areas.
(3) Against this background of the recognized need for bison protection were frequent changes in the ideas for best management. These varied from the idea of preserving the "pristine" dense forest while at the same time maximally limiting the access of people to the forest. These were deemed to be key requirements necessary for the survival of a bison. Then there was the idea that that commercial, "rational" logging under the control of reliable personnel could not harm European bison, and such "rational" logging would be beneficial to the Forest itself. Fortunately for the survival of the Forest, periods of commercial logging were relatively short, and the logging not too intense.
(4) In the entire long nineteenth century, almost the entire BPF regenerated naturally—and forestry specialists kept repeating that there was little need to artificially improve this. This was because natural forest regenerates in a natural way and needs human intervention only to a small extent (mainly to repair the effects of erroneous forest management or too intense grazing pressure from wild or domestic ungulates in the last appanage period 1888–1915).
(5) Cattle pasturing in BPF was limited to areas adjacent to forest villages, and these areas under cattle pasture were subject to frequent changes by forestry administration. Such changes were used by the administration as a disciplinary approach for villages suspected of poaching and other misdemeanours in the Forest.
(6) In the nineteenth century and into the early twentieth century, the forestry administration noted several outbreaks of nun moth, bark beetle, and other problem insects. These were considered to be harmful to Norway spruce (though this was not the most highly valued tree species of the period). Attempts at fighting outbreaks turned out to be time-consuming, costly and had little or no effect. In the archival documentation, there was no evidence of long-term negative consequences of insect outbreaks for forest regeneration.
(7) Forestry specialists of the long nineteenth century considered BPF interesting not only as the last habitat of European bison, but as an example of a pristine forest. As soon as the special governmental agency to manage state forests was created in the Russian Empire (1843), BPF became one of the first areas where a forest inventory team was sent. One of the unique features of BPF observed in this period was a high number of culturally modified trees connected with

traditional, non-timber utilization of forest by local communities. The presence of such trees, considered "damaged", to some extent hindered attempts at the quick introduction of modern forest management in BPF.

(8) In addition to winter supplementary feeding of European bison and prohibition of unauthorized hunting (both introduced under the Polish administration in the royal period), the Russian administration of BPF has introduced other measures of "rational" European bison management: eradication of all predators, confinement of bison to the forested area of BPF, and creating artificial meadows inside the Forest to improve feeding conditions.

(9) European bison were counted annually during the period 1809–1915, however there are doubts about the reliability of monitoring methods before the mid-1890s, when three parallel ways of counting were introduced. The reliable growth of bison numbers was observed at the 1890s, with an increased budget for game management.

(10) While the first scientific reports on European bison and the species' behaviour and food preferences were published in the late eighteenth (Gilibert 1781a, b), and early nineteenth century (Bojanus 1825; Górski 1829), the systematic study of bison and BPF's biodiversity was started only in the late nineteenth to early twentieth century. Expeditions by botanists Karol Drymmer, Franciszek Ksawery Błoński, and Antoni Ejsmond in the late 1880s, Józef Paczoski in the late 1890s, and a European bison expedition under the leadership of Professor Nikolai Kulagin between 1906 and 1908, etc., brought significant development in the understanding of unique features of BPF. The latter expedition, was the first veterinary study on European bison, and documented the significant rates of infection of European bison by parasites, connected with high mortality in late winter-early spring. Nowadays, parasite infestations are noted as one of the main threats to the bison population in BPF (Krasińska and Krasiński 2013).

(11) After the first Imperial hunt of 1860, the forestry administration of BPF gradually introduced measures of "rational" game management: reintroduction of red deer, introduction of fallow deer, and the "refreshment of blood" of BPF's roe deer. After 1888, significant resources were allocated annually to enriching the hunting potential of the Forest. Animals were brought to BPF from different regions of Europe, supplementary feeding expanded both in the period covered and quantity of fodder; and predators were thoroughly exterminated. (In the case of the brown bear this policy resulted in total eradication of the species in 1870s). The increase in game numbers led to concerns about natural regeneration of the Forest and overall wellbeing of European bison in the early twentieth century.

(12) During the entire long nineteenth century, BPF administration employed increasingly intensive measures to limit traditional forest use by local peasant communities. These activities were cattle pasturing in the forest, the chopping resinous pieces of wood from the pine tree trunks, traditional beekeeping and the stripping lime bark, etc. Most of these forest uses were considered not only outdated but also harmful for "rational" forestry, aimed at maximizing the production of timber. Despite that, peasants successfully resisted these directives.

They protested, sabotaged, or directly violated the orders of the local administration, allowing for preservation of some of traditional forest practices well into the twentieth century.

(13) The Forest, its natural features (like the presence of European bison, amount of dead wood, diversity of animal and plant species, lack of visible human intervention in the majority of the area), and local inhabitants served as sources of inspiration for numerous works of art in the long nineteenth century. Depicted in paintings, drawings, sketches and photographs, described in memoirs, novels, popular articles, and other literary works, BPF became widely known as the last pristine forest of lowland Europe.

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